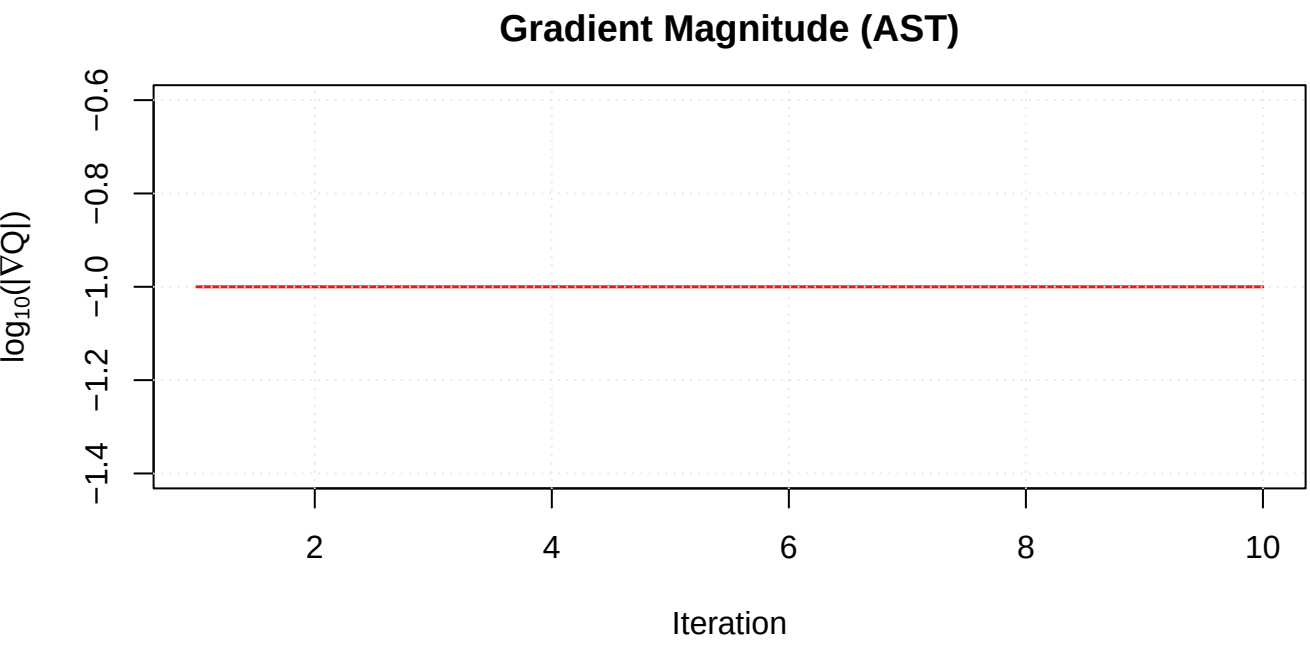
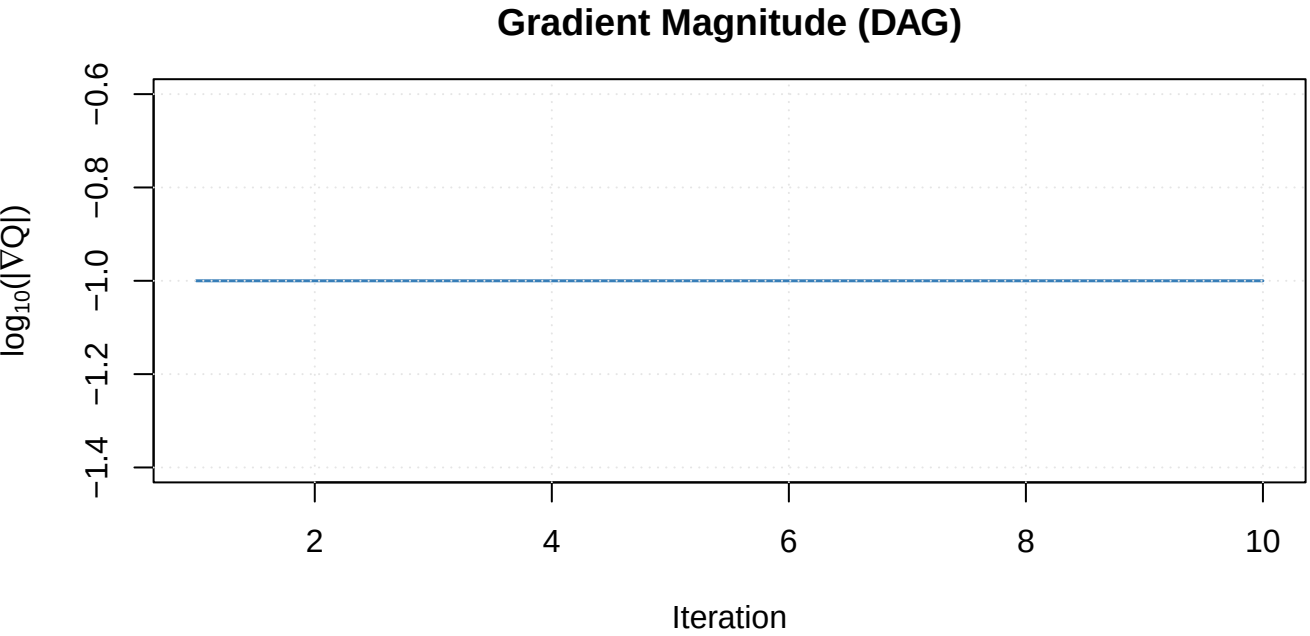
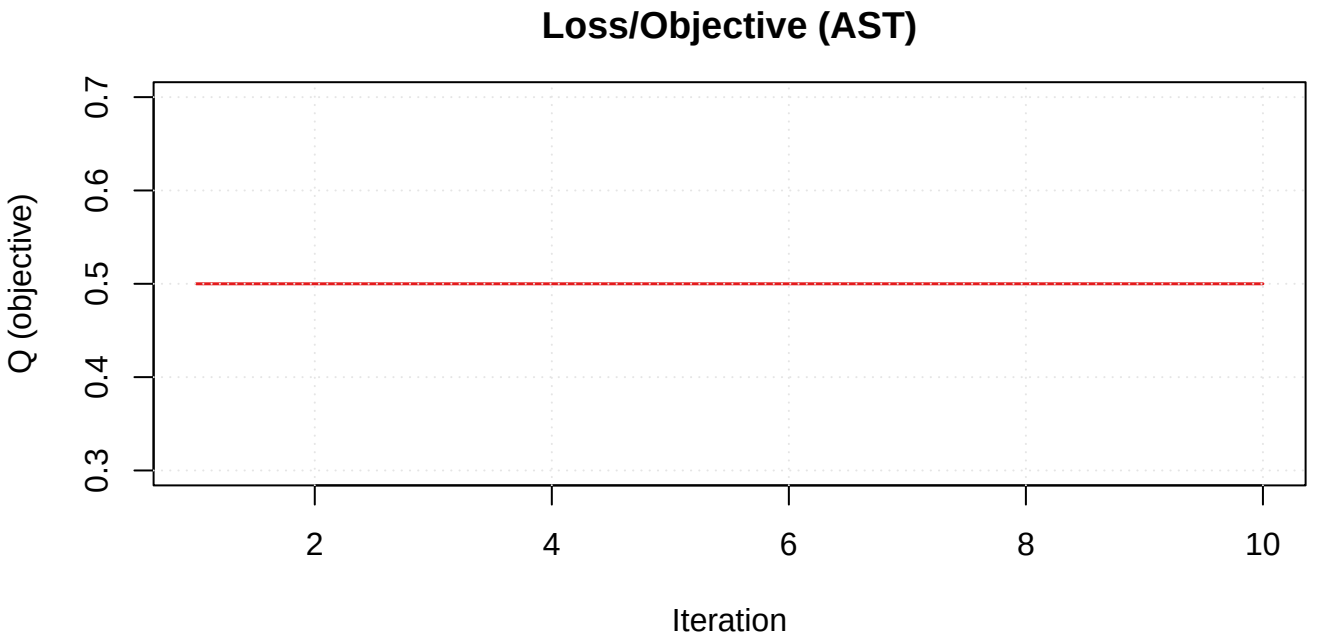
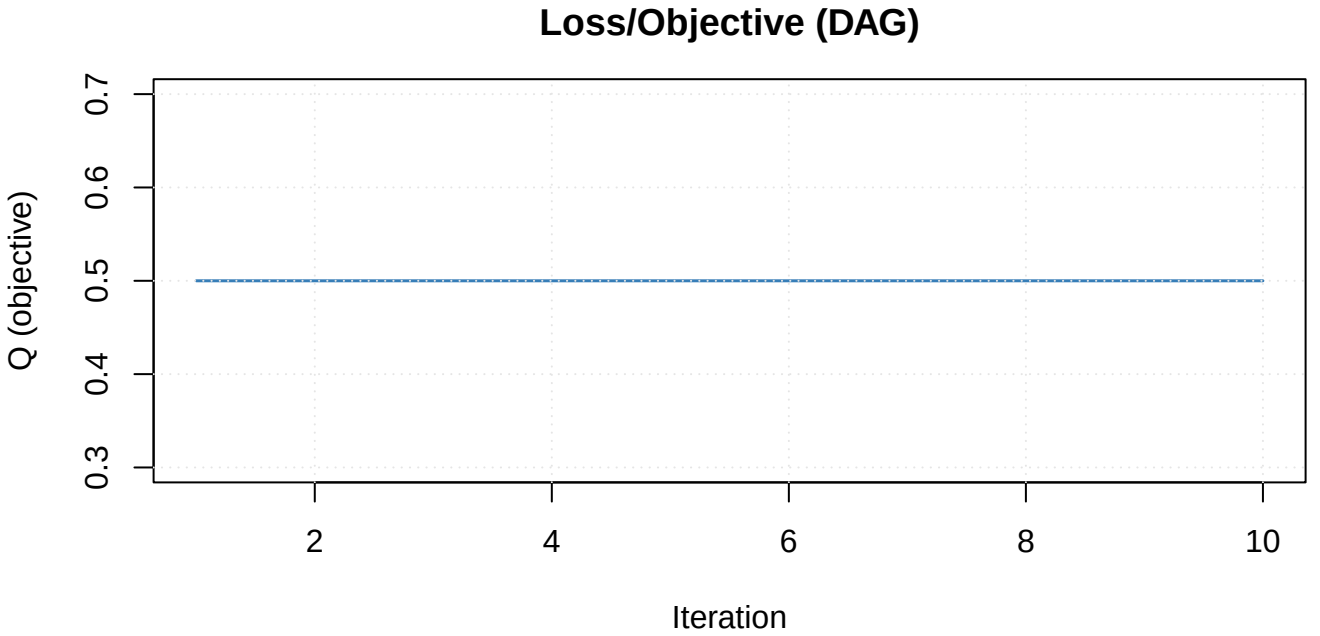
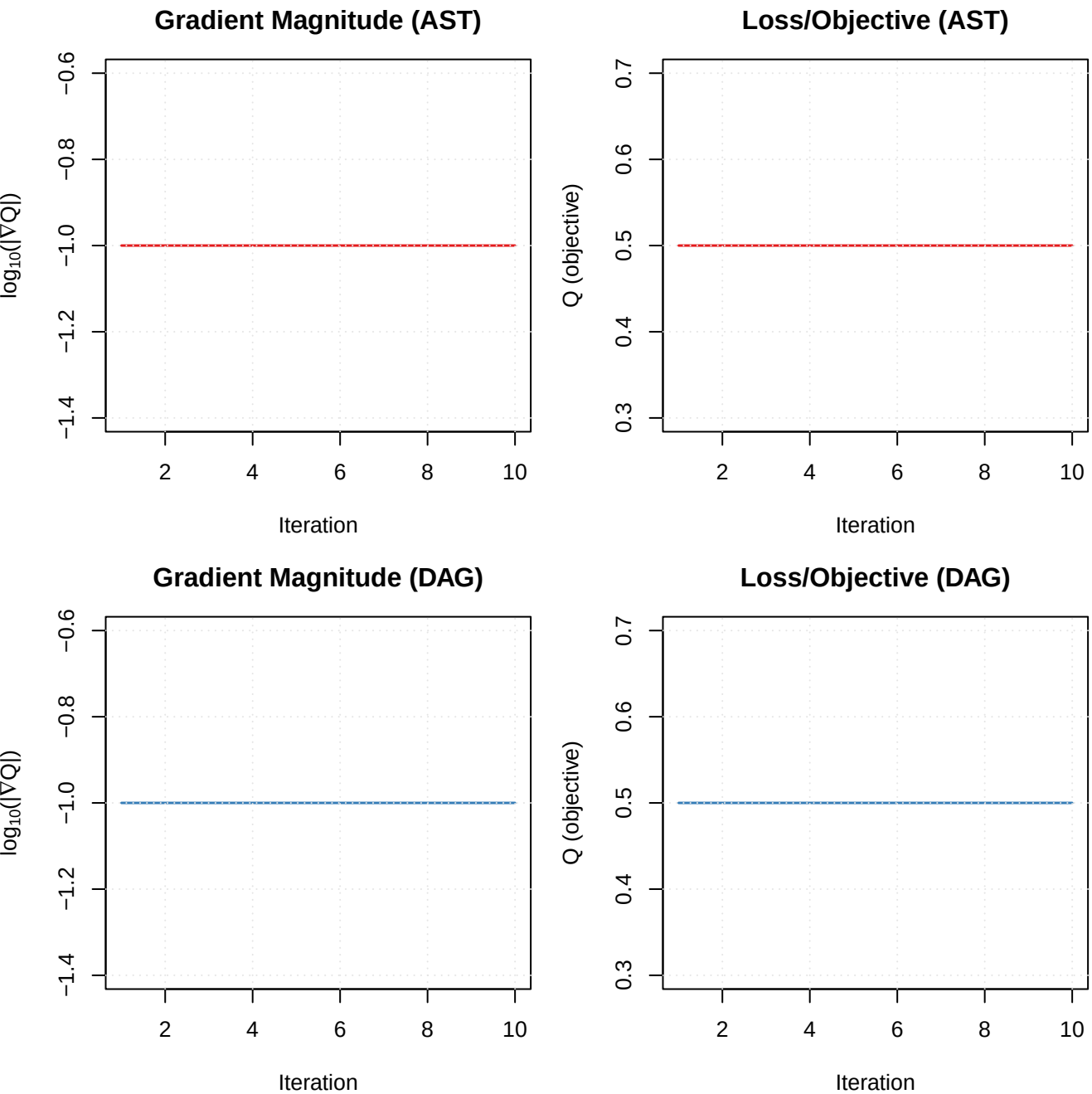
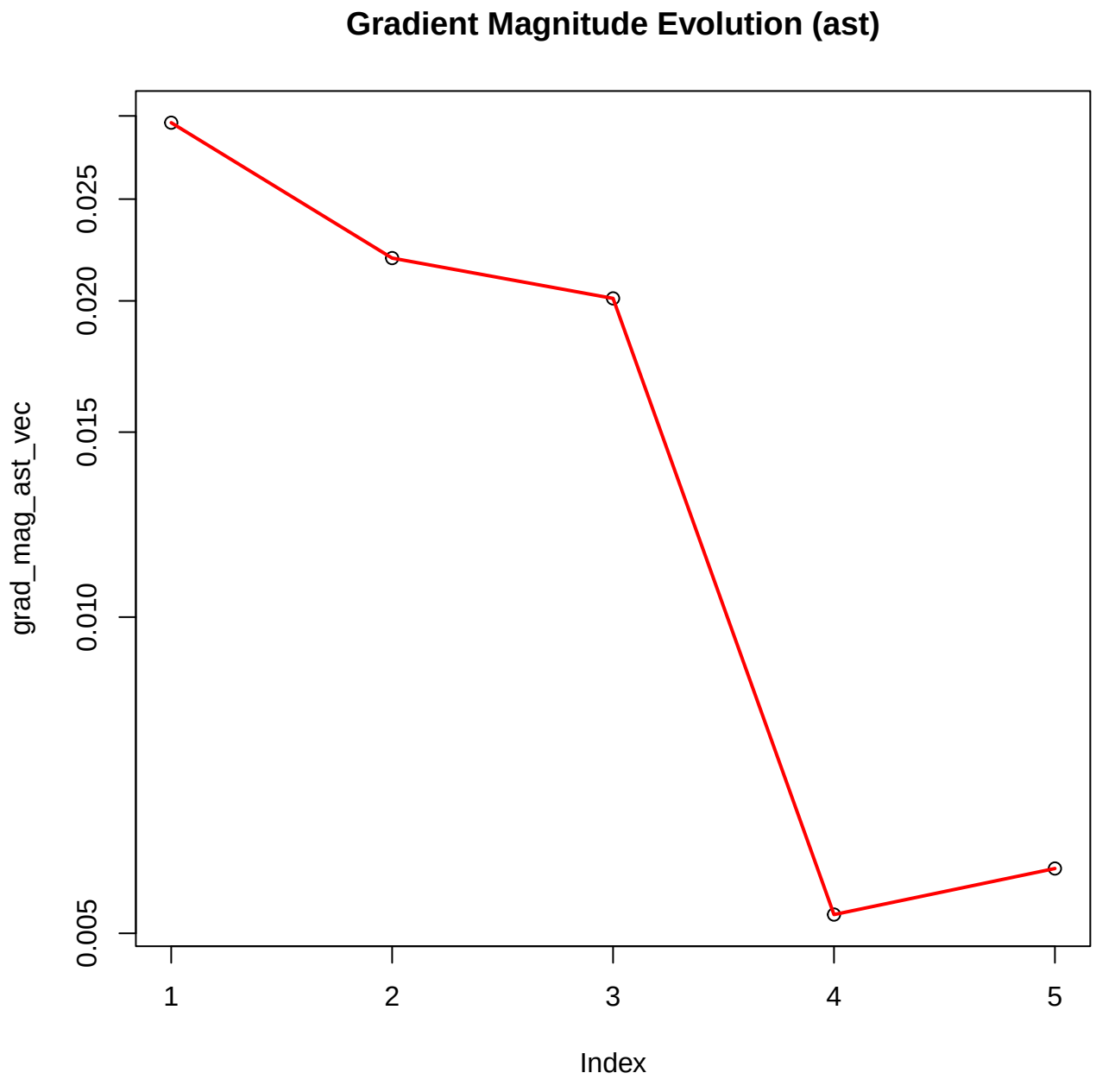

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

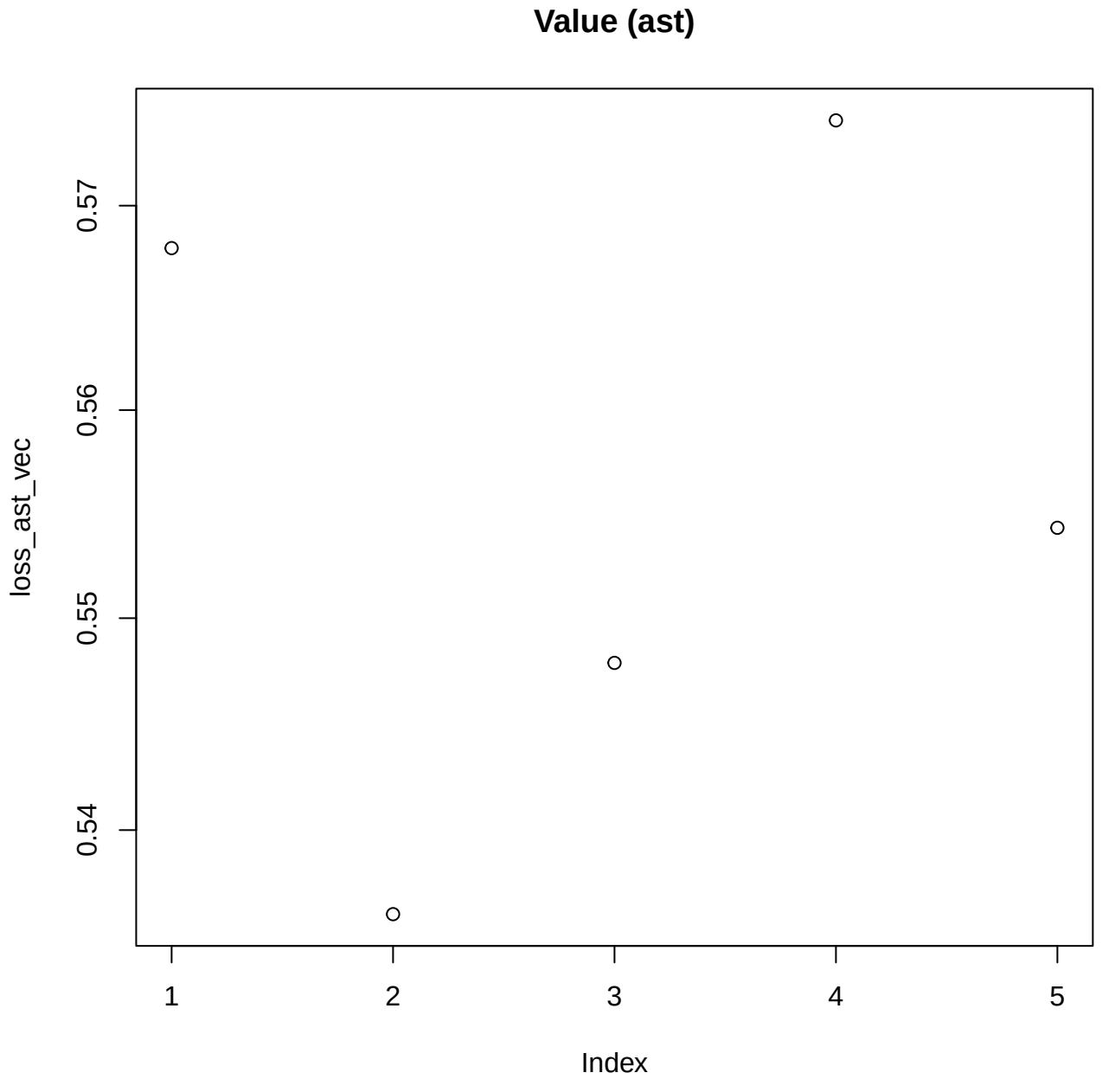

Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

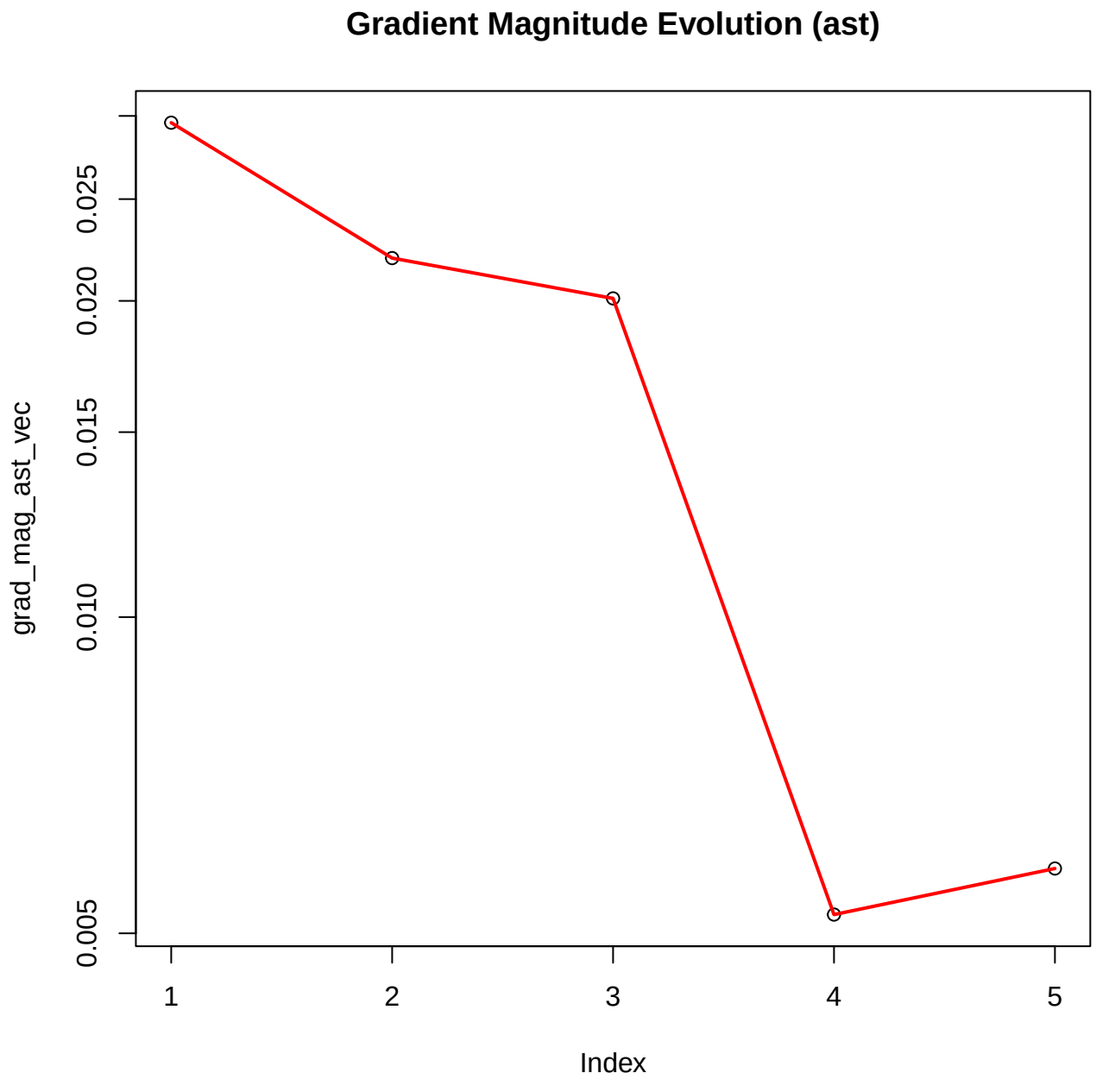

Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

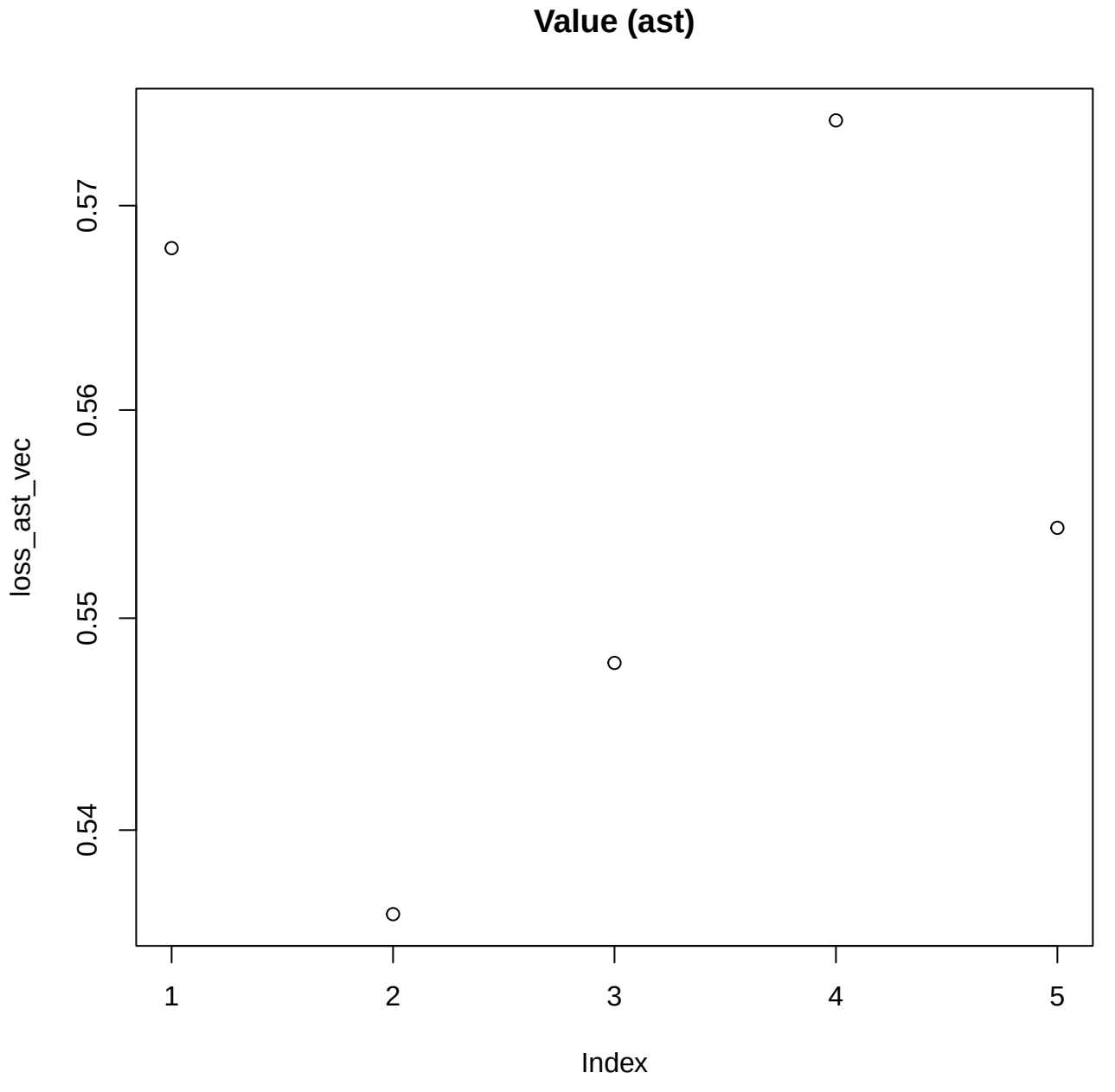

Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

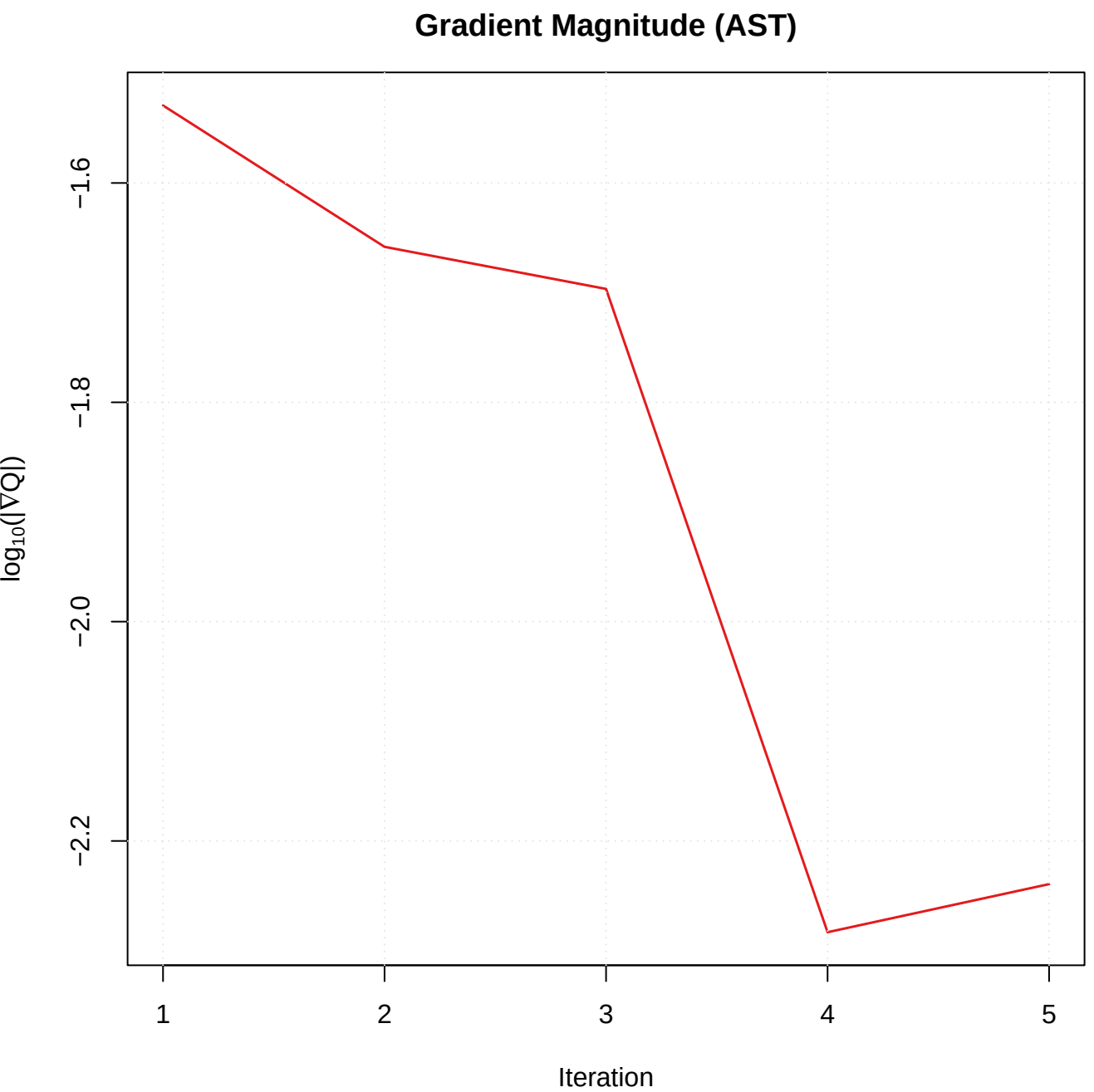

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration

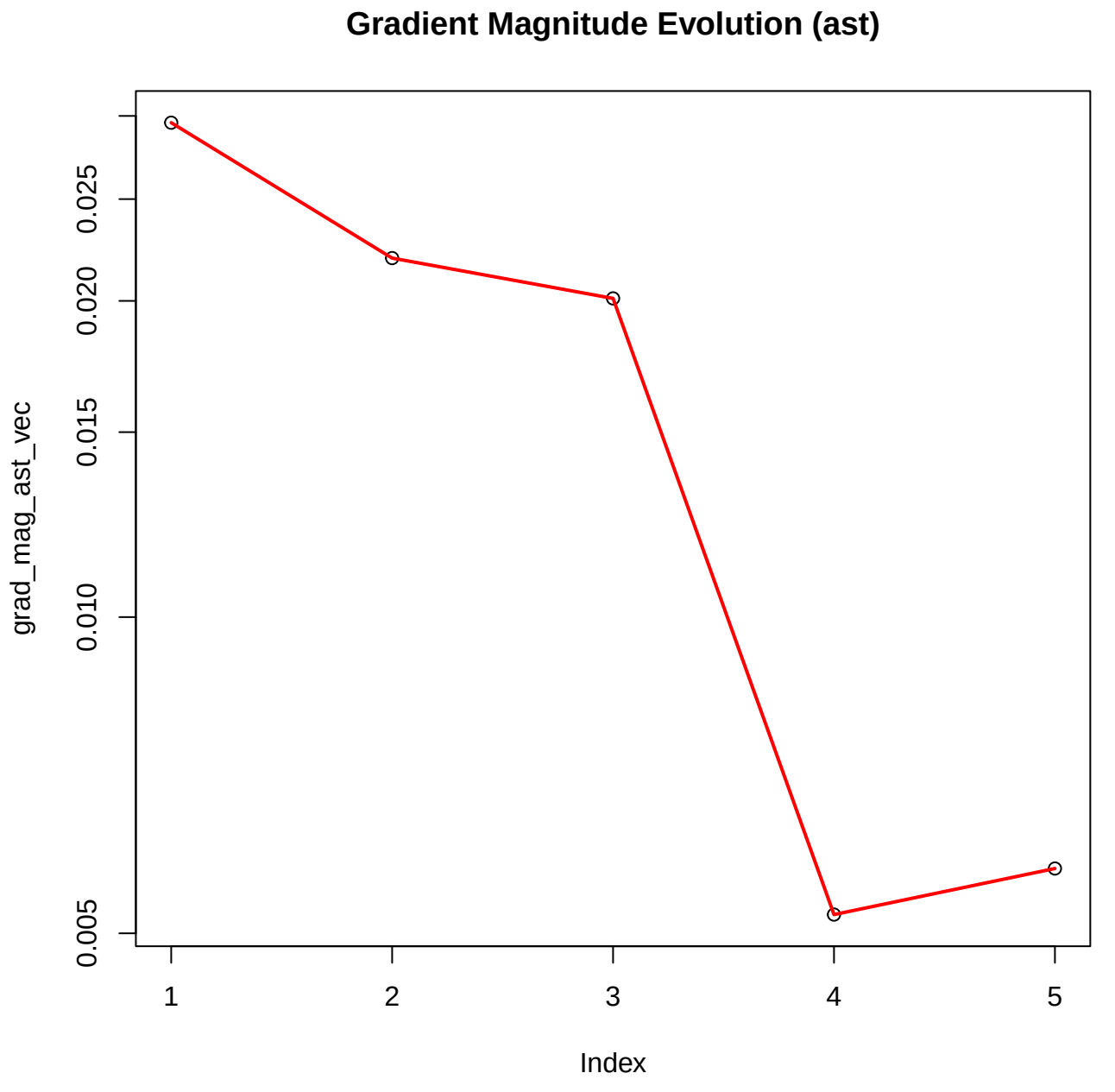

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

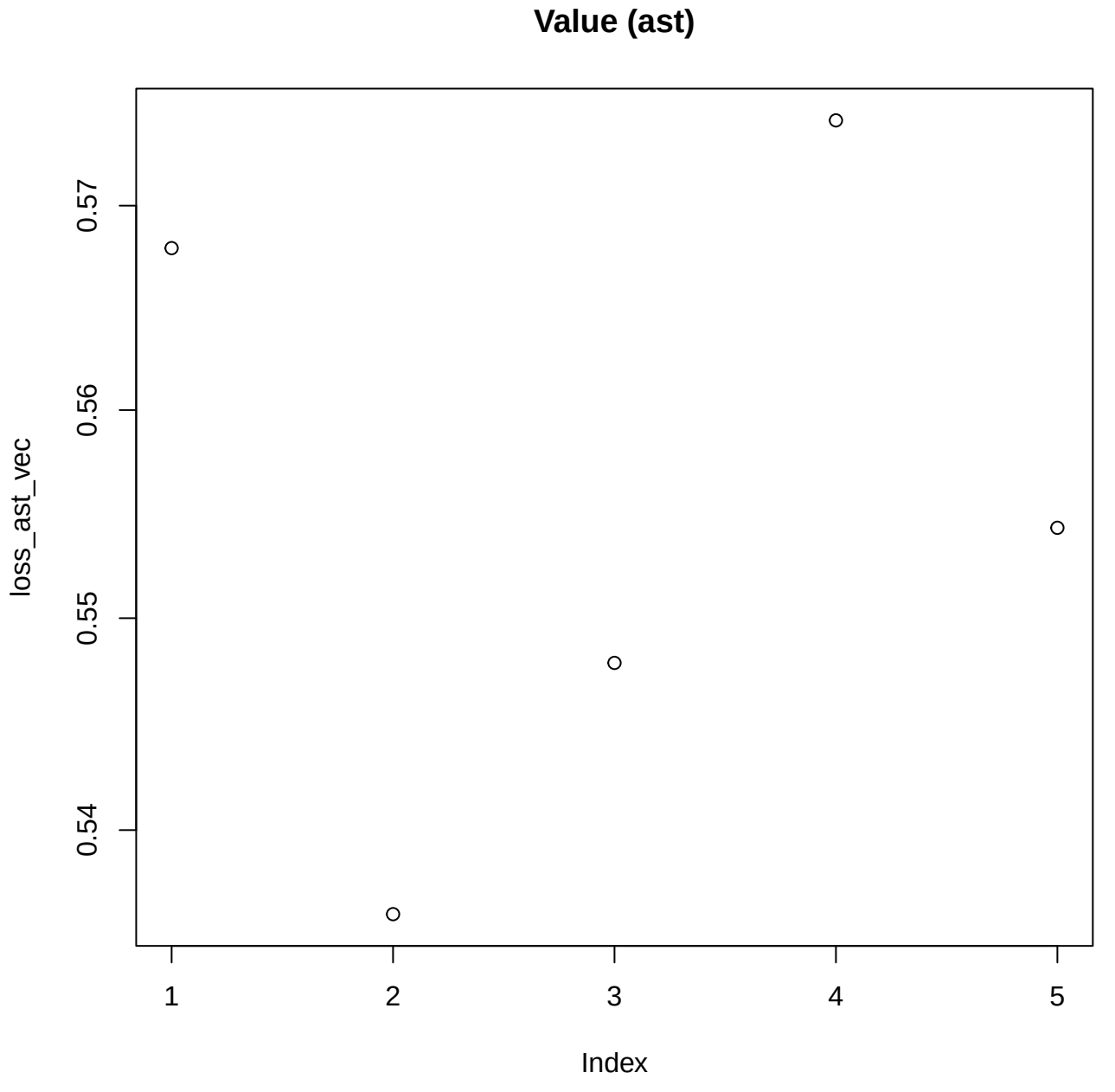

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

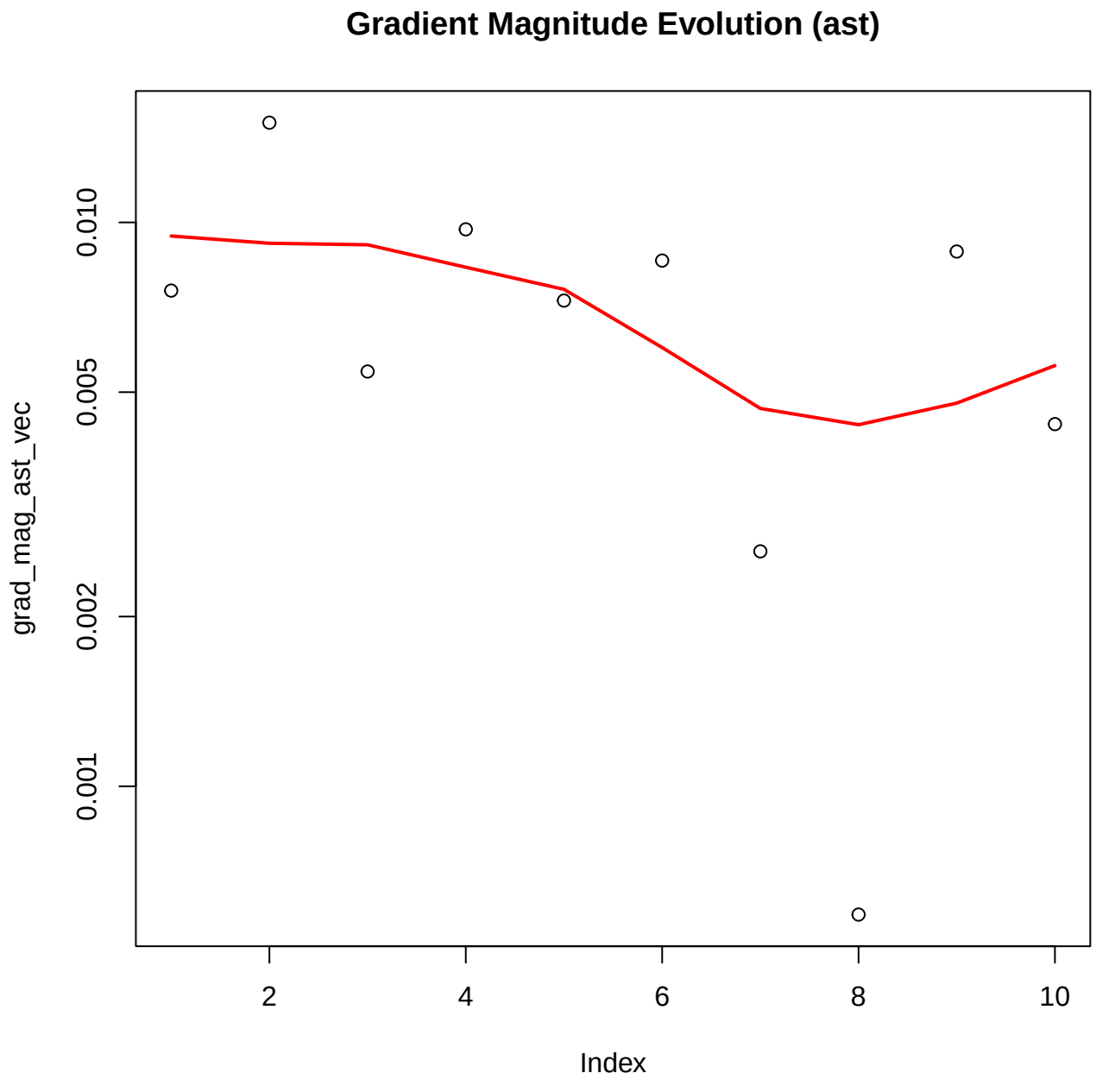

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

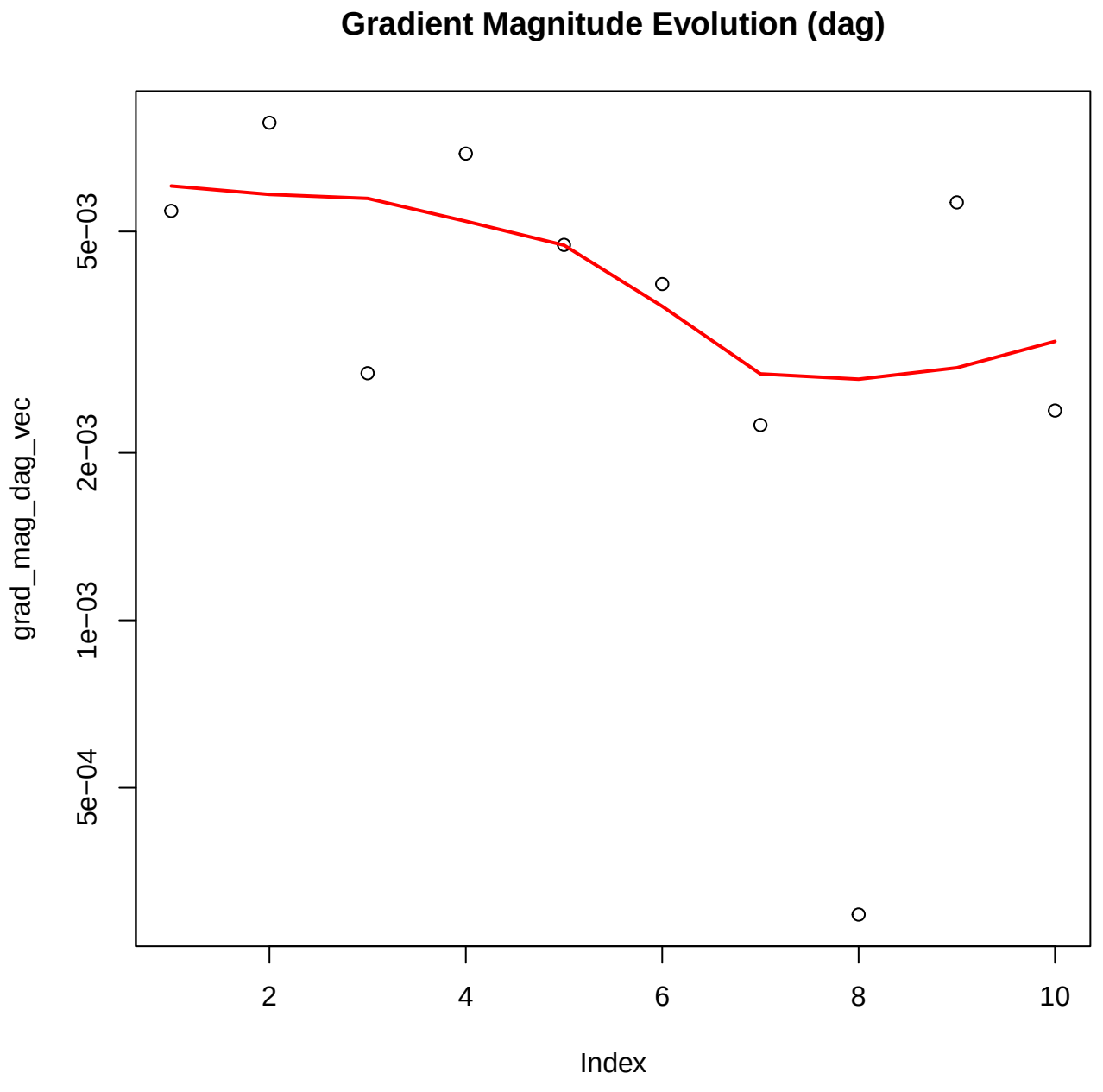

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

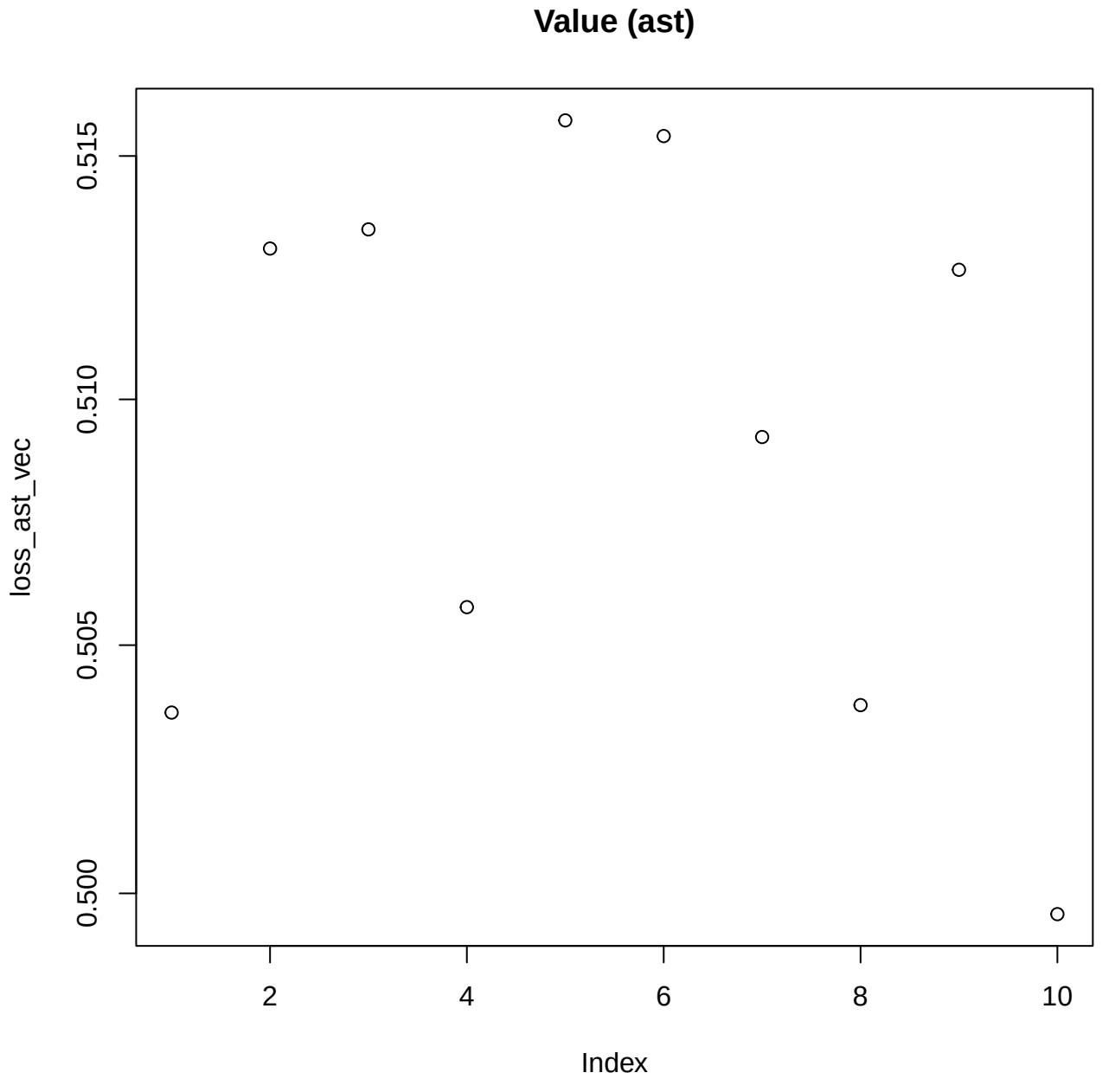

Gradient Magnitude (AST)
−1.6
−1.8
−2.0
−2.2
log10(|∇Q|)
1
2
3
4
5
Iteration

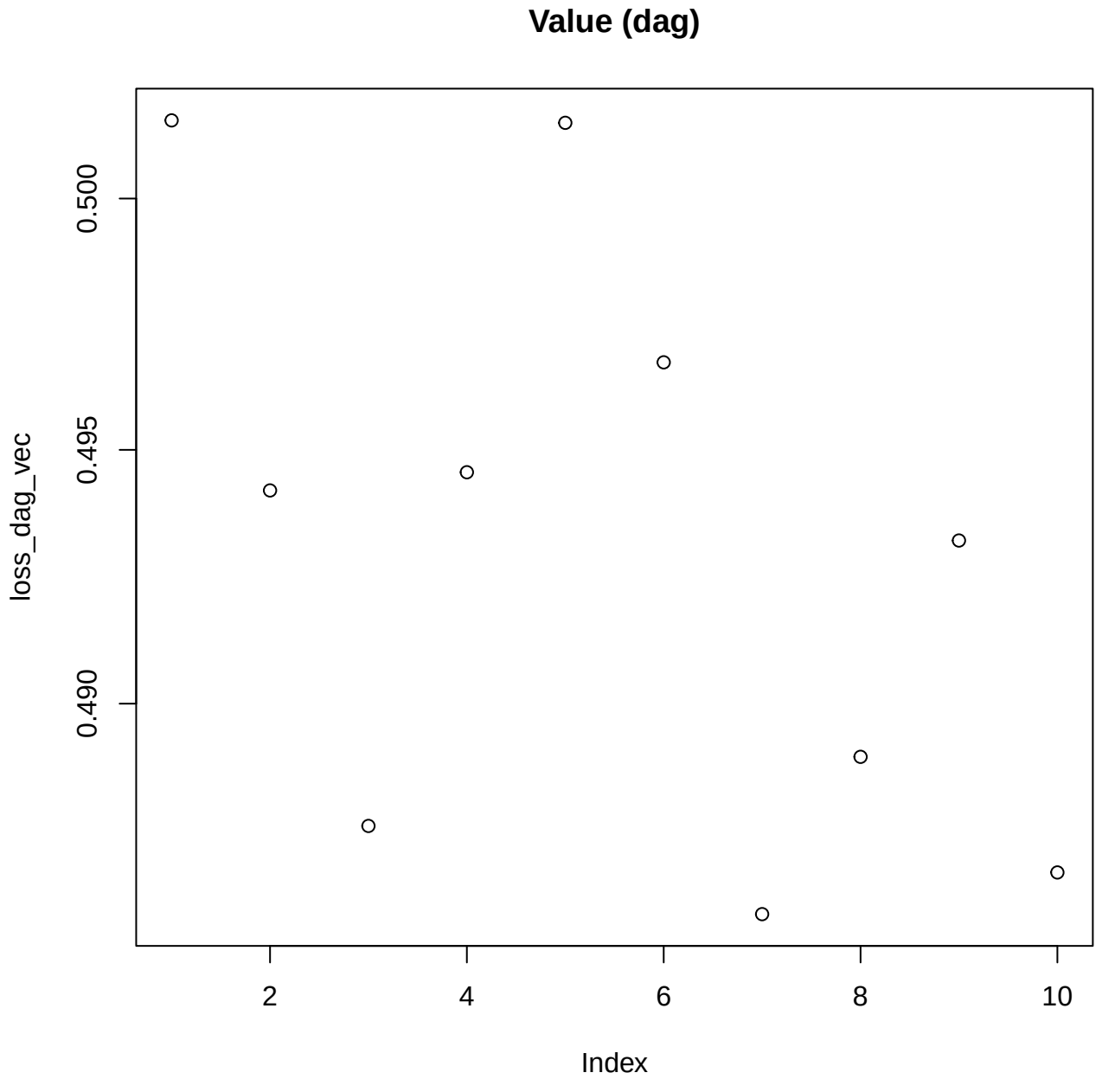

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

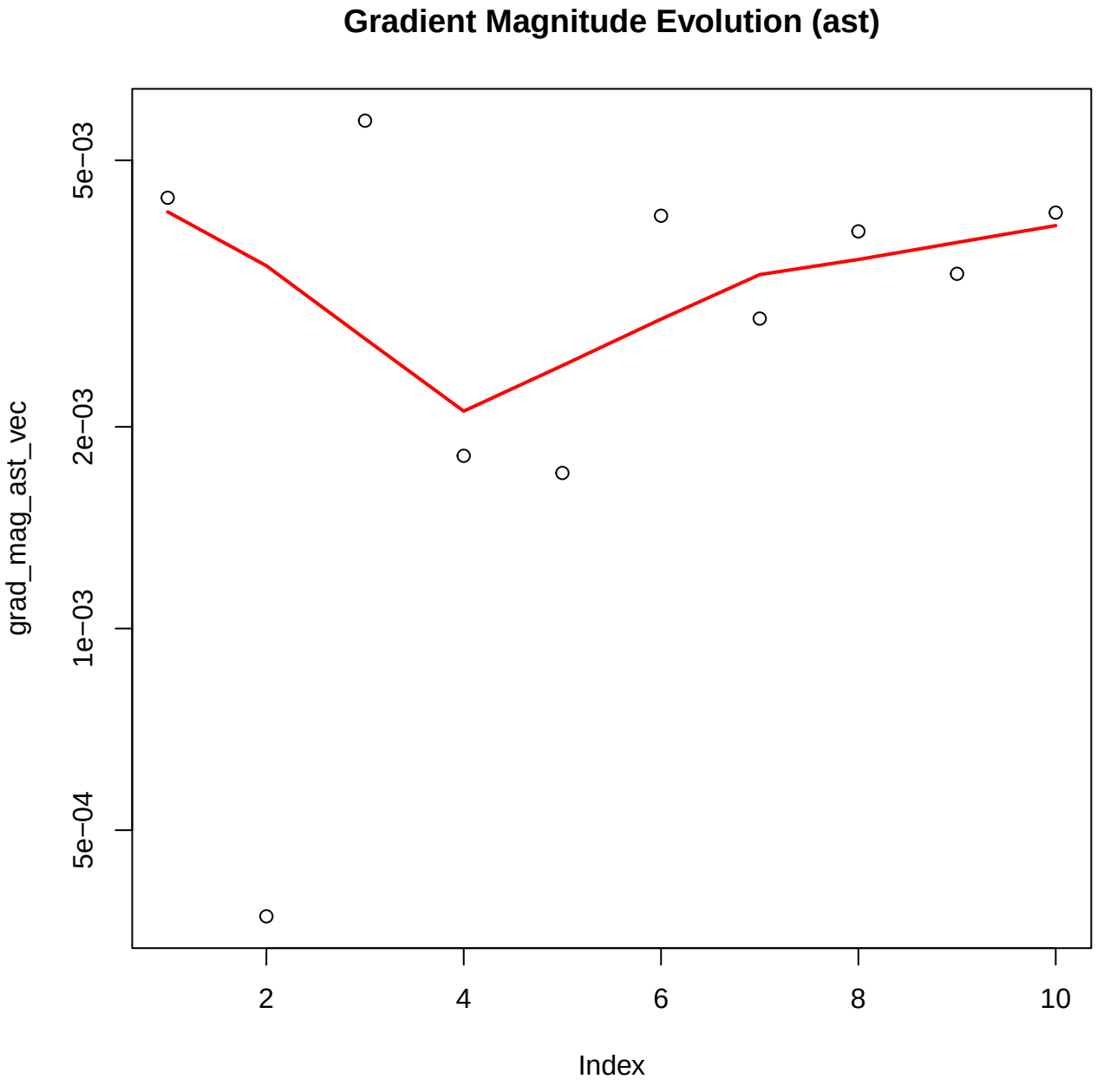

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

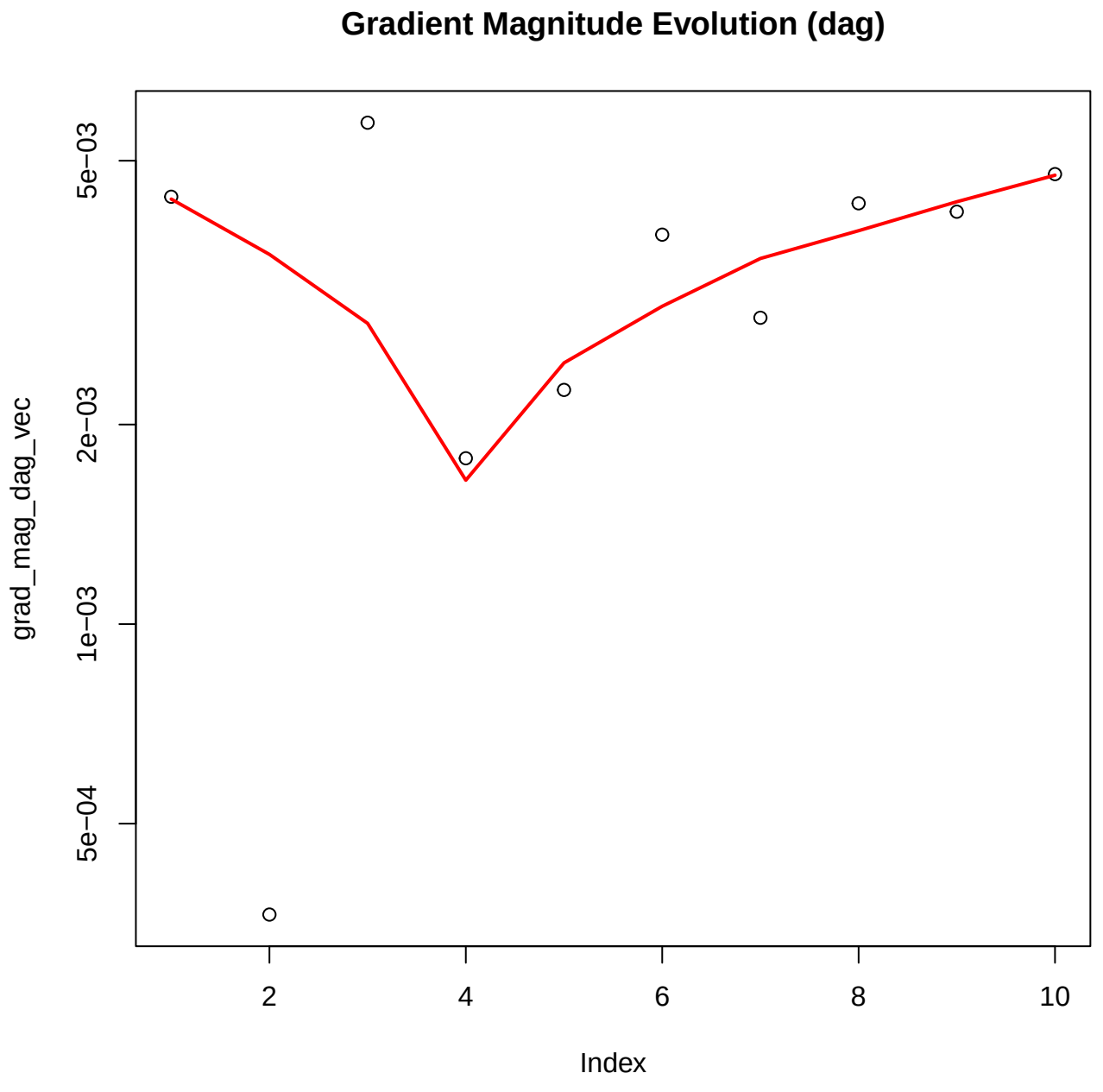

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.001
0.002
0.005
0.010
2
4
6
8
10
Index

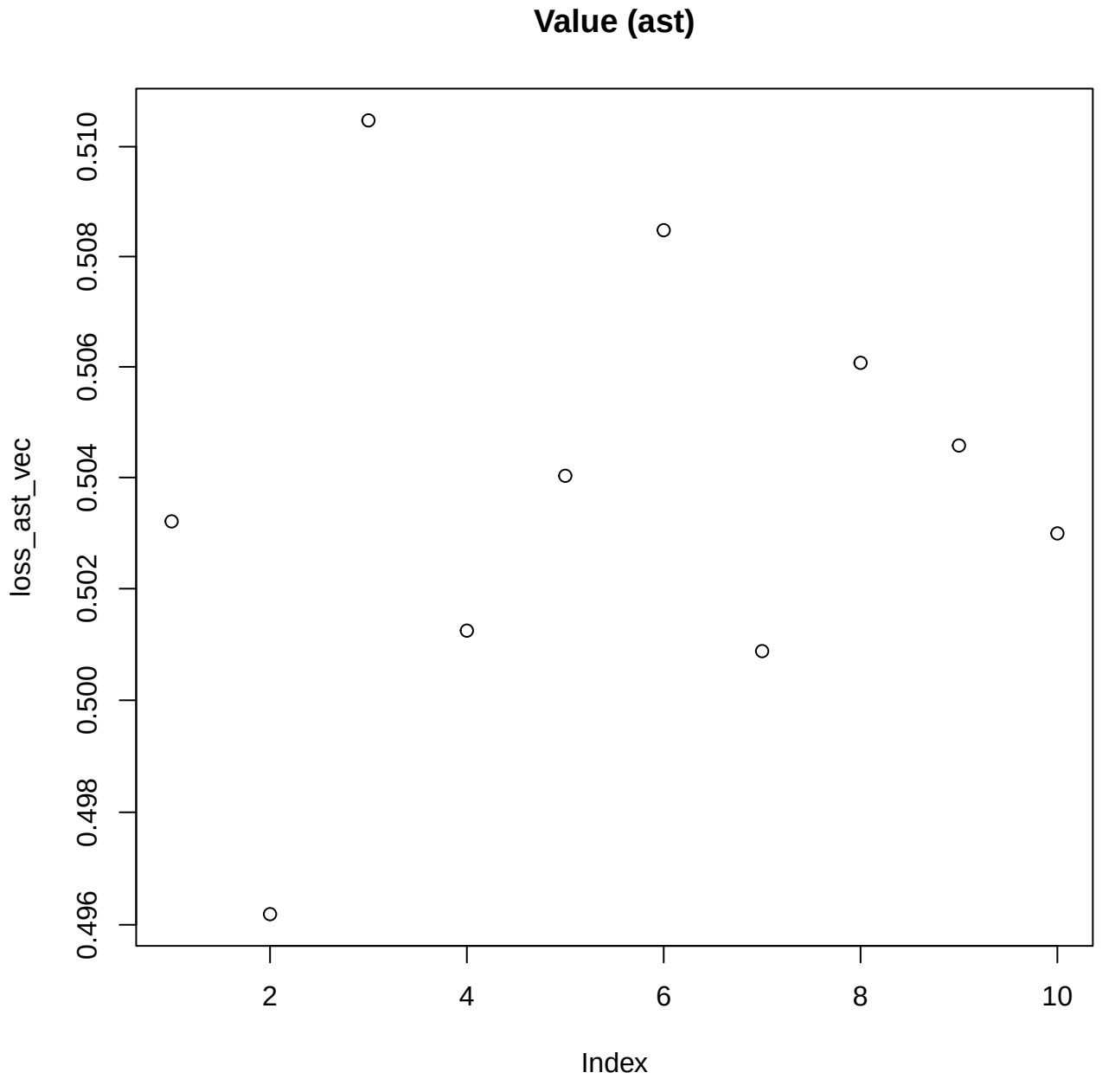

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

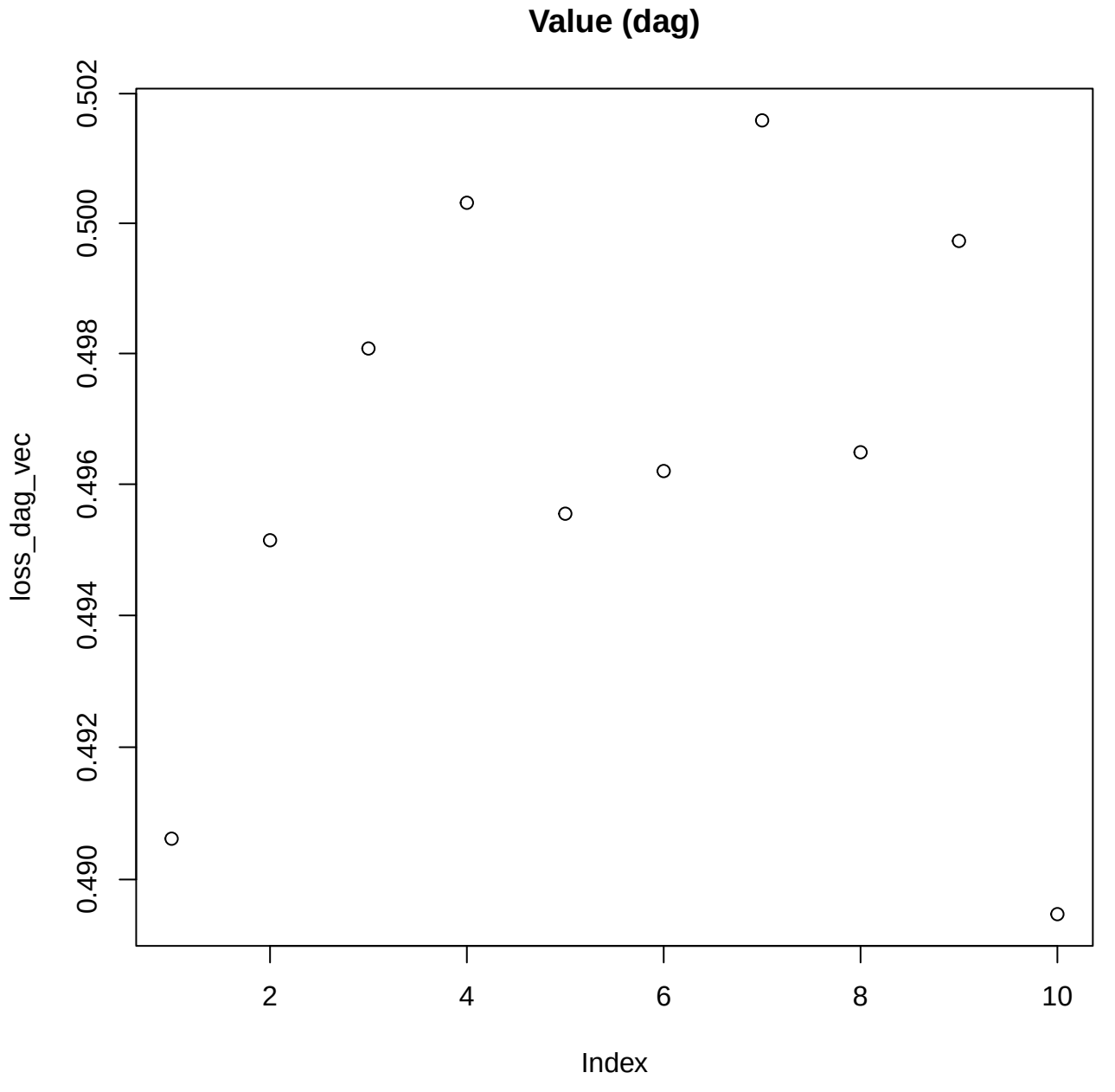

Value (ast)
loss_ast_vec
0.500
0.505
0.510
0.515
2
4
6
8
10
Index

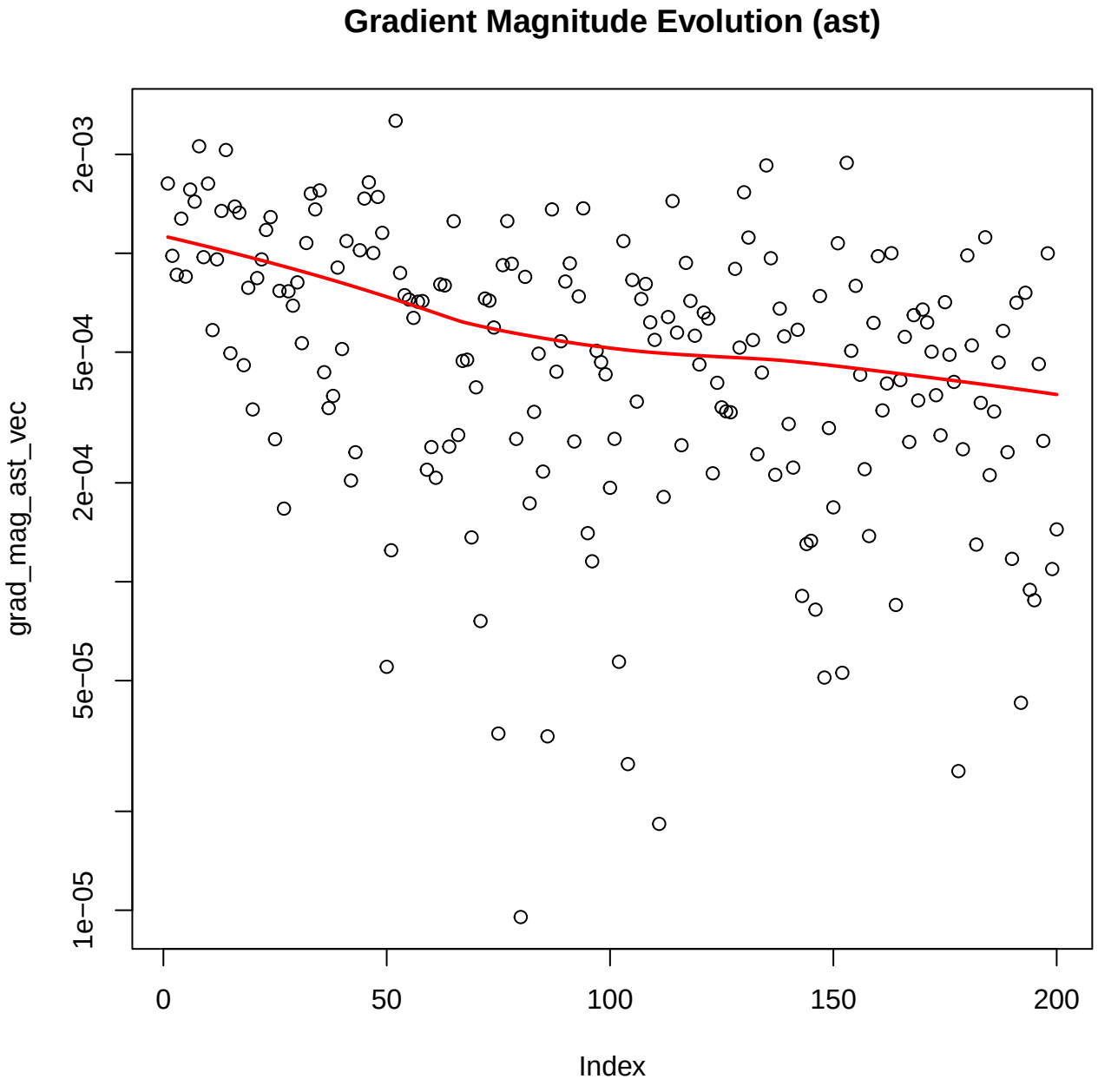

Value (dag)
loss_dag_vec
0.490
0.495
0.500
2
4
6
8
10
Index

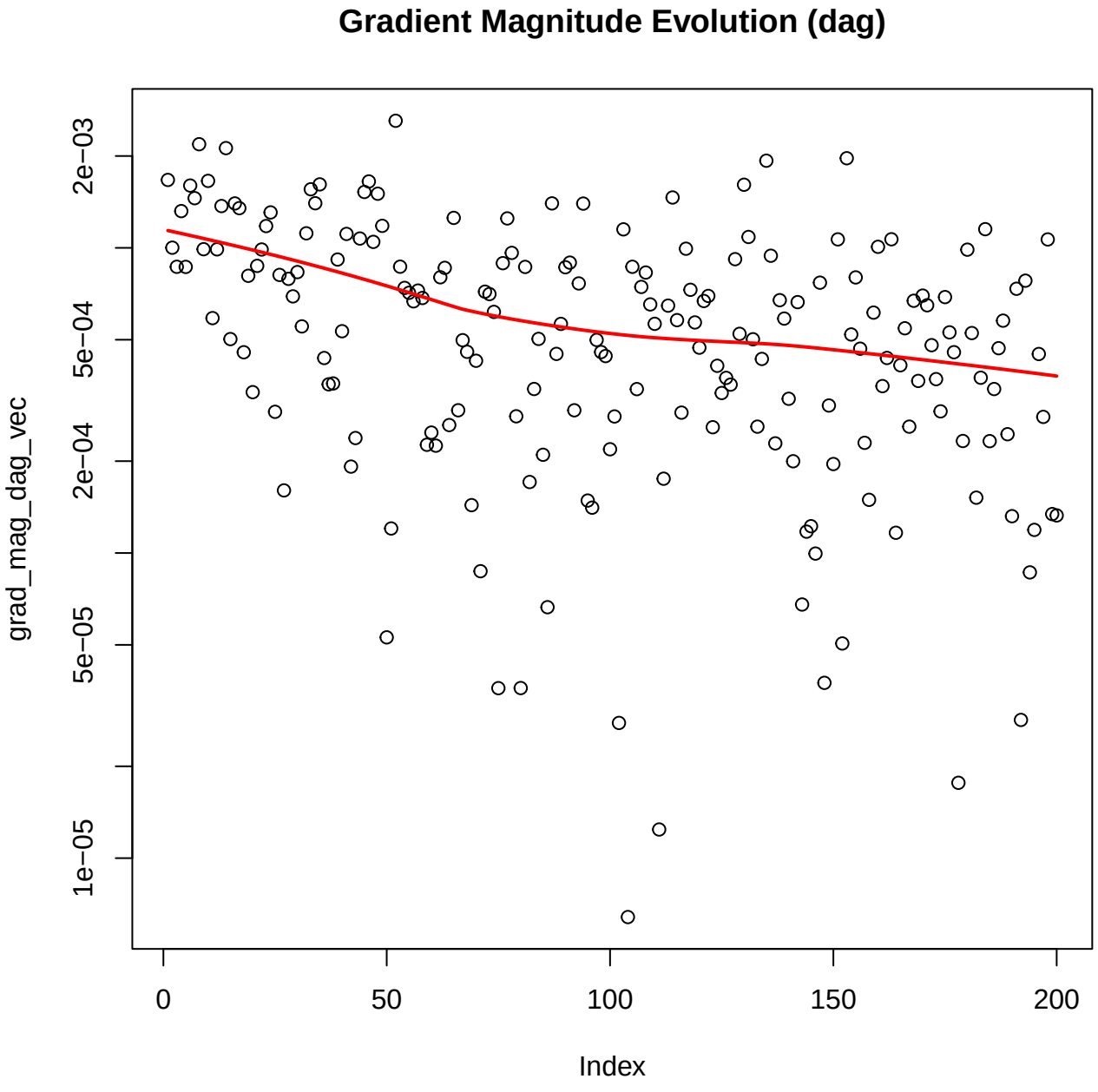

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

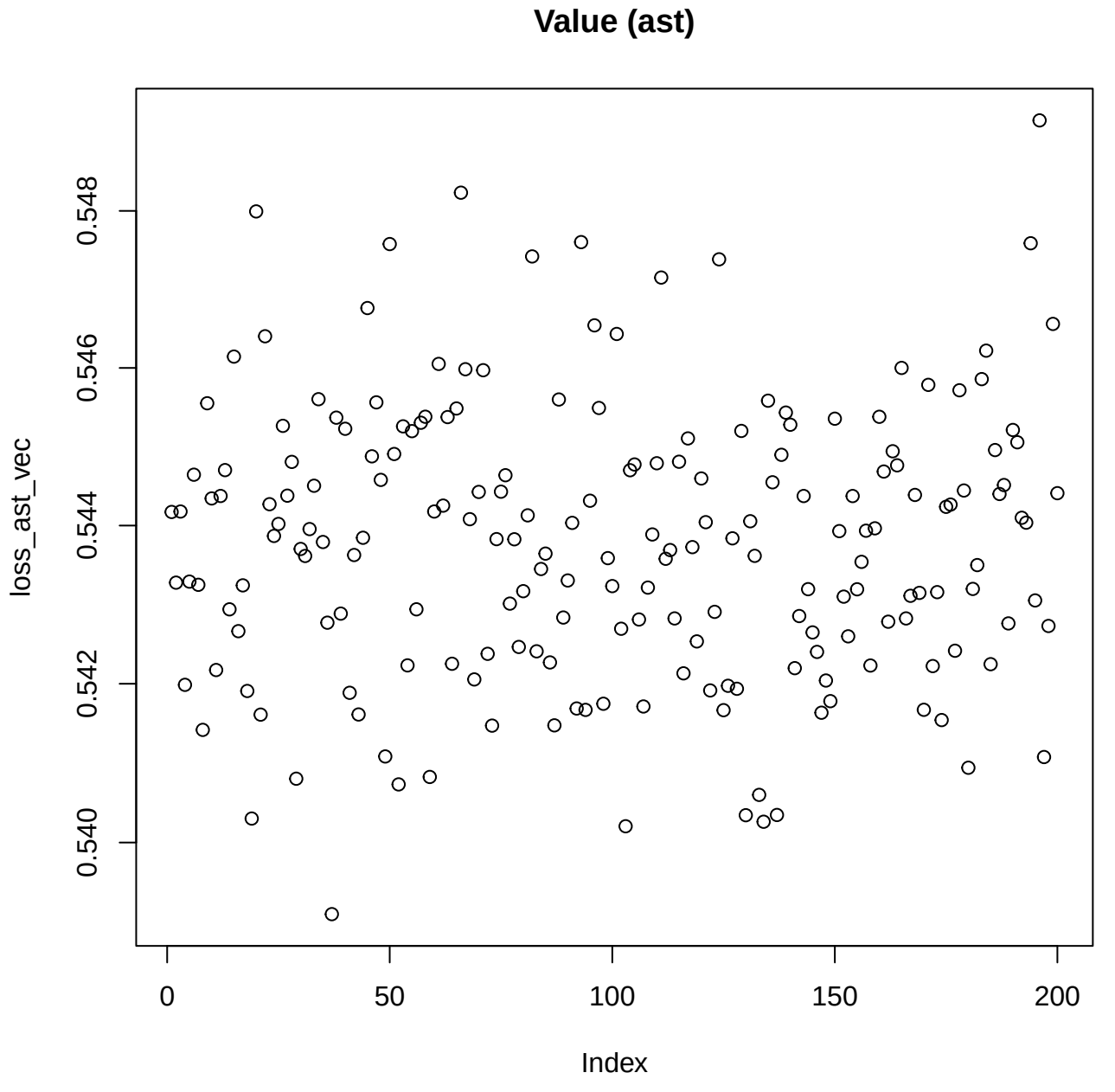

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

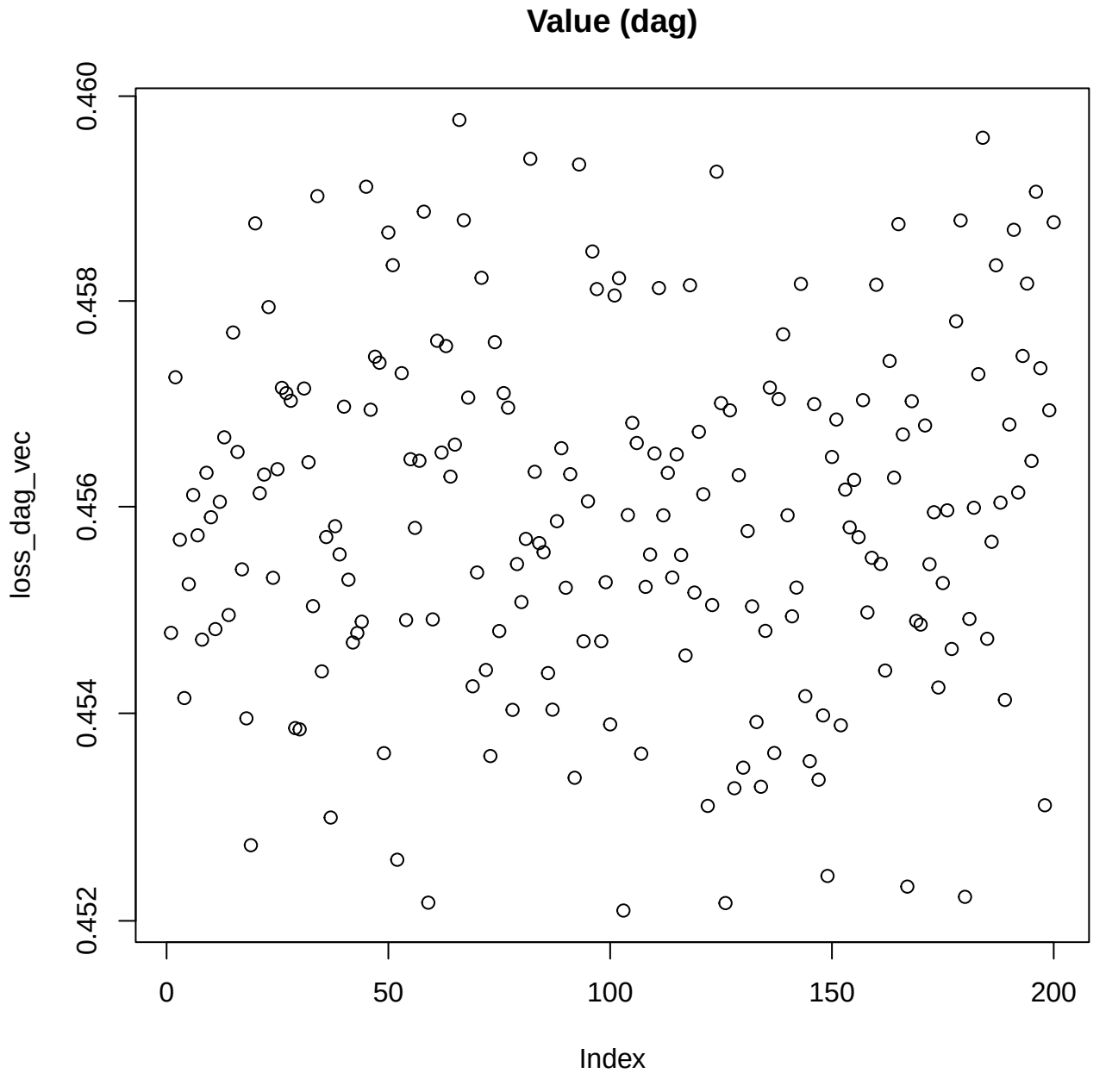

Value (ast)
loss_ast_vec
0.496
0.498
0.500
0.502
0.504
0.506
0.508
0.510
2
4
6
8
10
Index

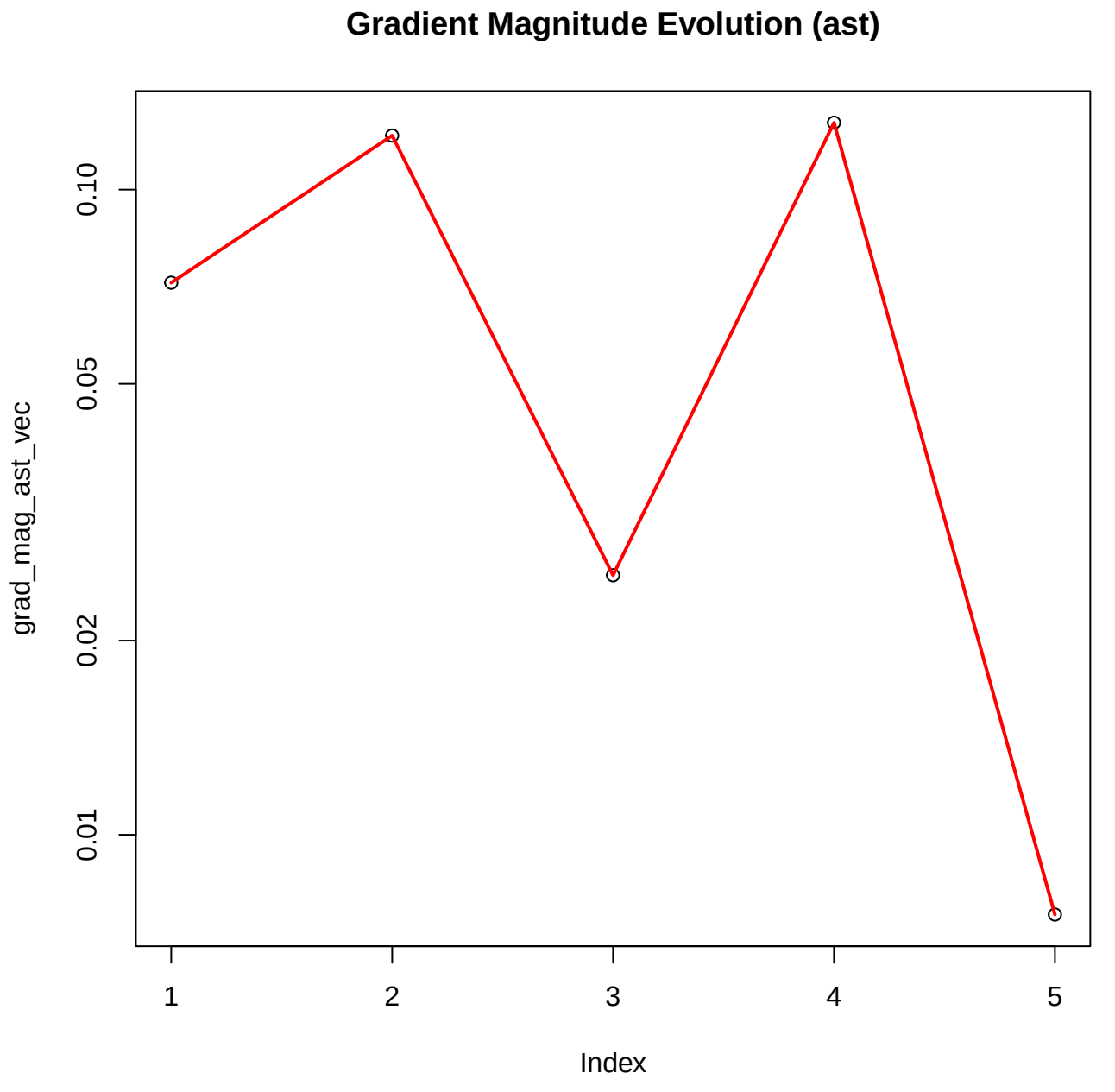

Value (dag)
loss_dag_vec
0.490
0.492
0.494
0.496
0.498
0.500
0.502
2
4
6
8
10
Index

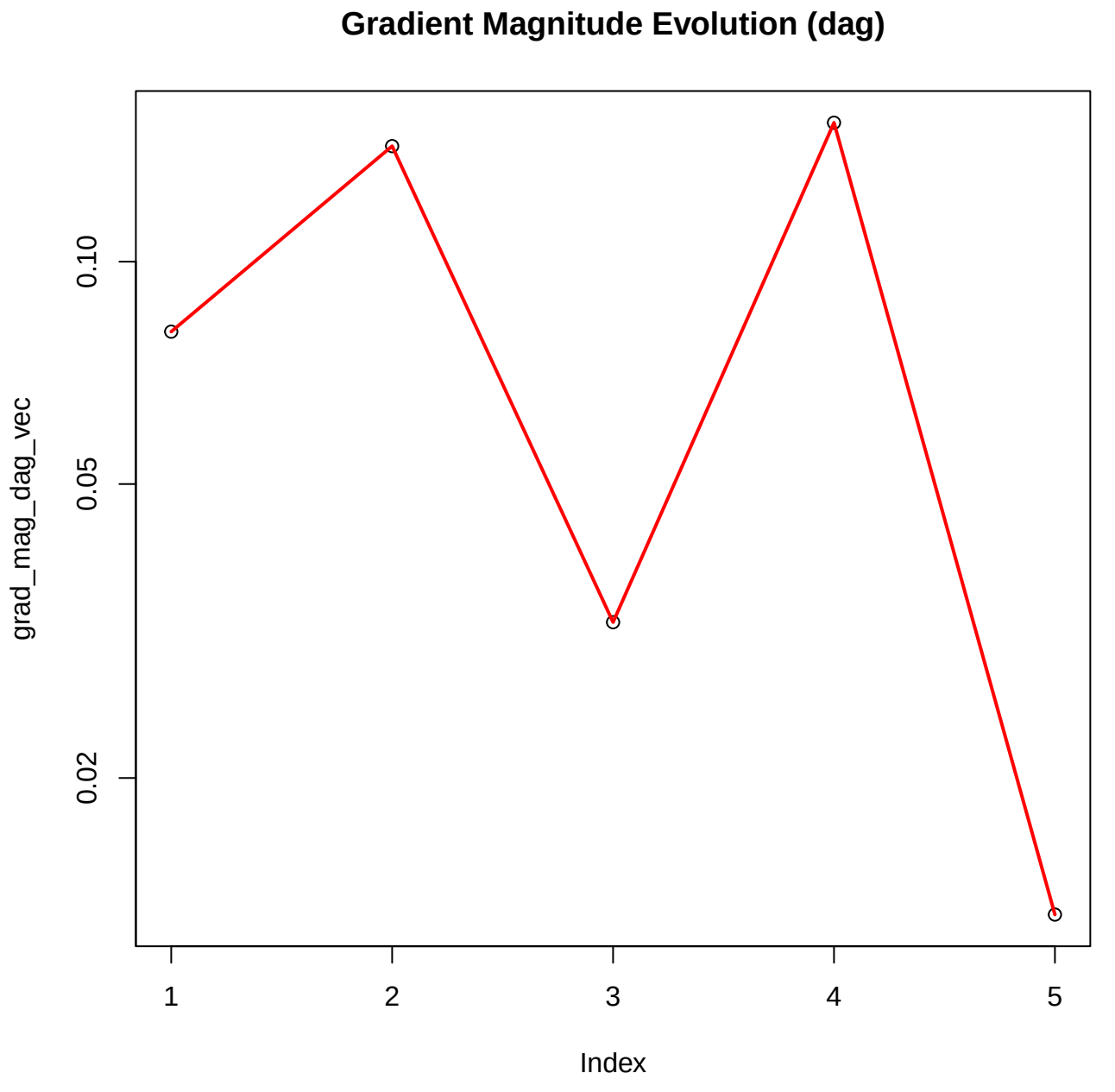

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

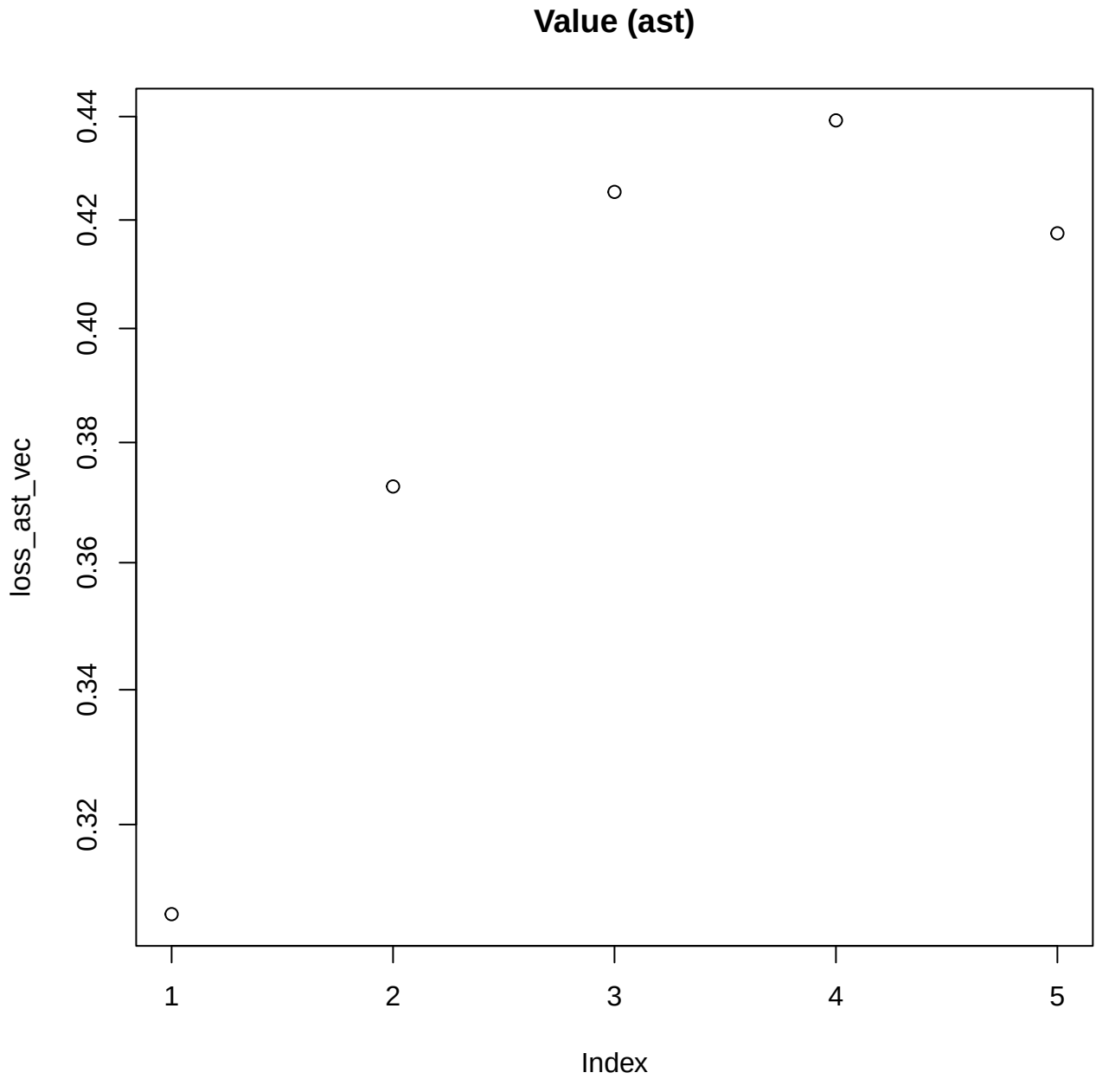

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index


Value (ast)
loss_ast_vec
0.540
0.542
0.544
0.546
0.548
0
50
100
150
200
Index


Value (dag)
loss_dag_vec
0.452
0.454
0.456
0.458
0.460
0
50
100
150
200
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
1
2
3
4
5
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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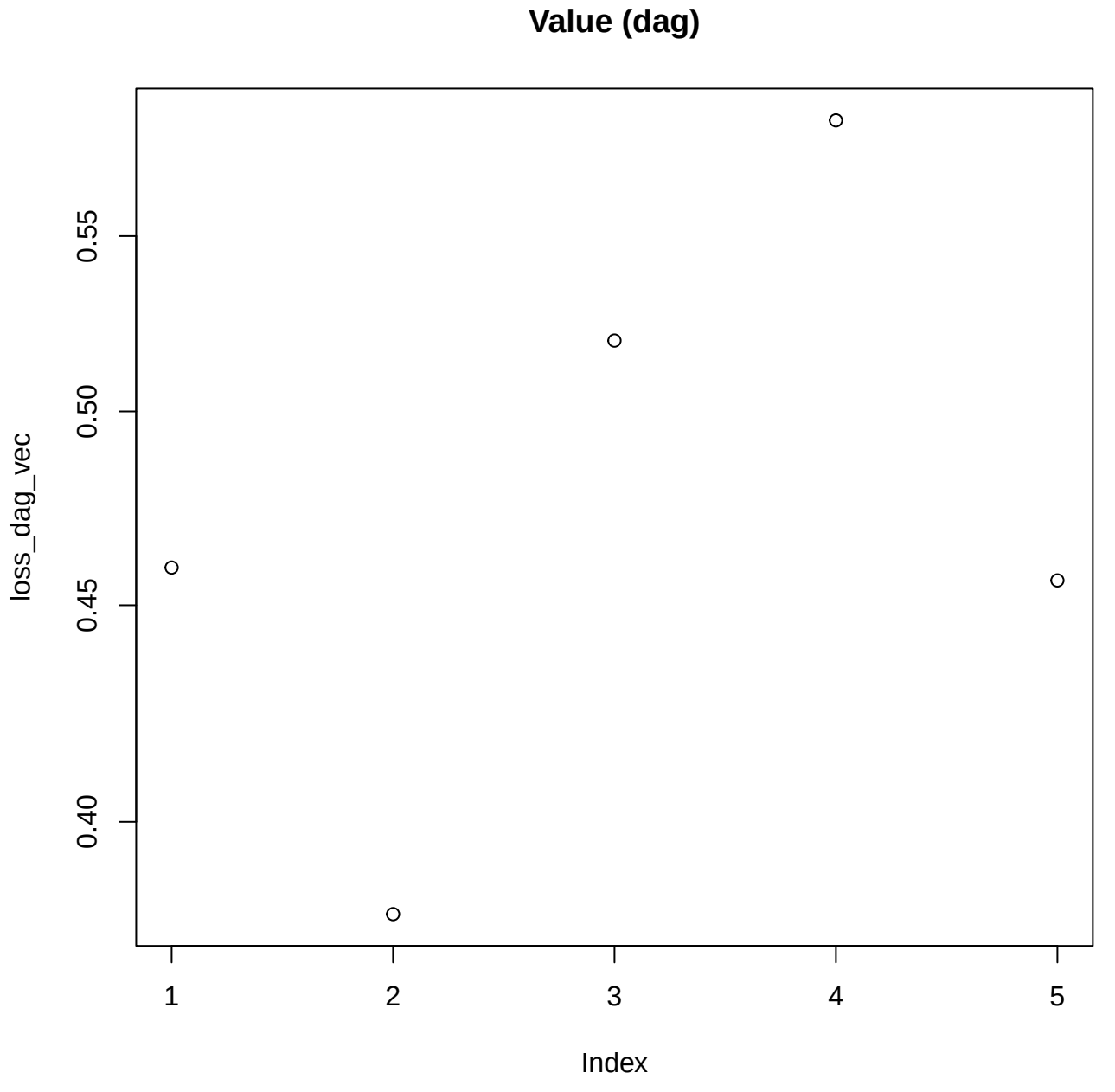
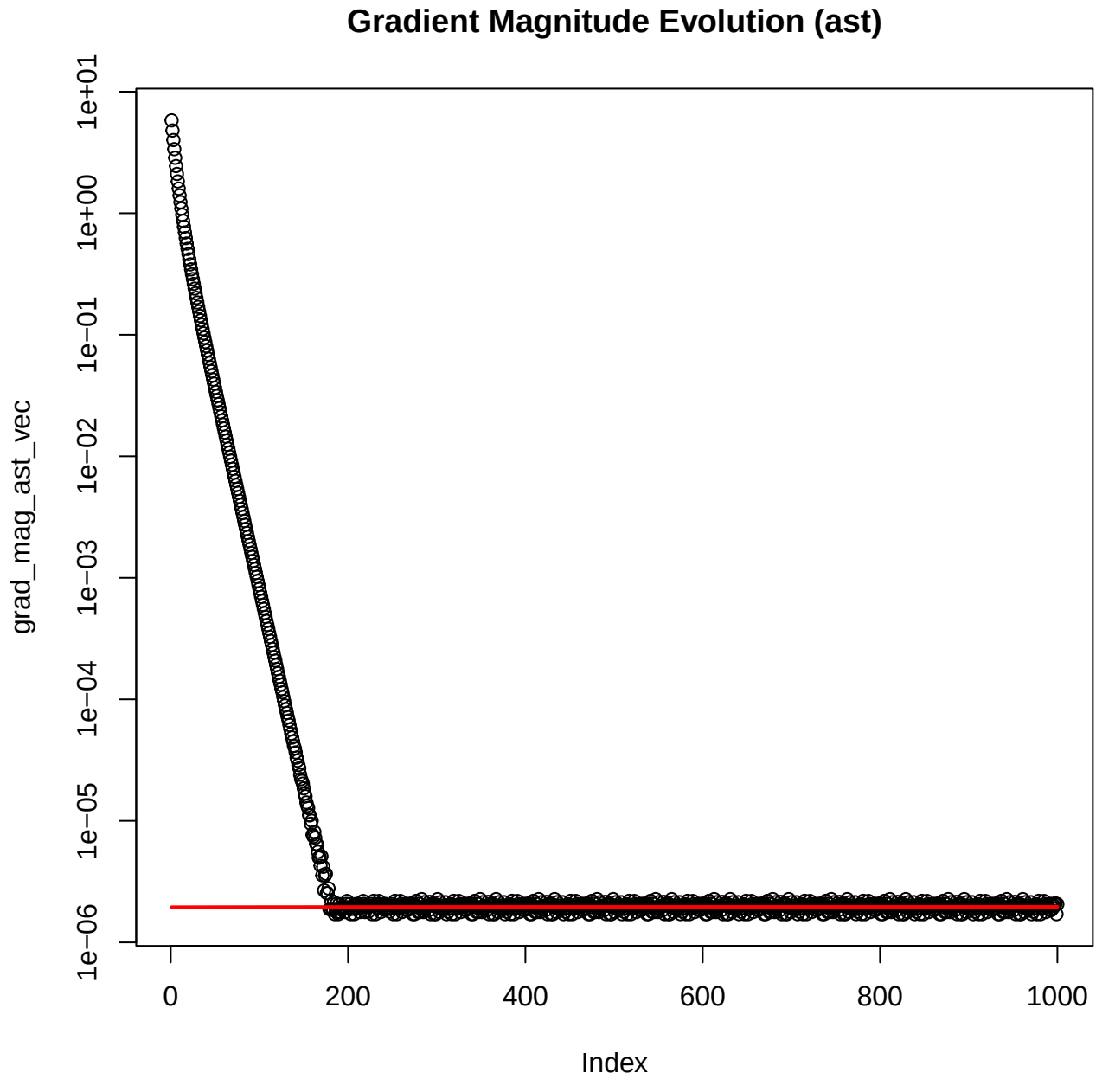
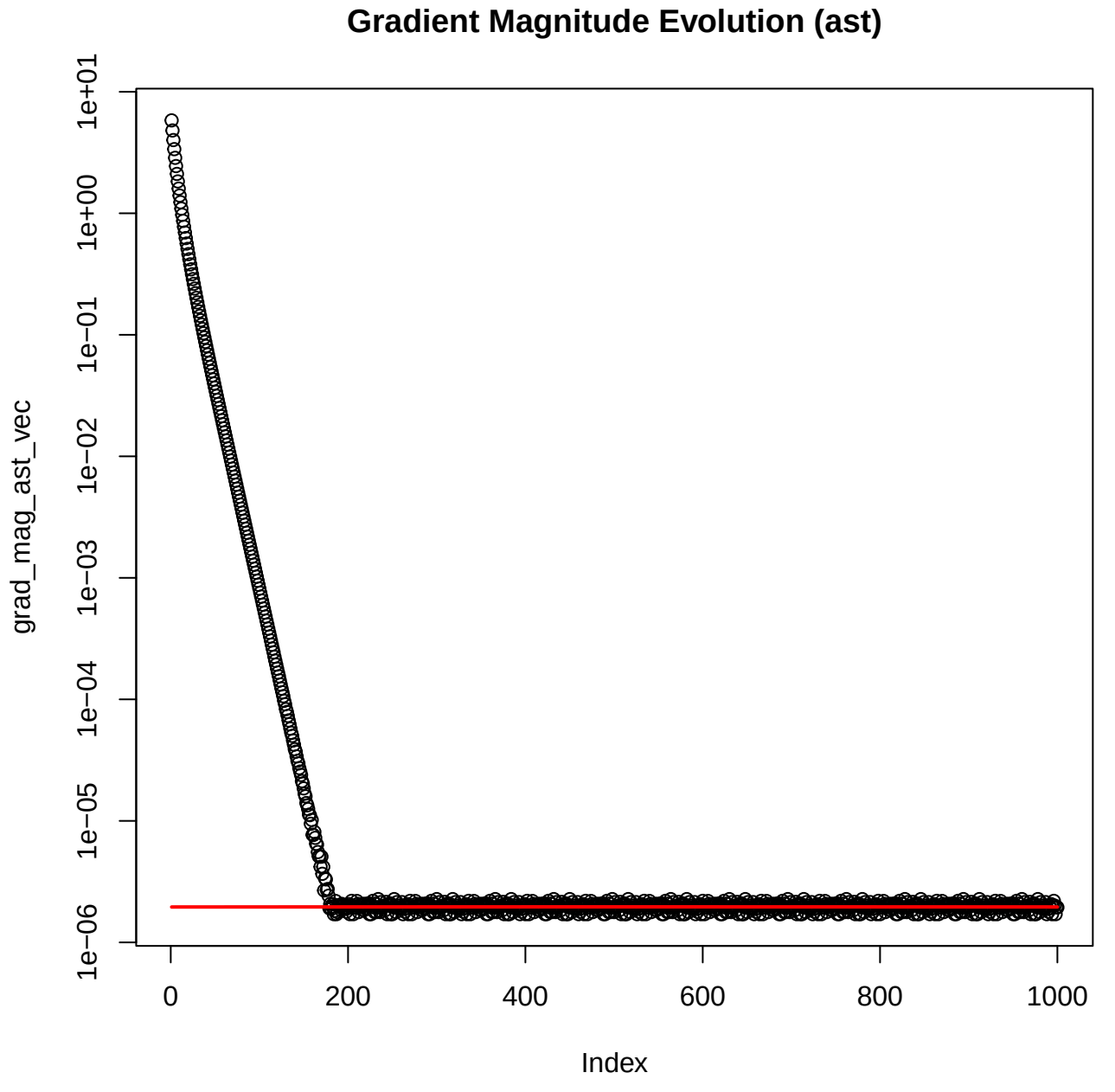
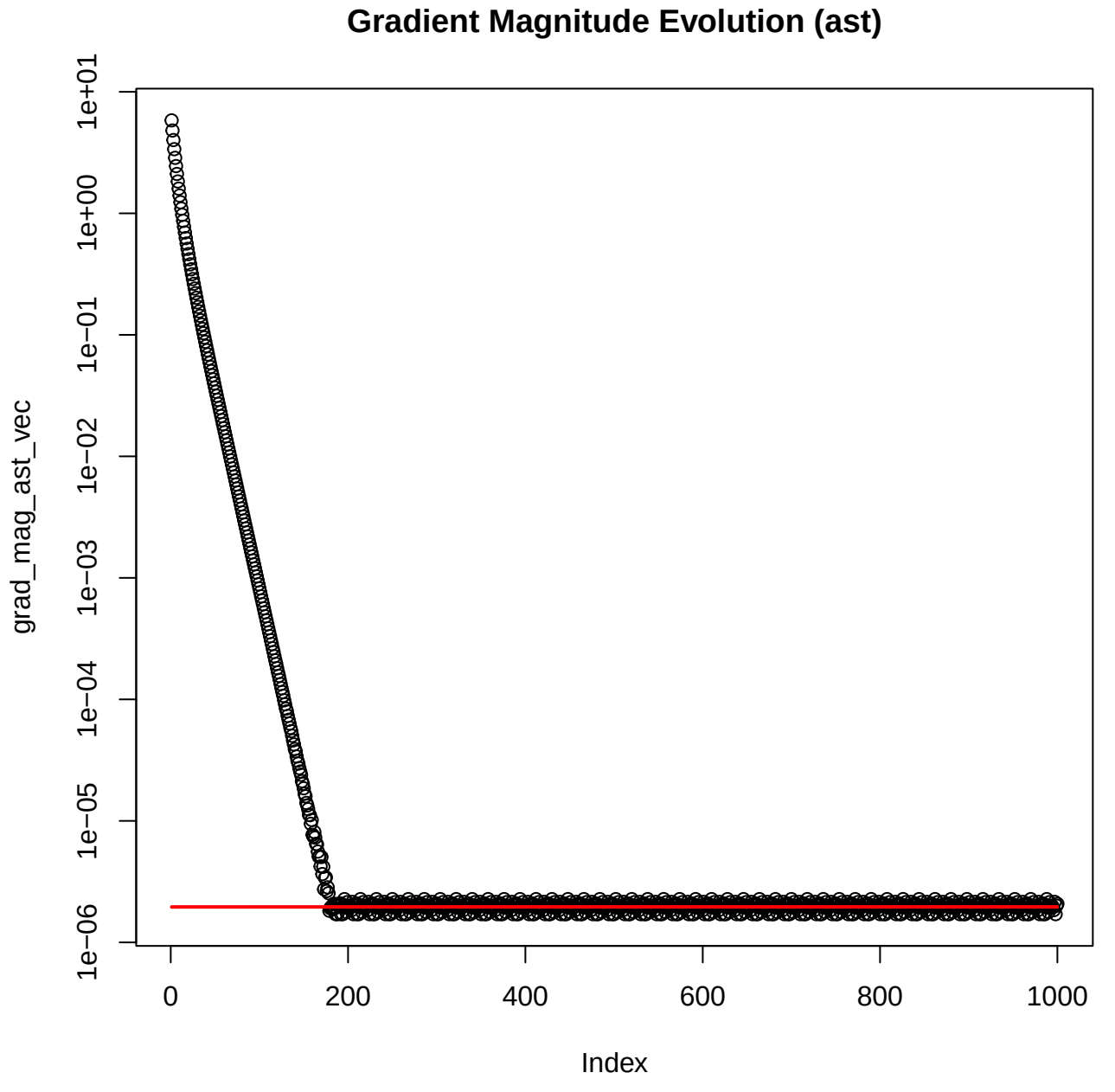
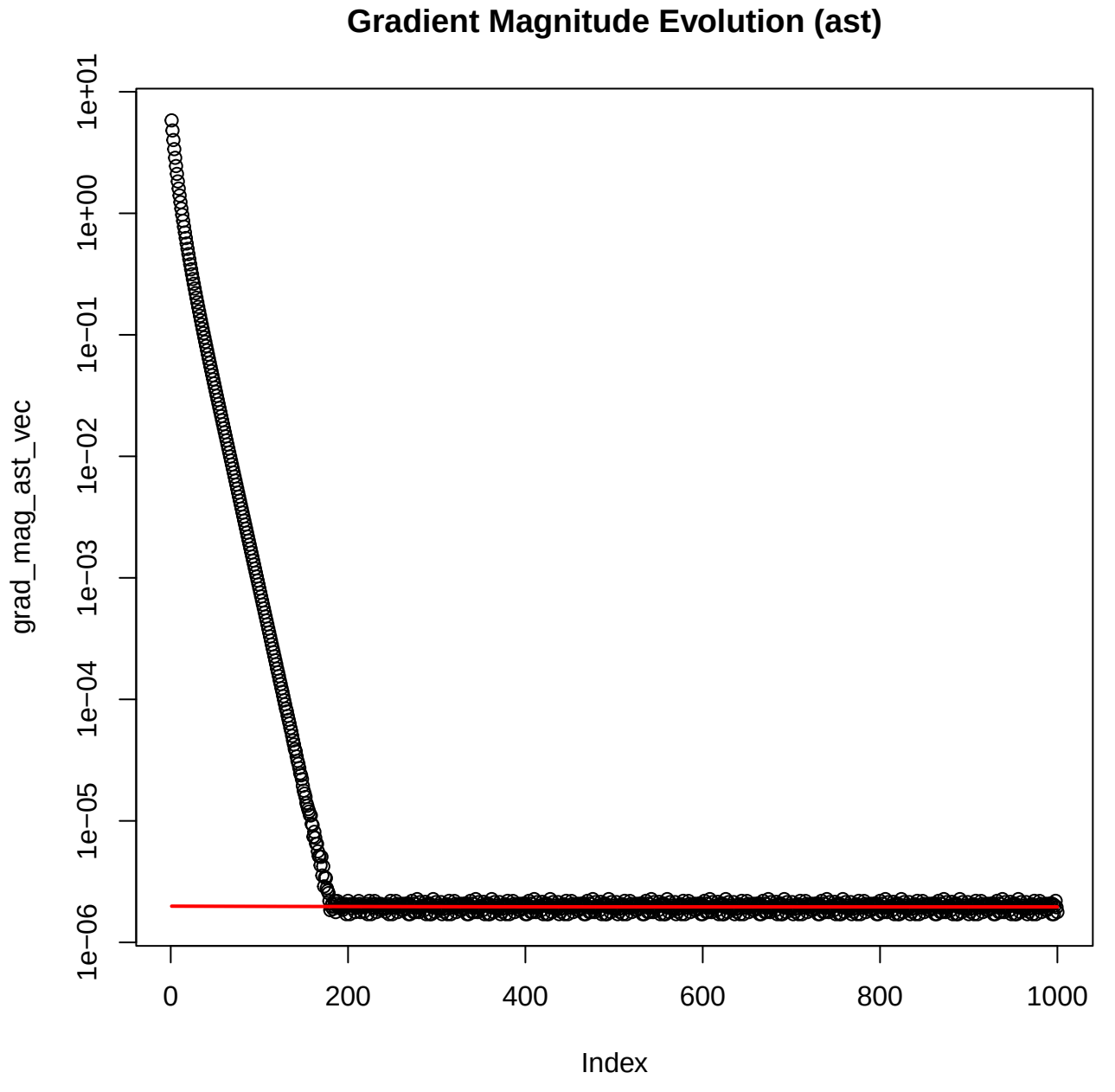
0.34
0.36
0.38
0.40
0.42
0.44
1
2
3
4
5
Index

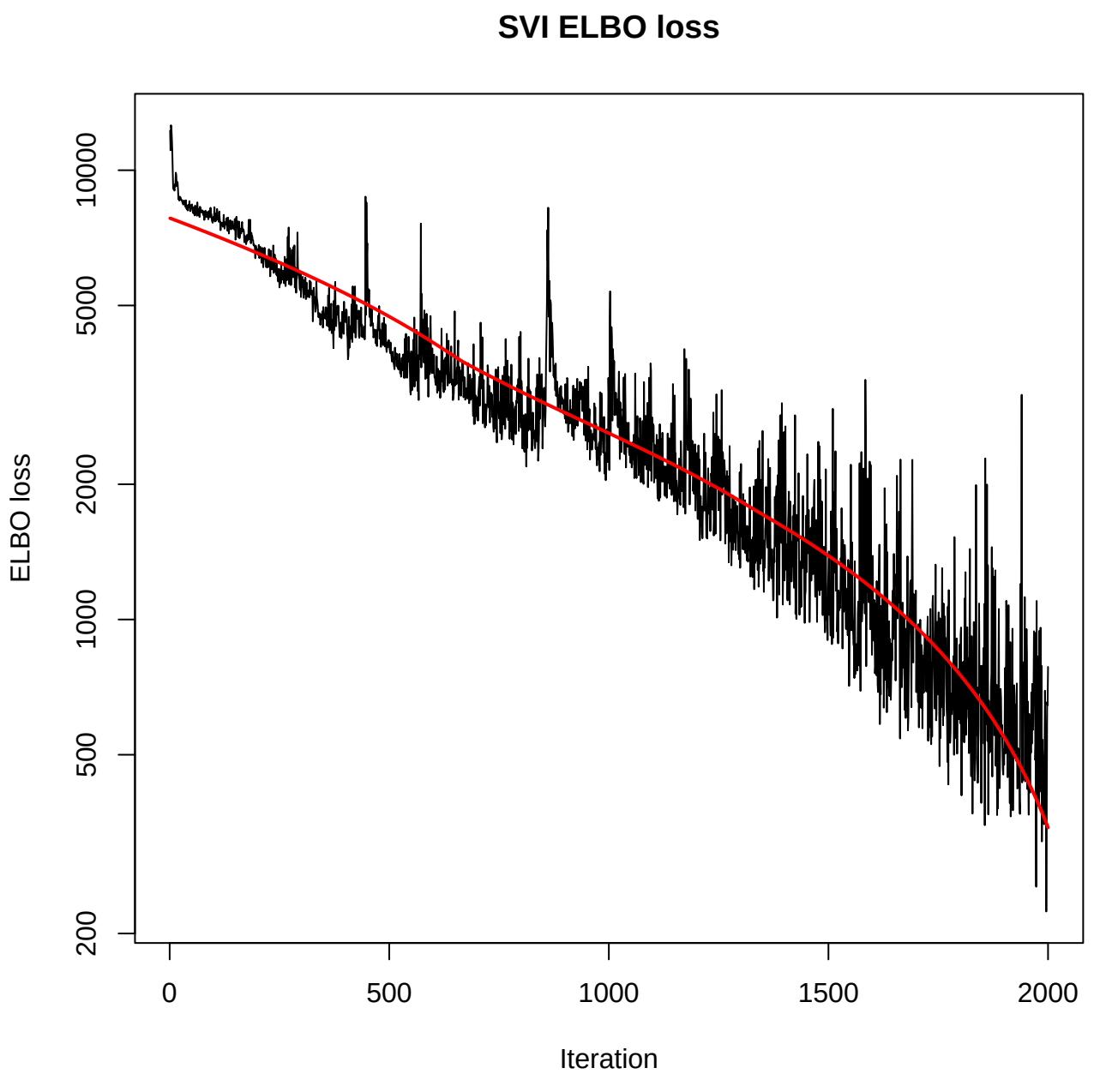

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

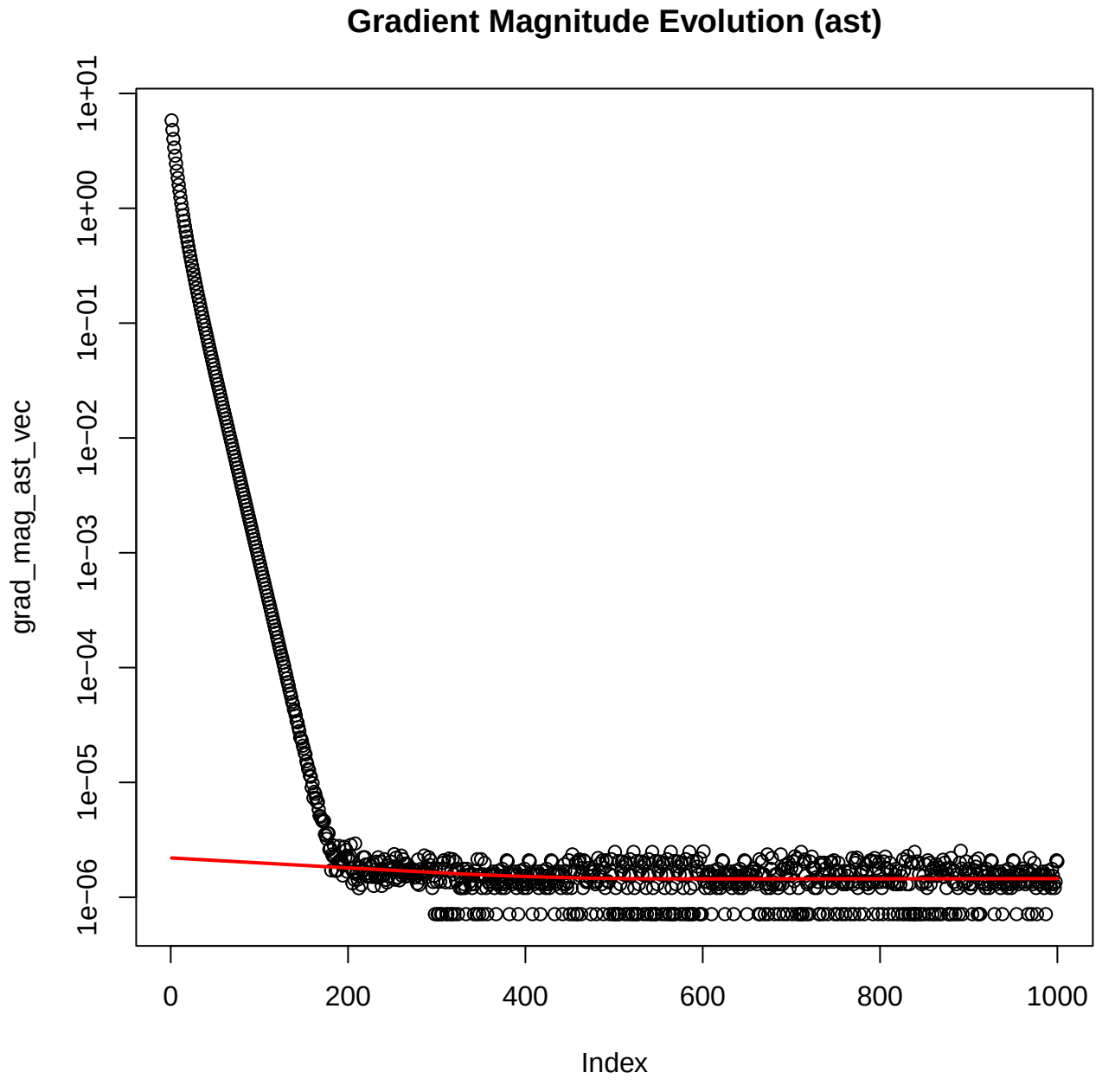

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

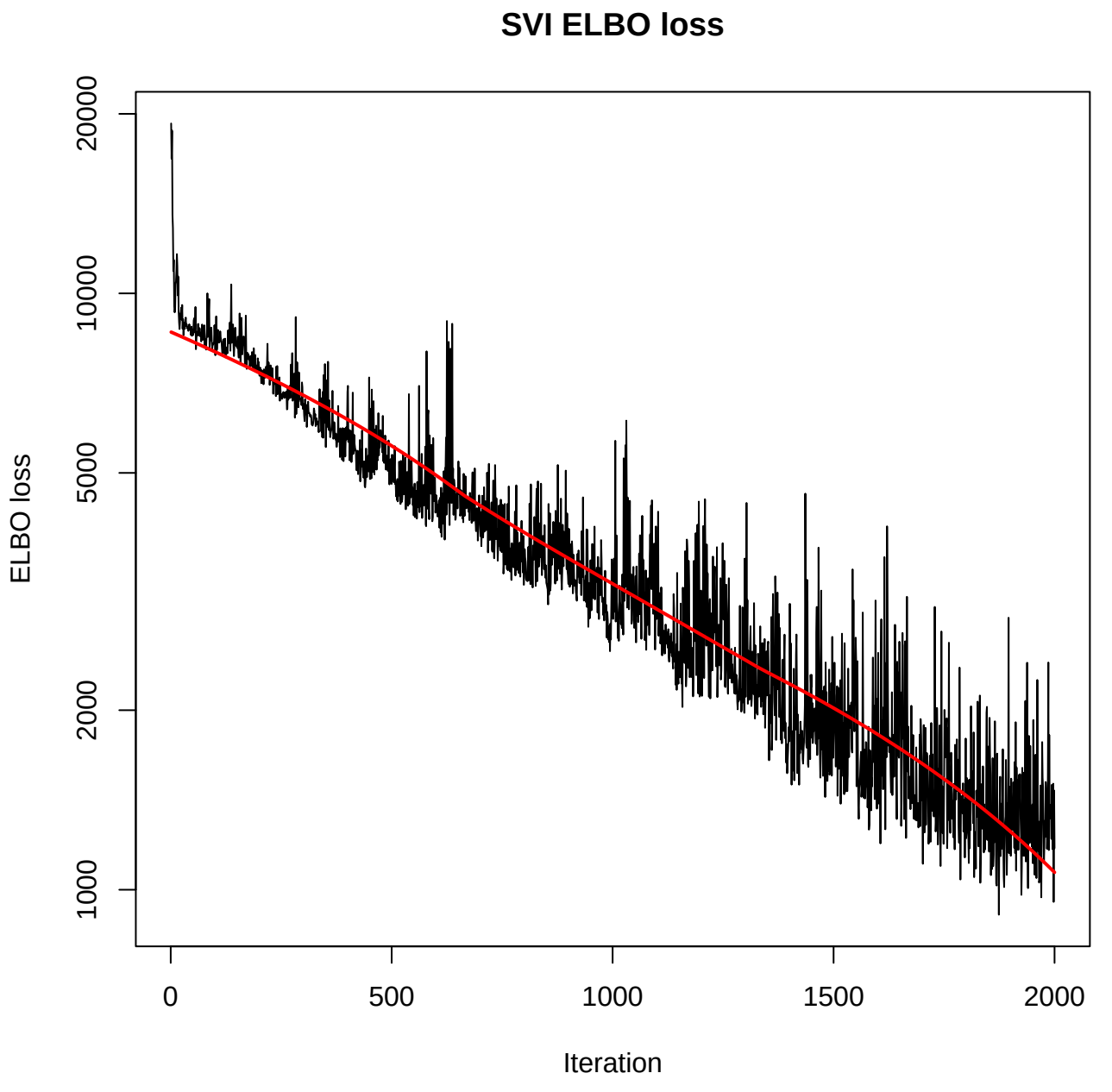

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

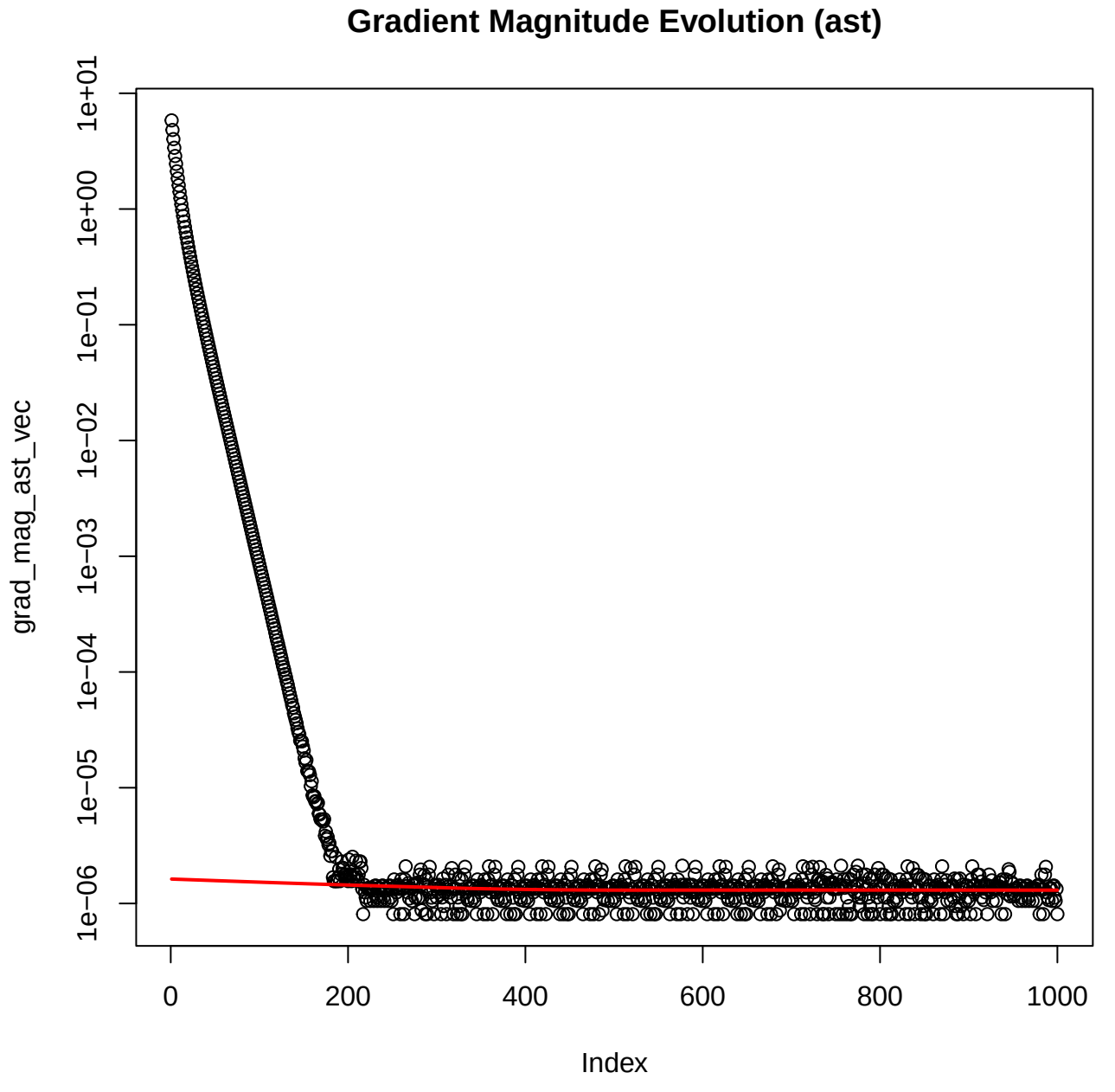

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

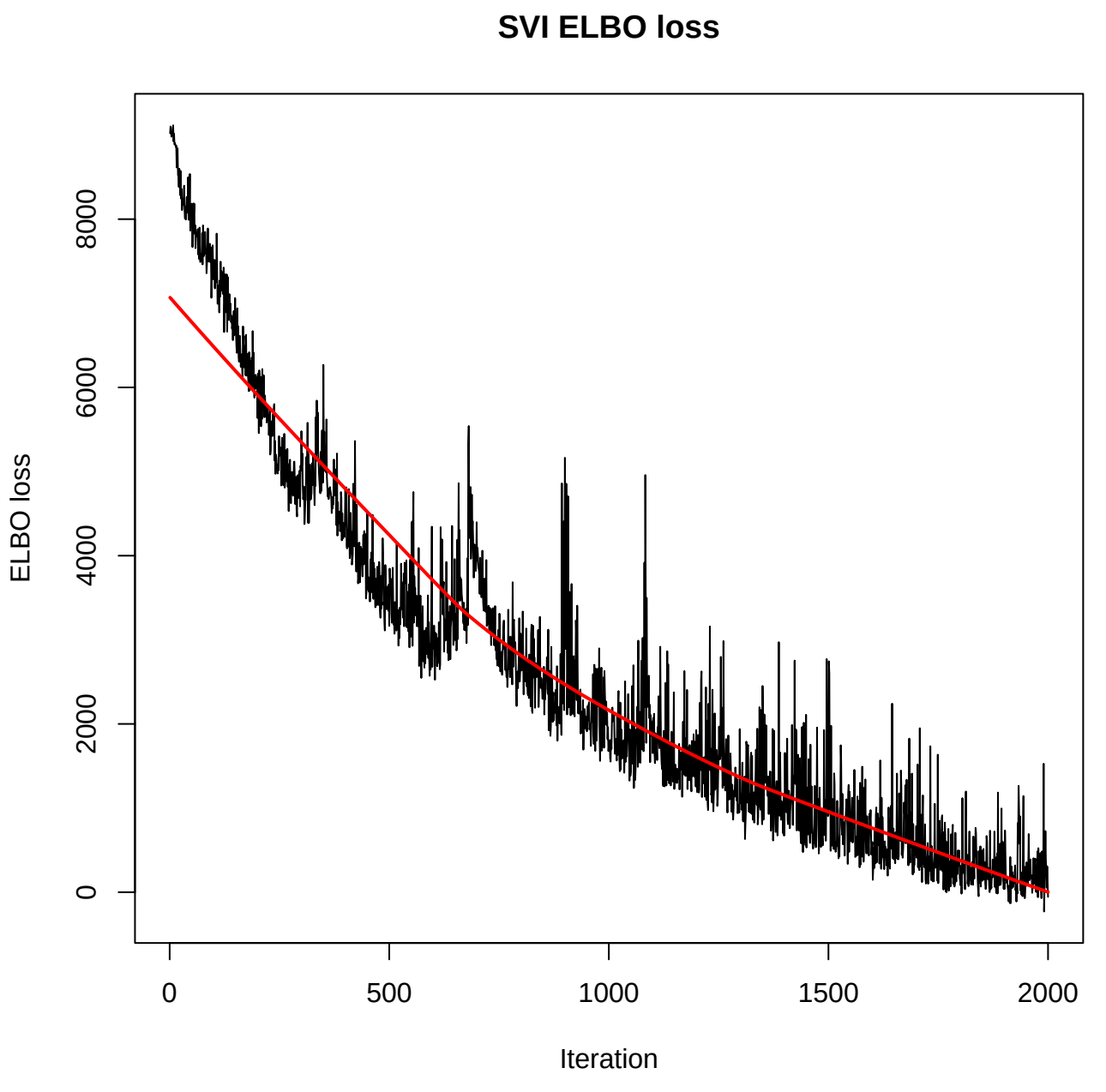

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

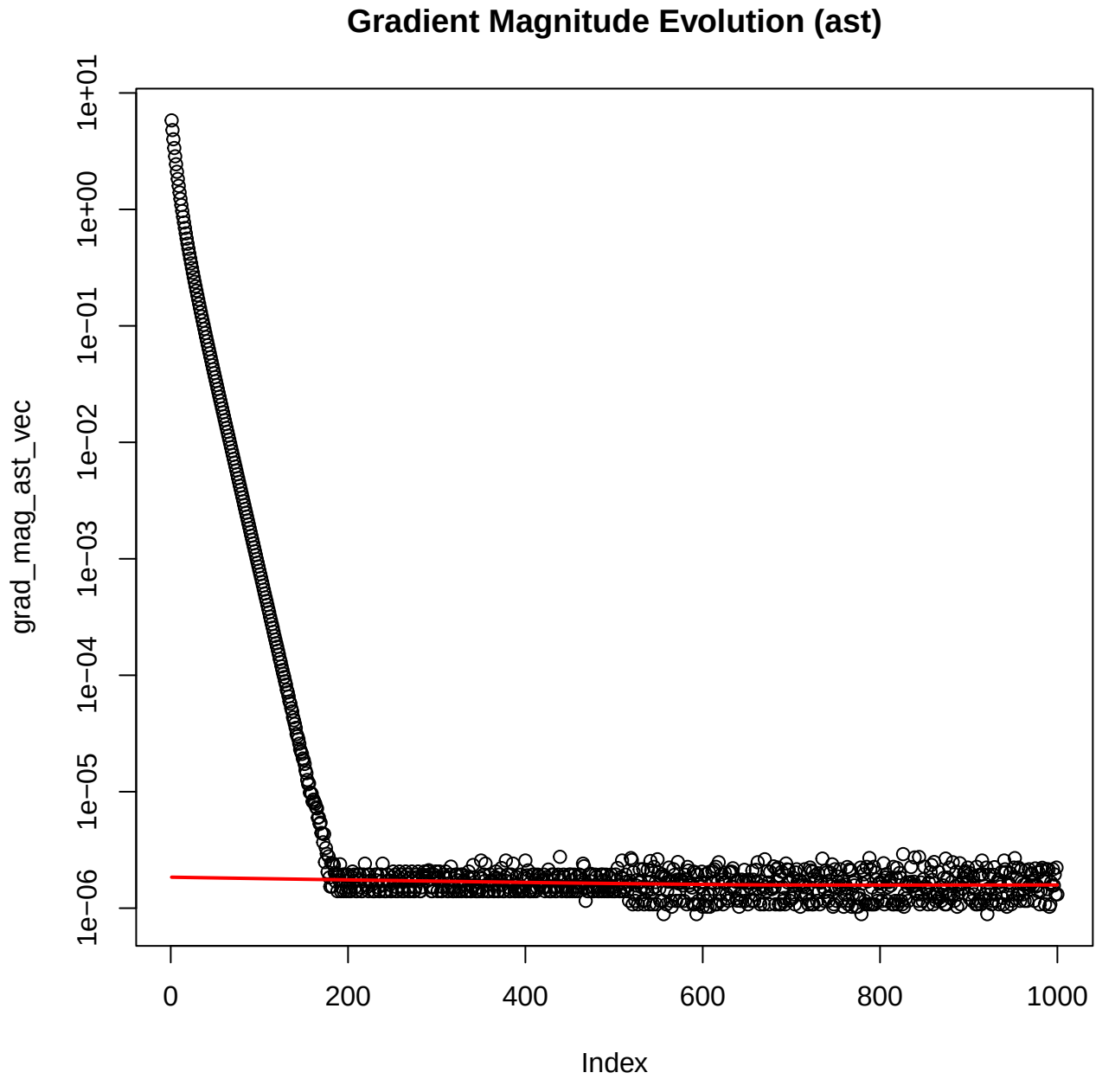

SVI ELBO loss
ELBO loss
10000
5000
2000
1000
500
200
0
500
1000
1500
2000
Iteration

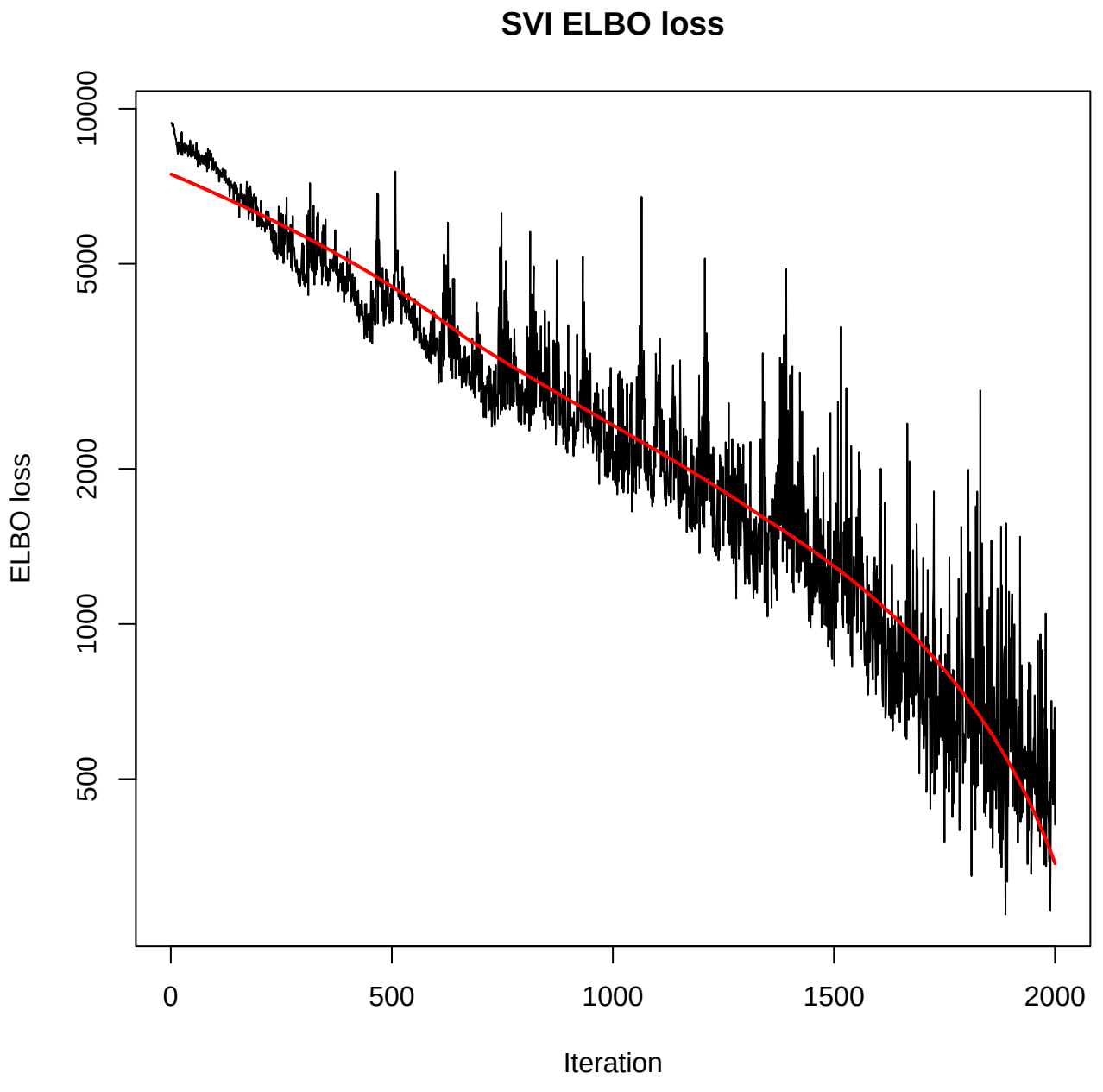

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
20000
10000
5000
2000
1000
0
500
1000
1500
2000
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
0
2000
4000
6000
8000
0
500
1000
1500
2000
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
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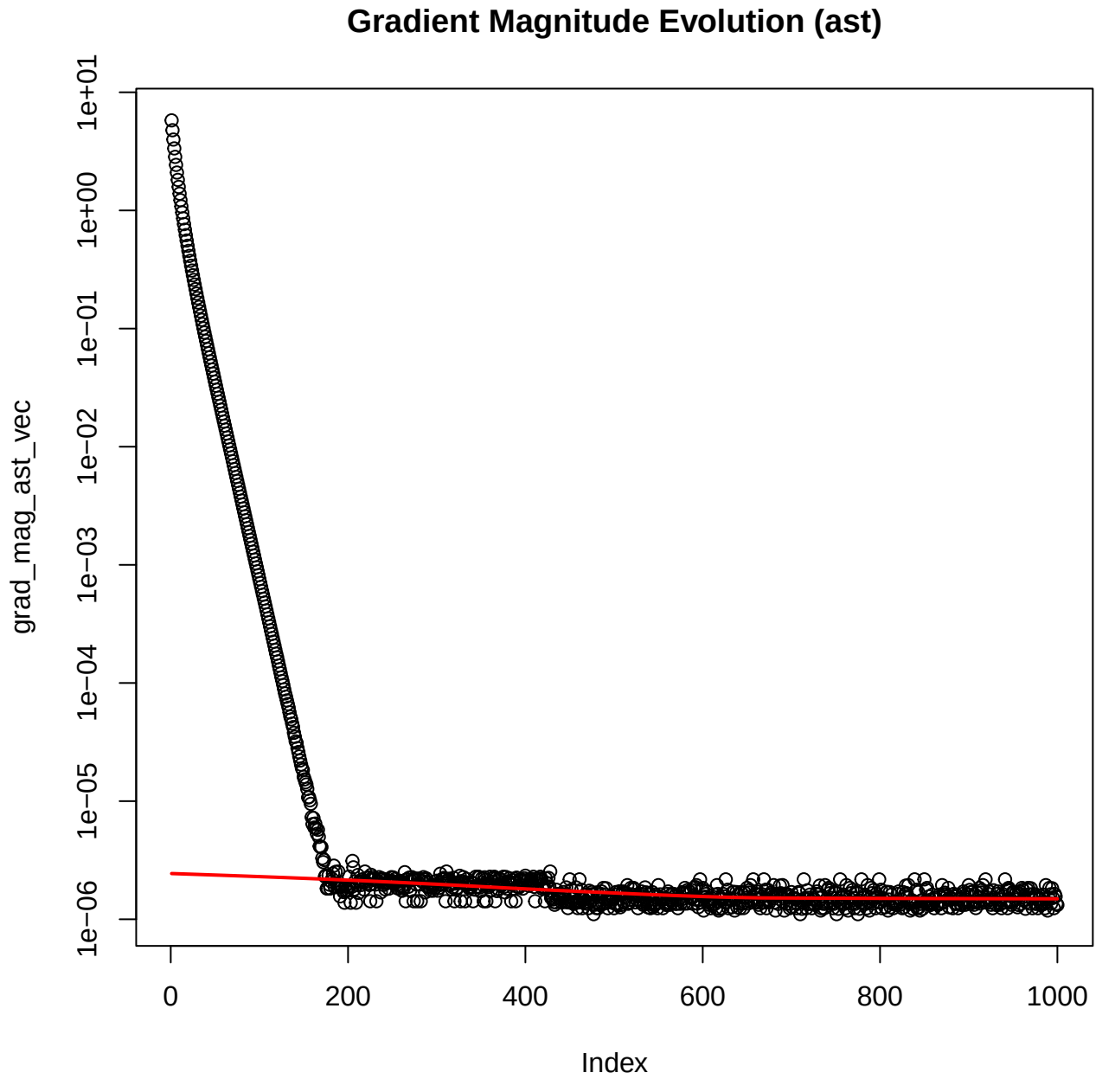
1500
2000
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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1e+00
1e−01
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1000
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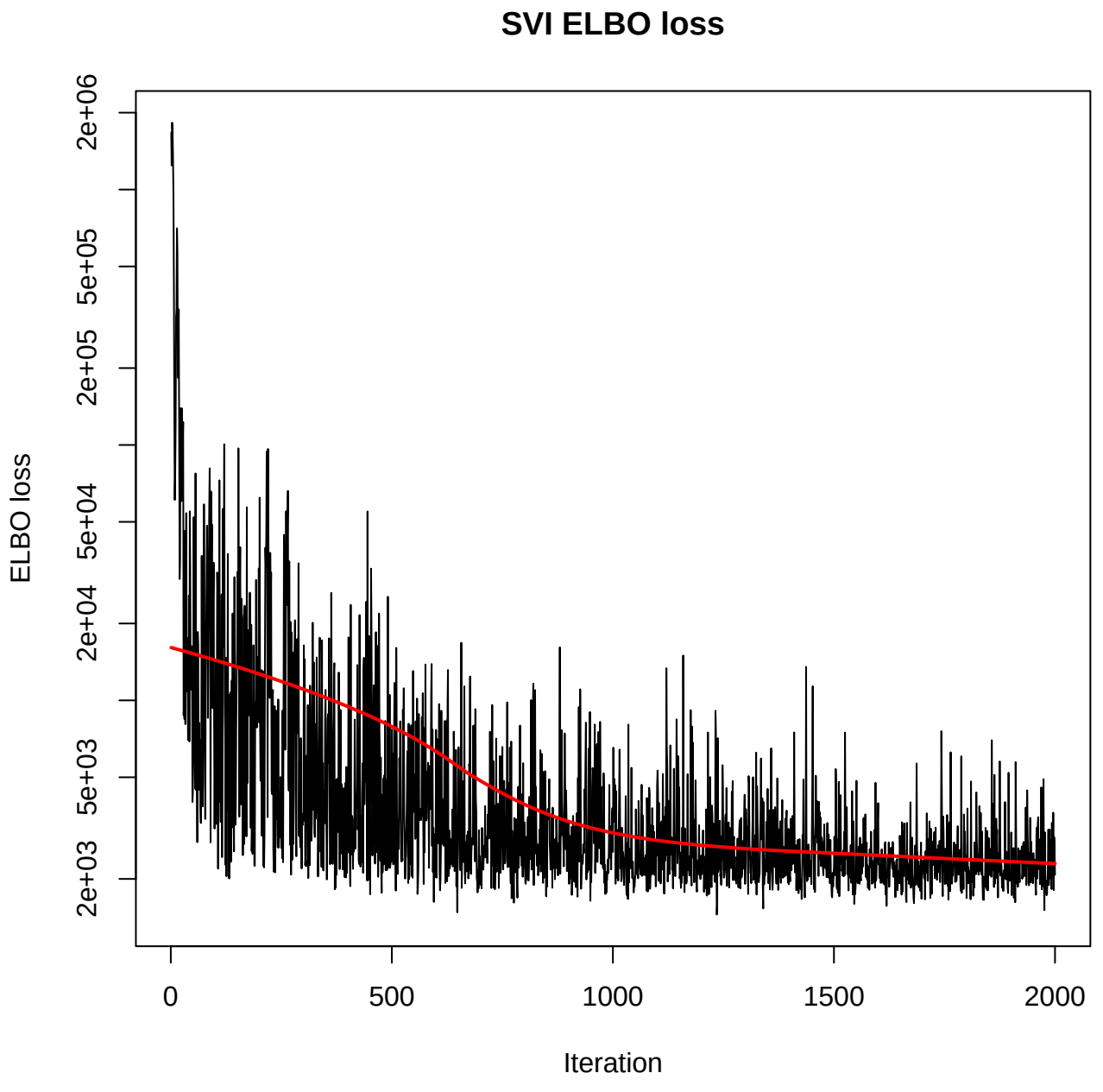
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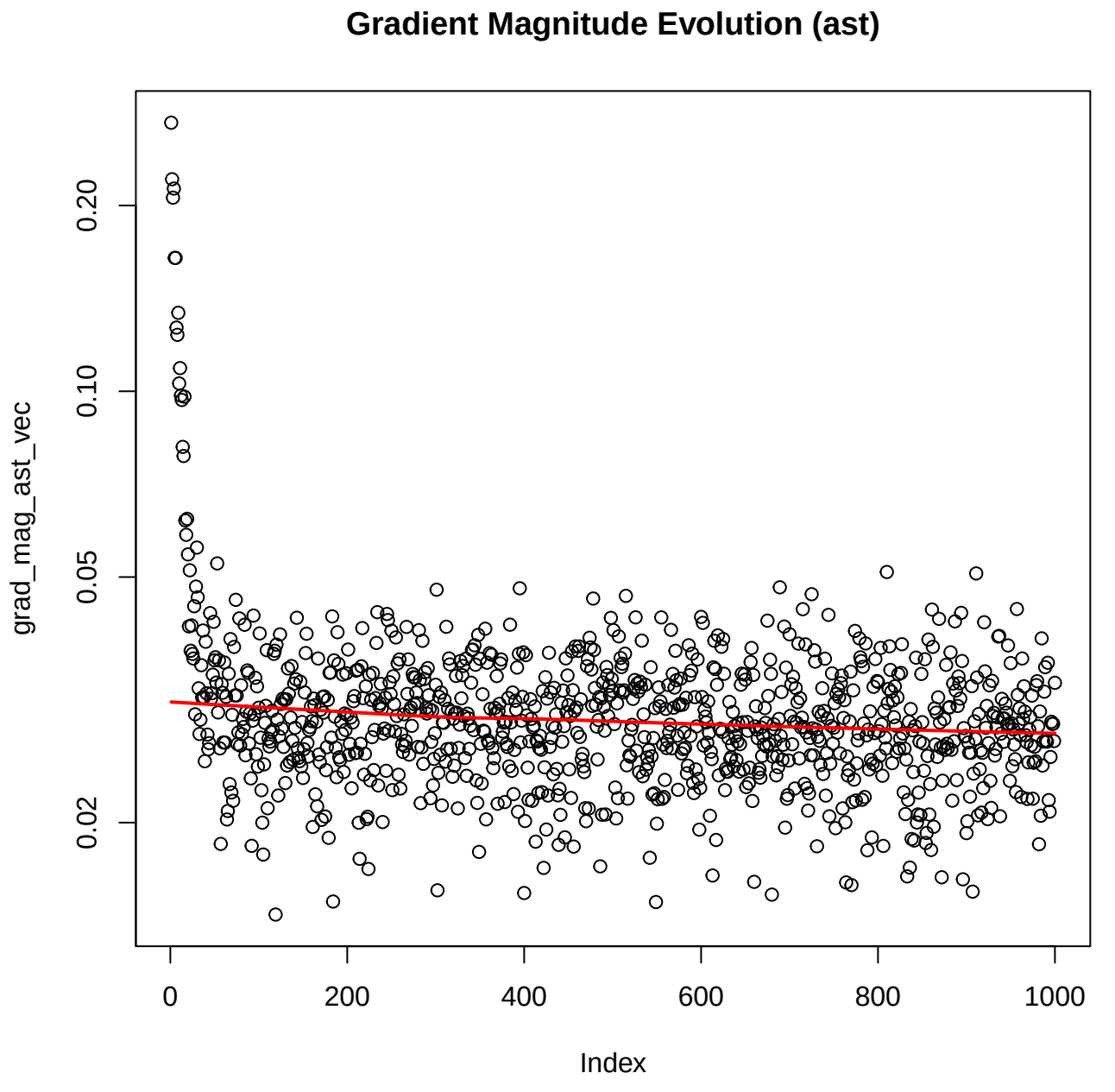
Iteration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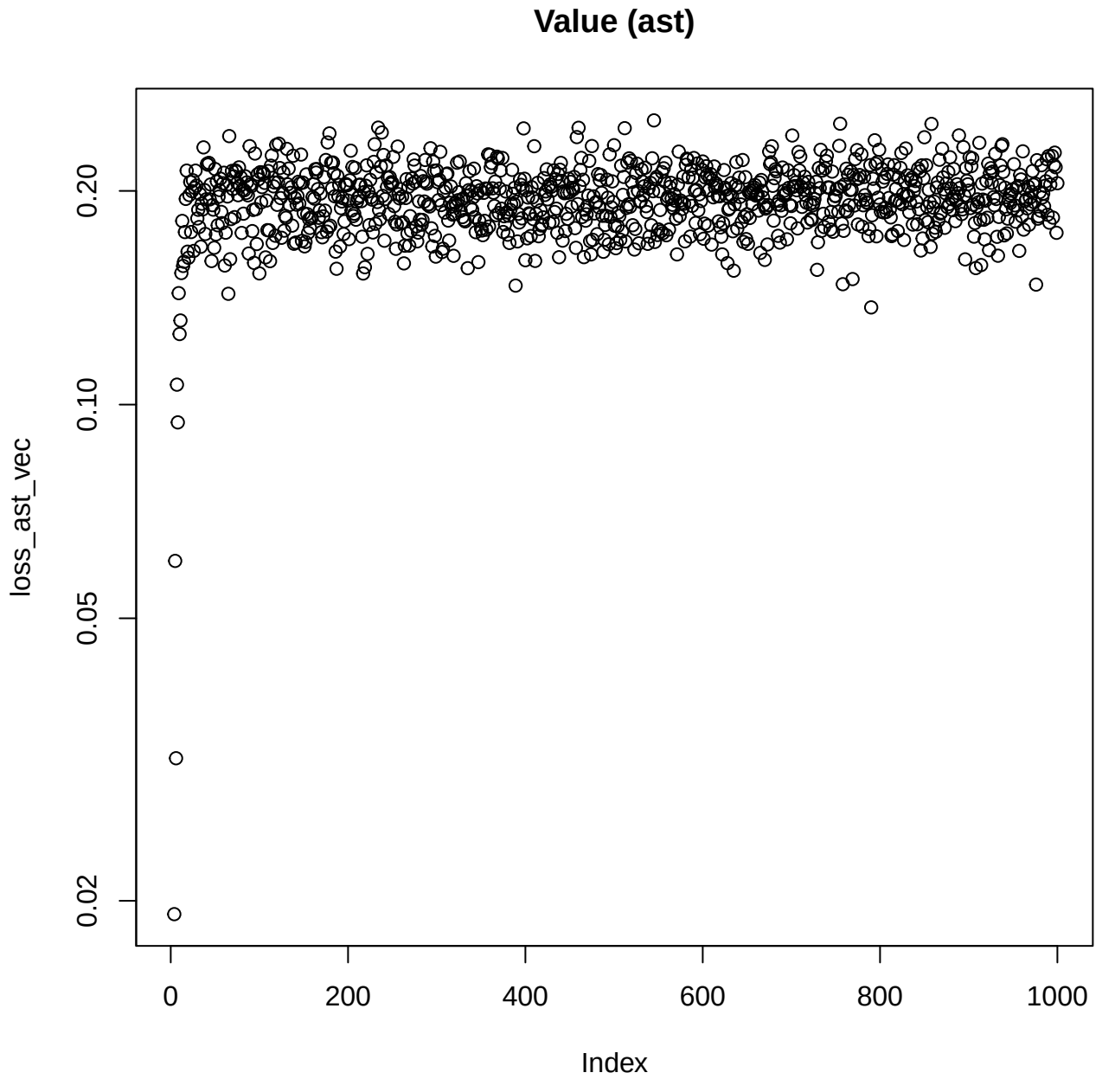
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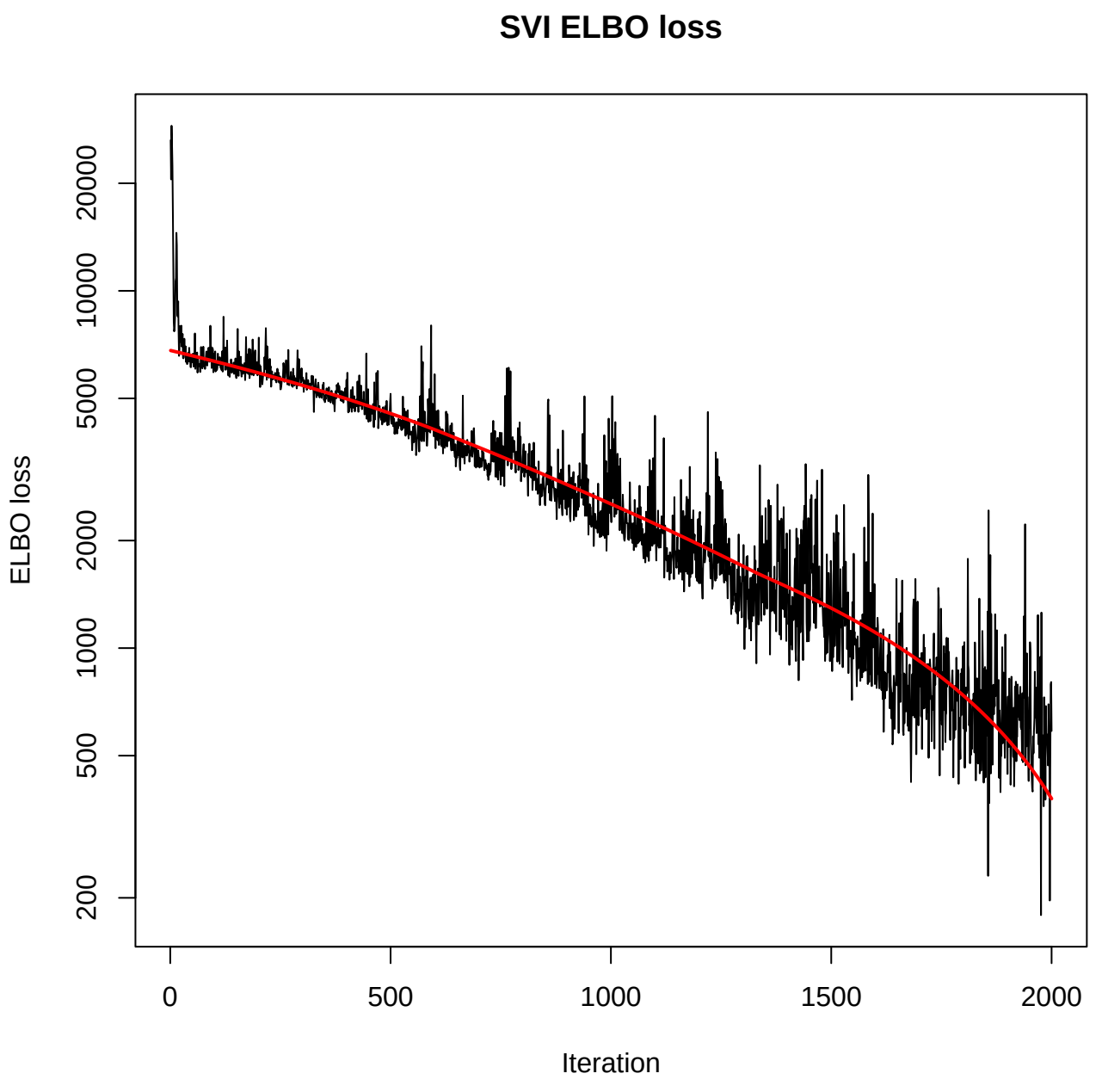
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ELBO loss
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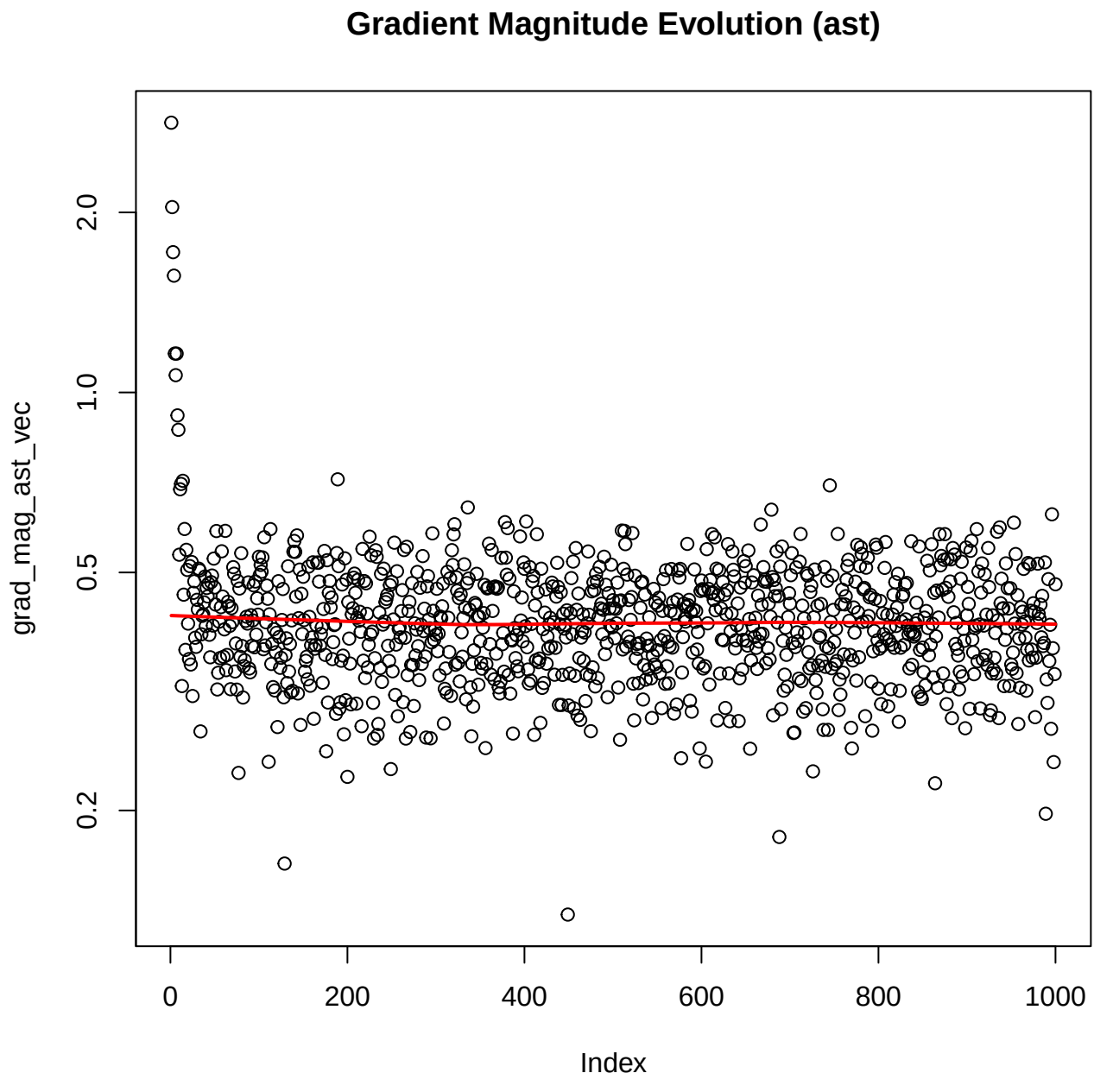
Iteration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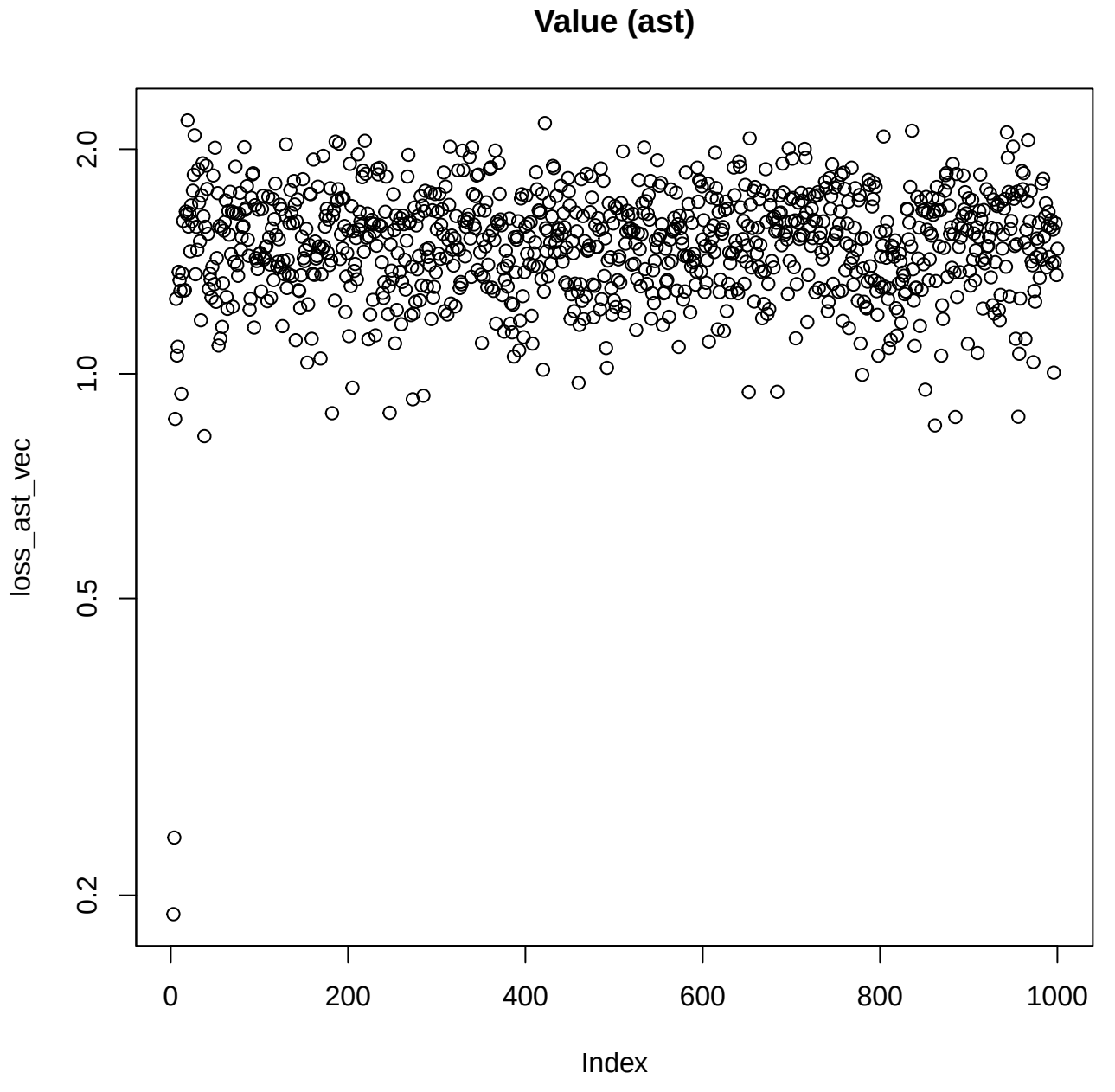
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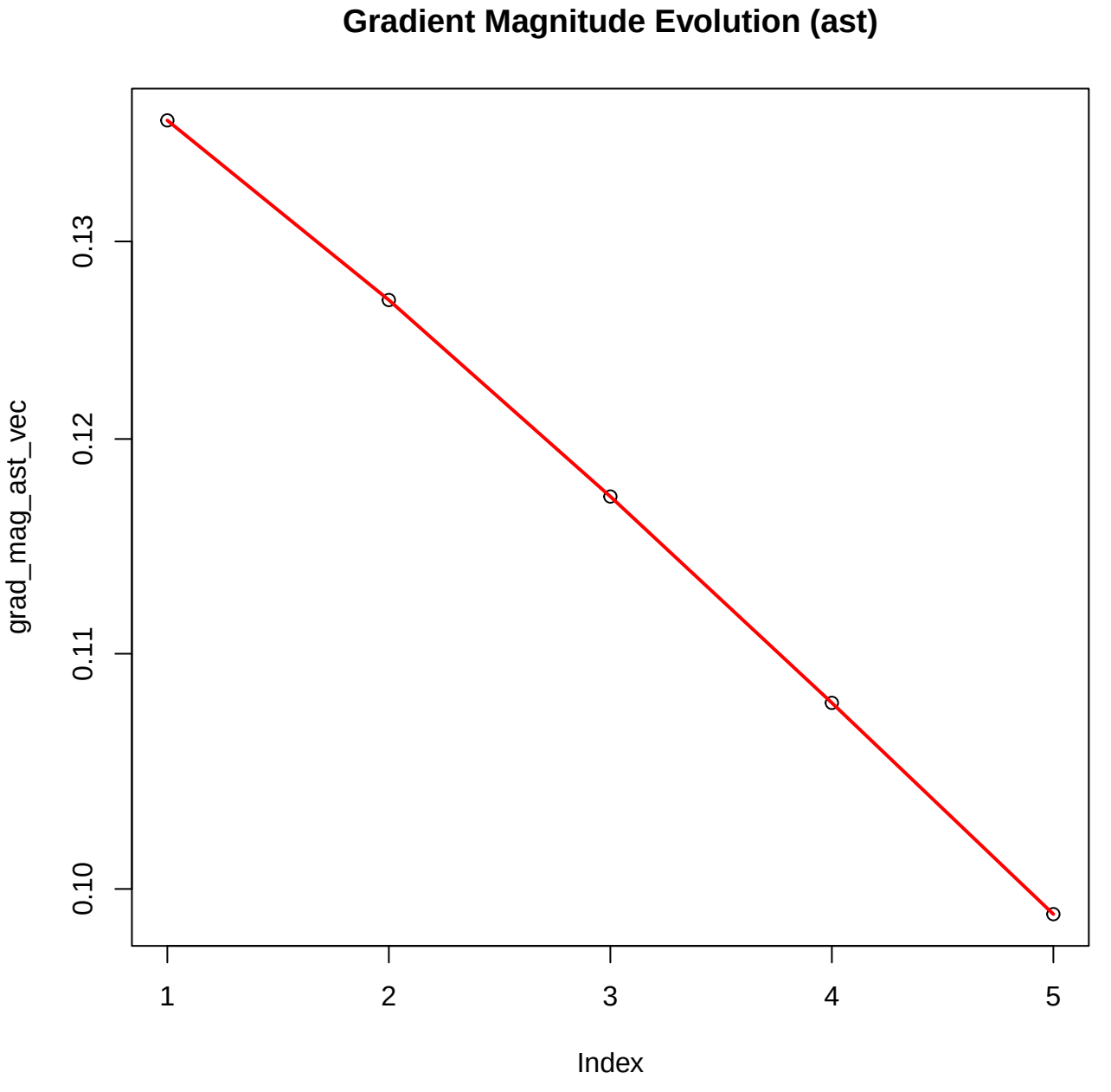
Gradient Magnitude Evolution (ast)
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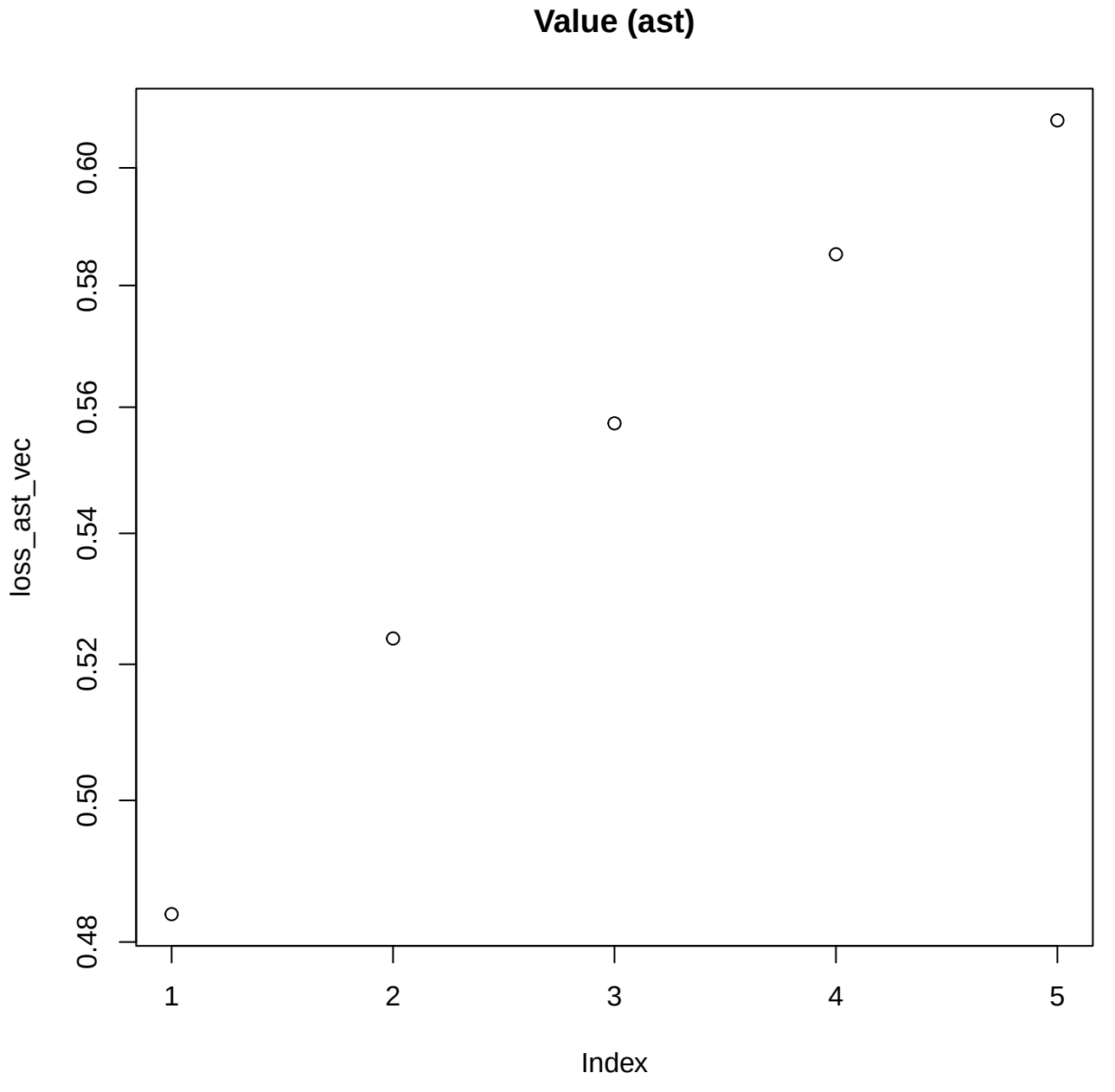
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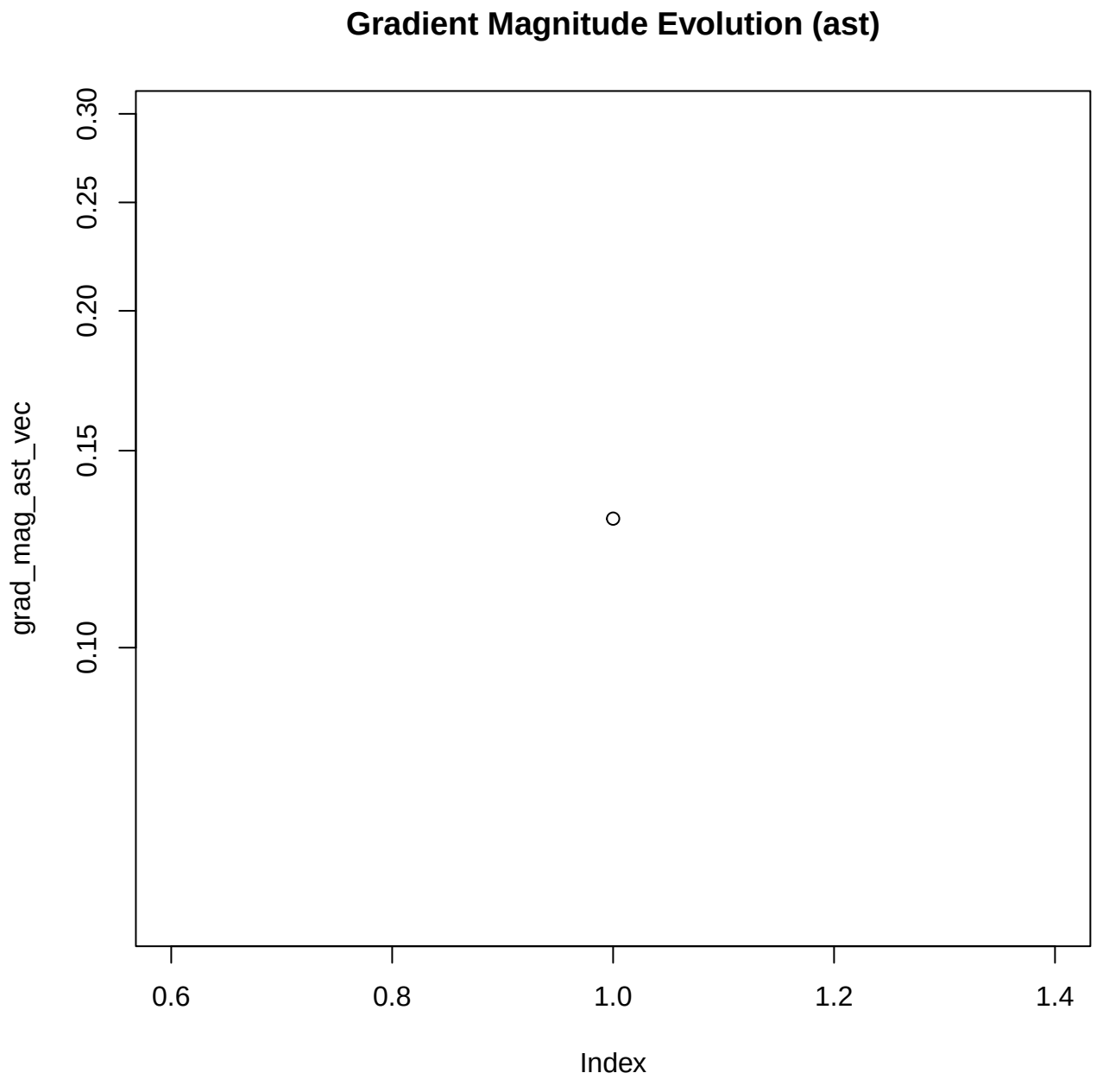
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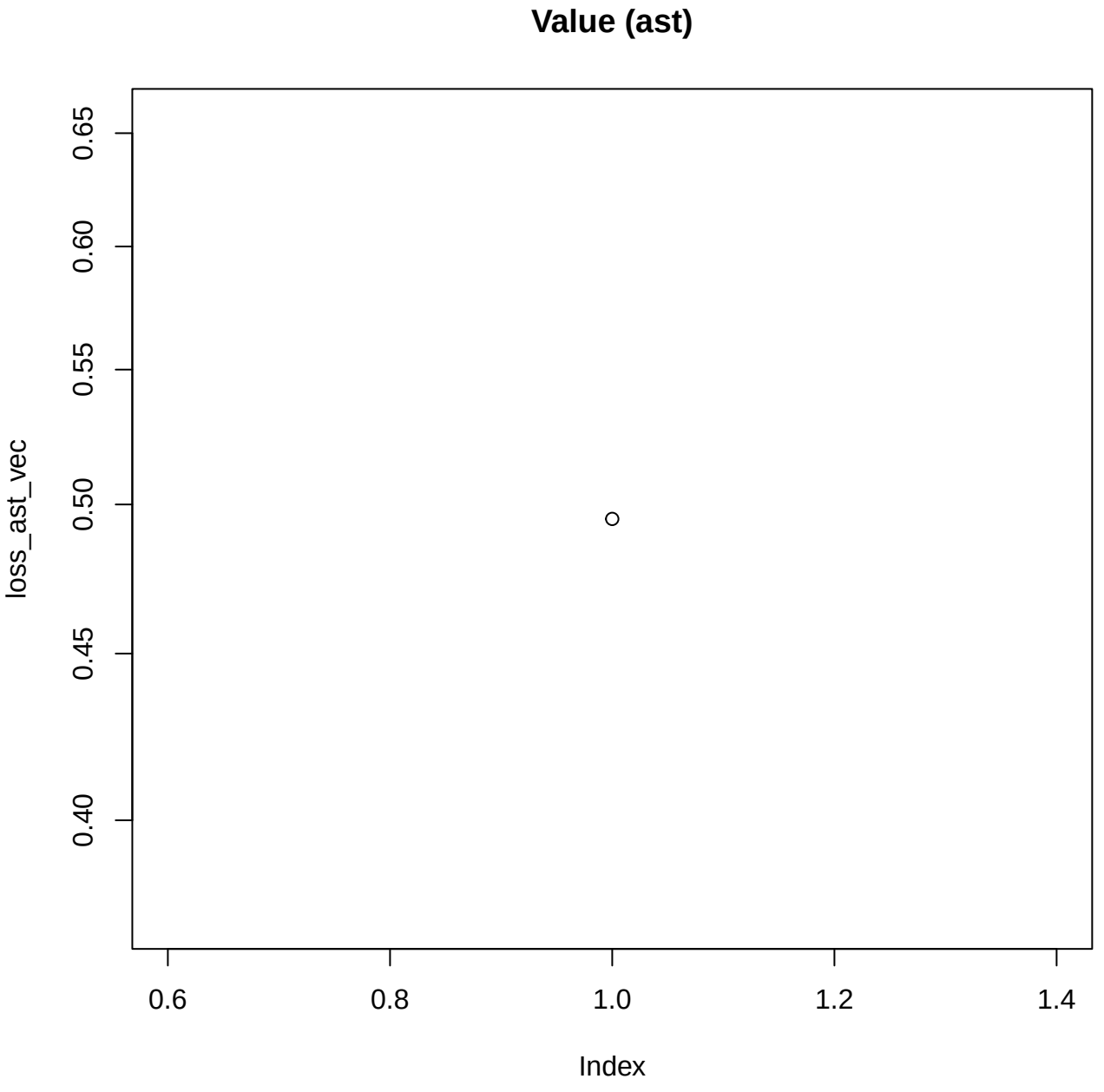
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loss_ast_vec
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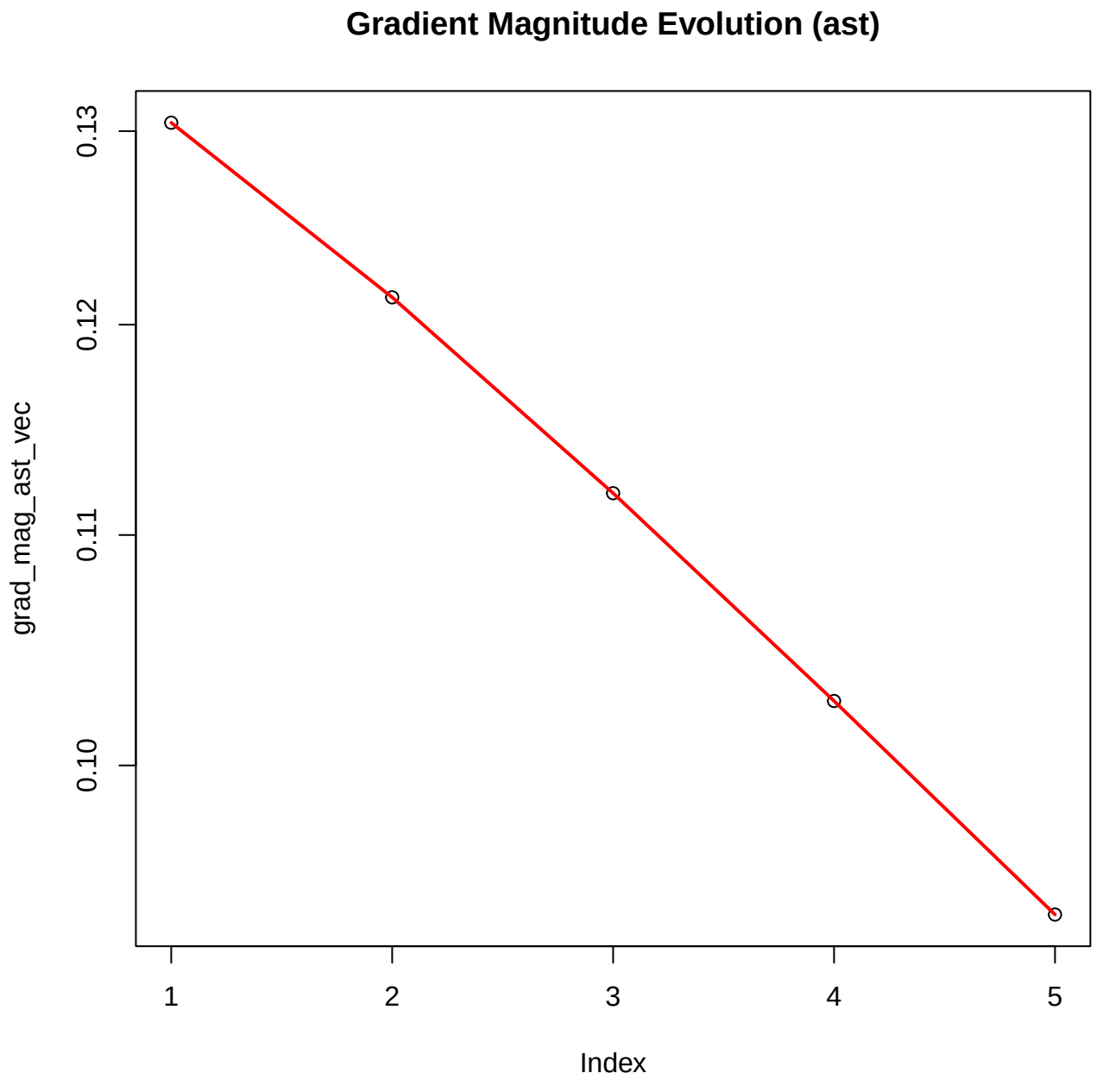
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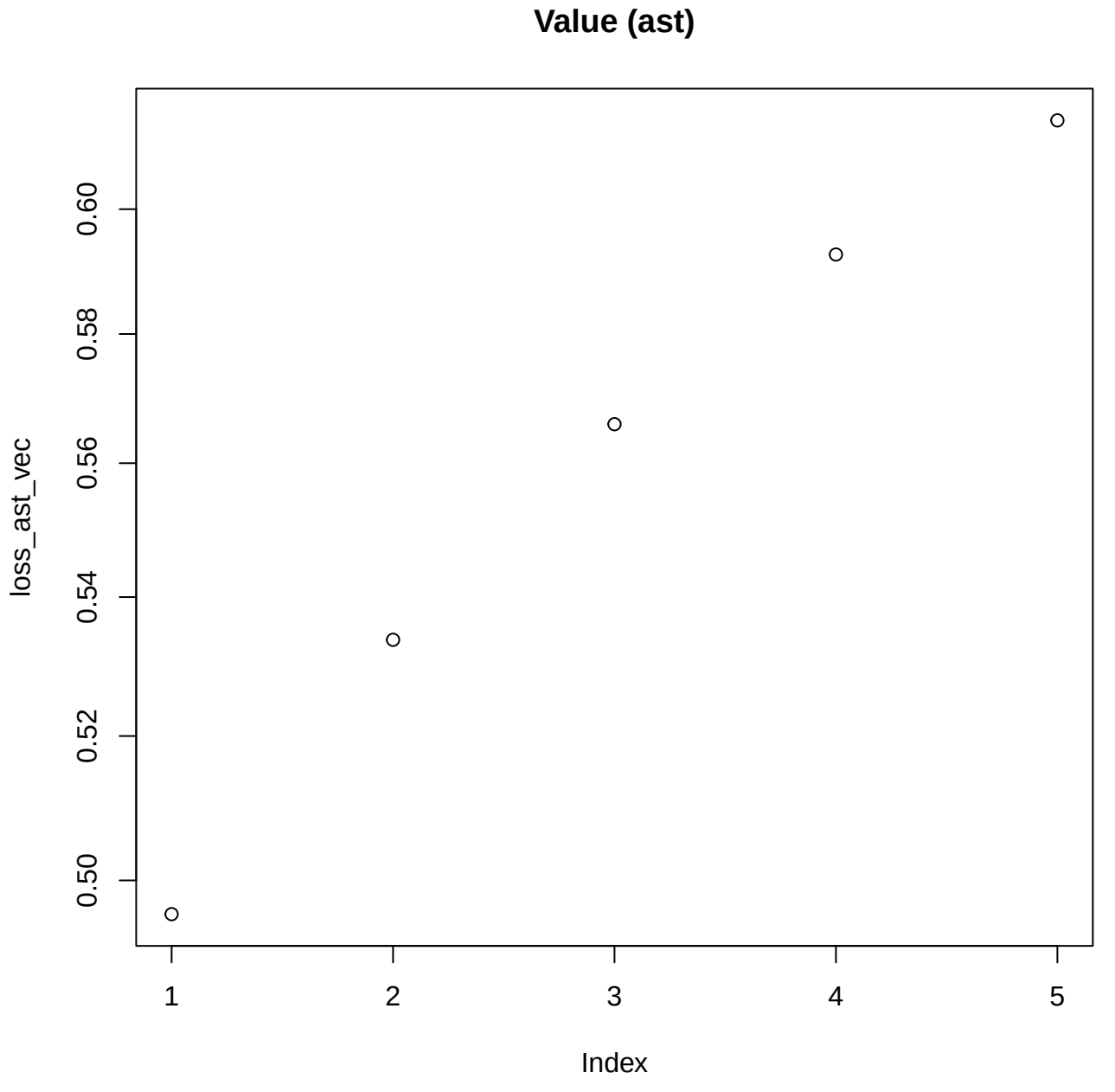
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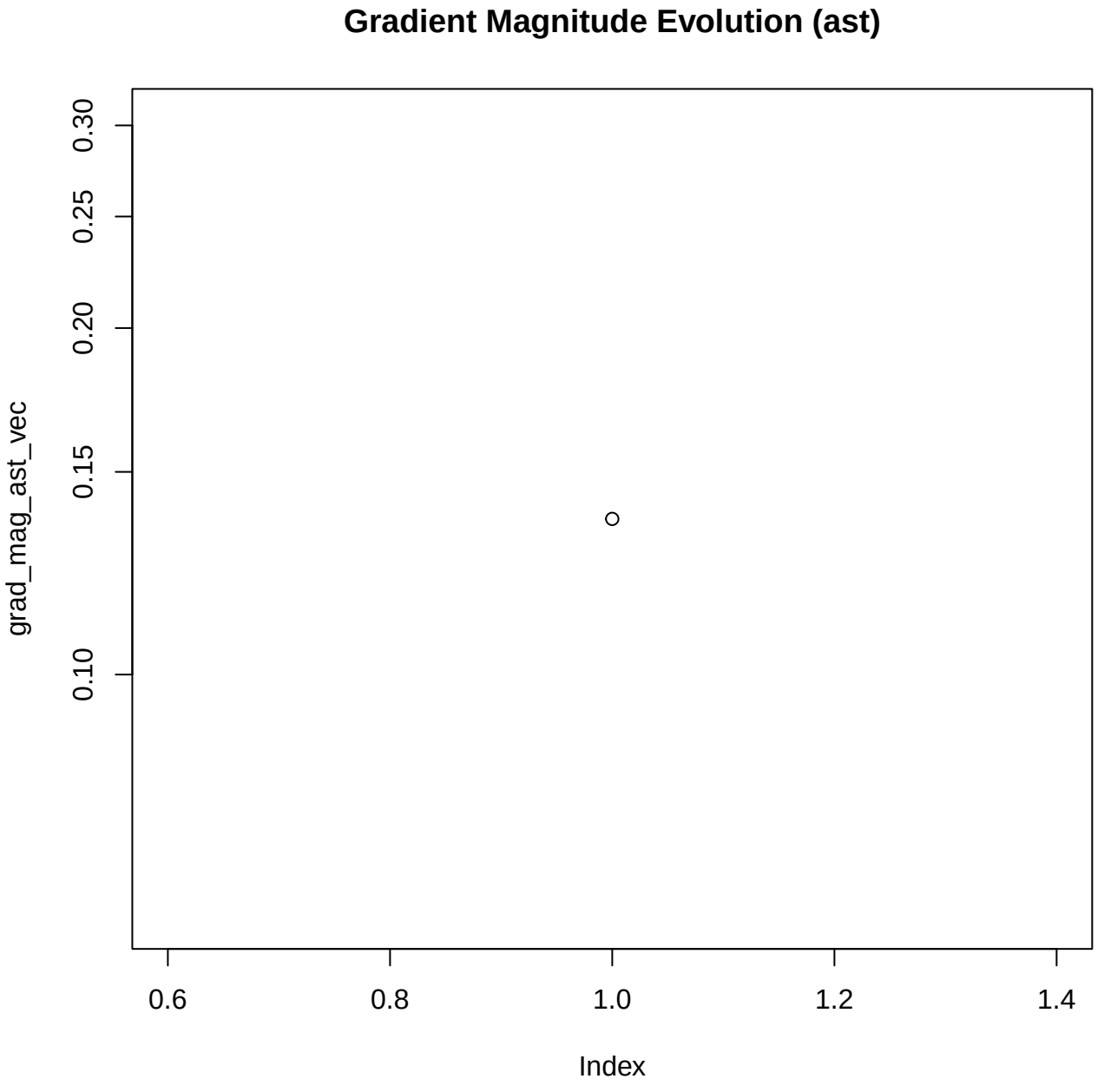
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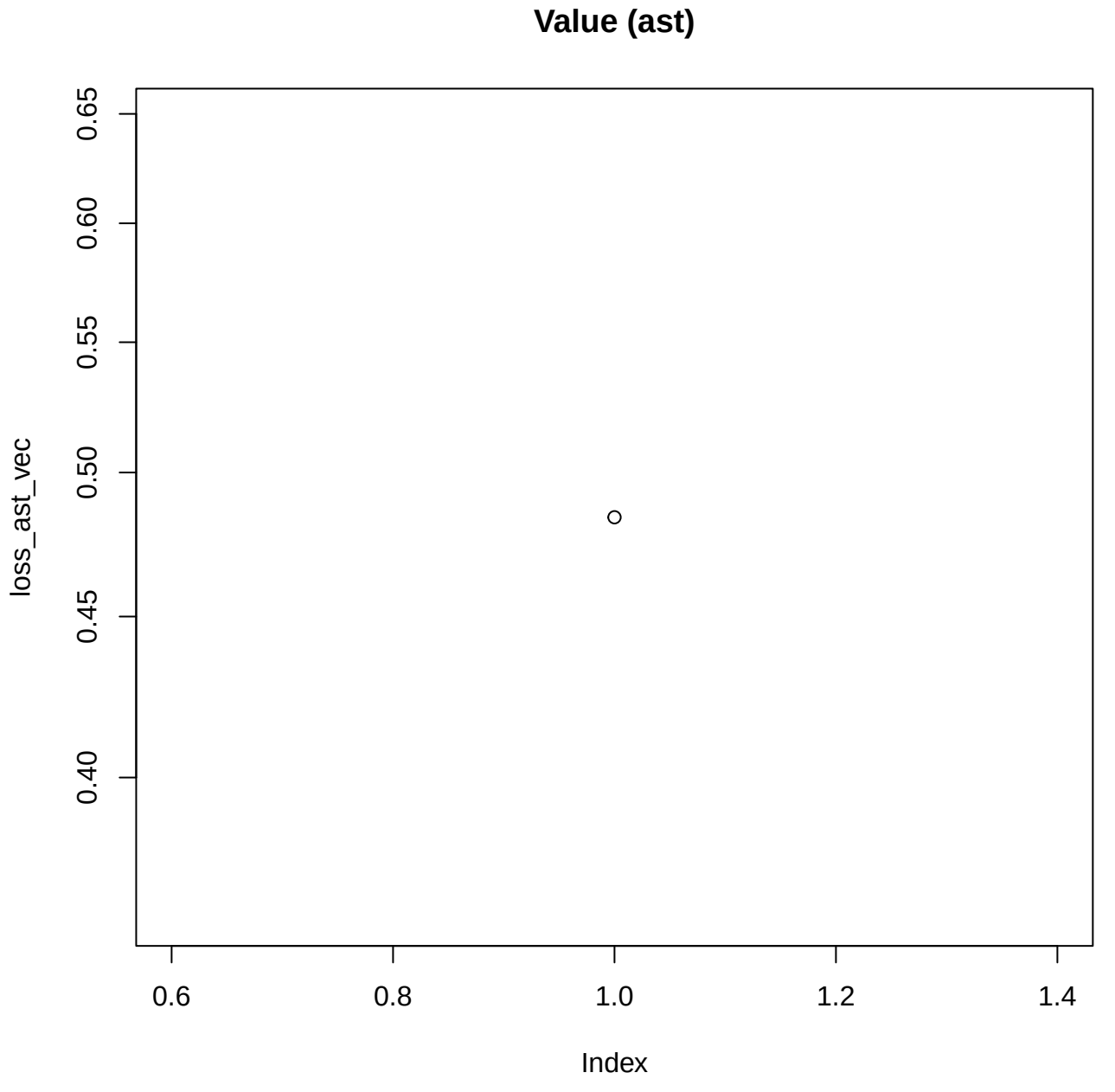
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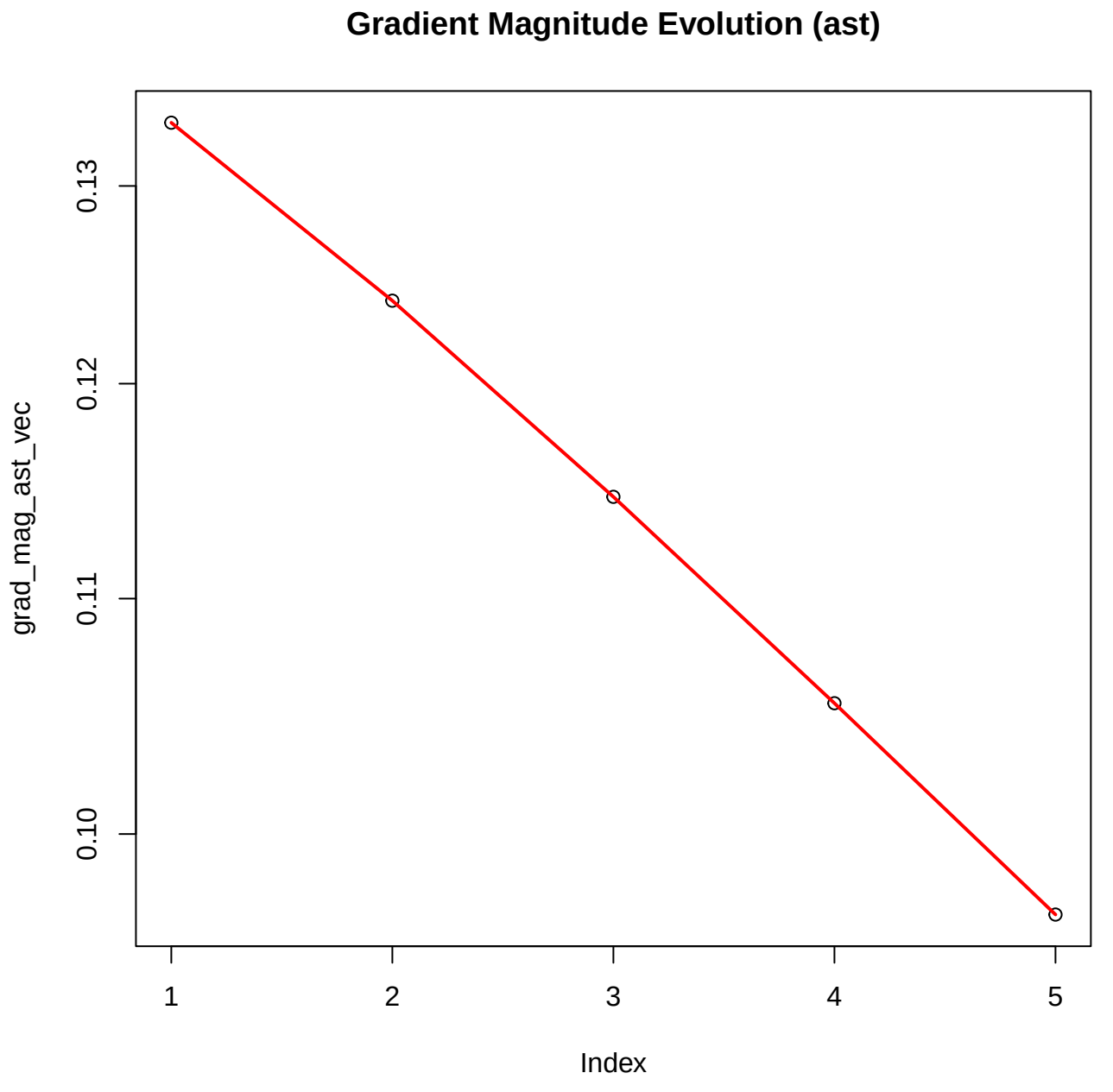
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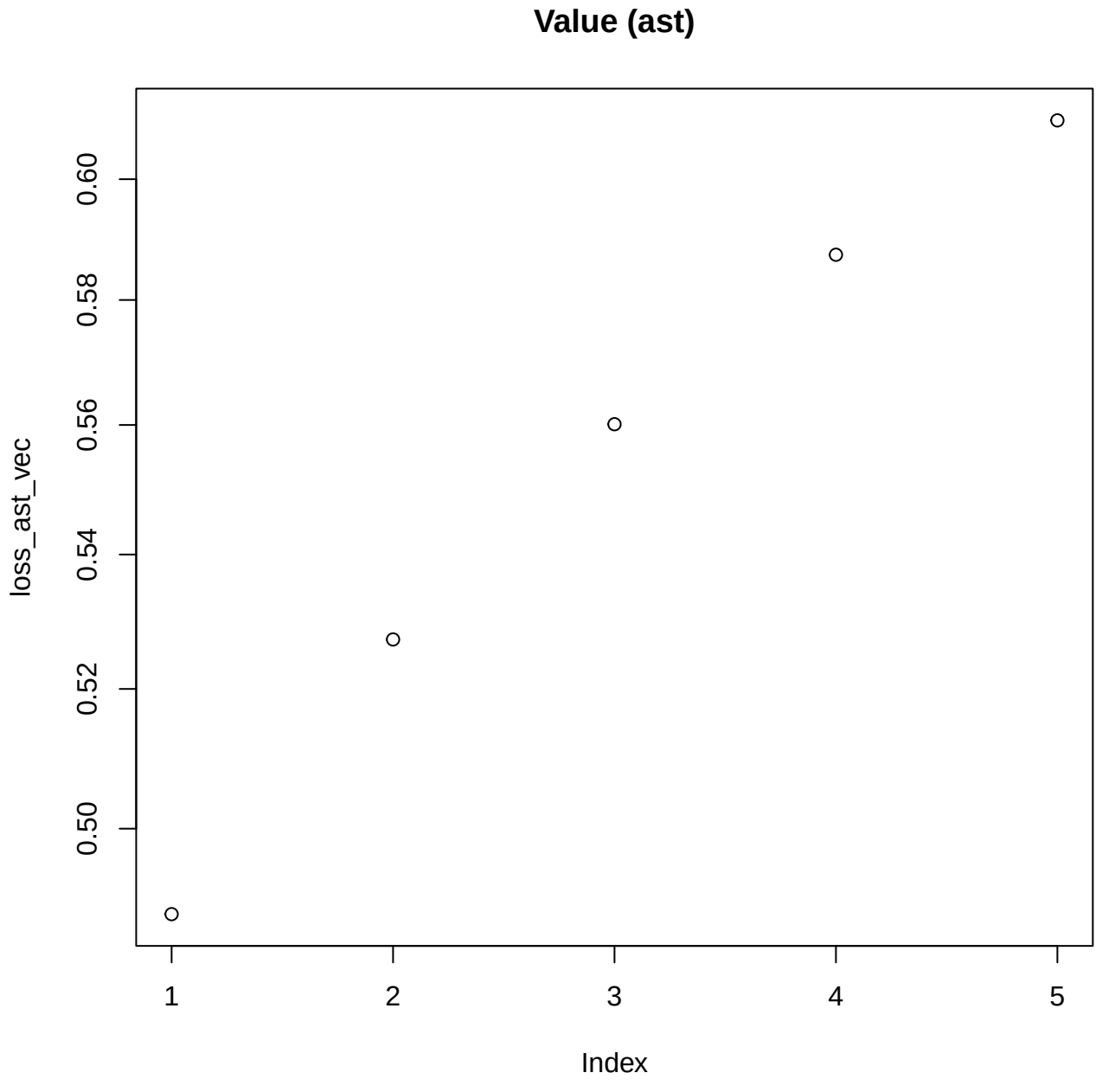
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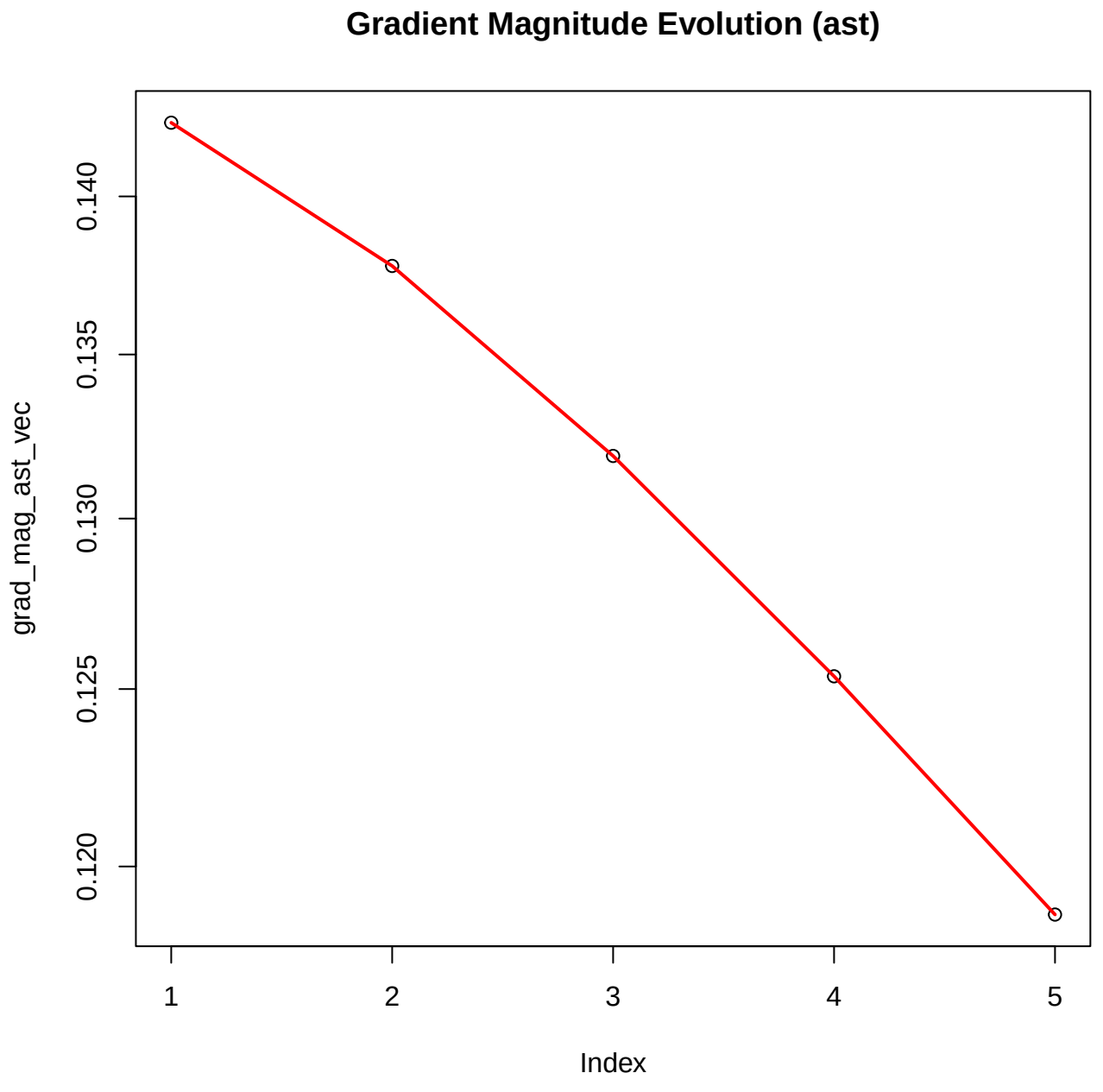
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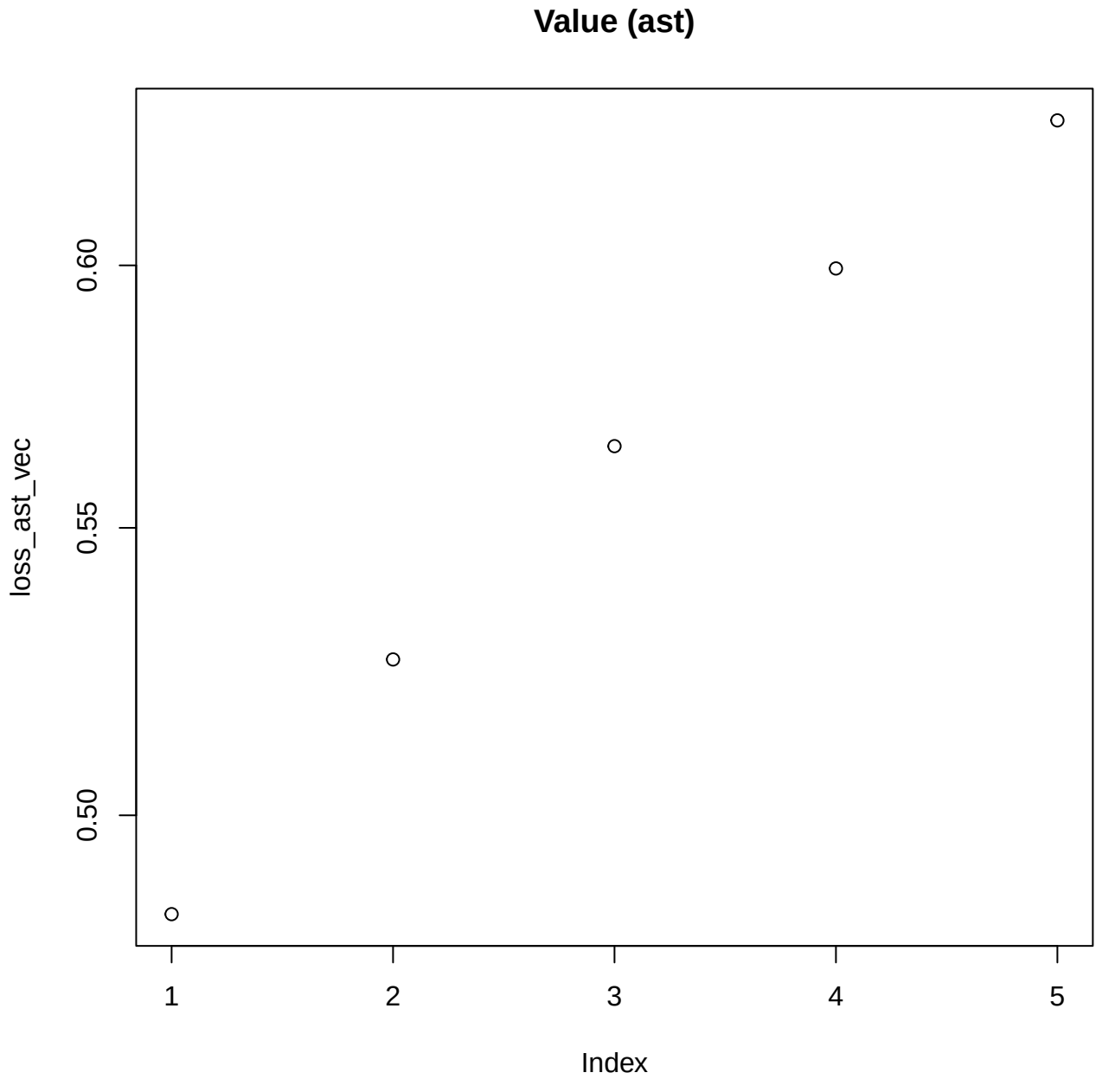
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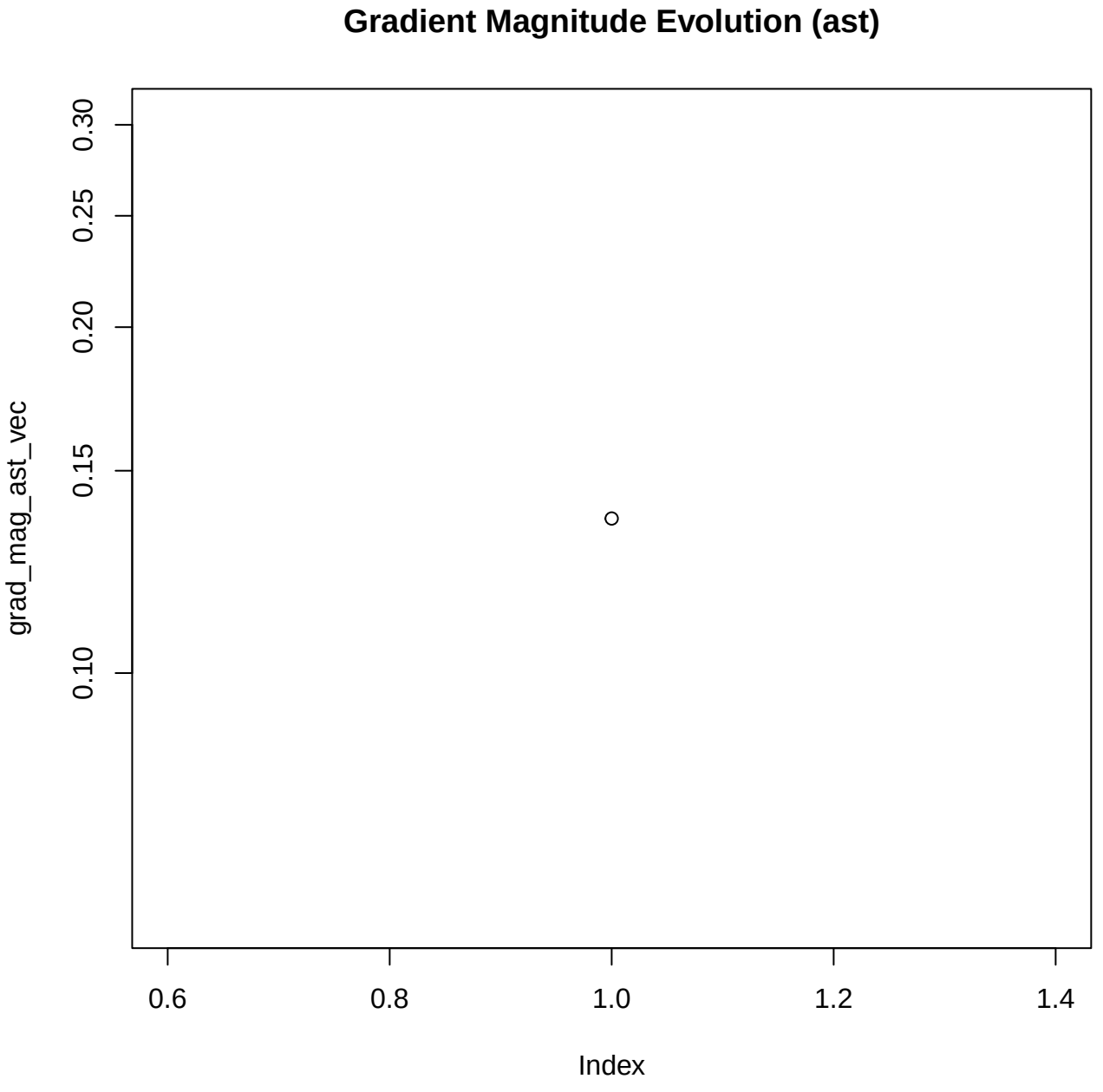
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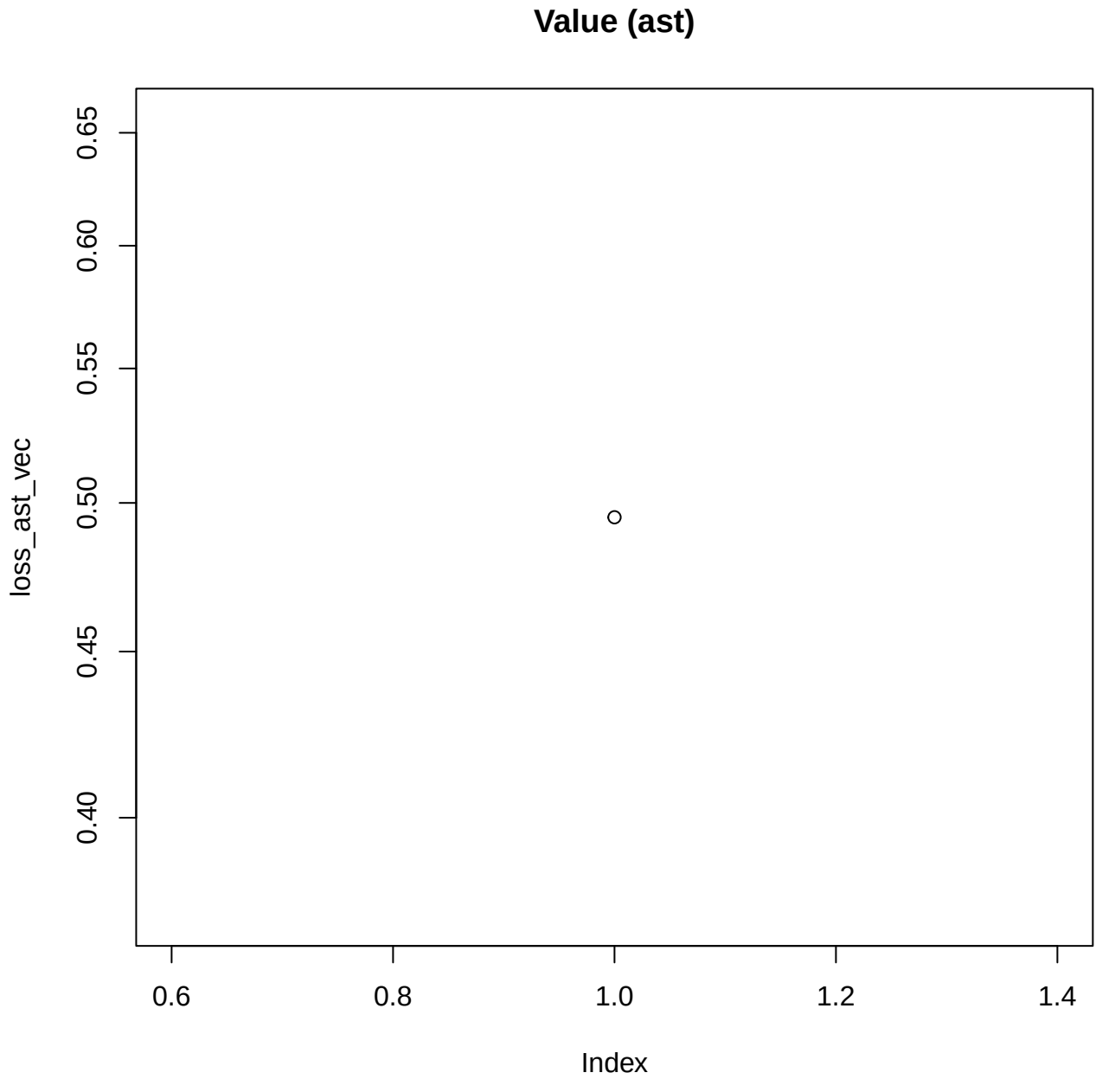
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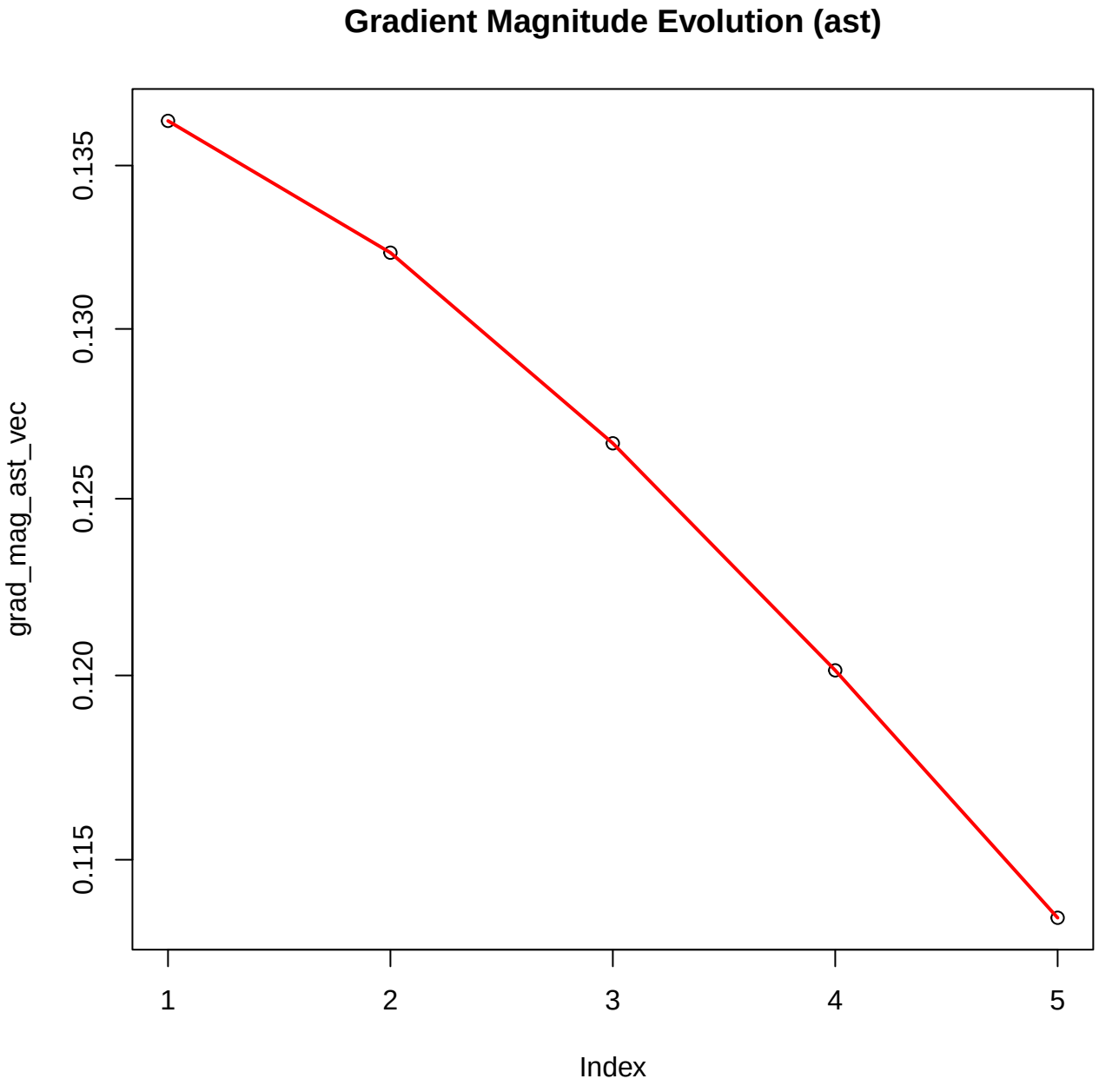
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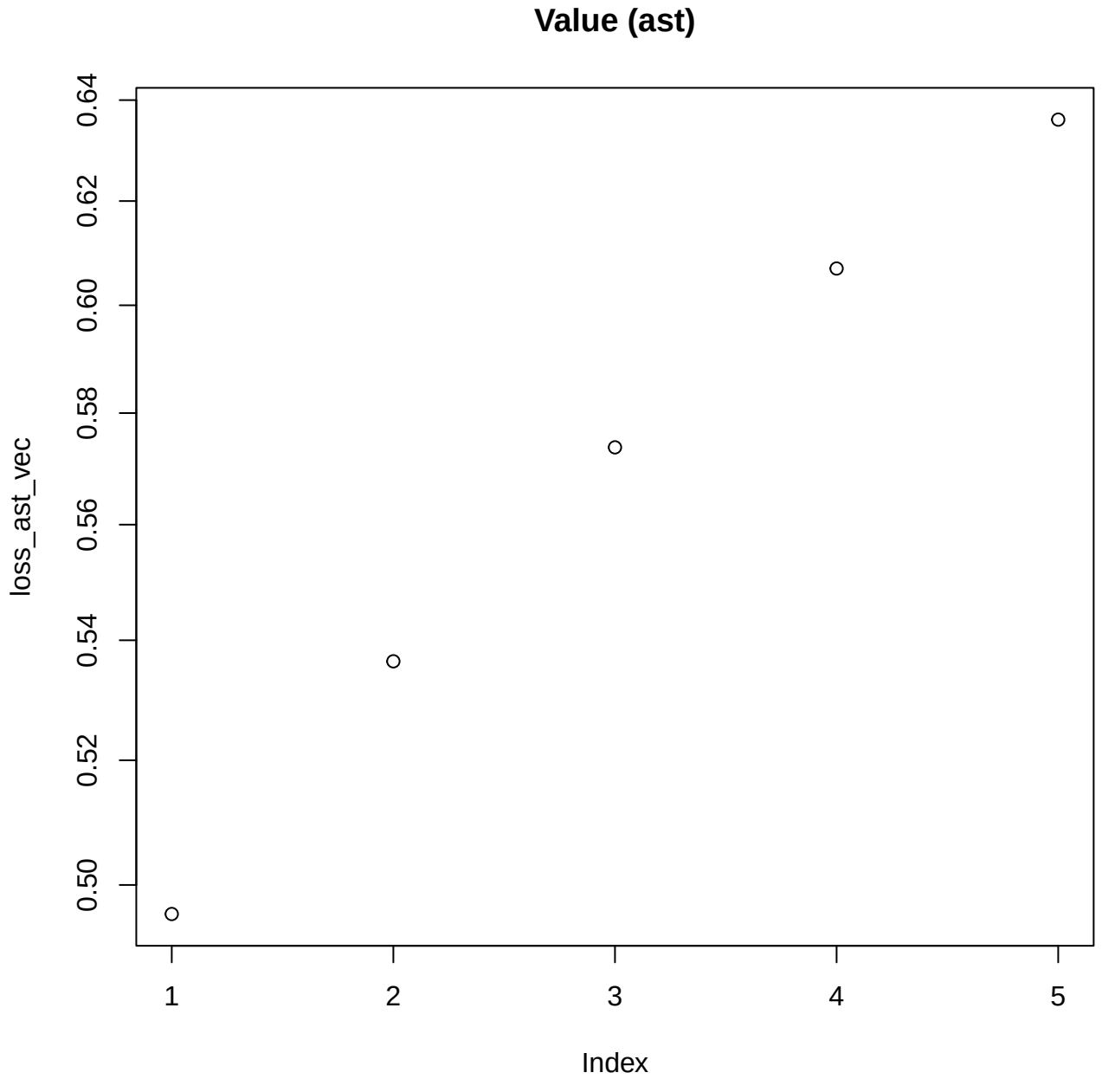
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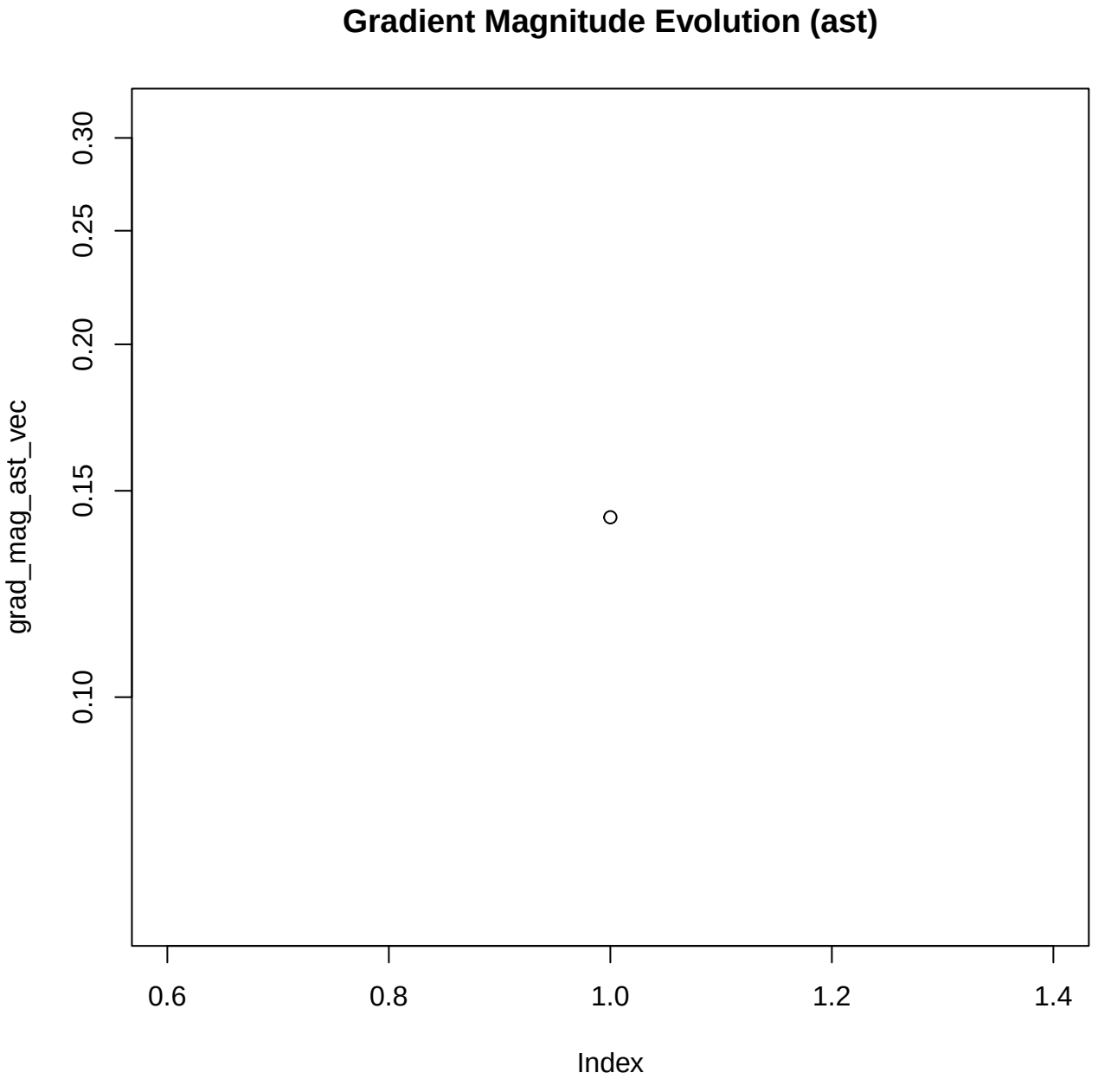
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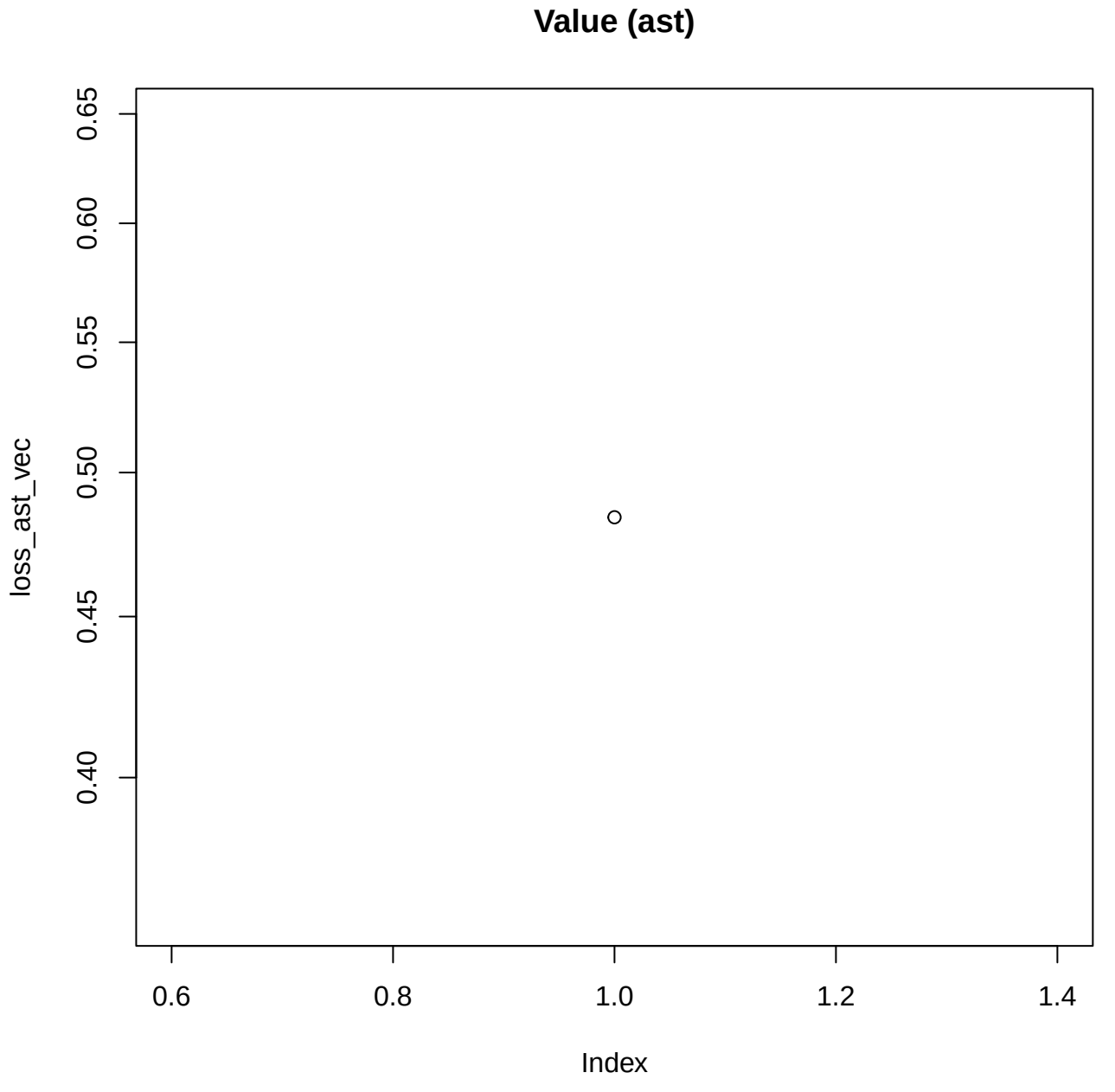
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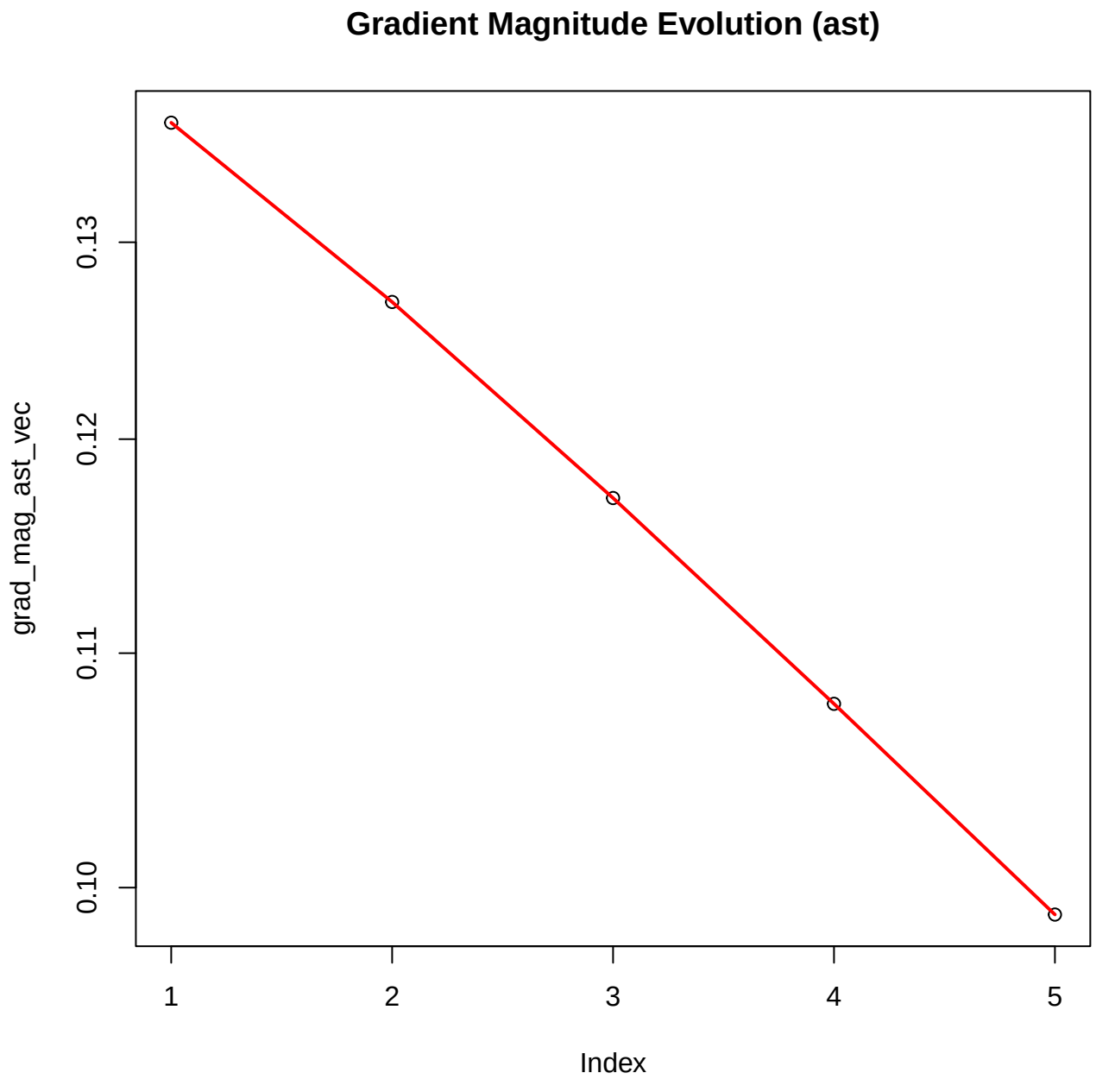
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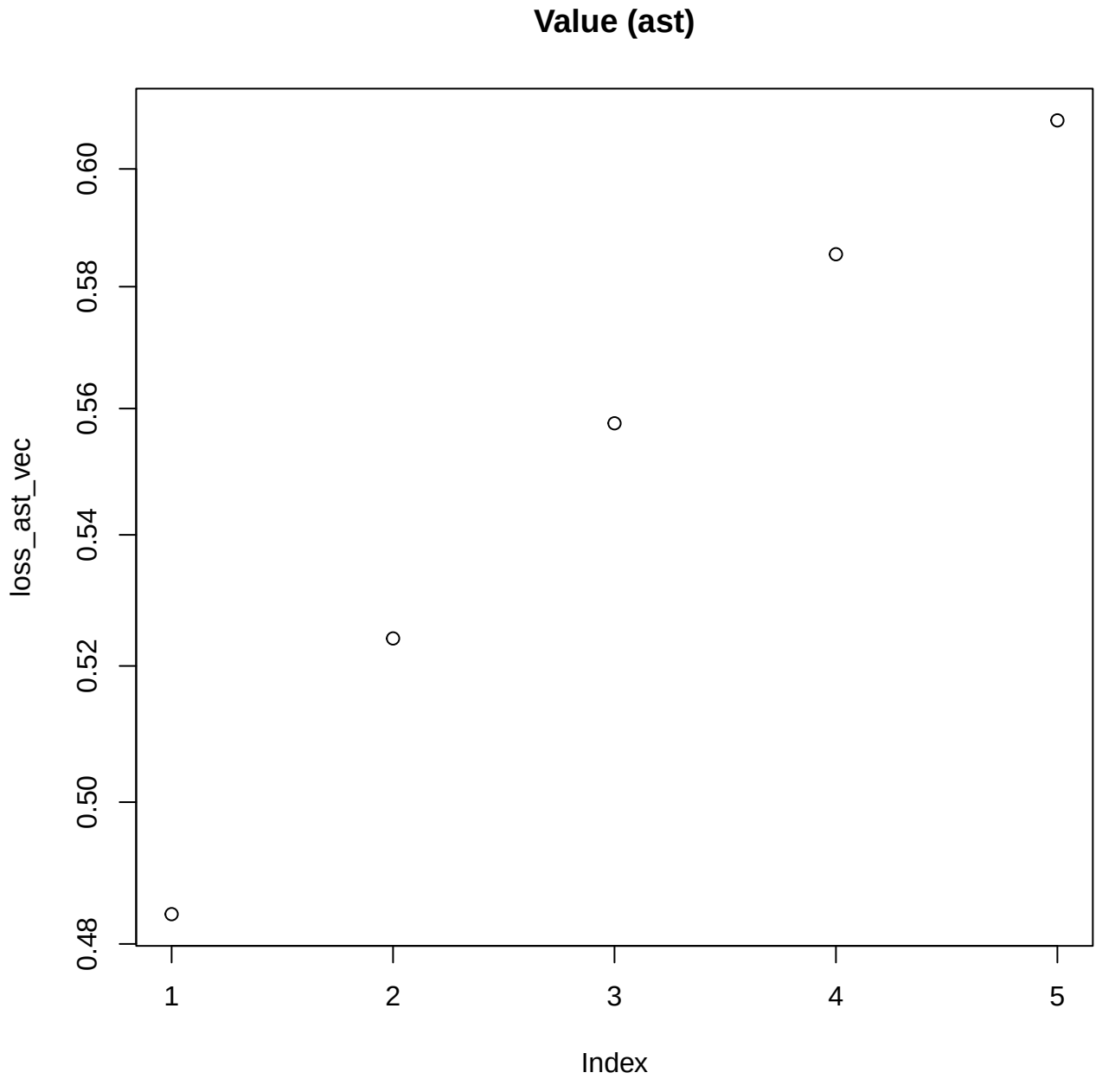
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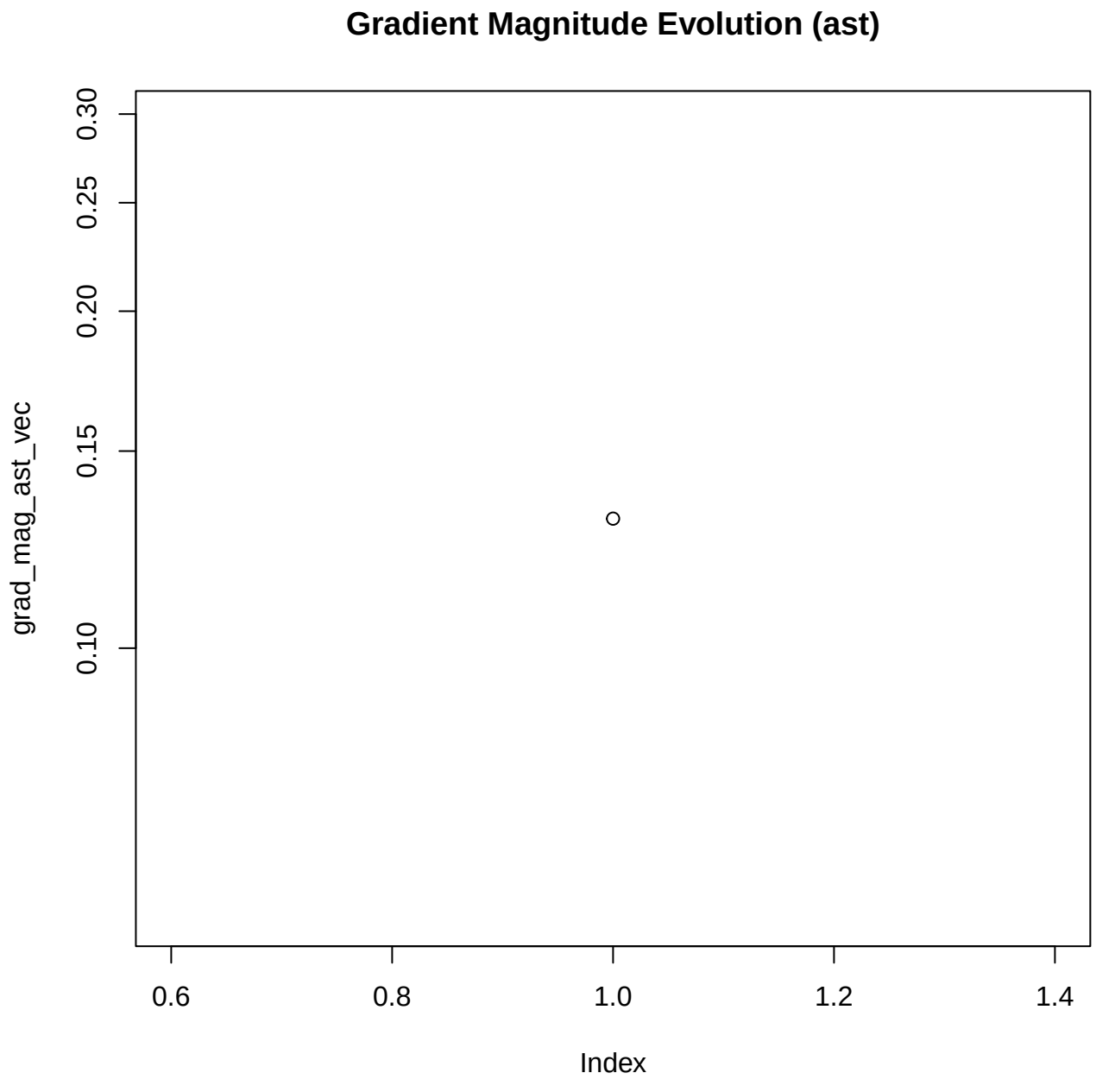
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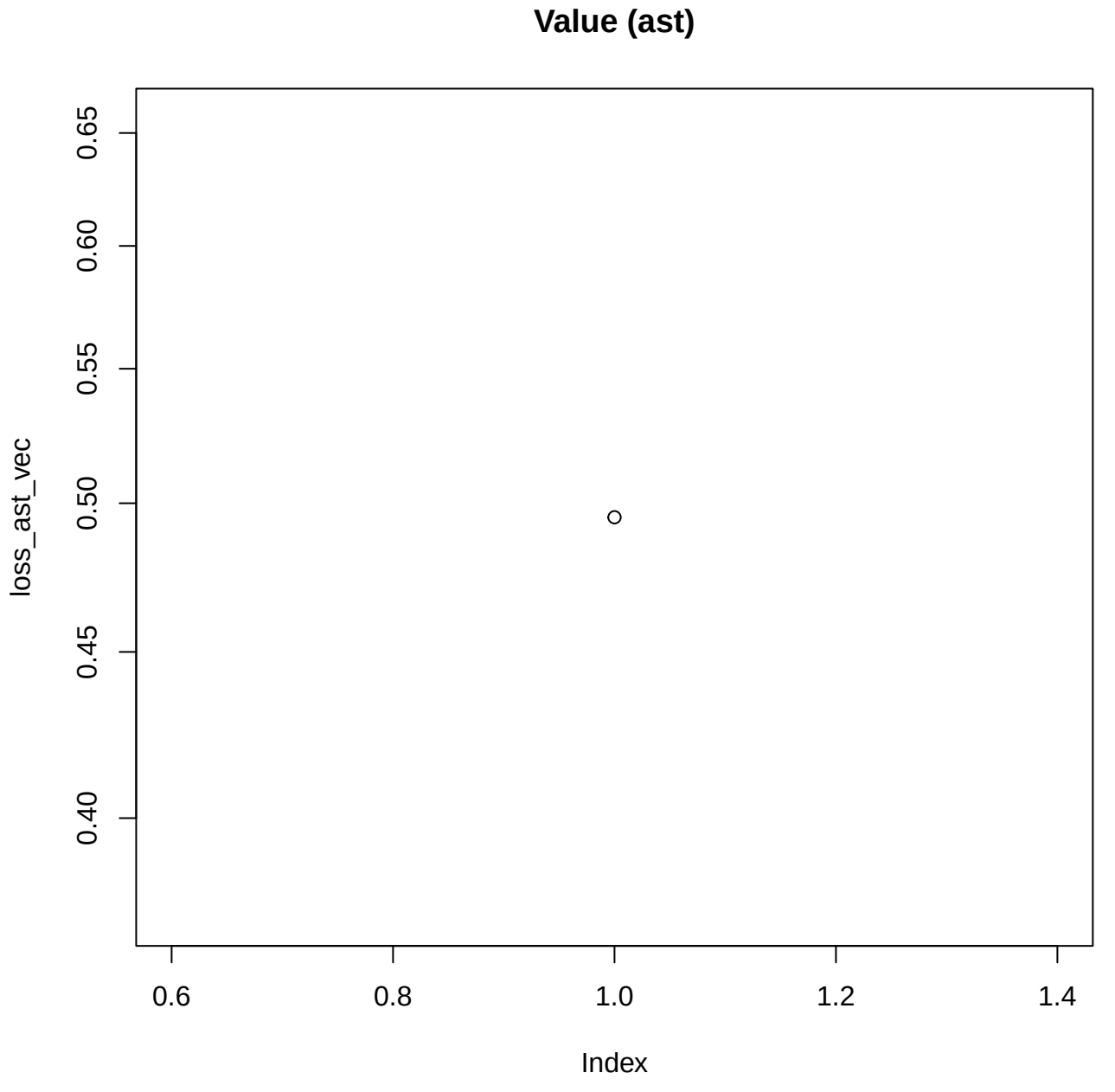
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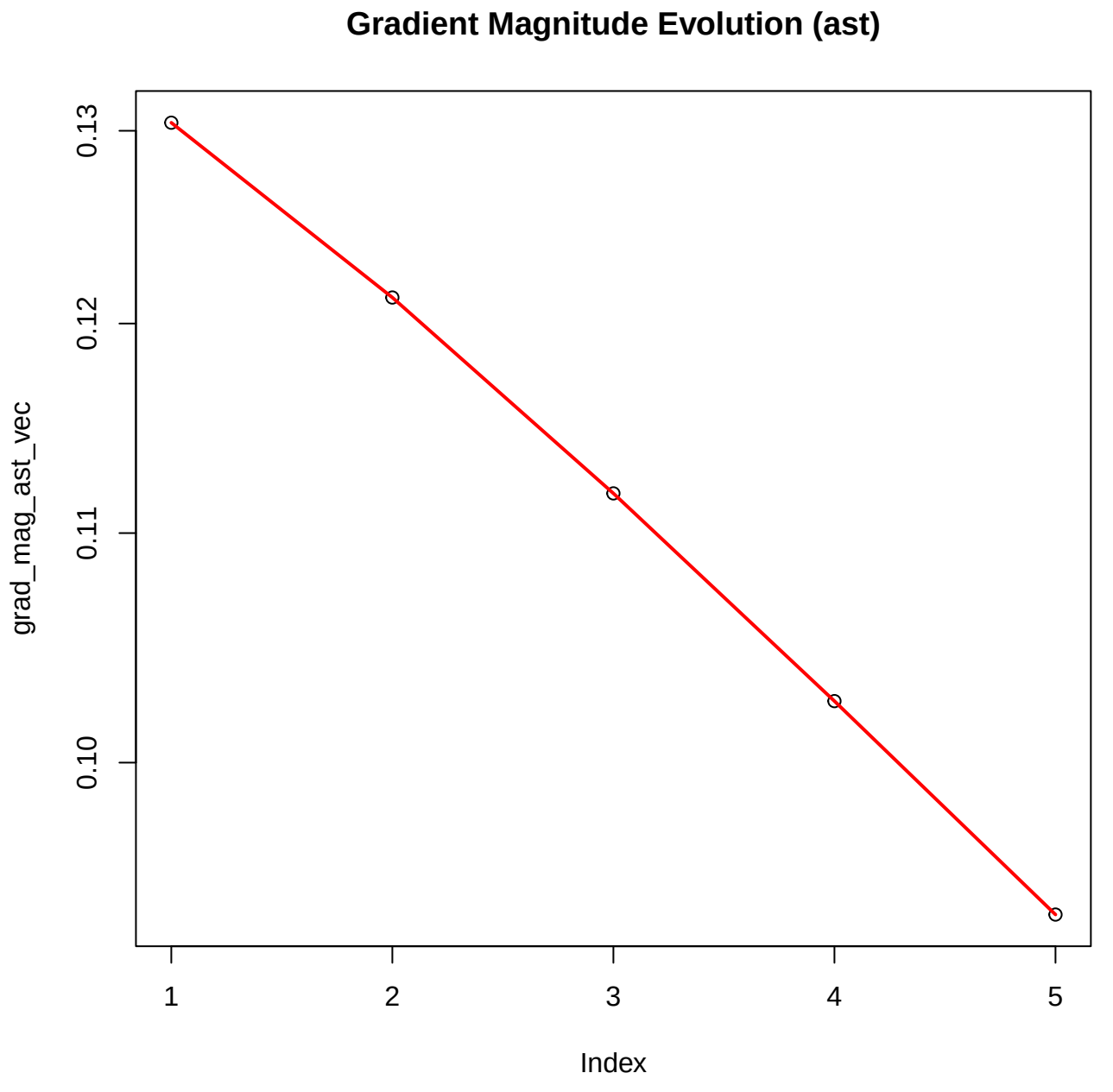
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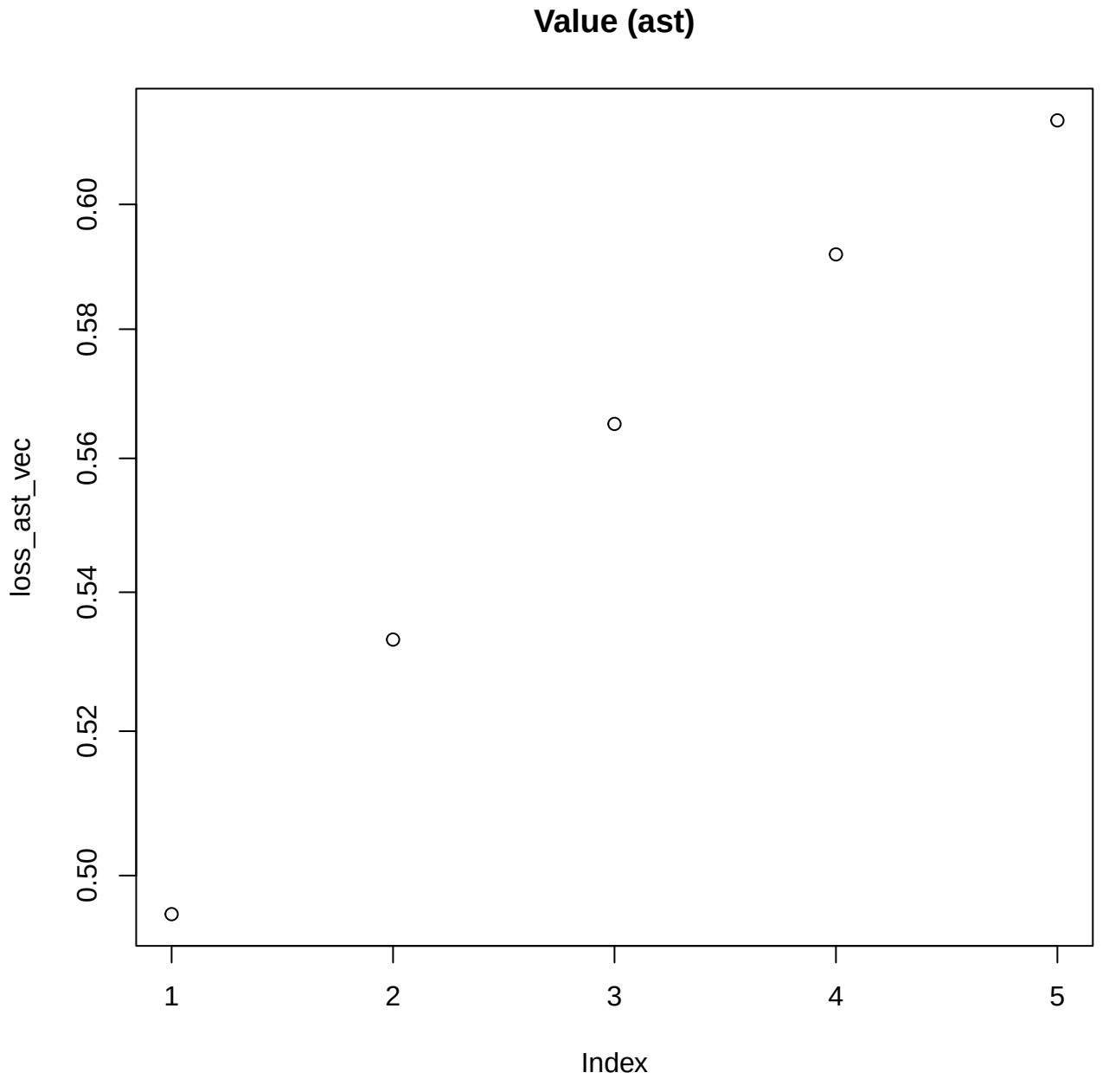
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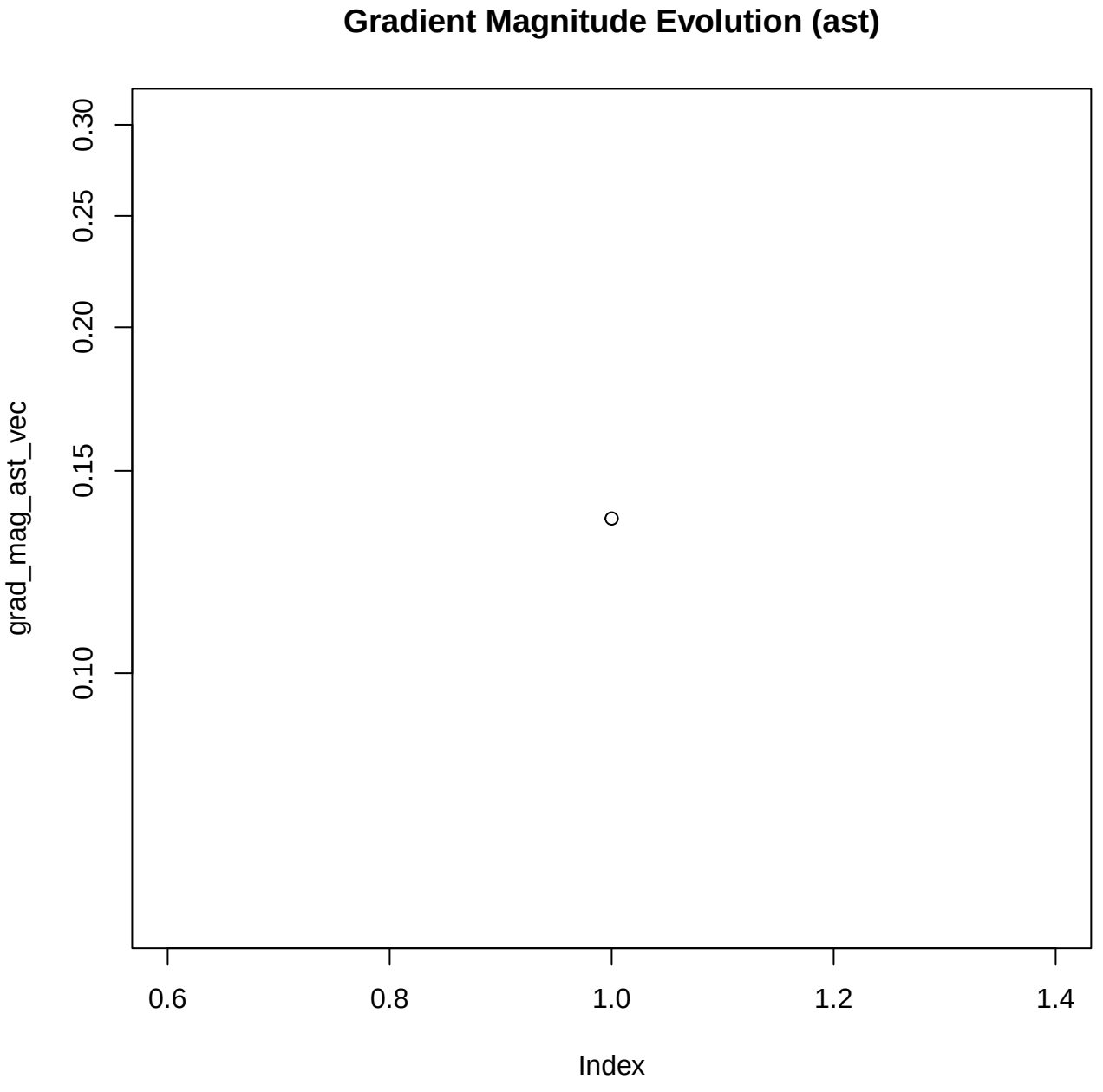
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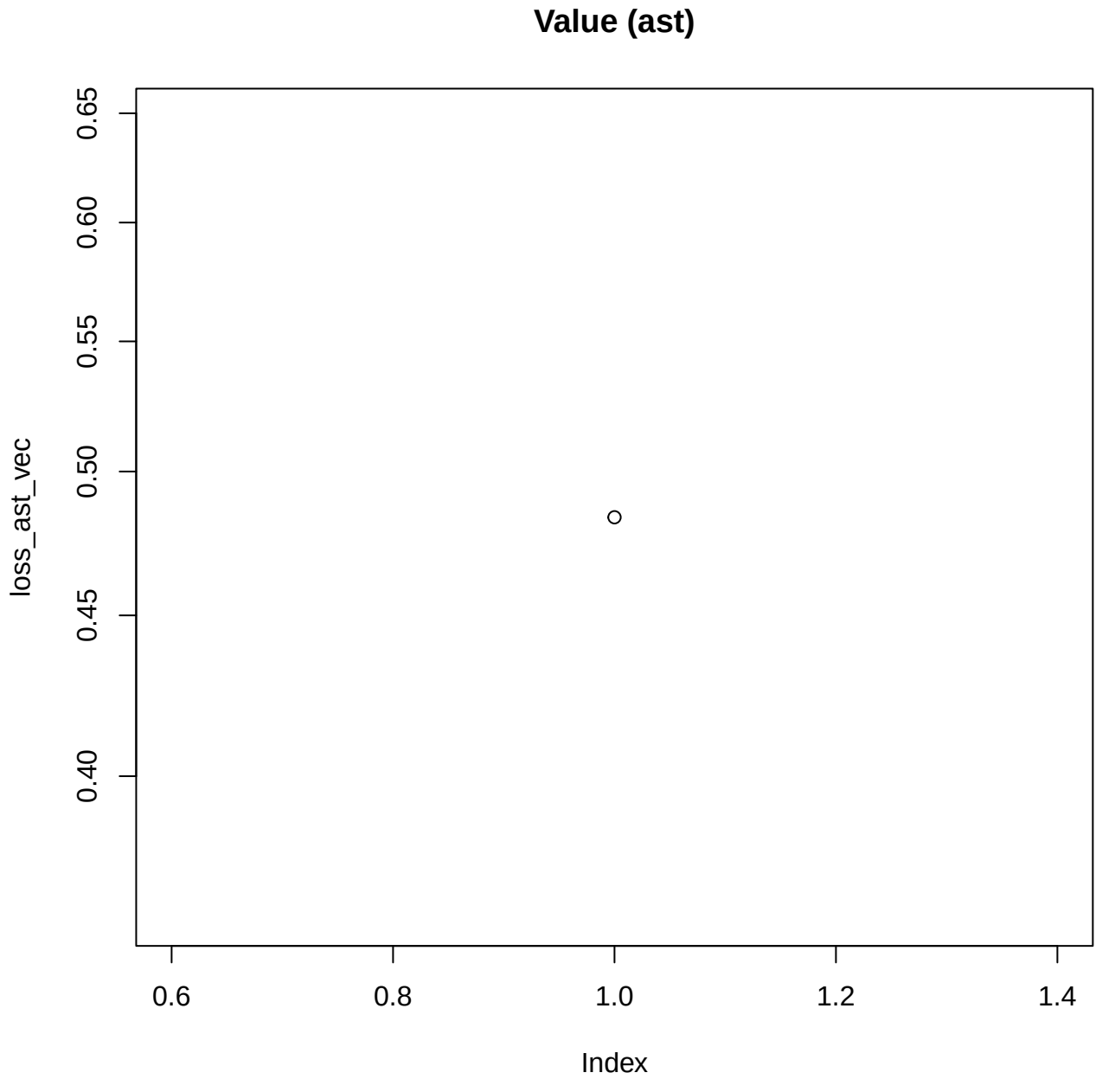
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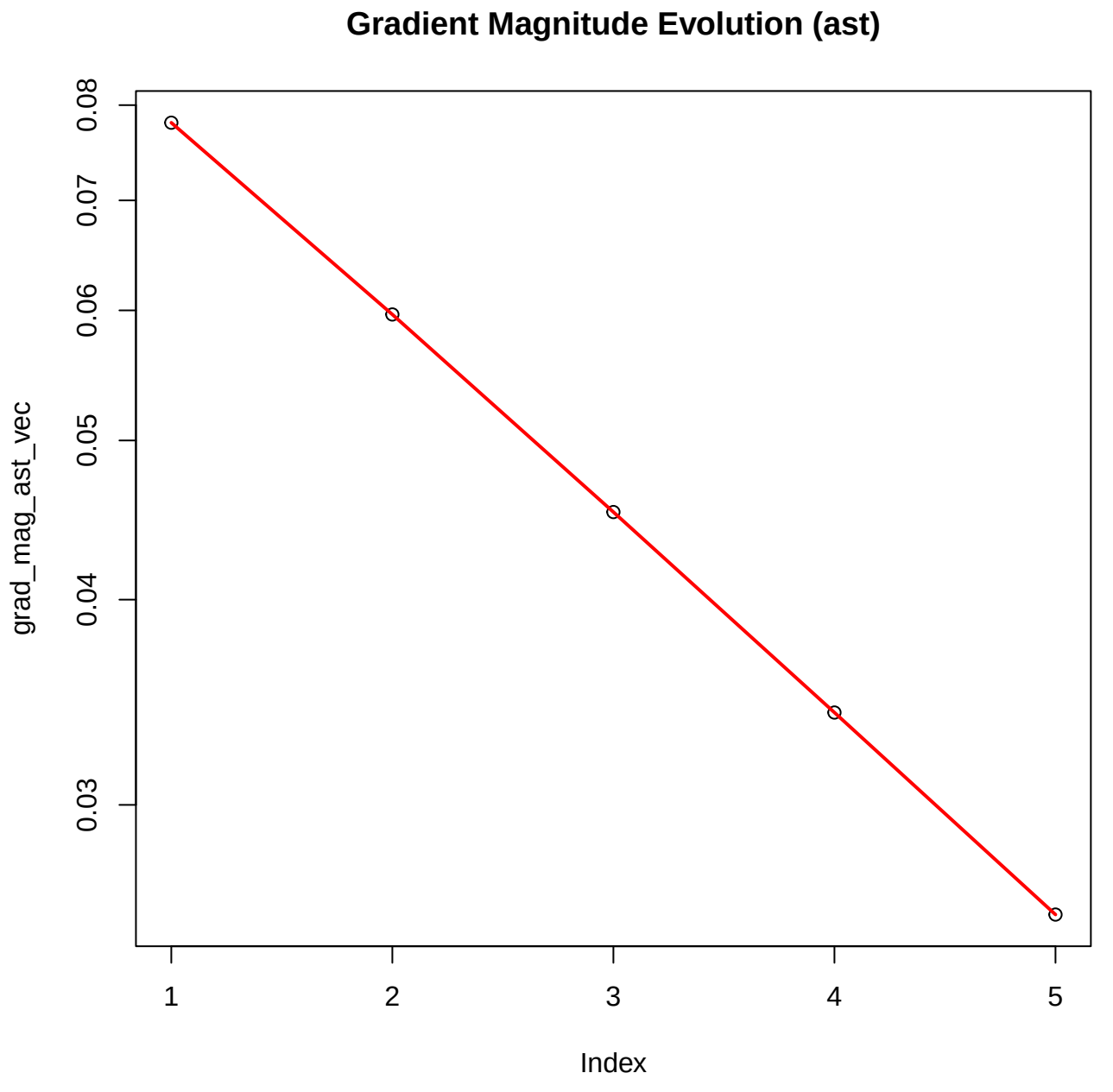
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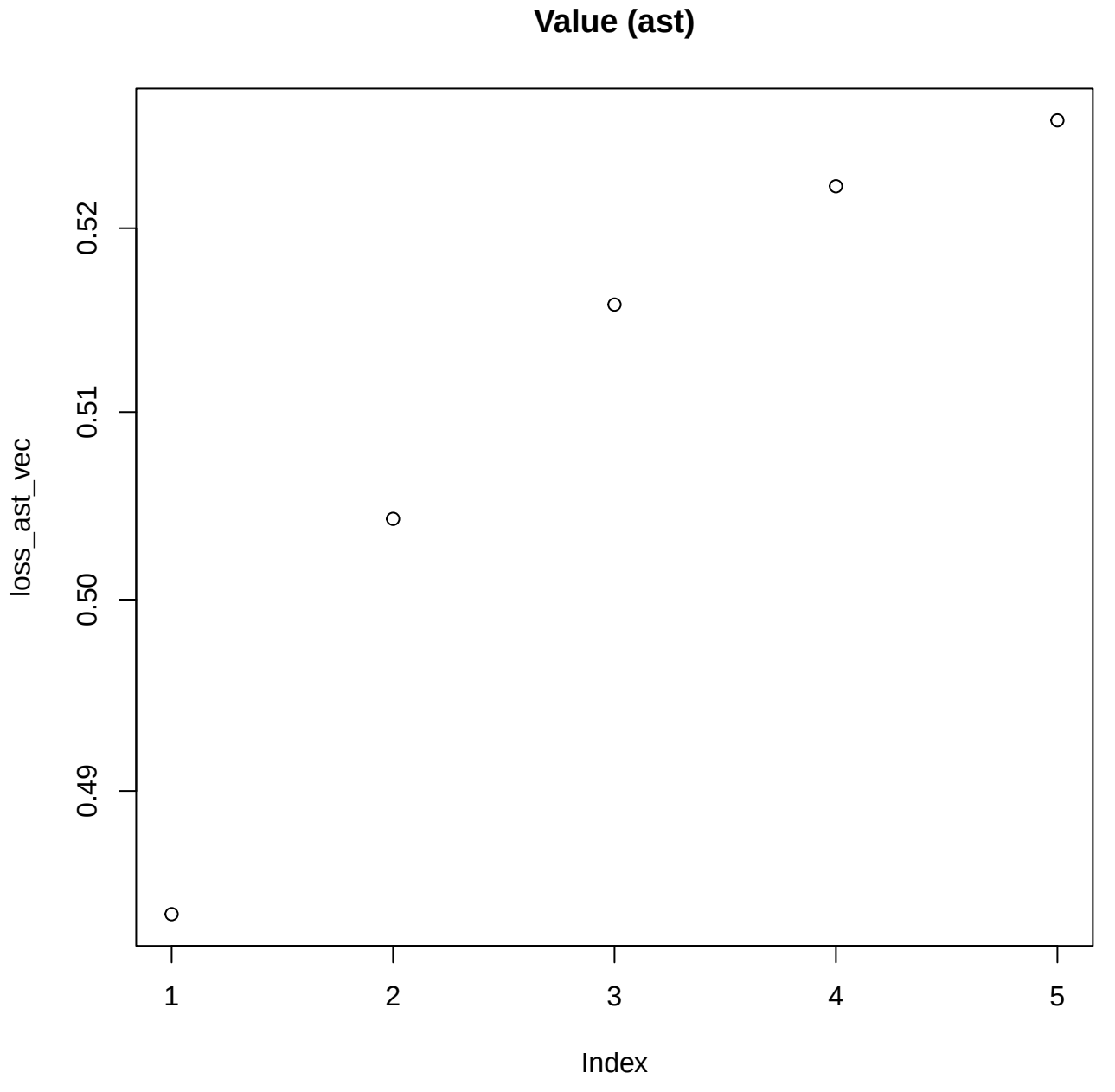
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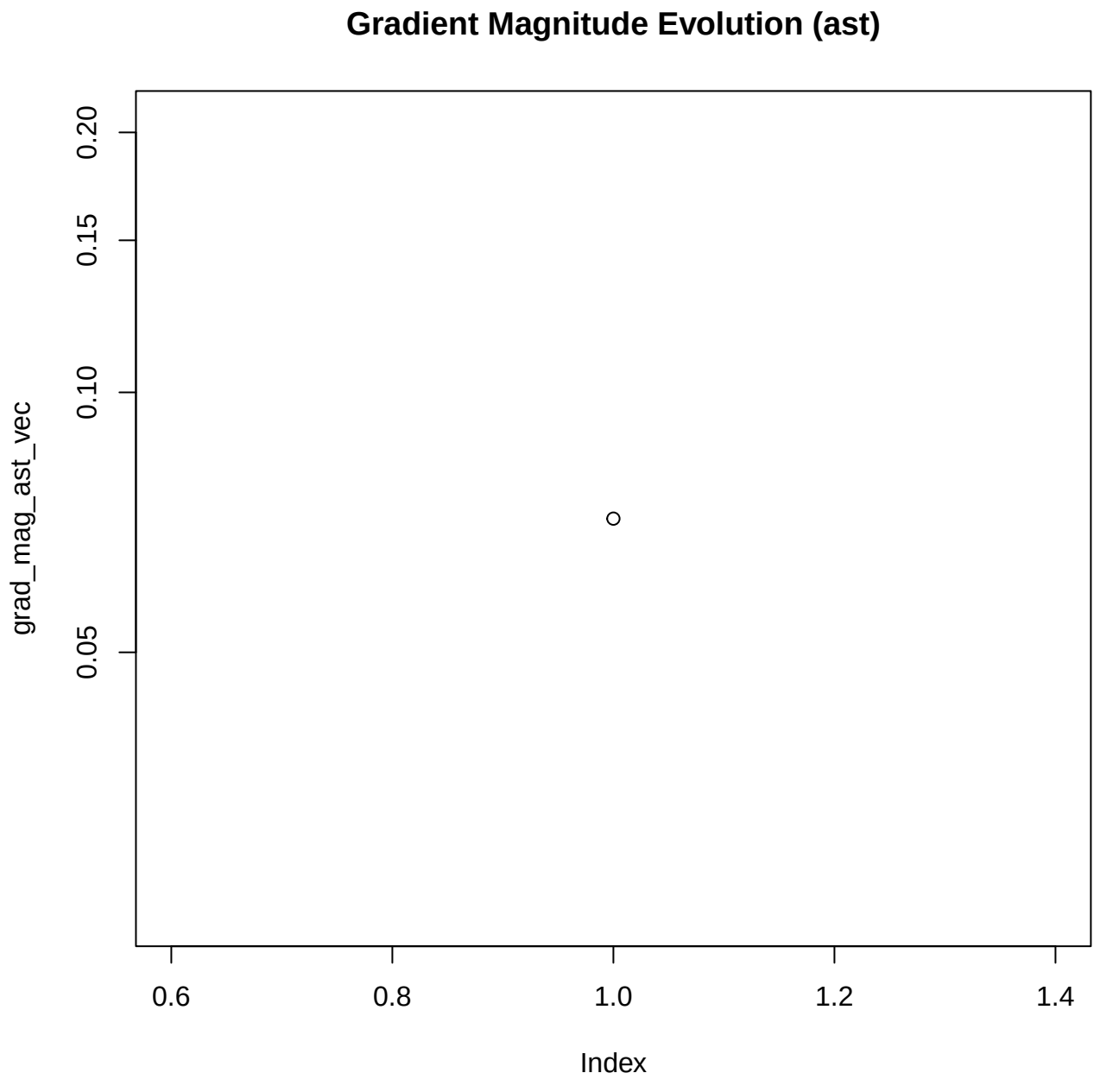
Gradient Magnitude Evolution (ast)
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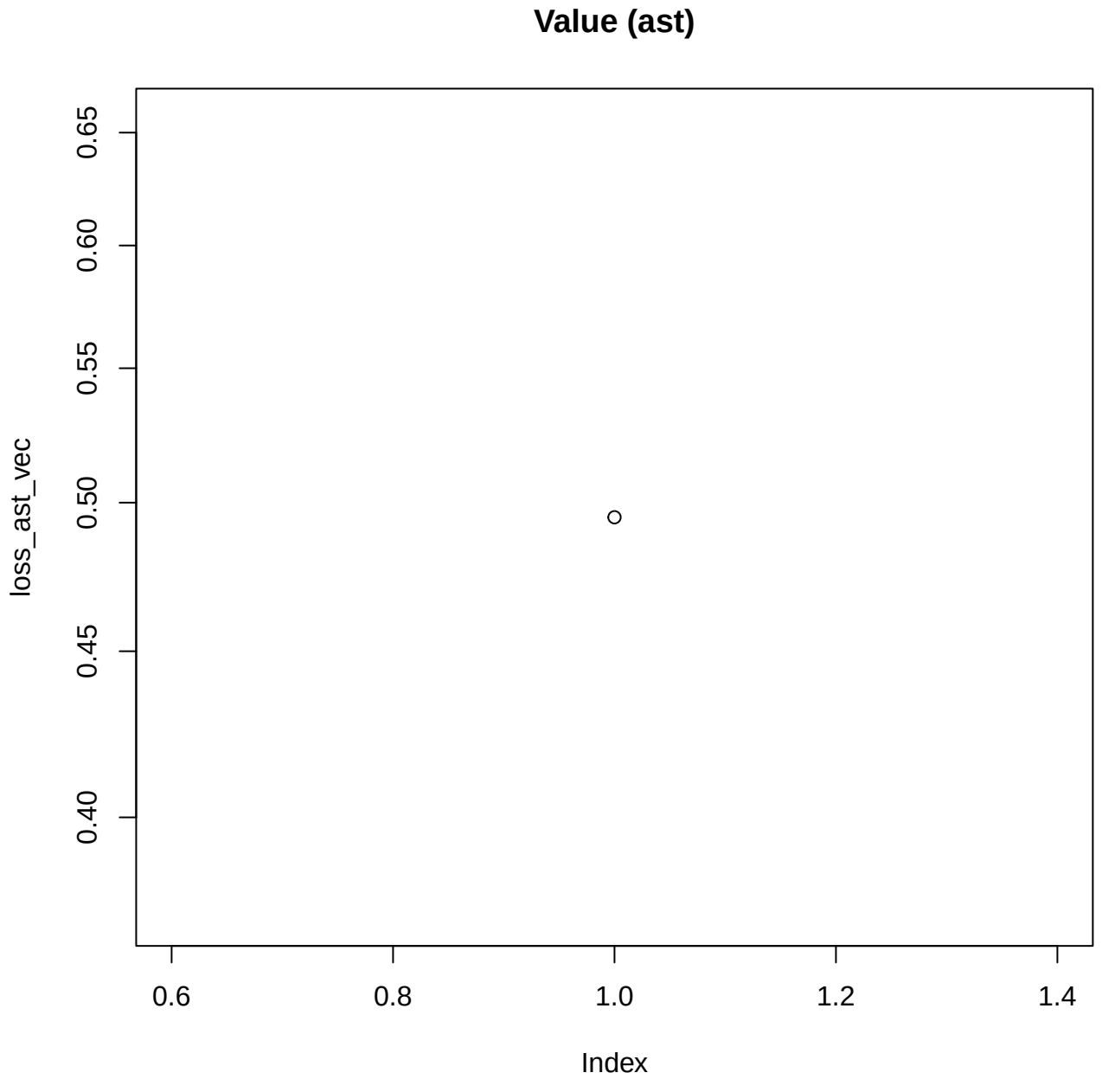
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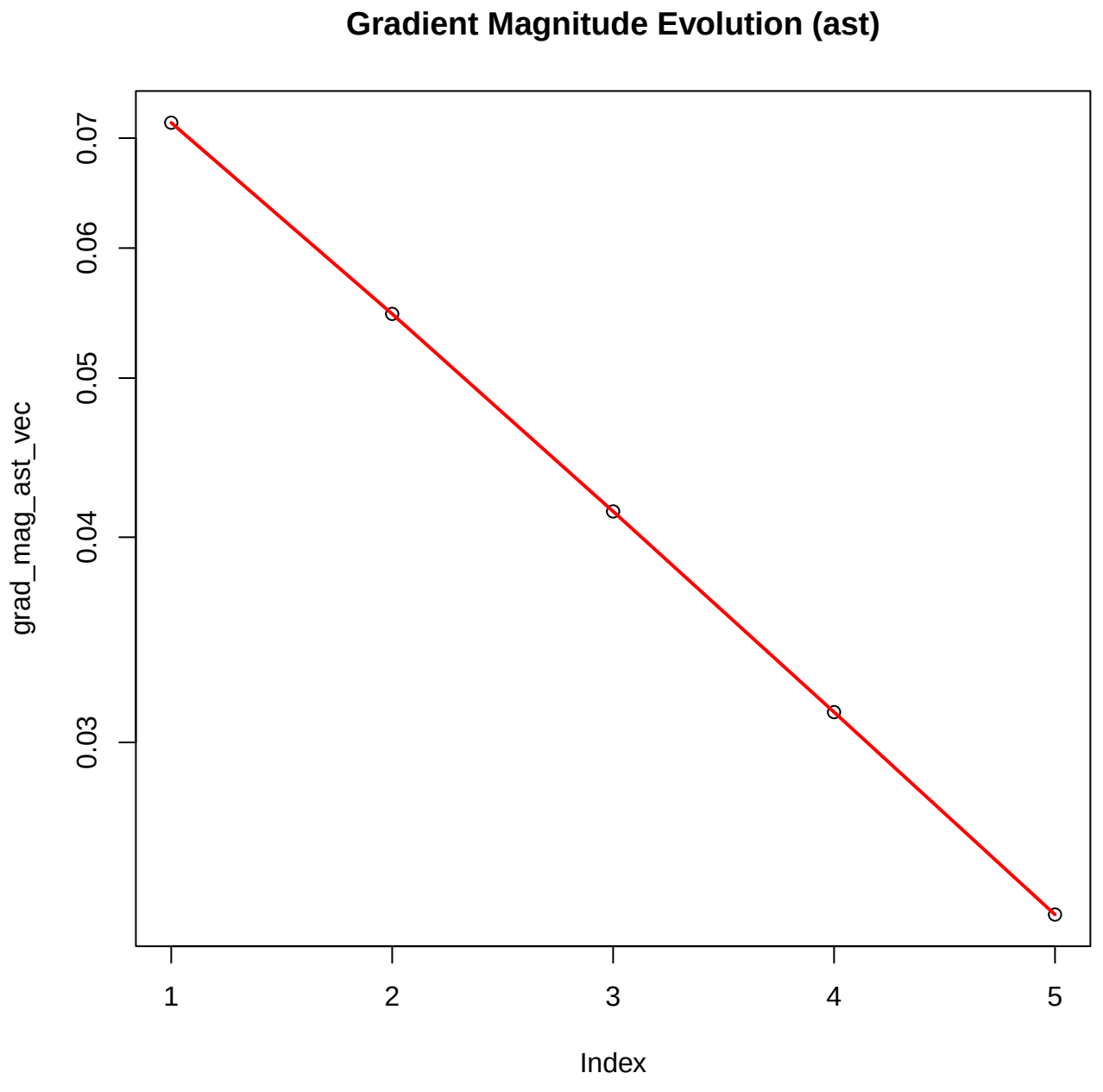
Gradient Magnitude Evolution (ast)
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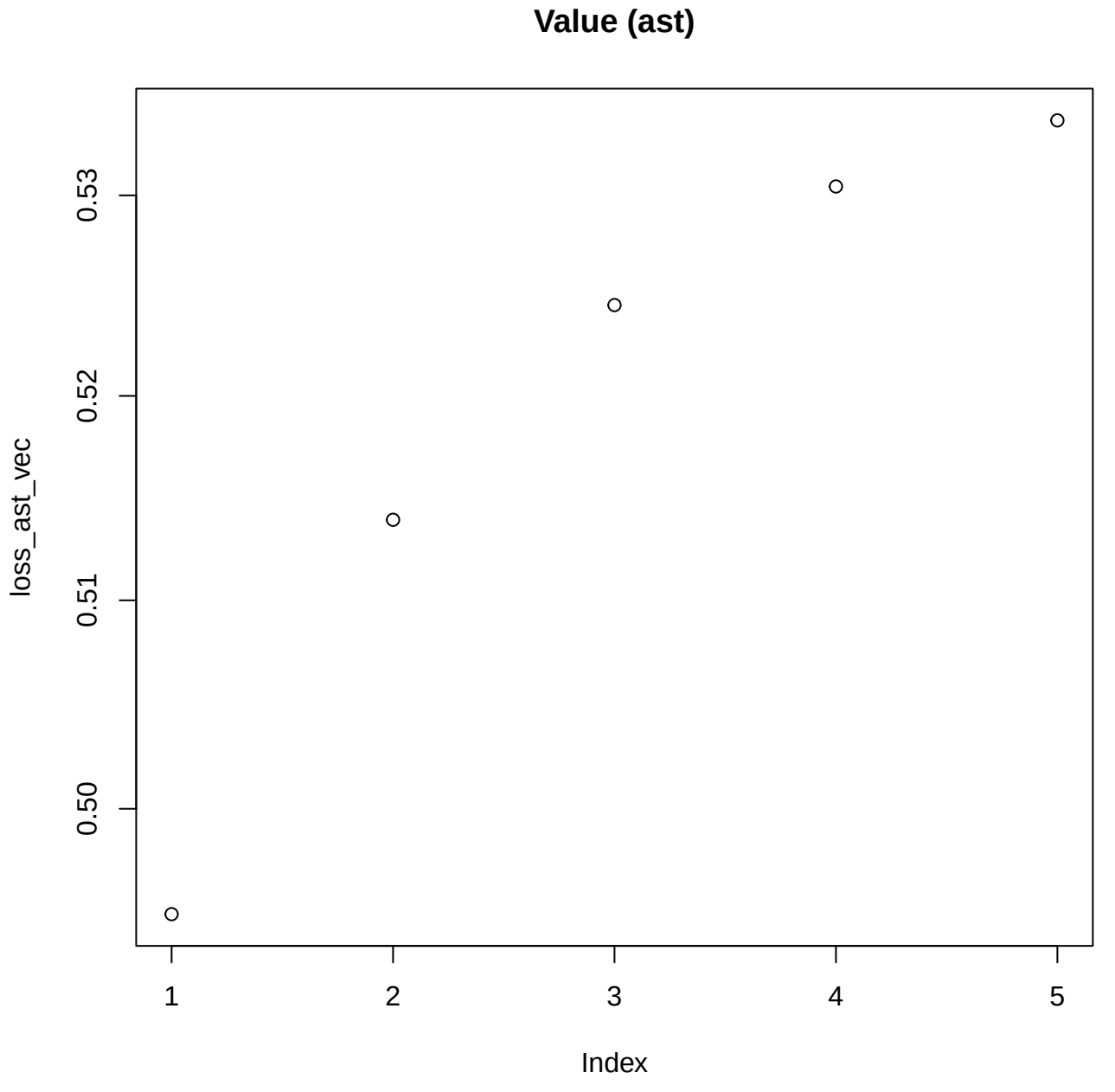
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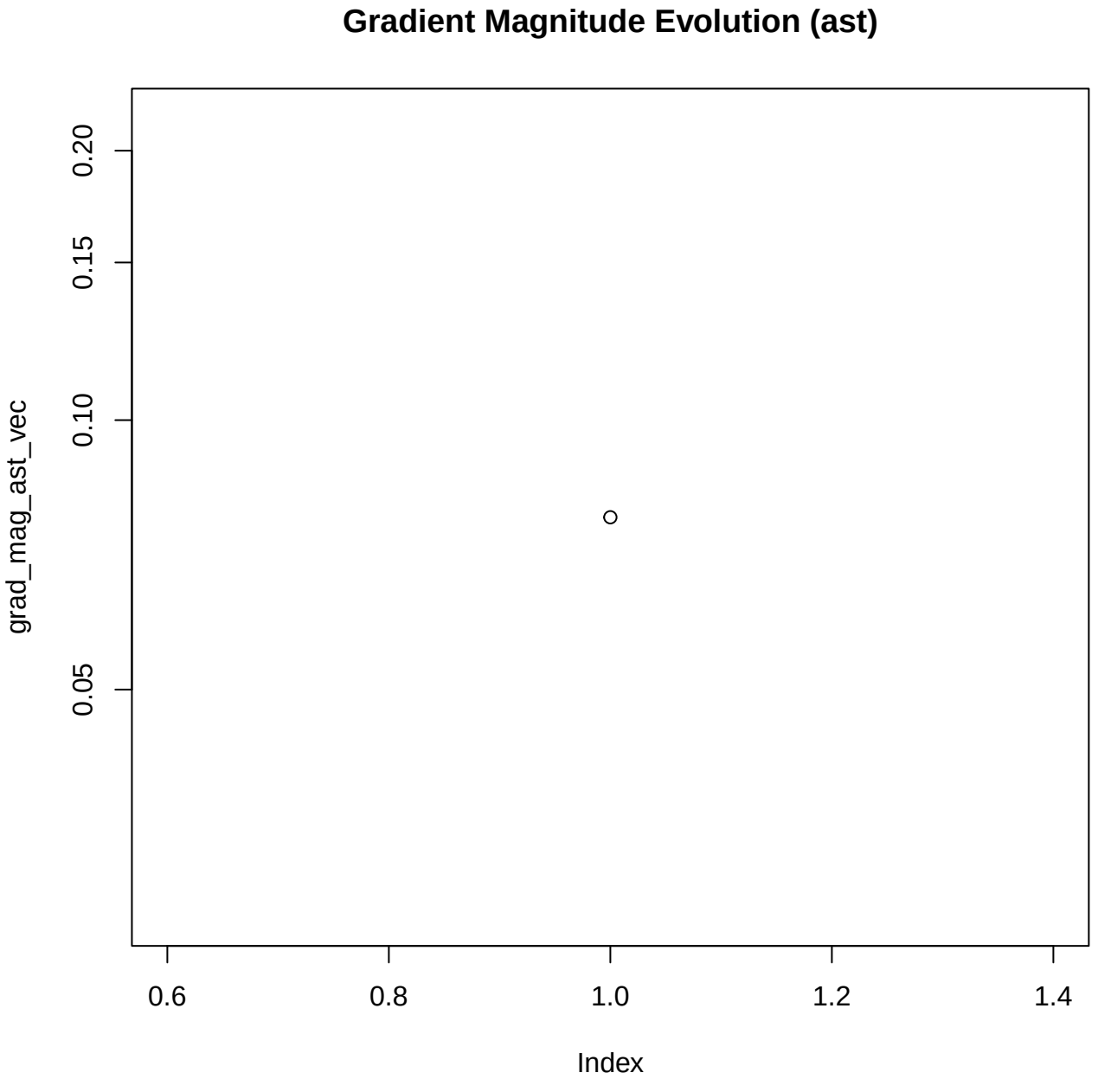
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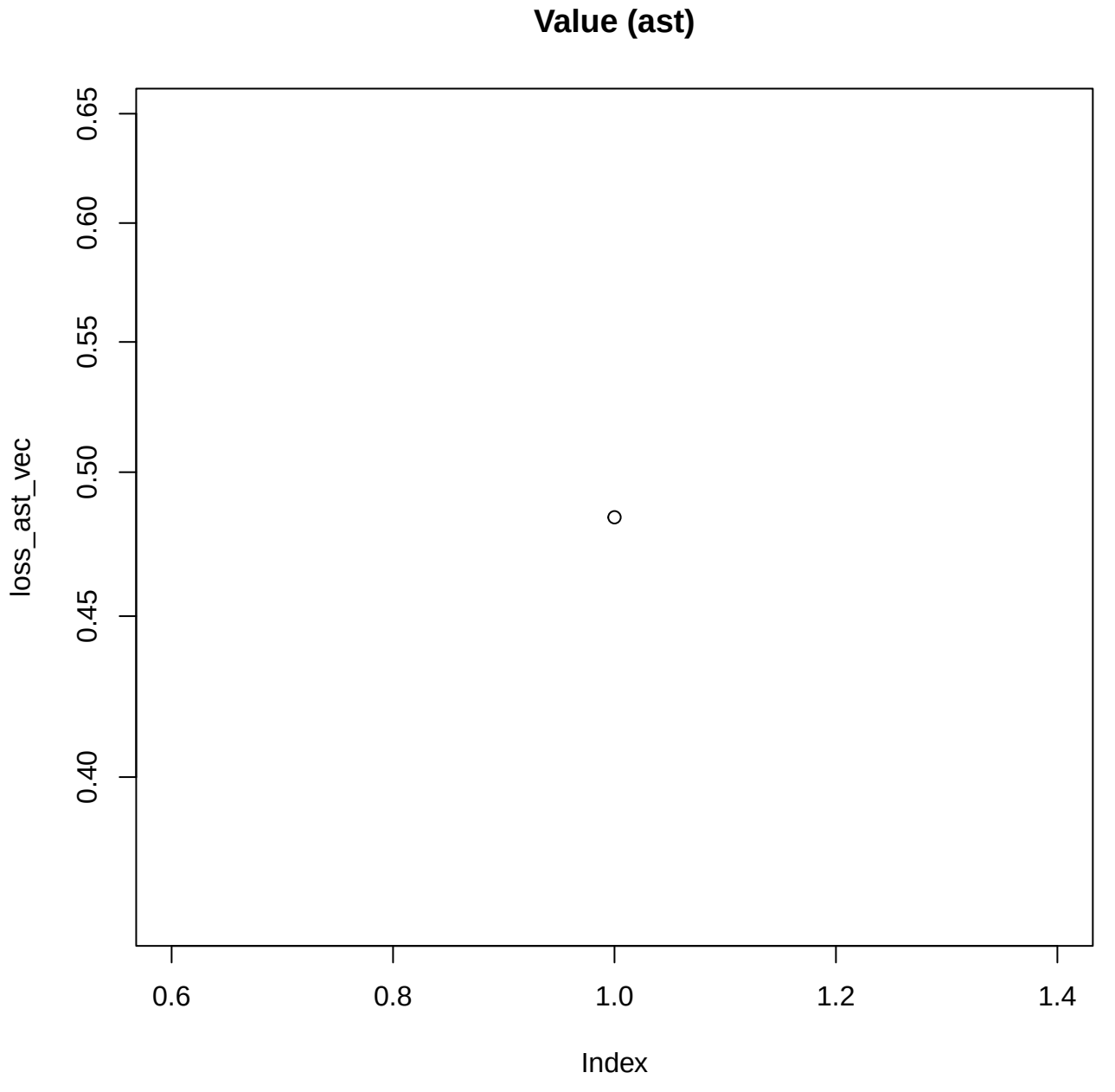
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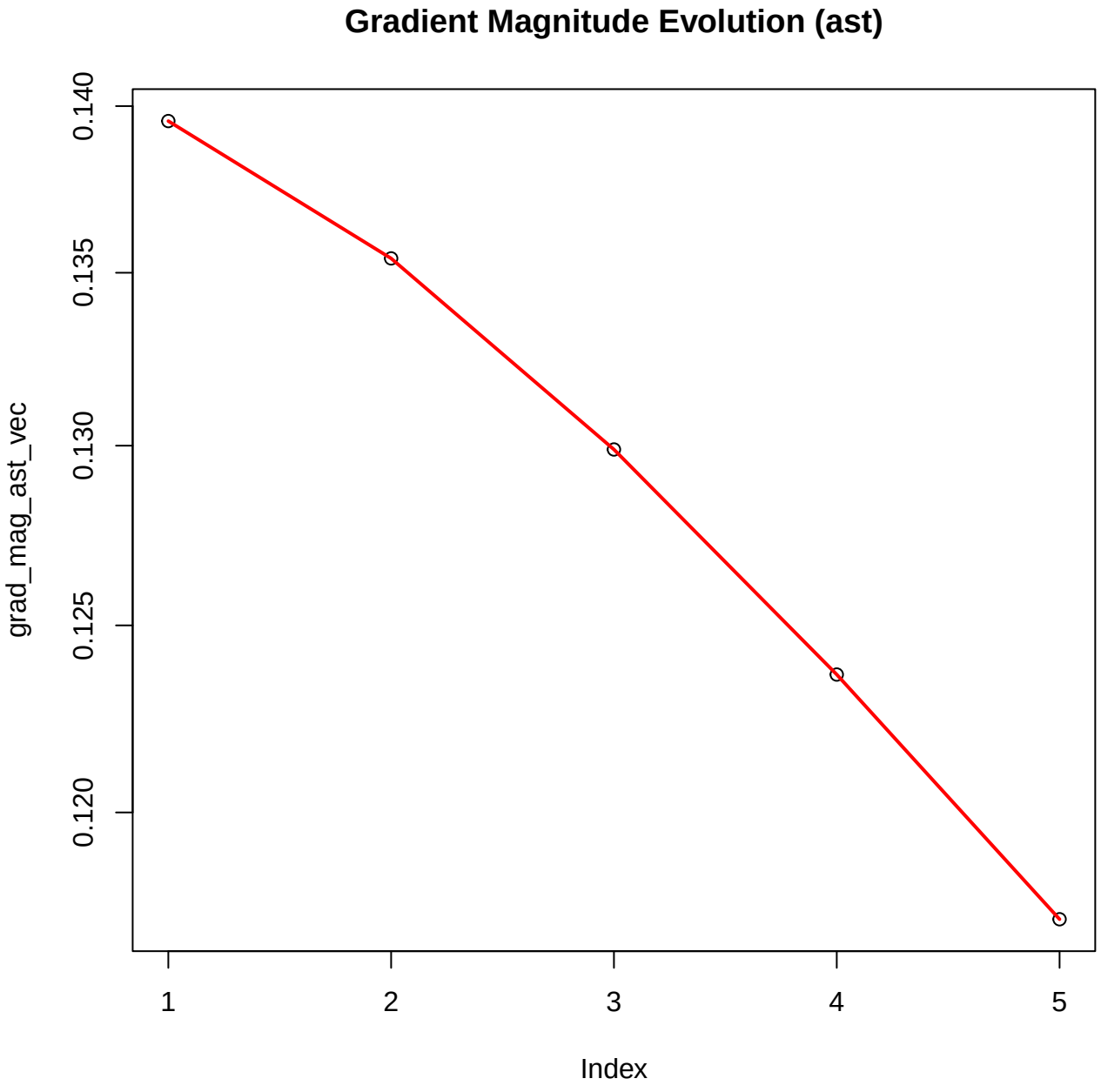
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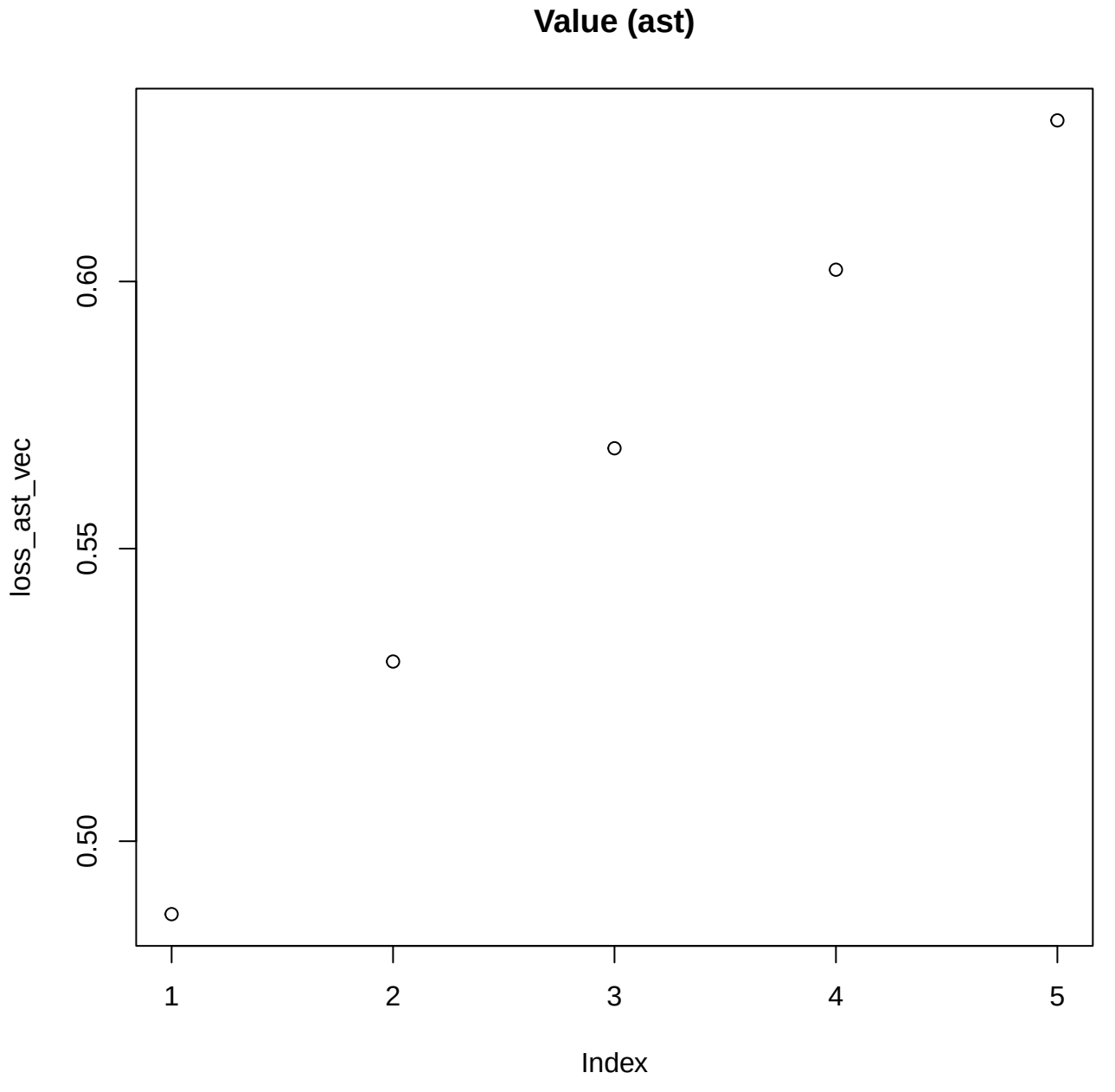
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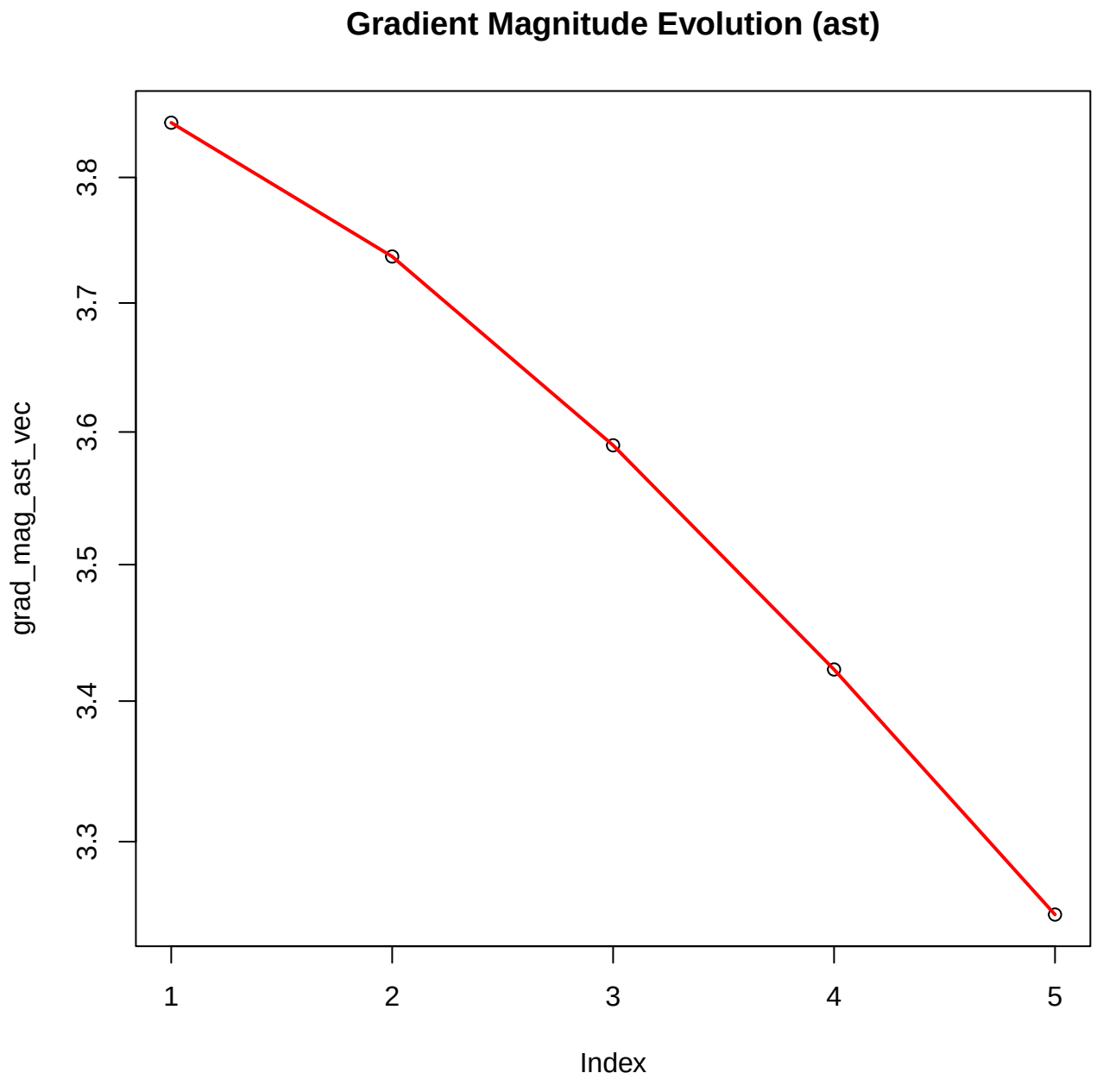
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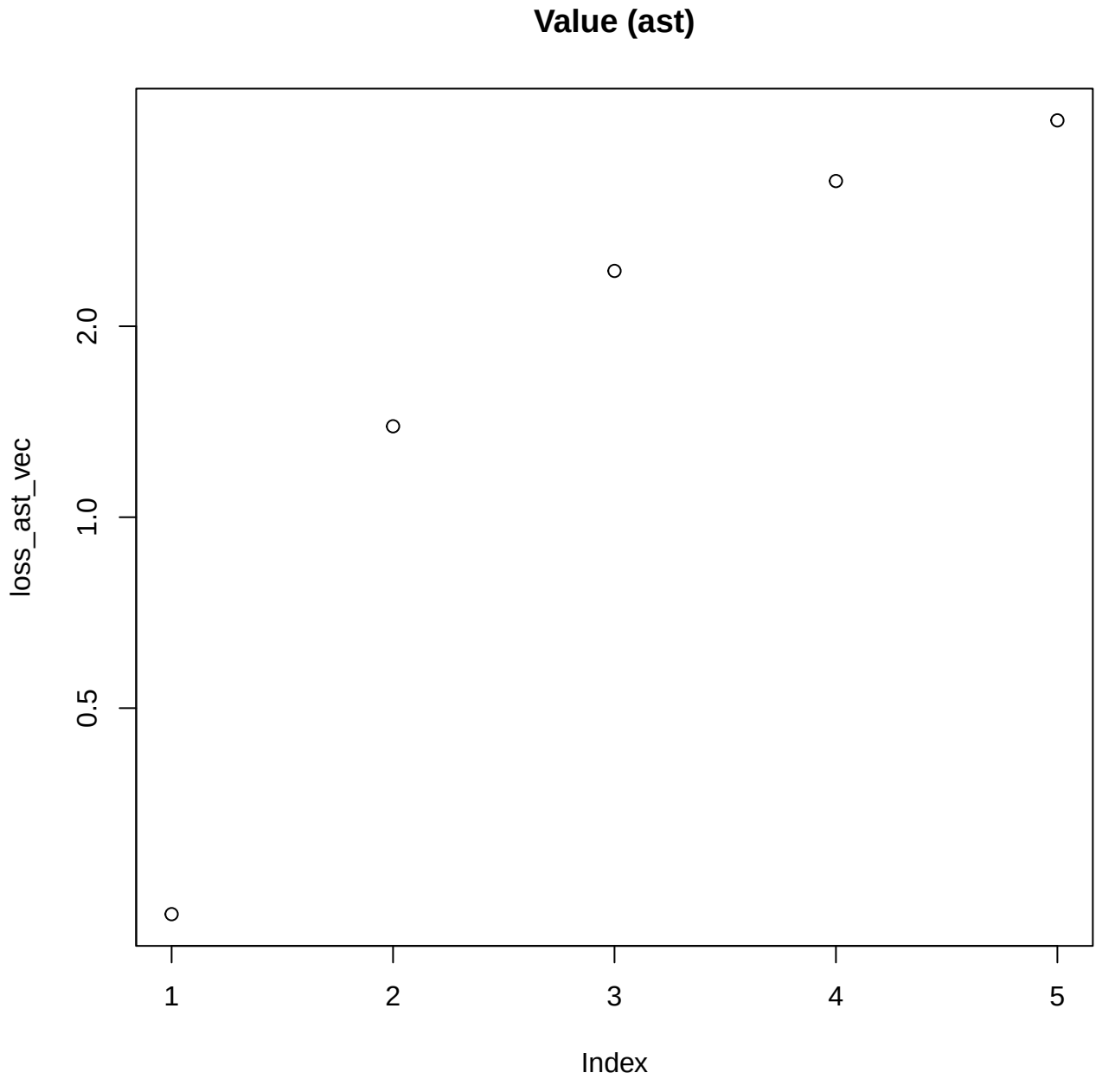
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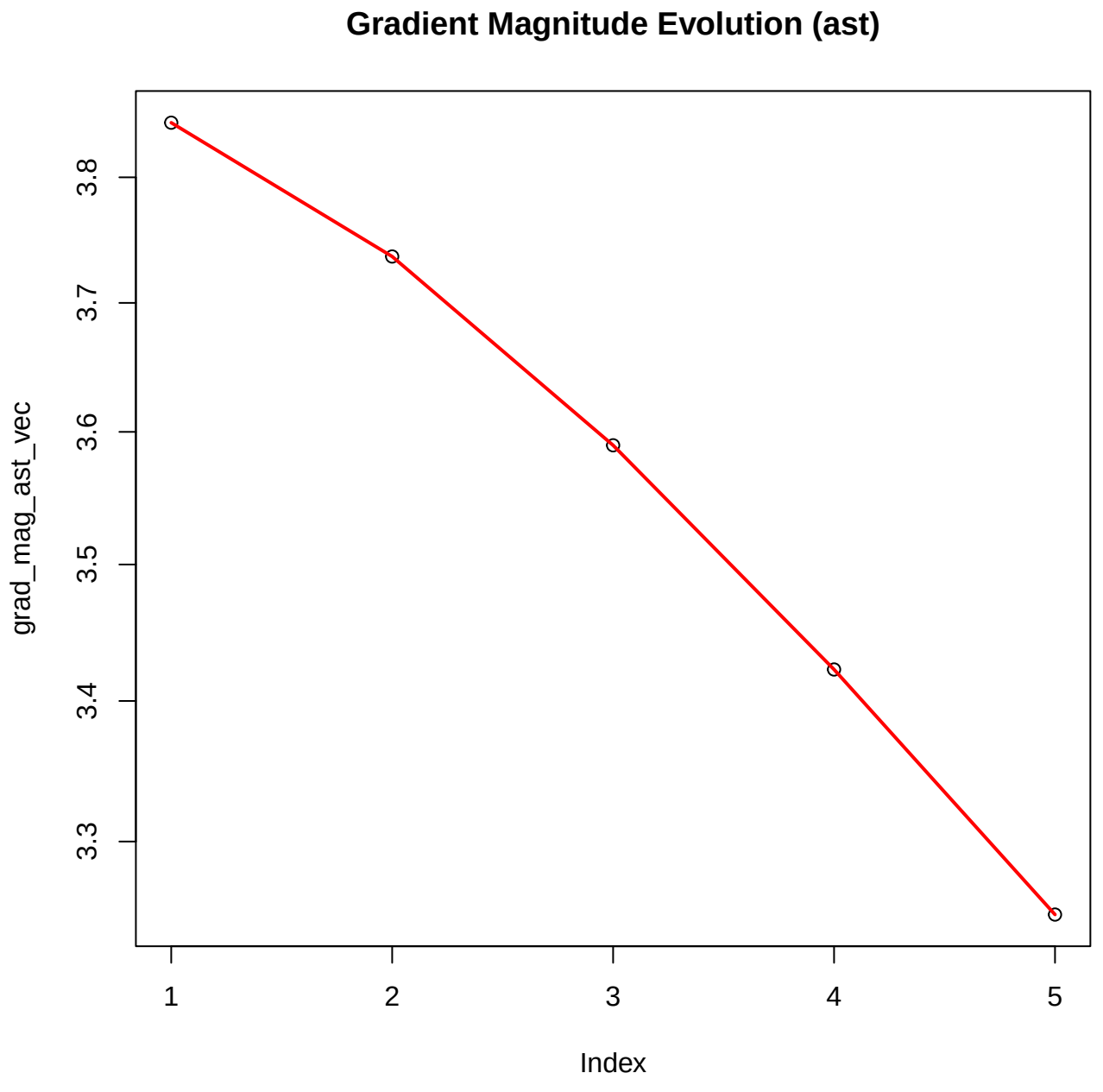
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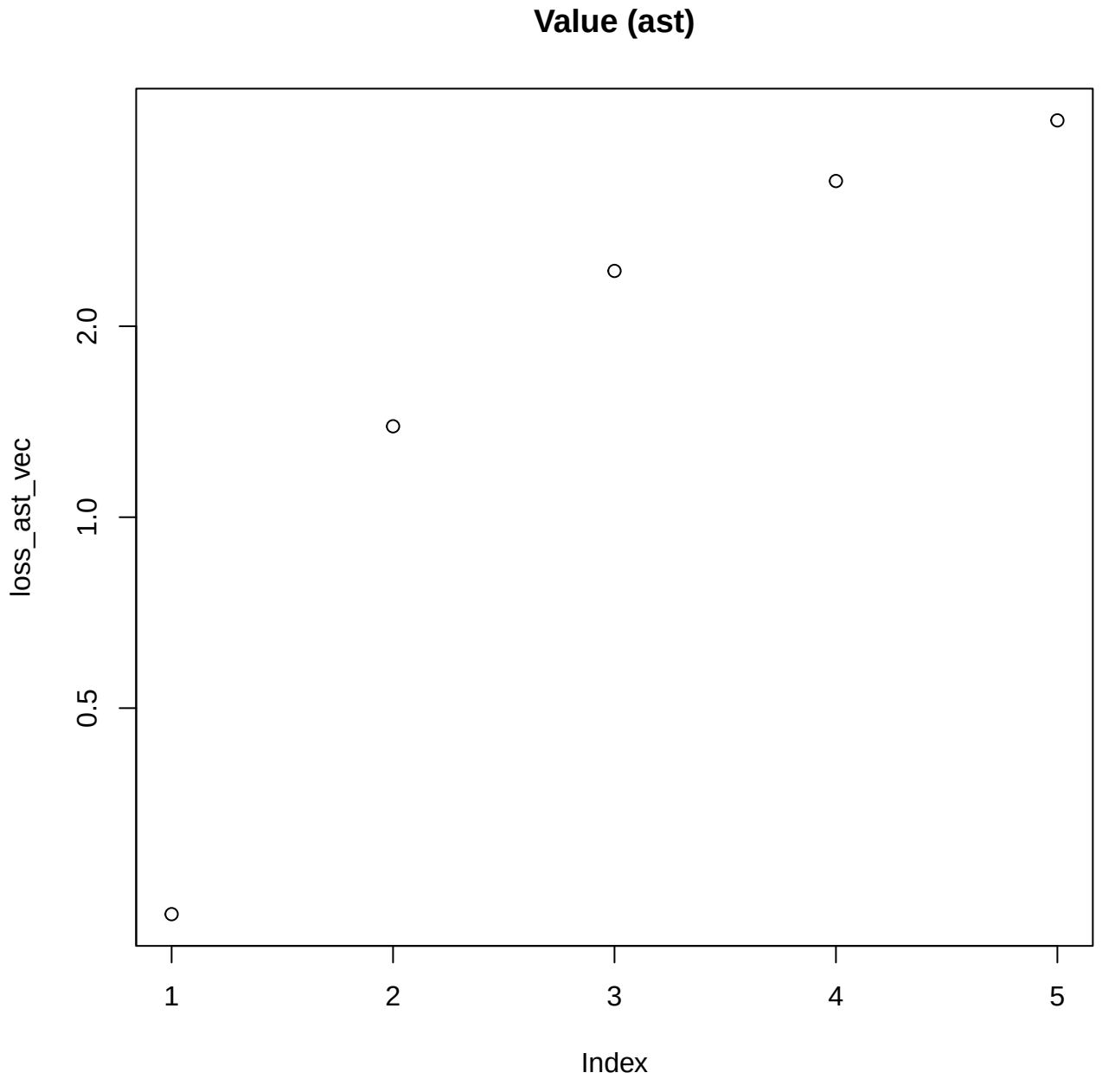
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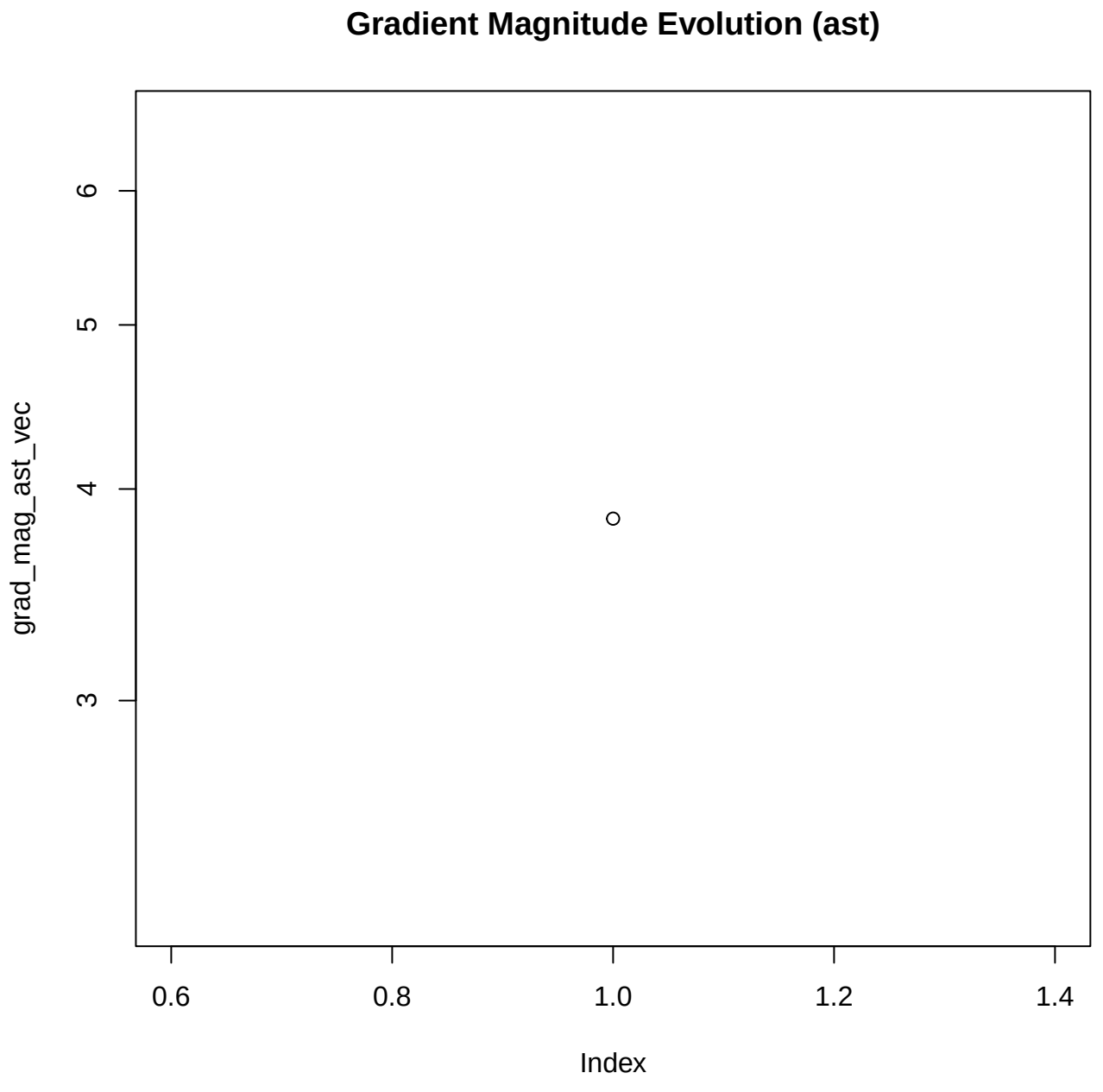
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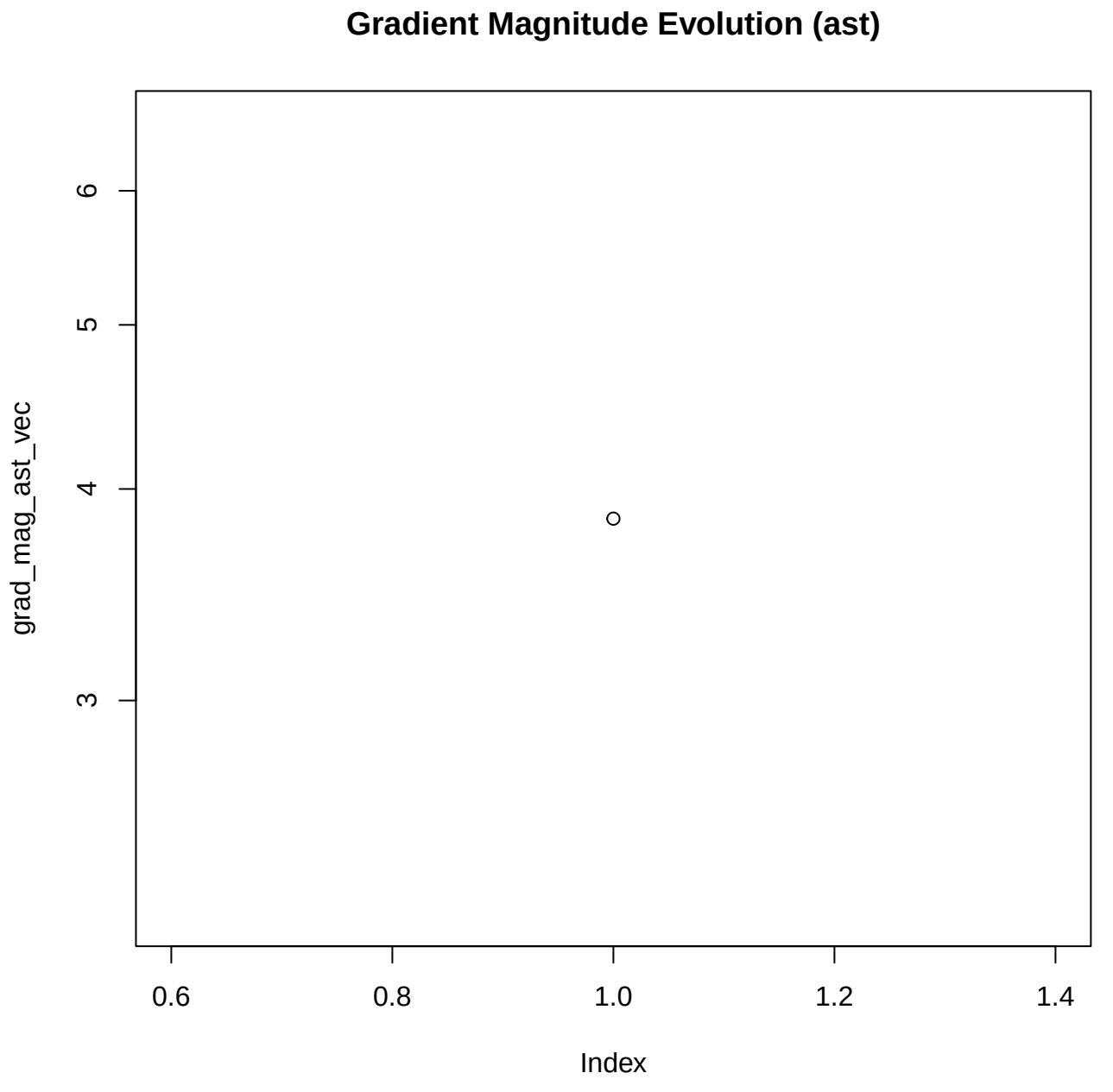
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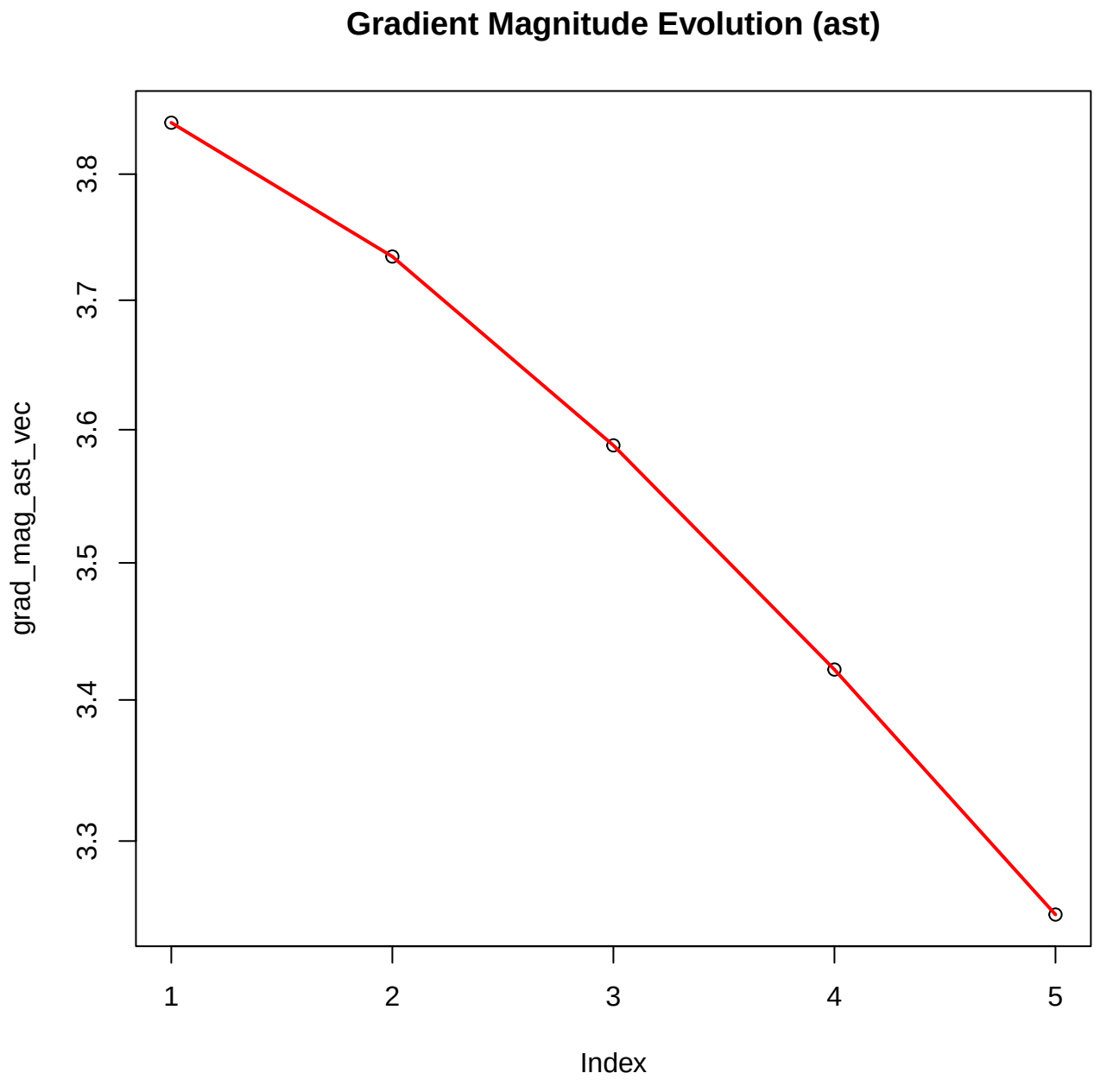
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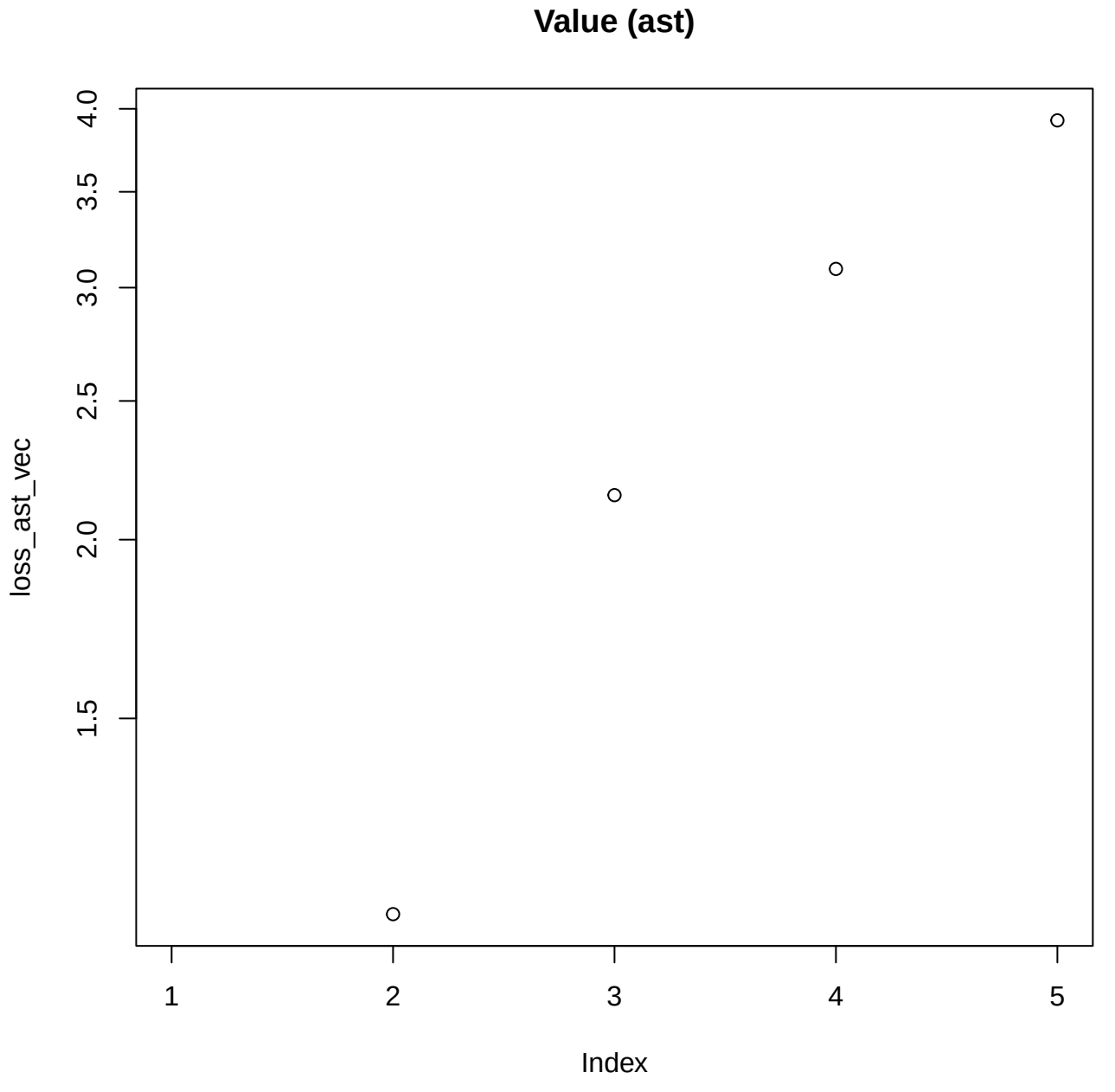
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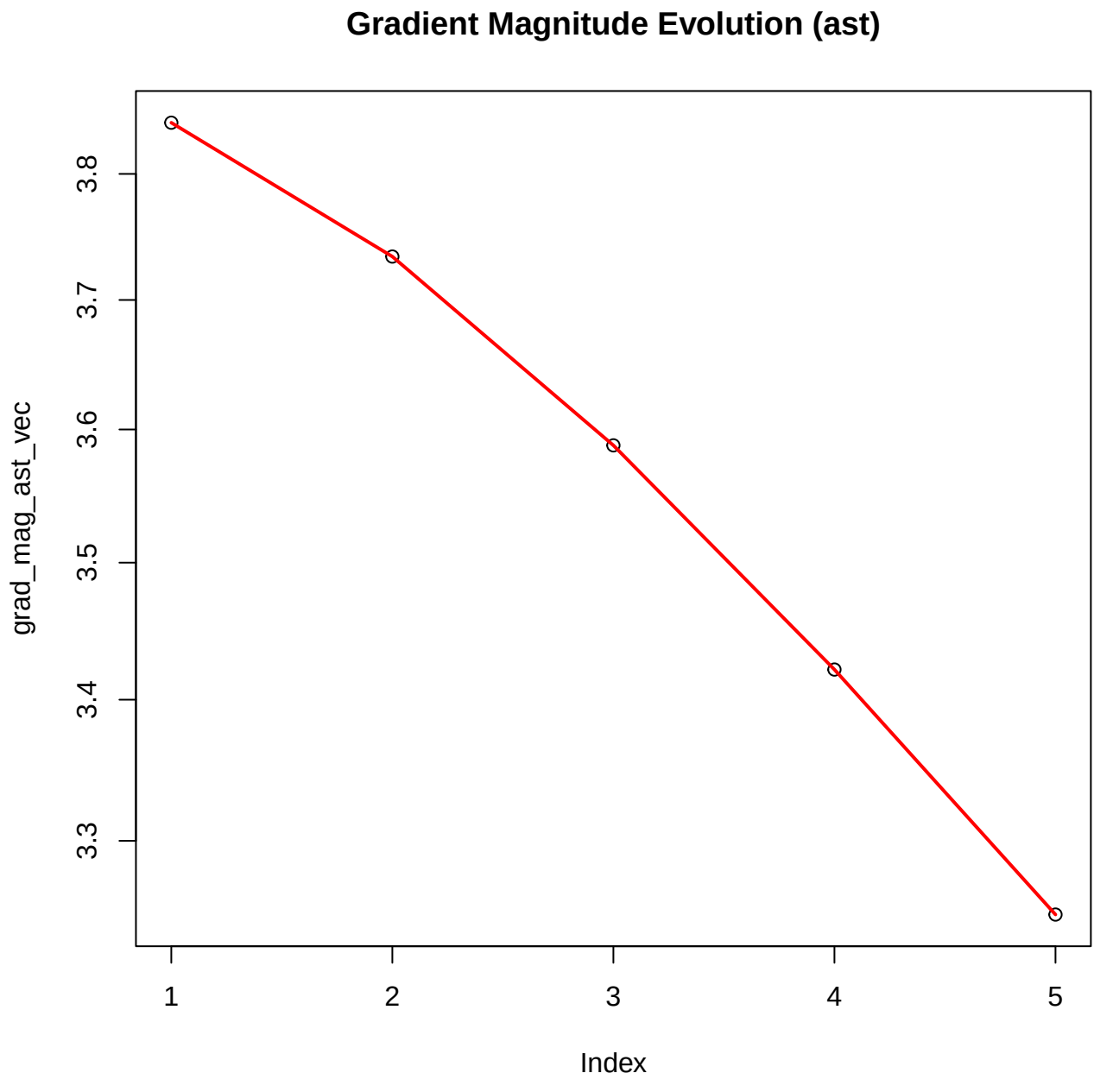
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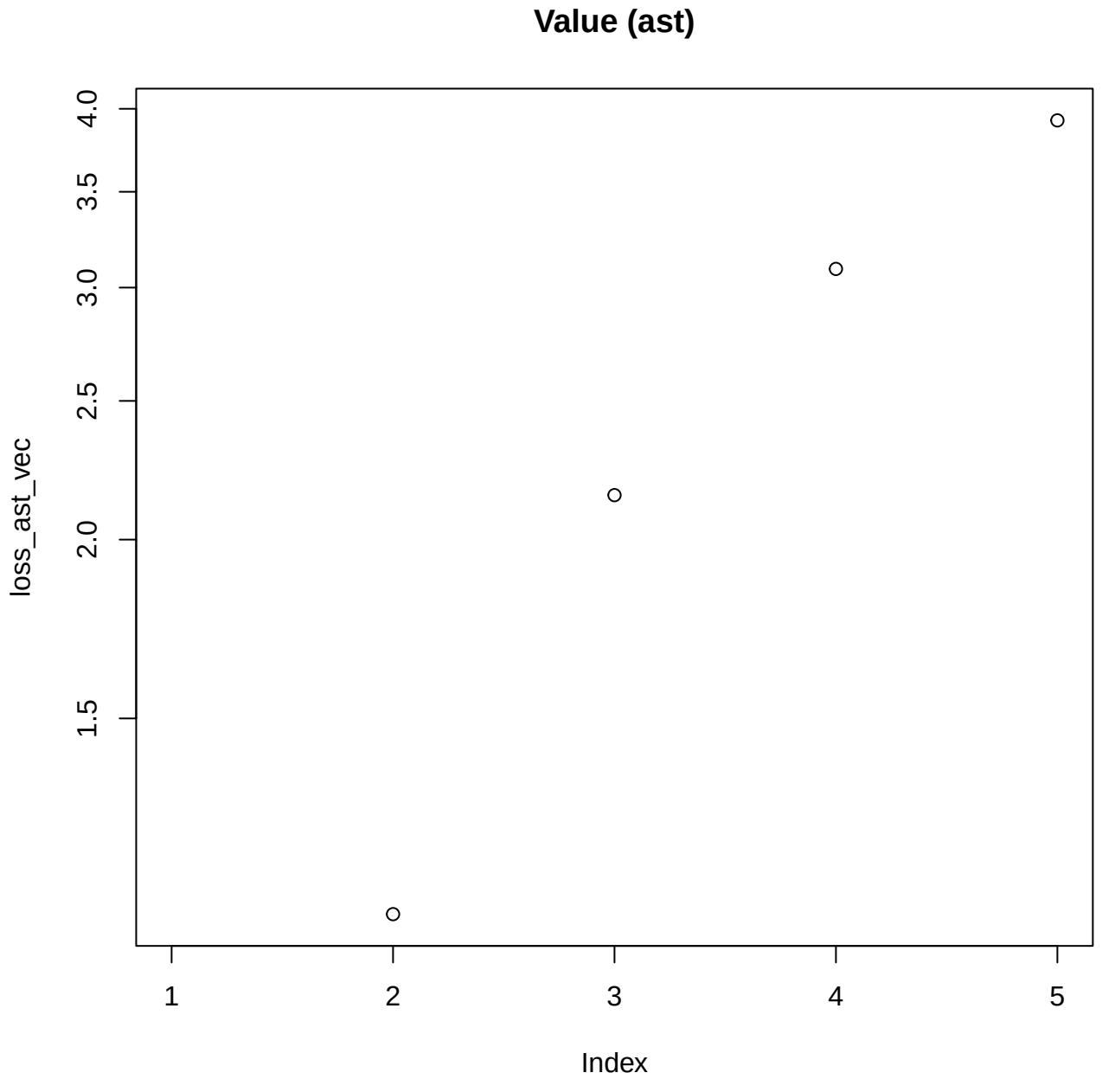
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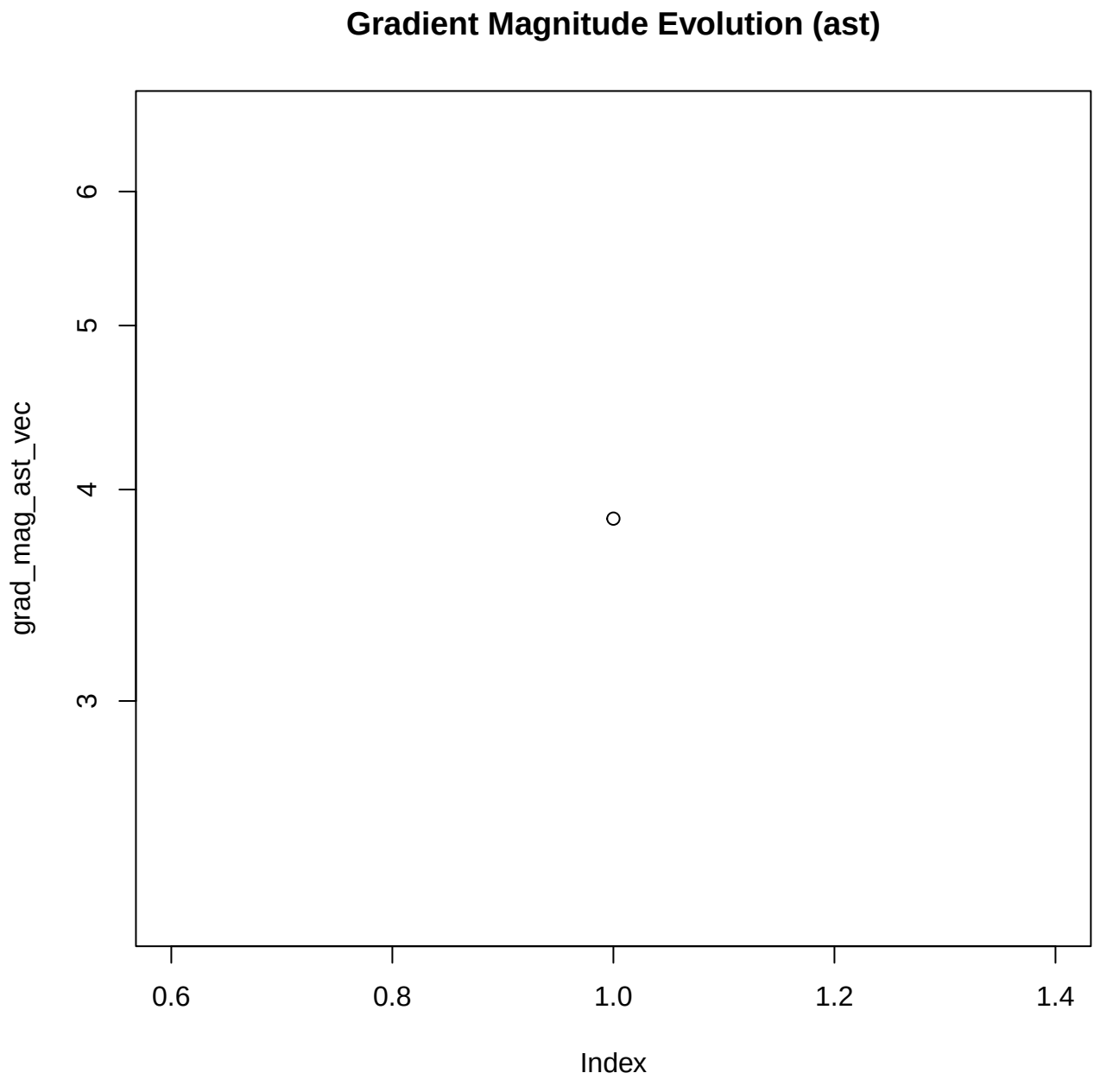
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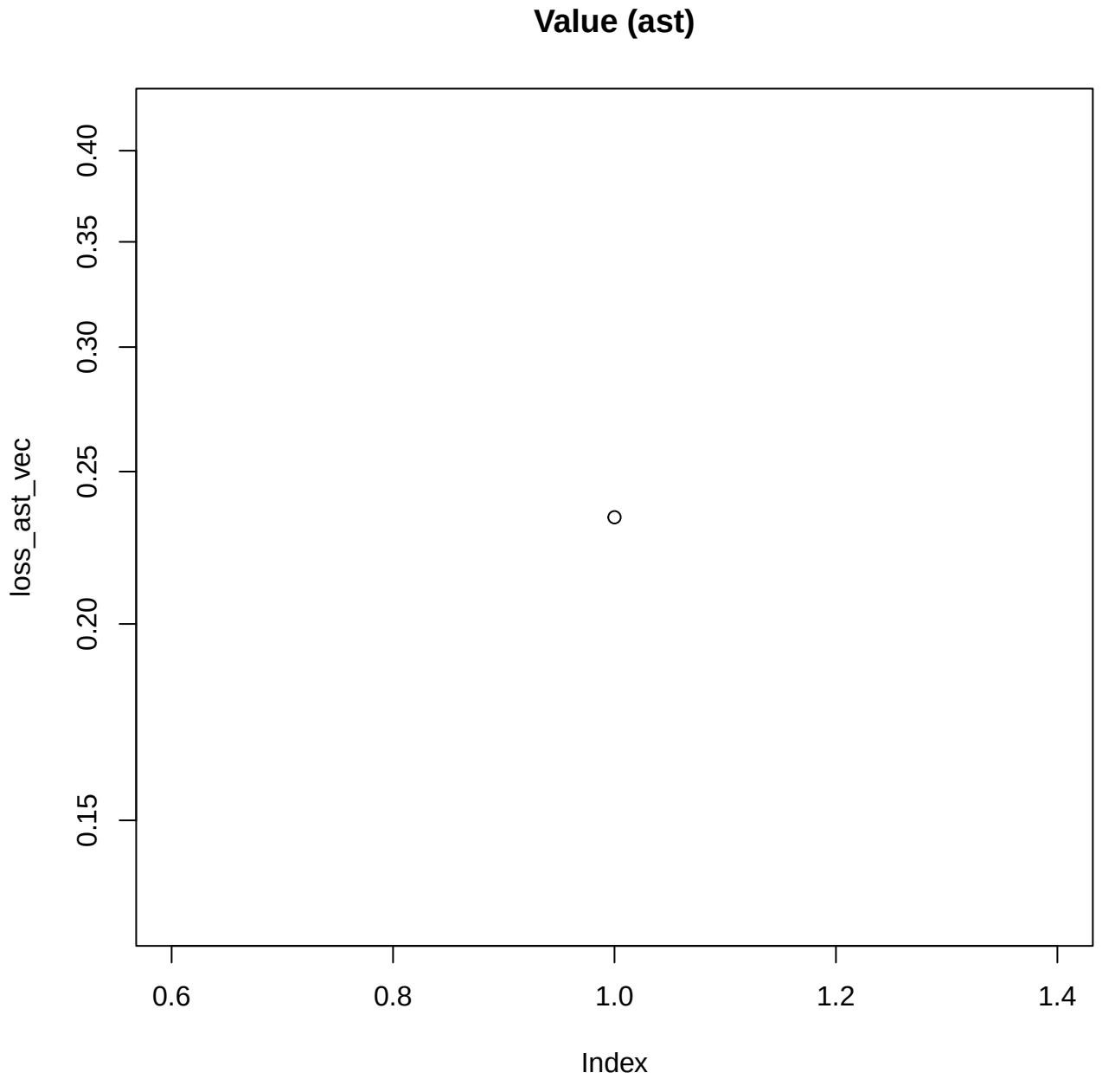
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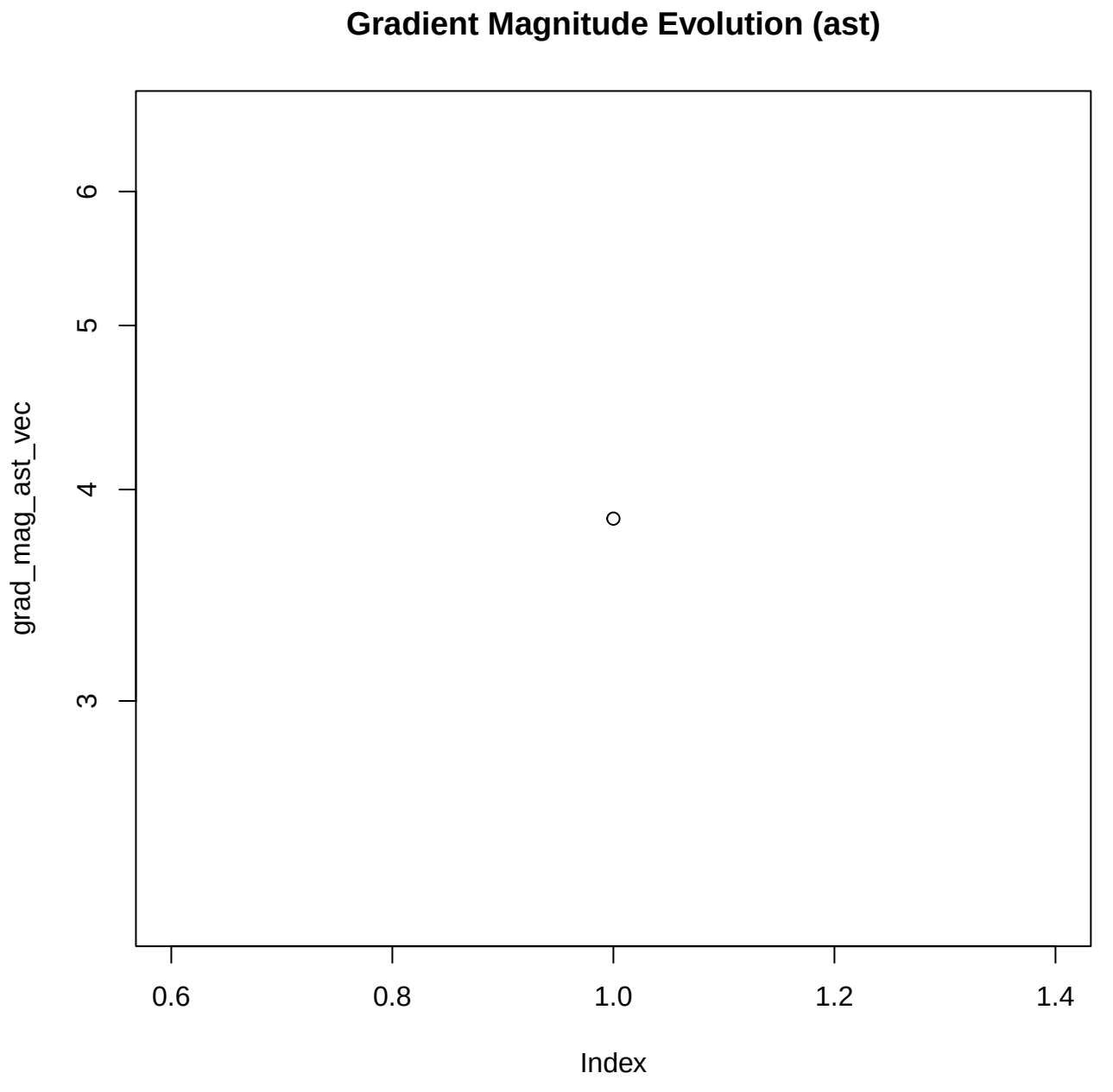
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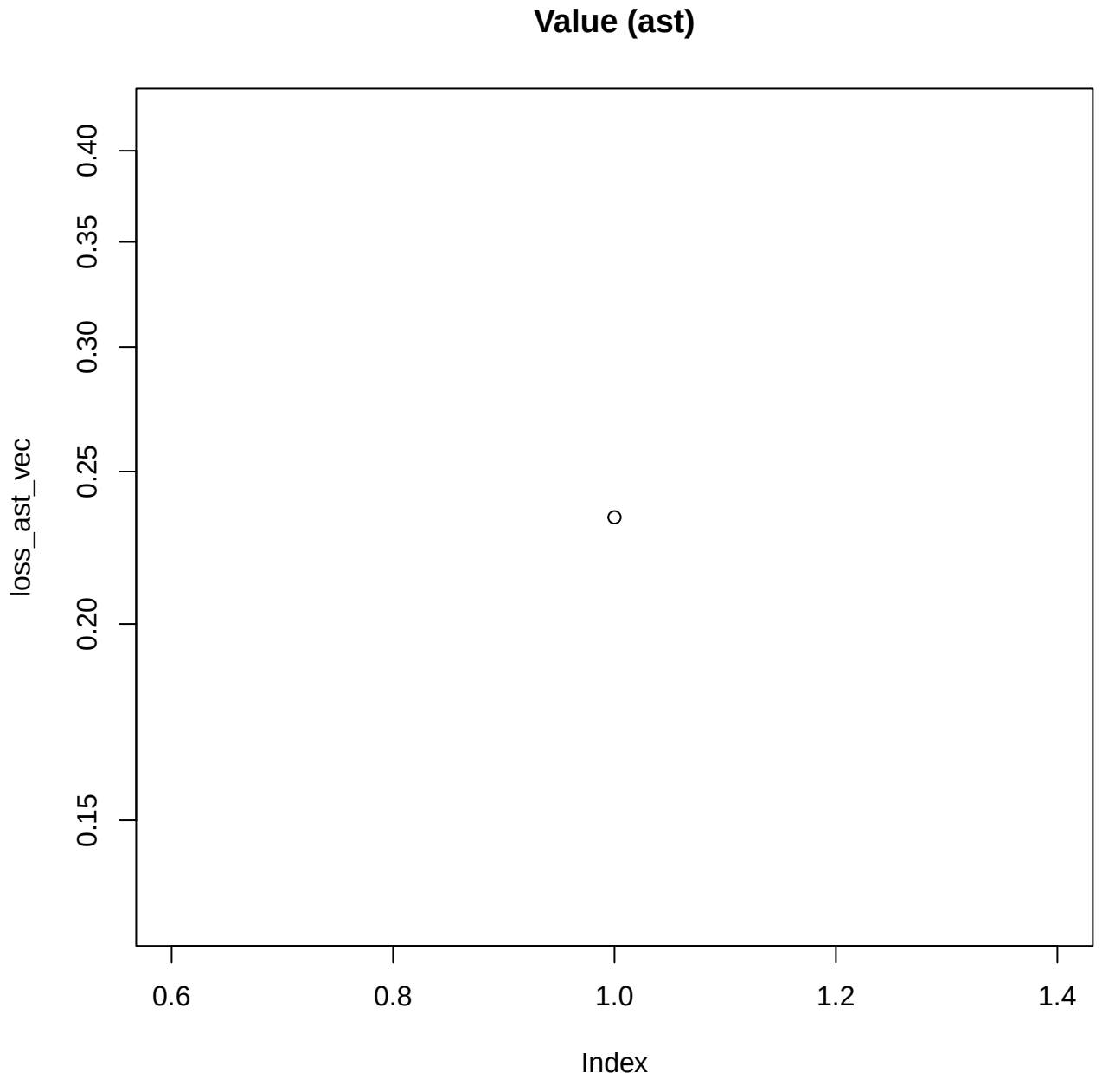
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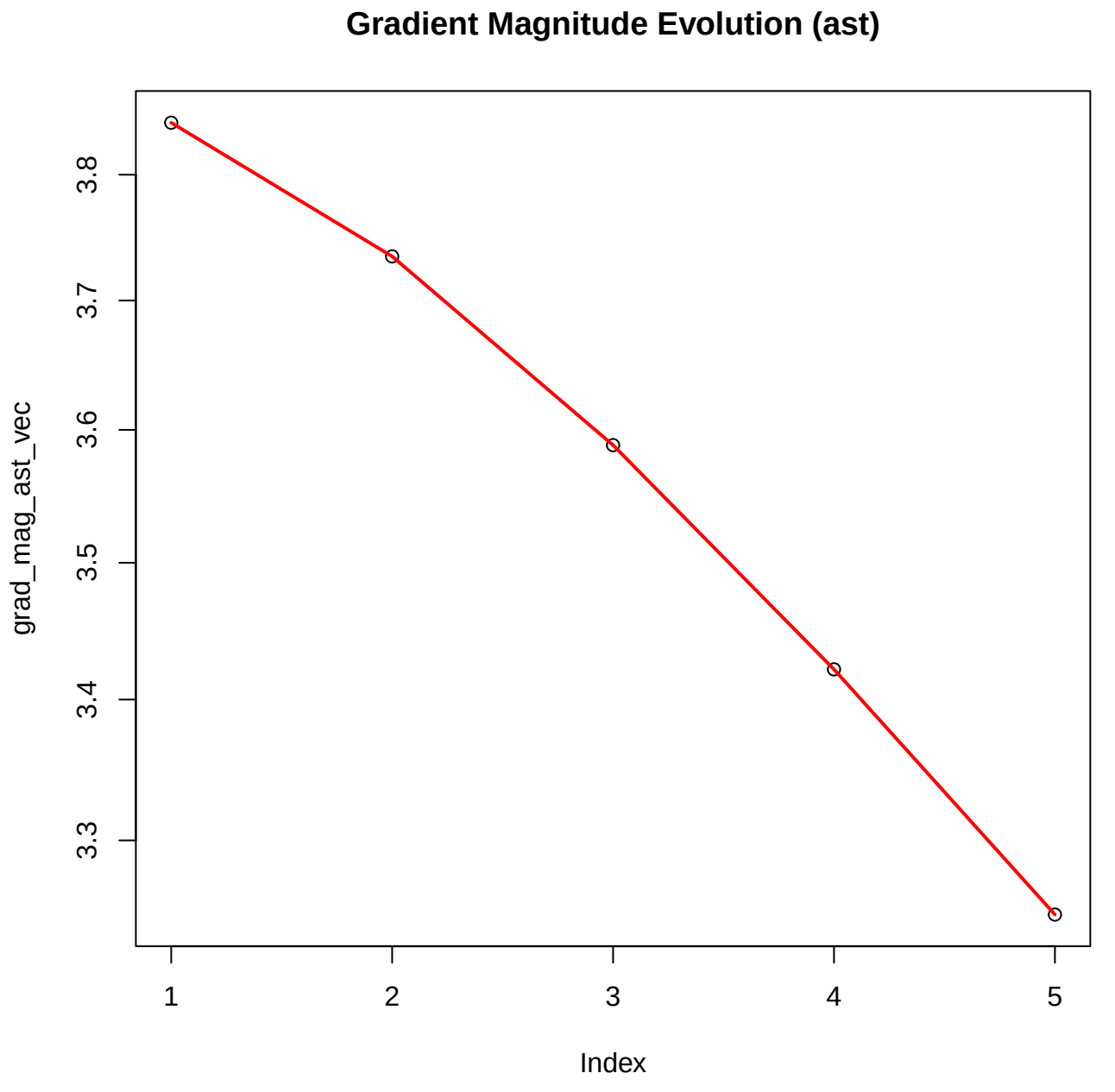
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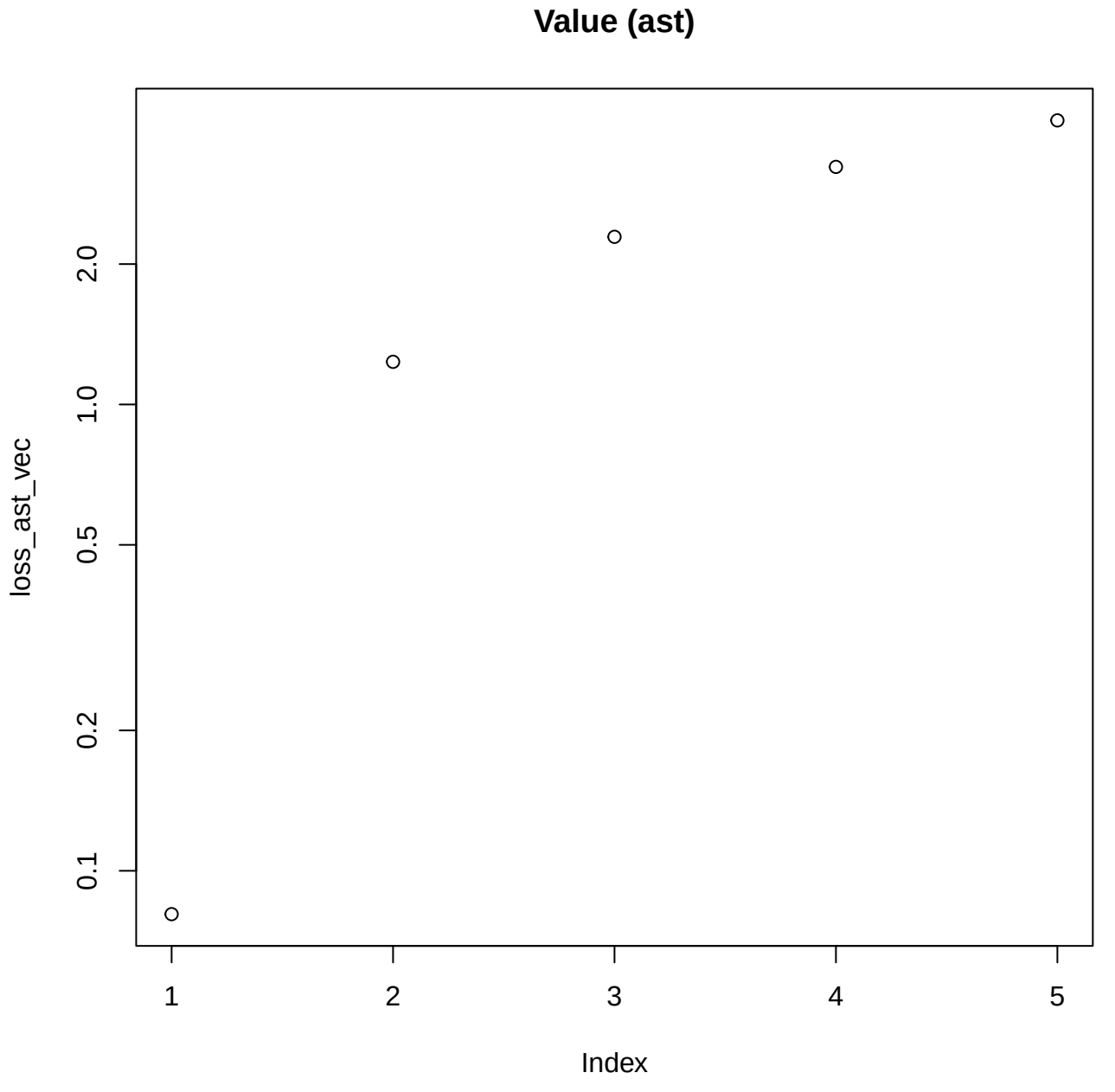
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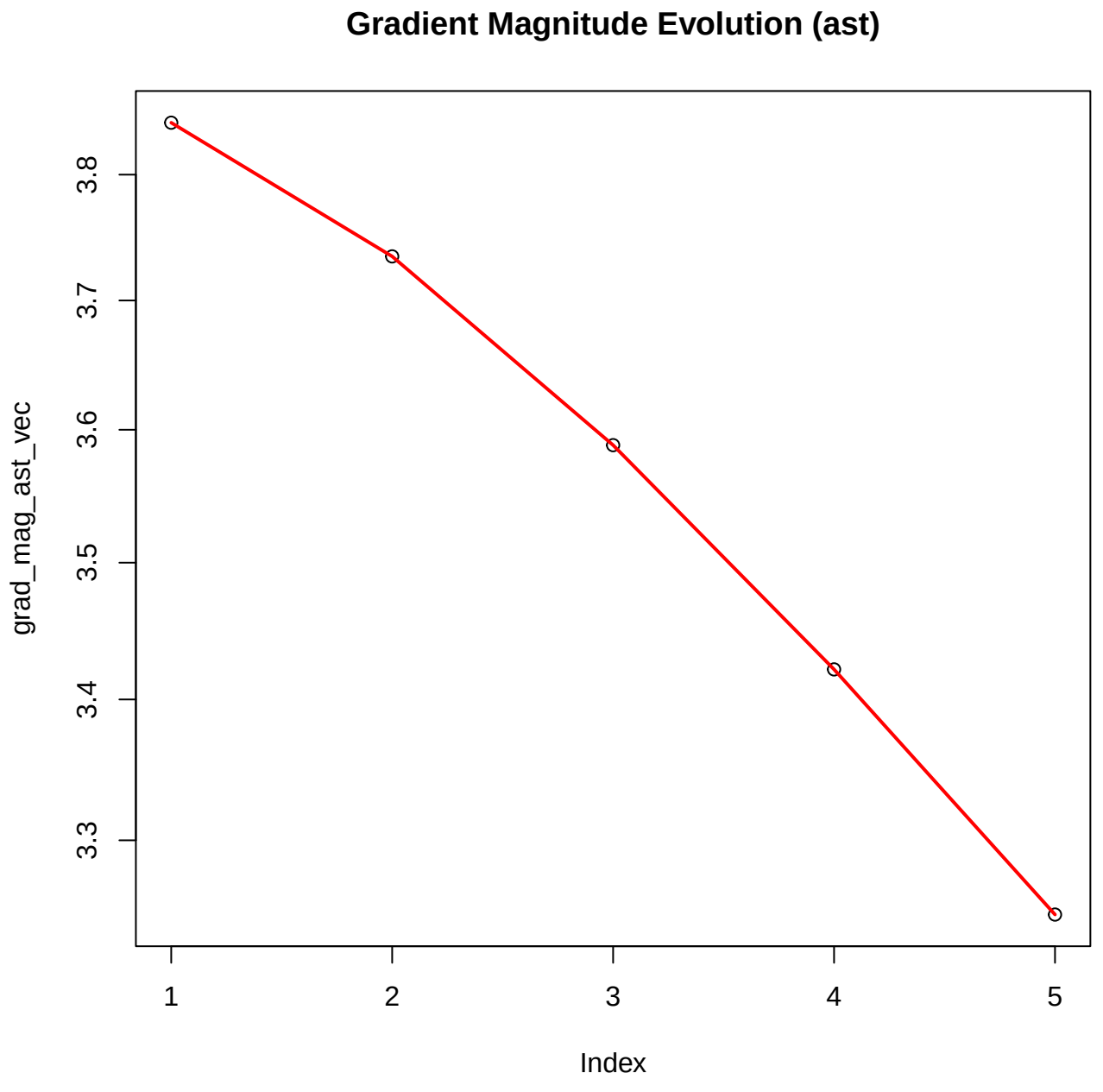
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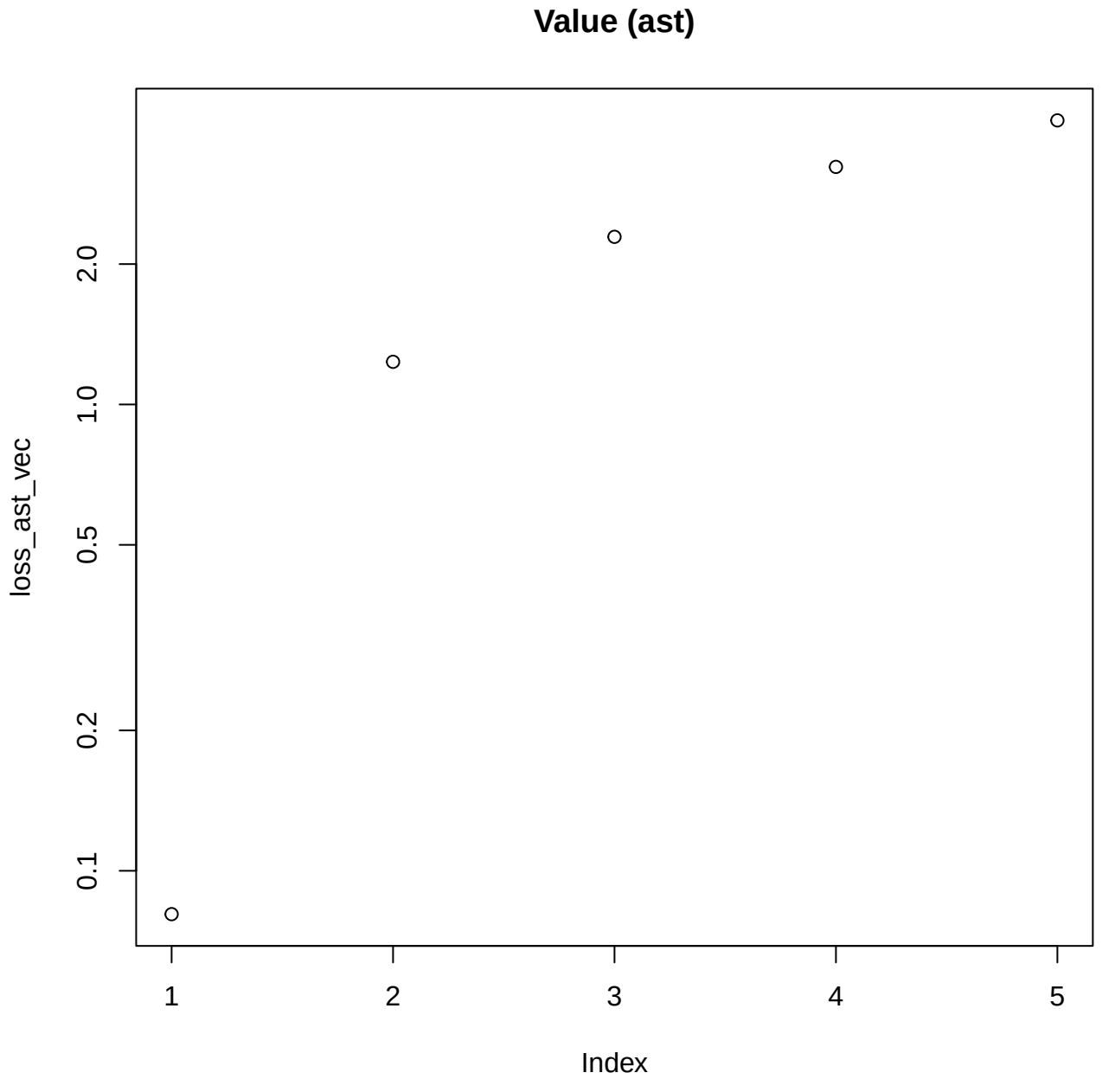
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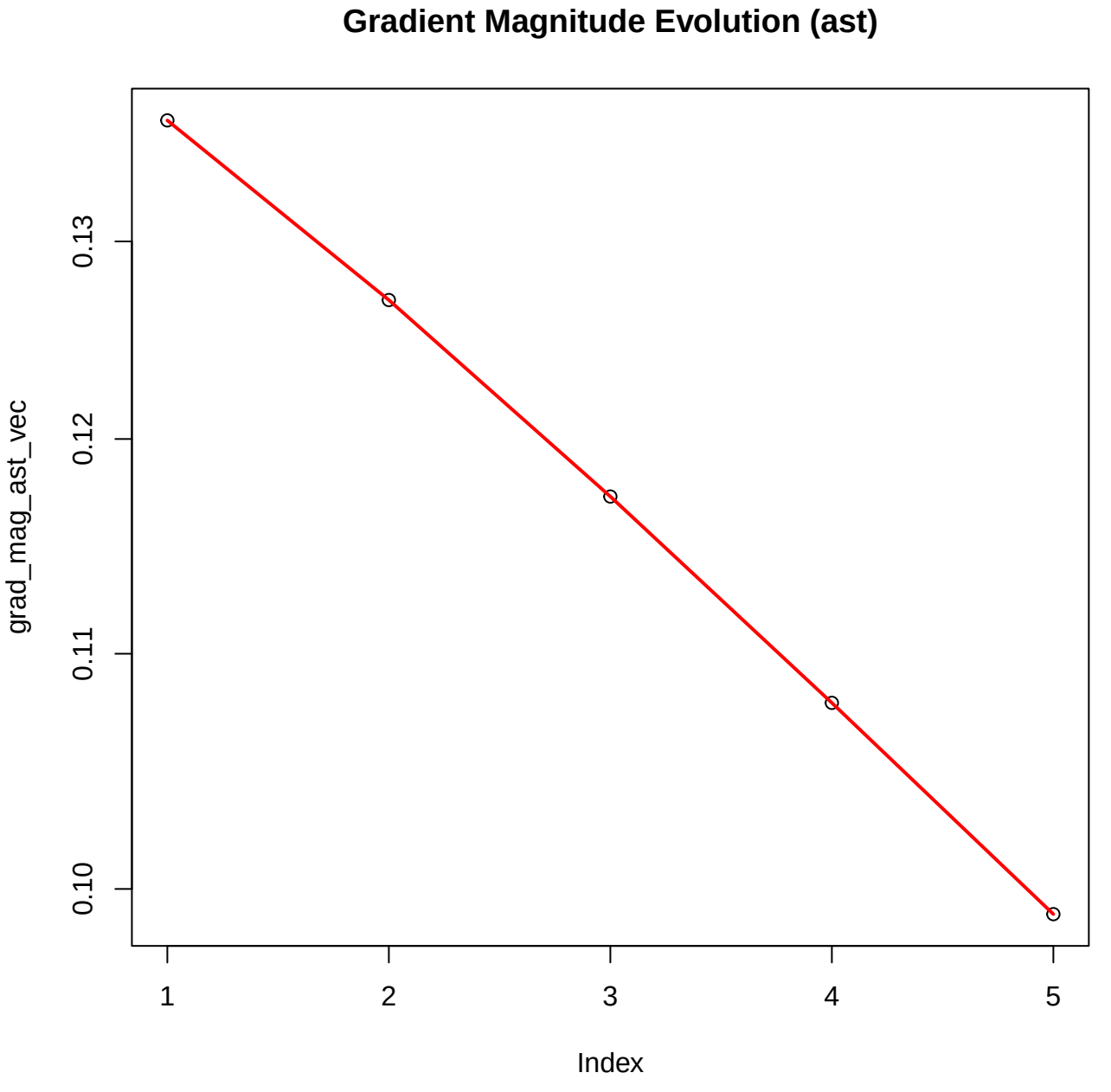
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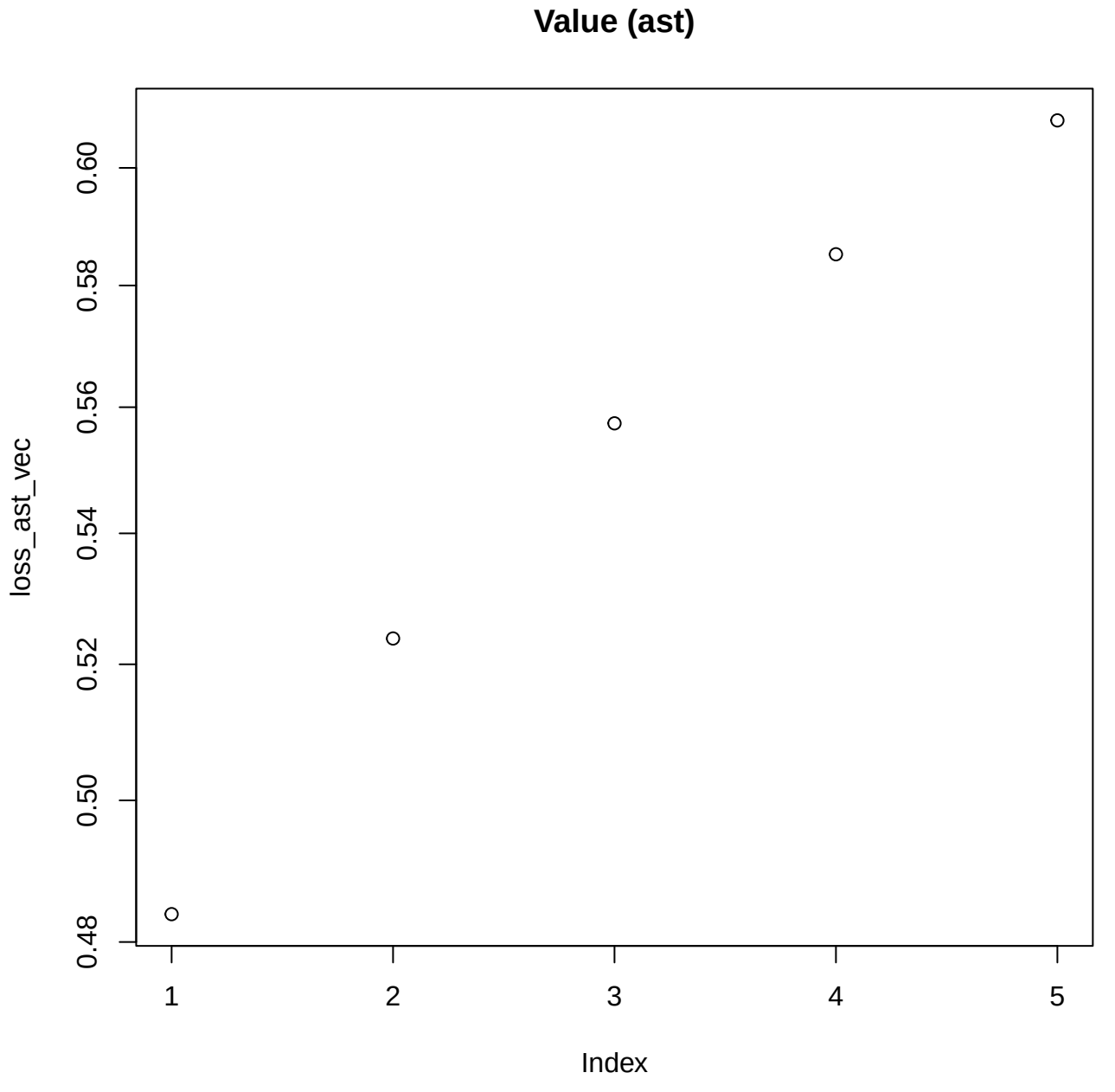
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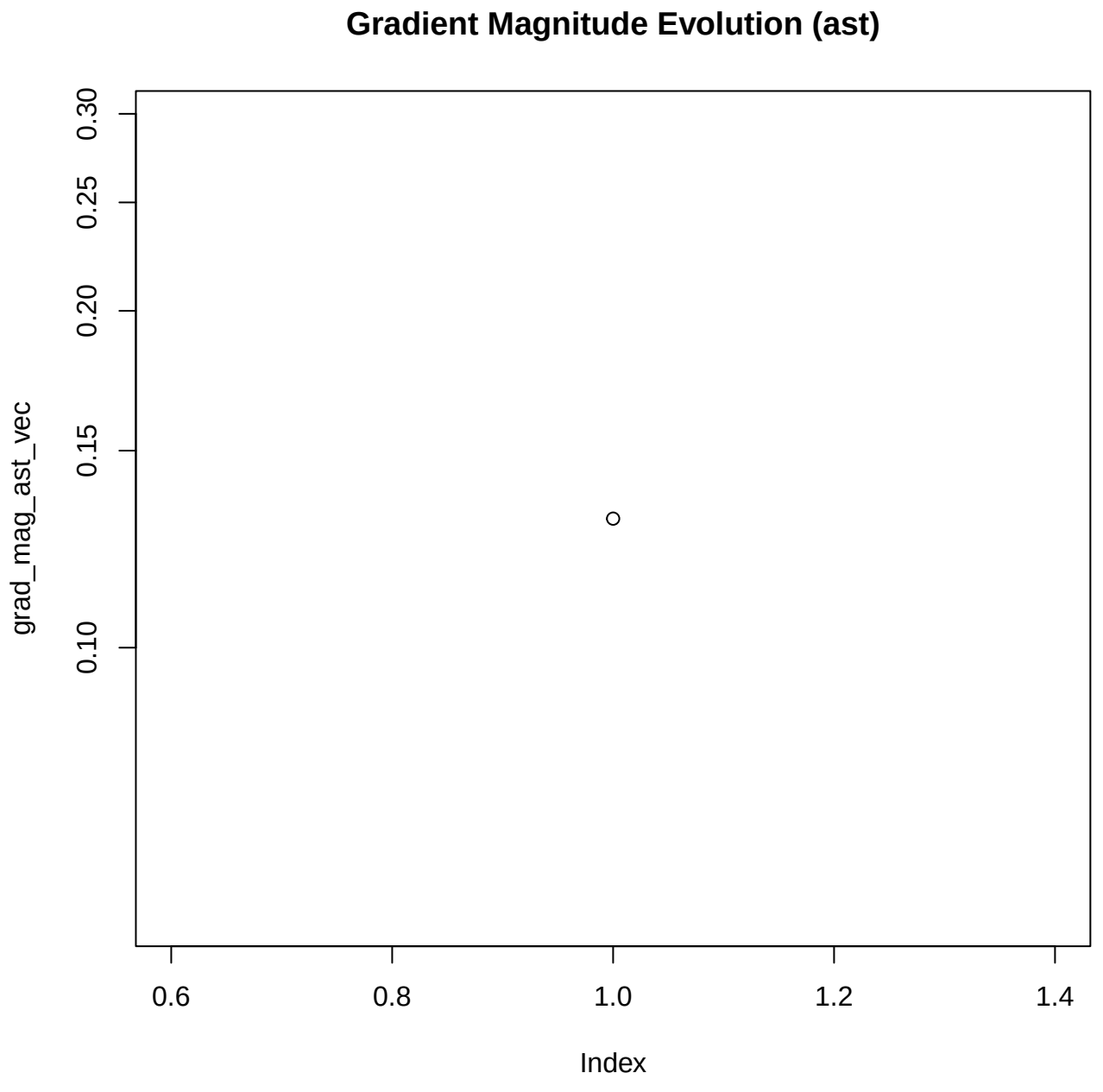
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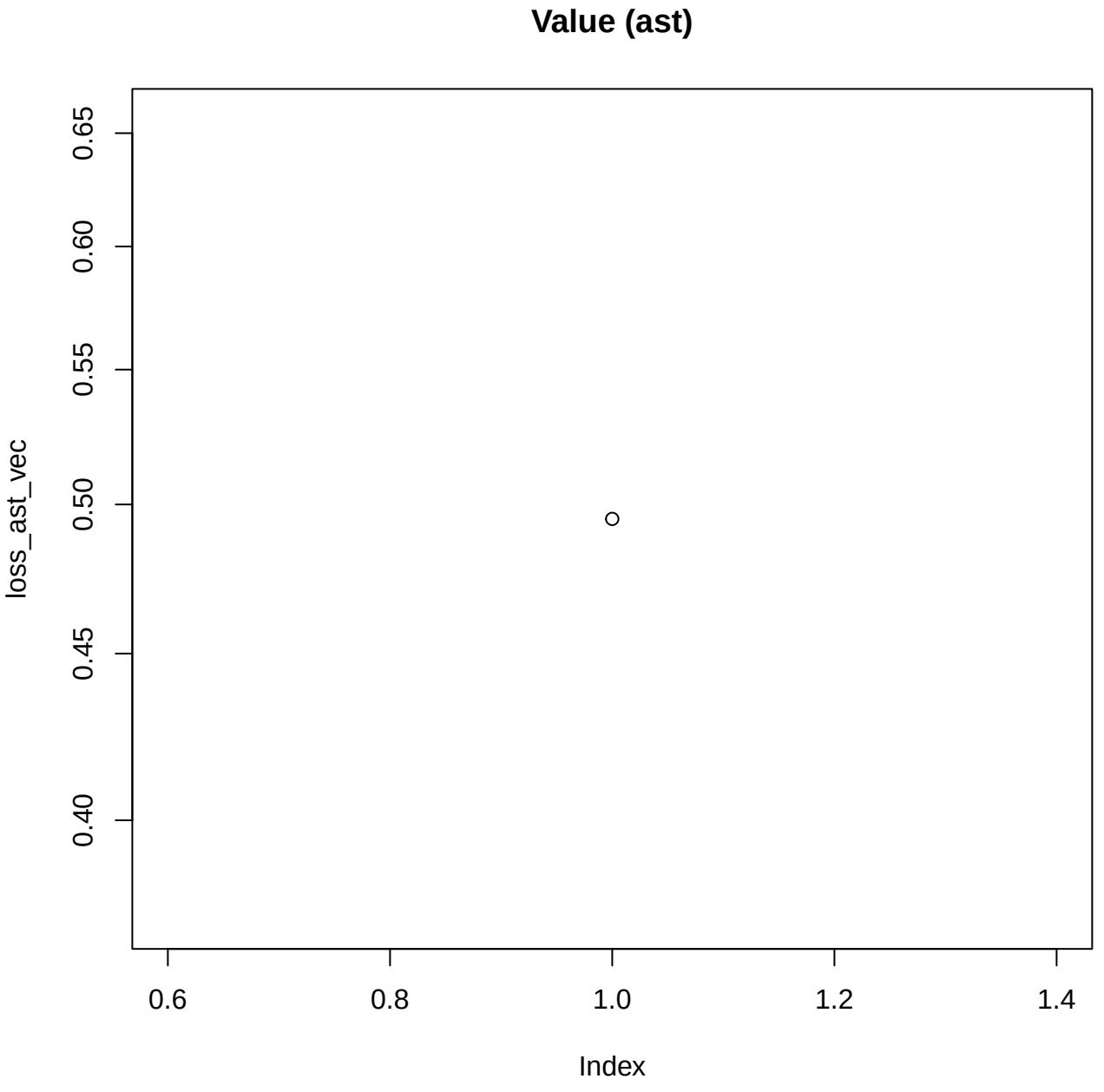
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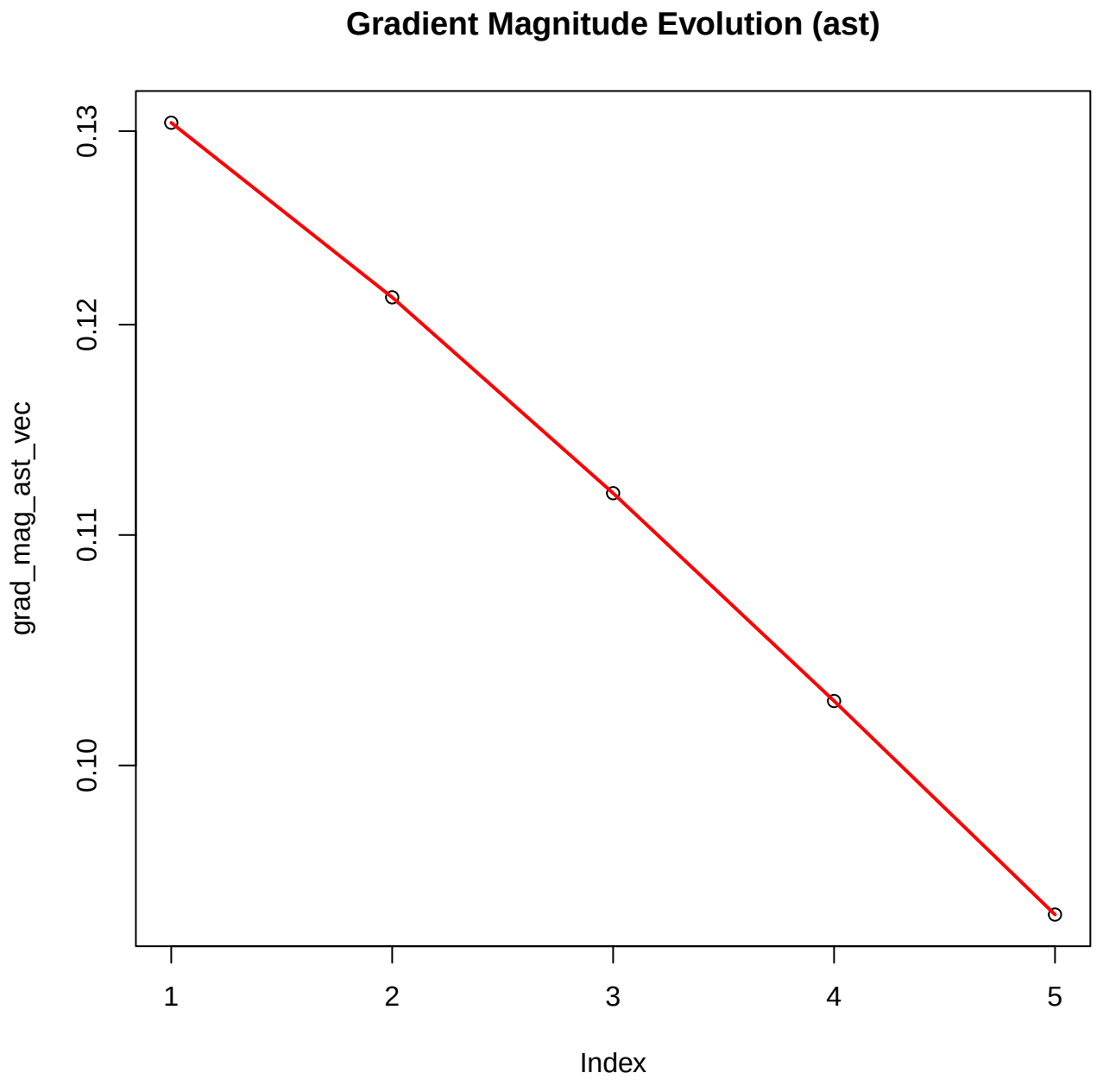
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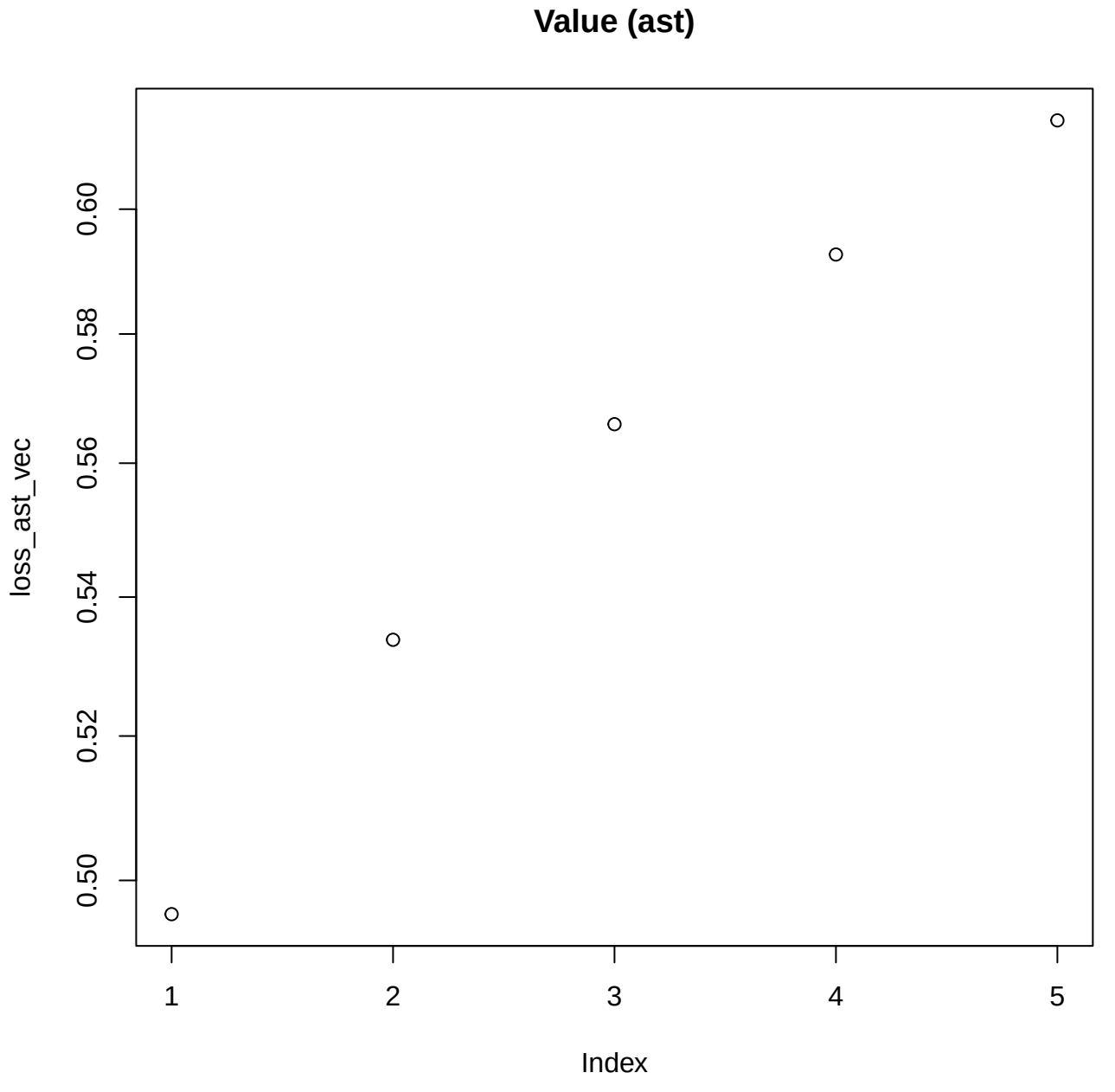
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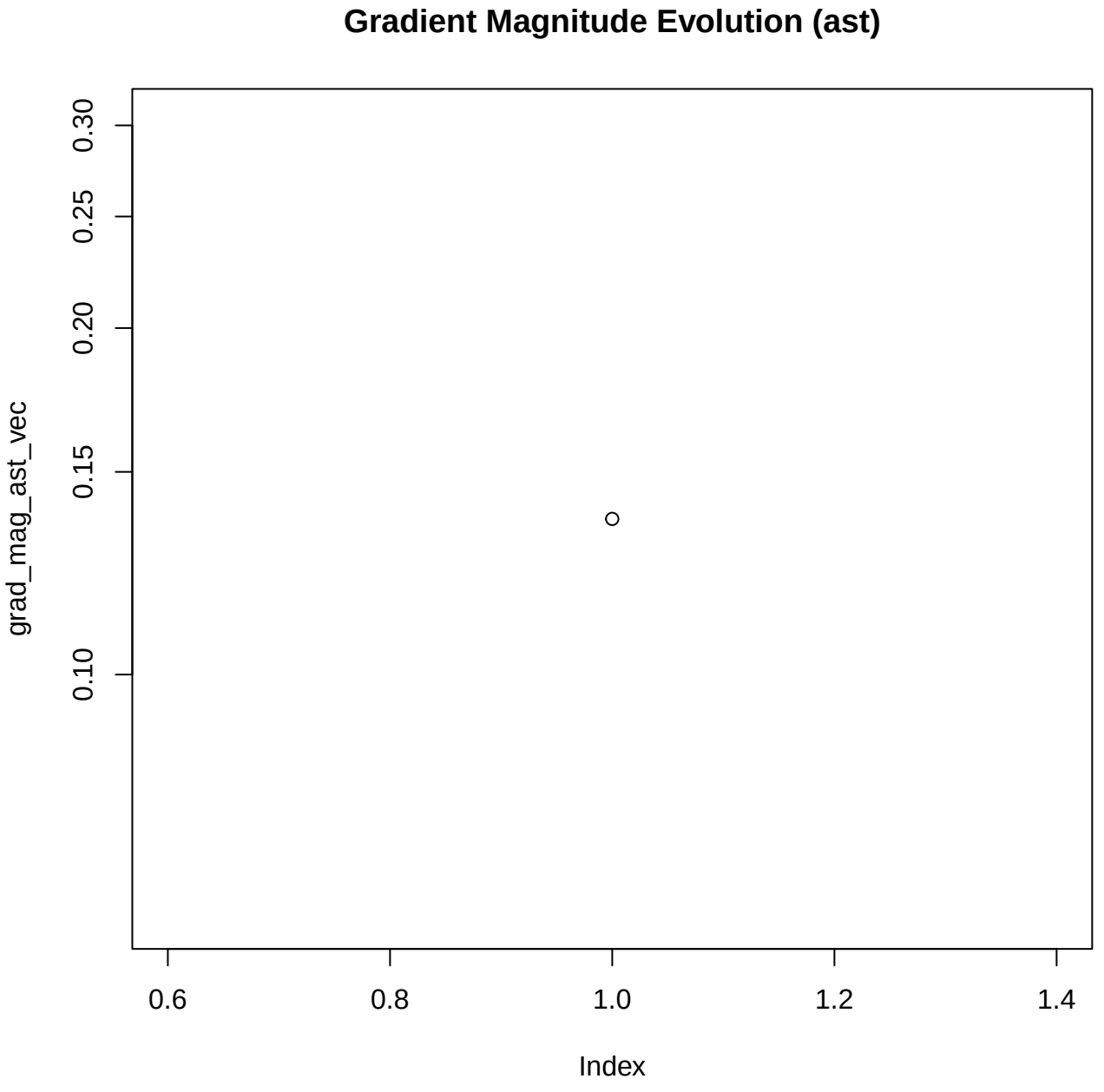
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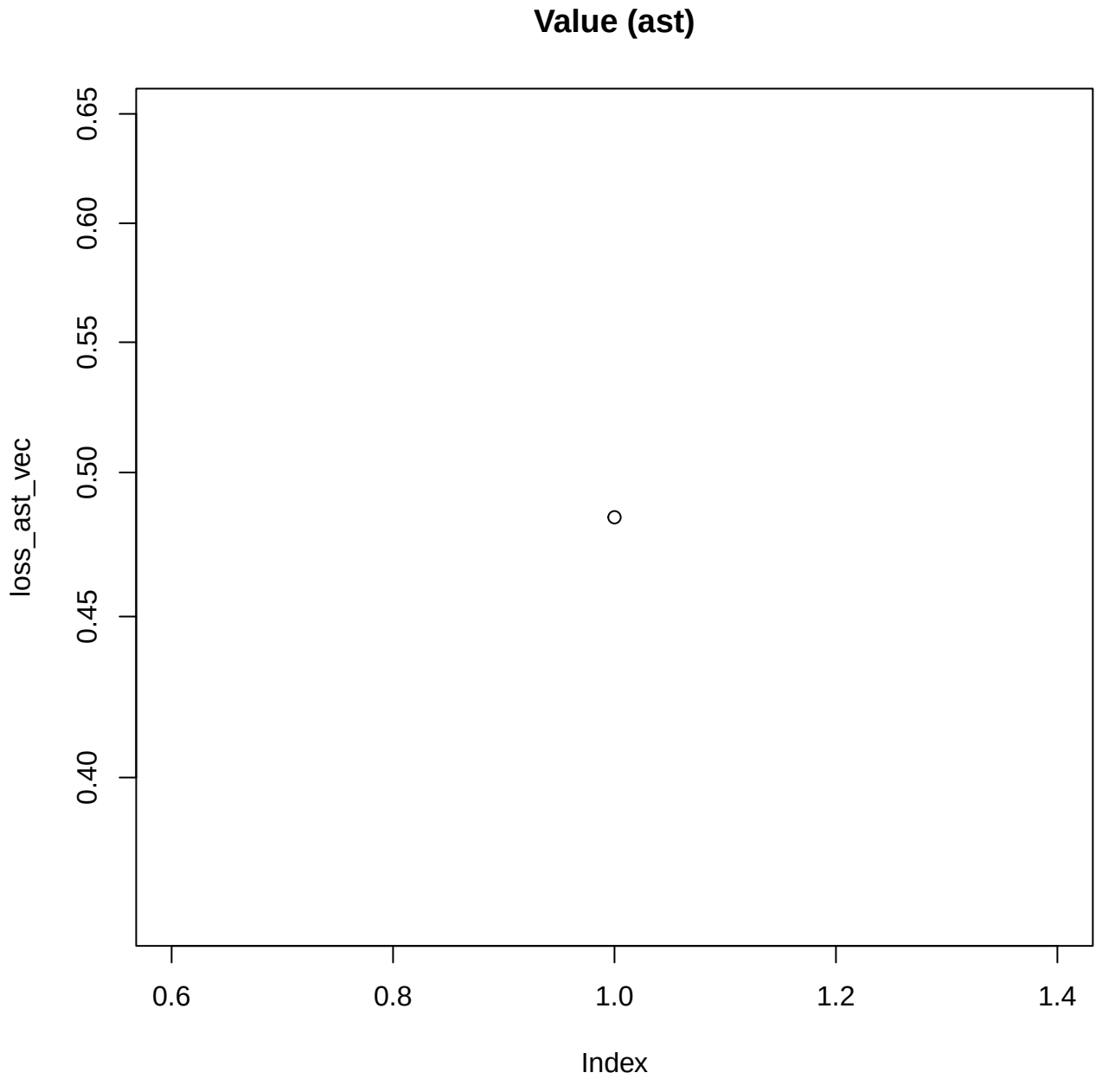
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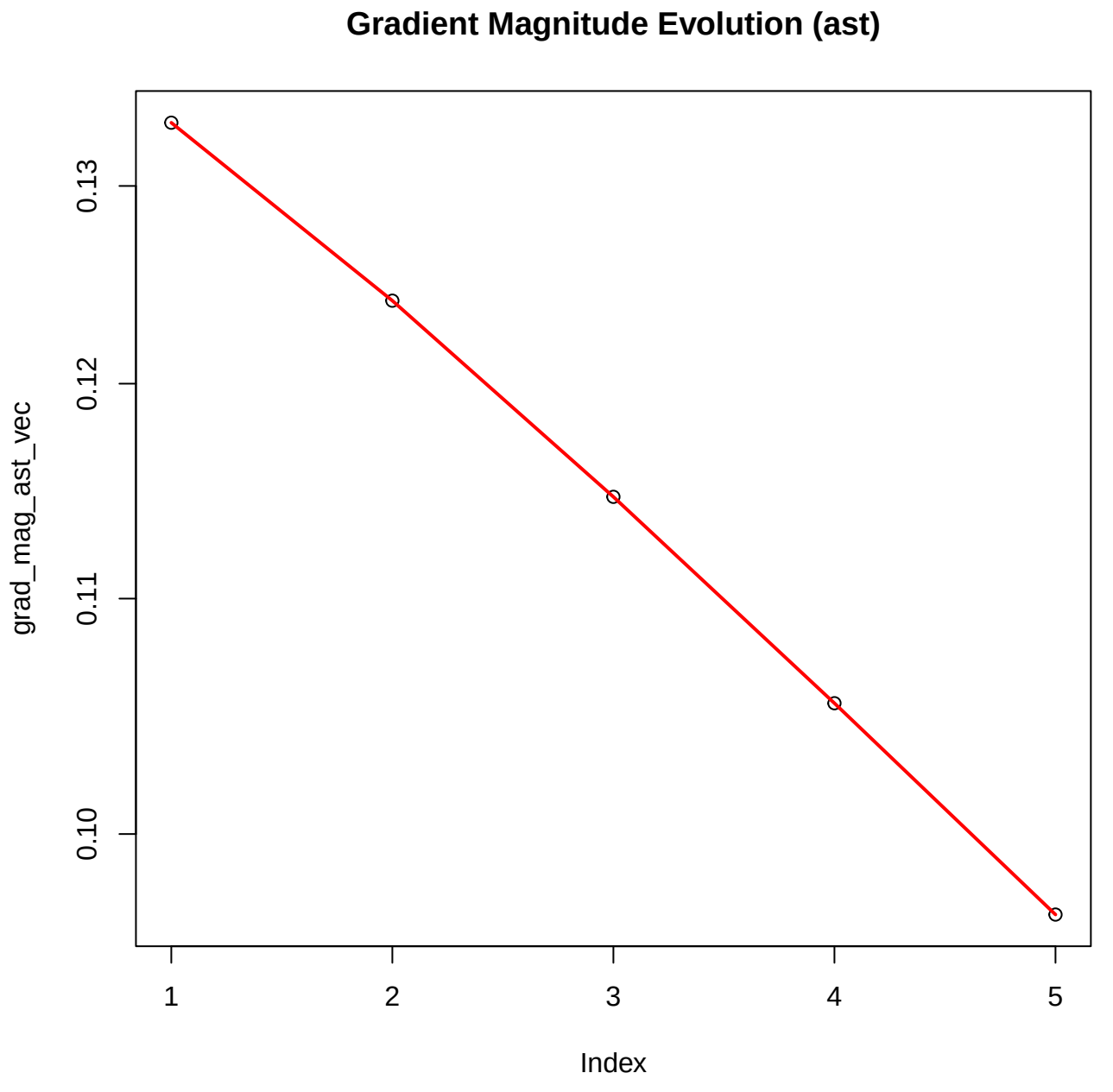
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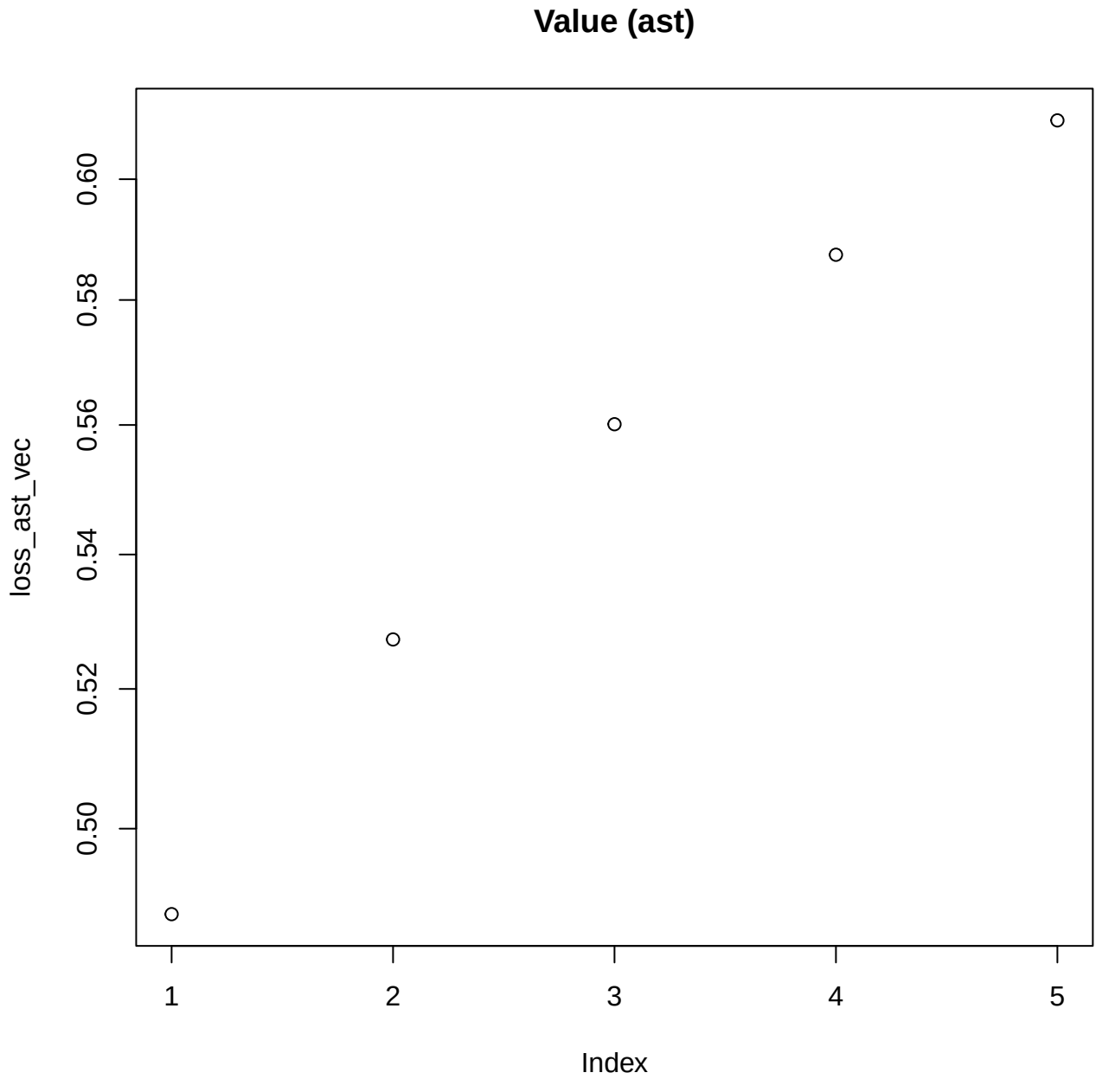
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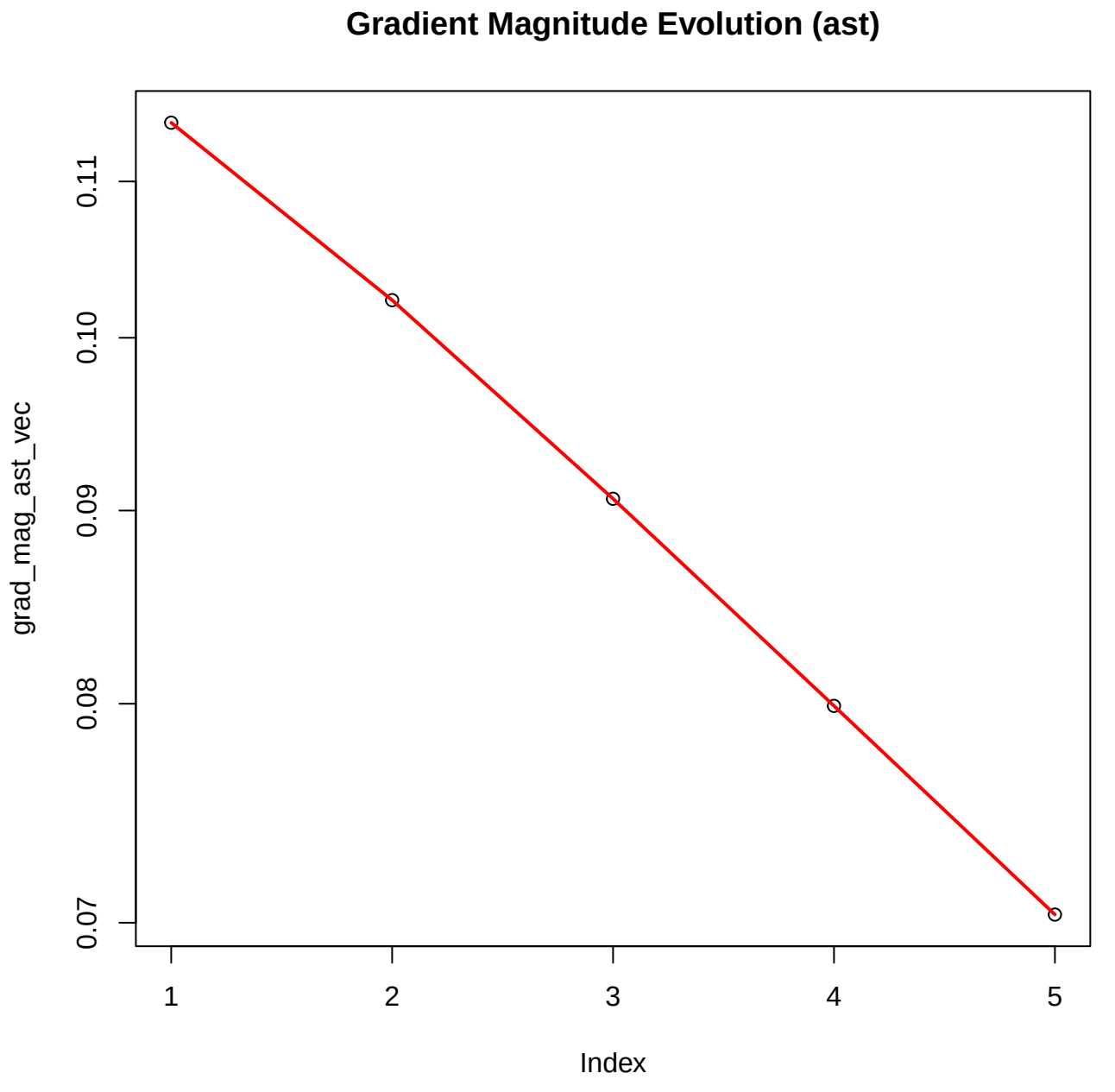
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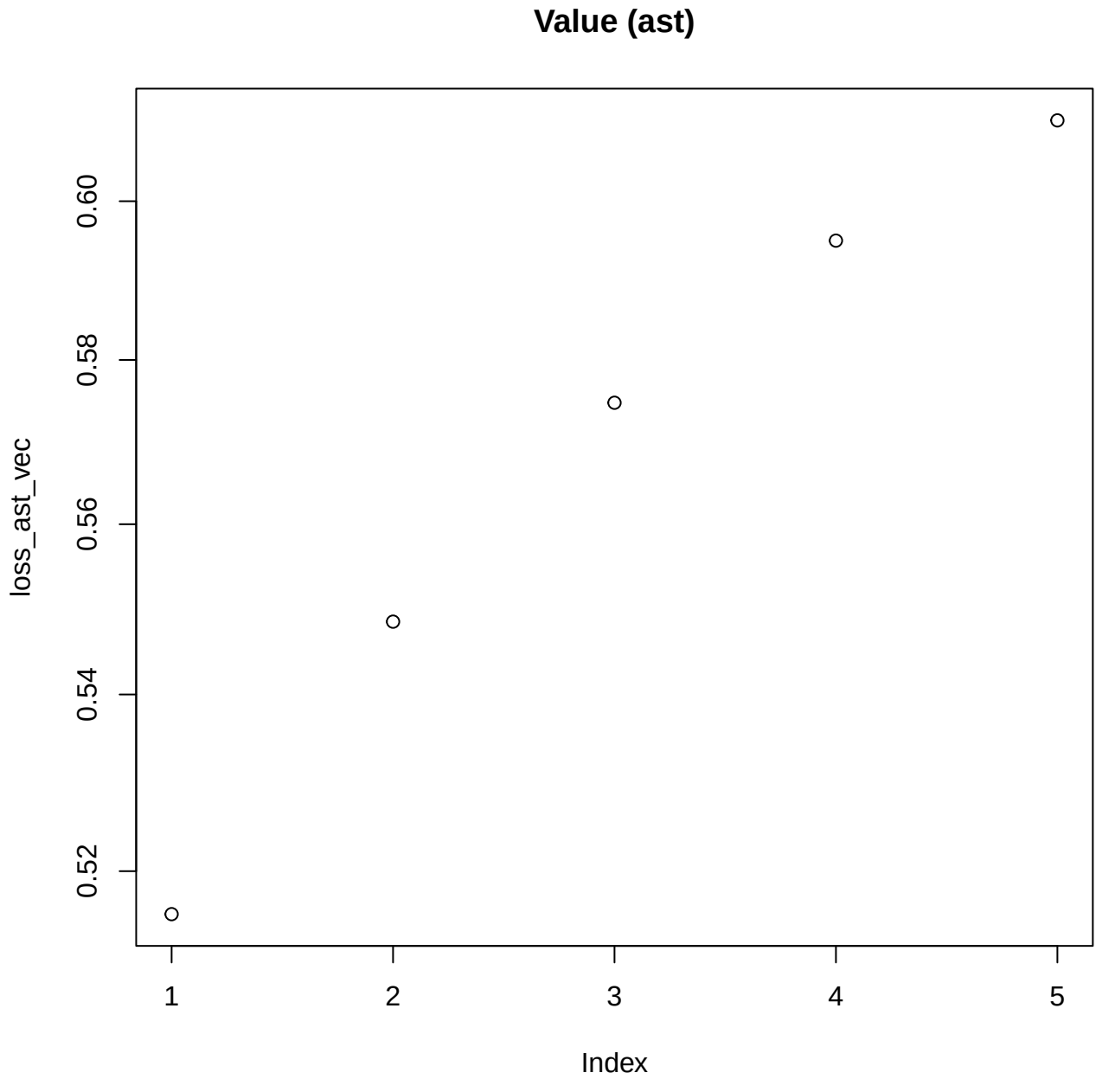
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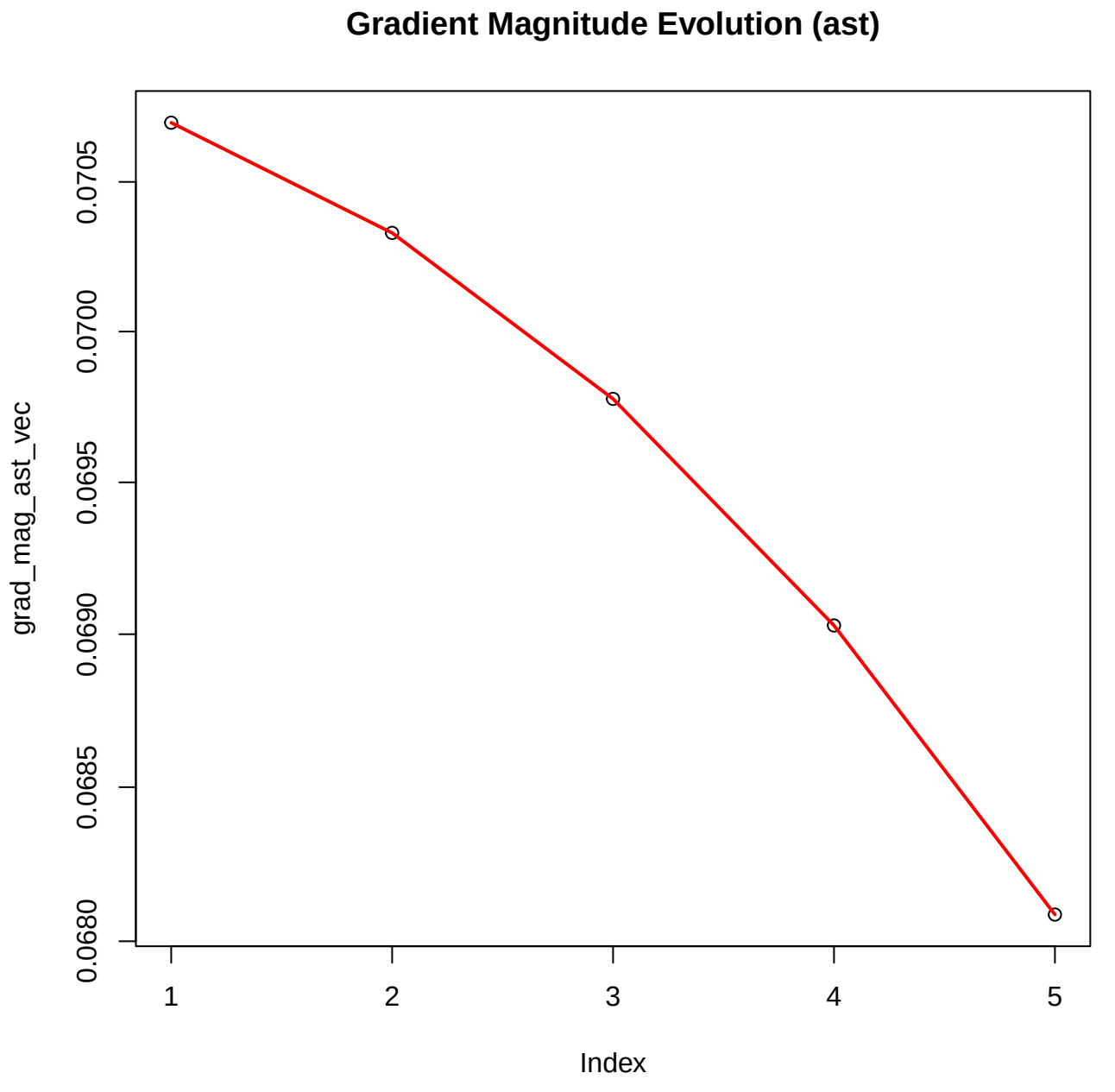
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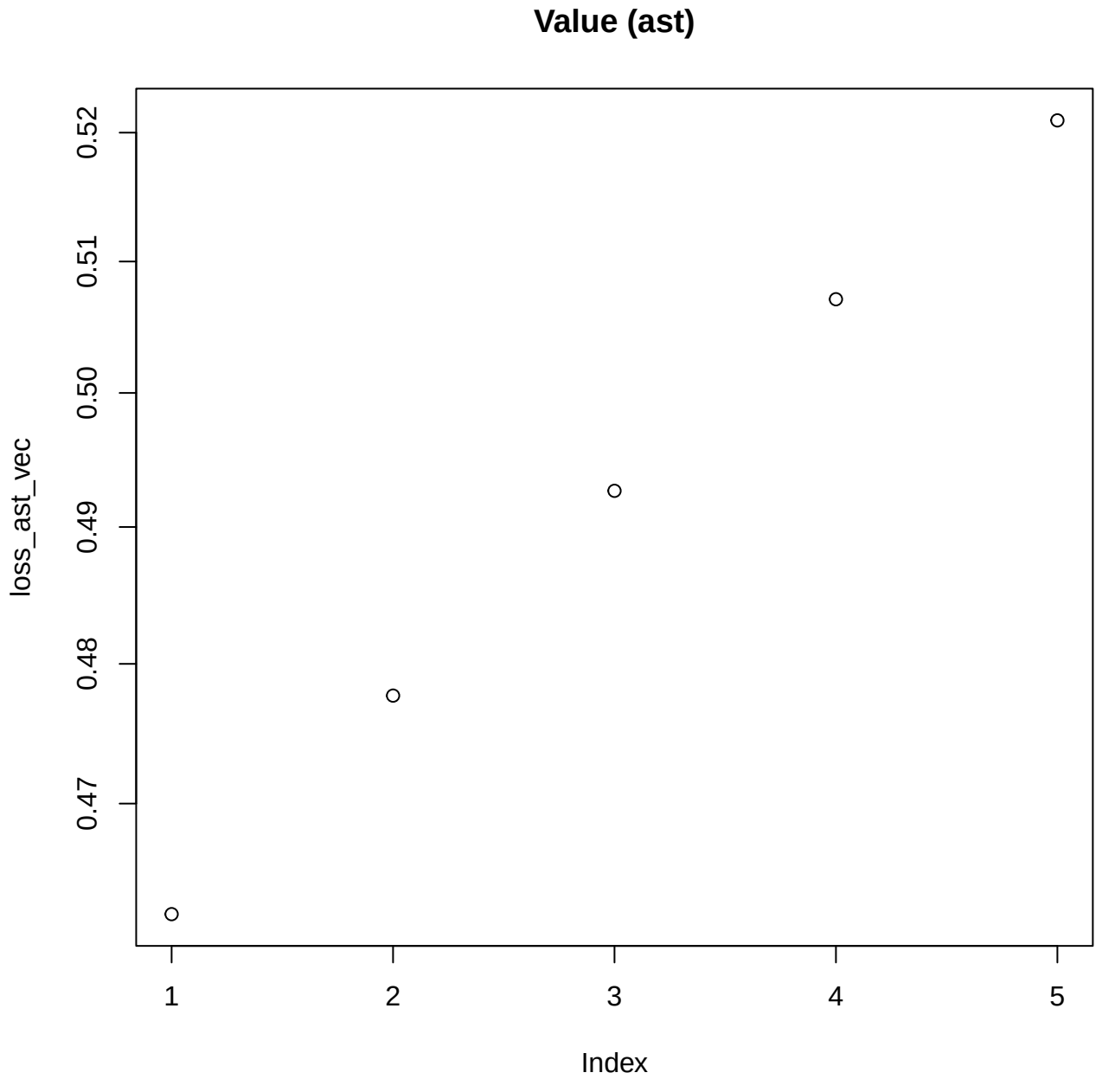
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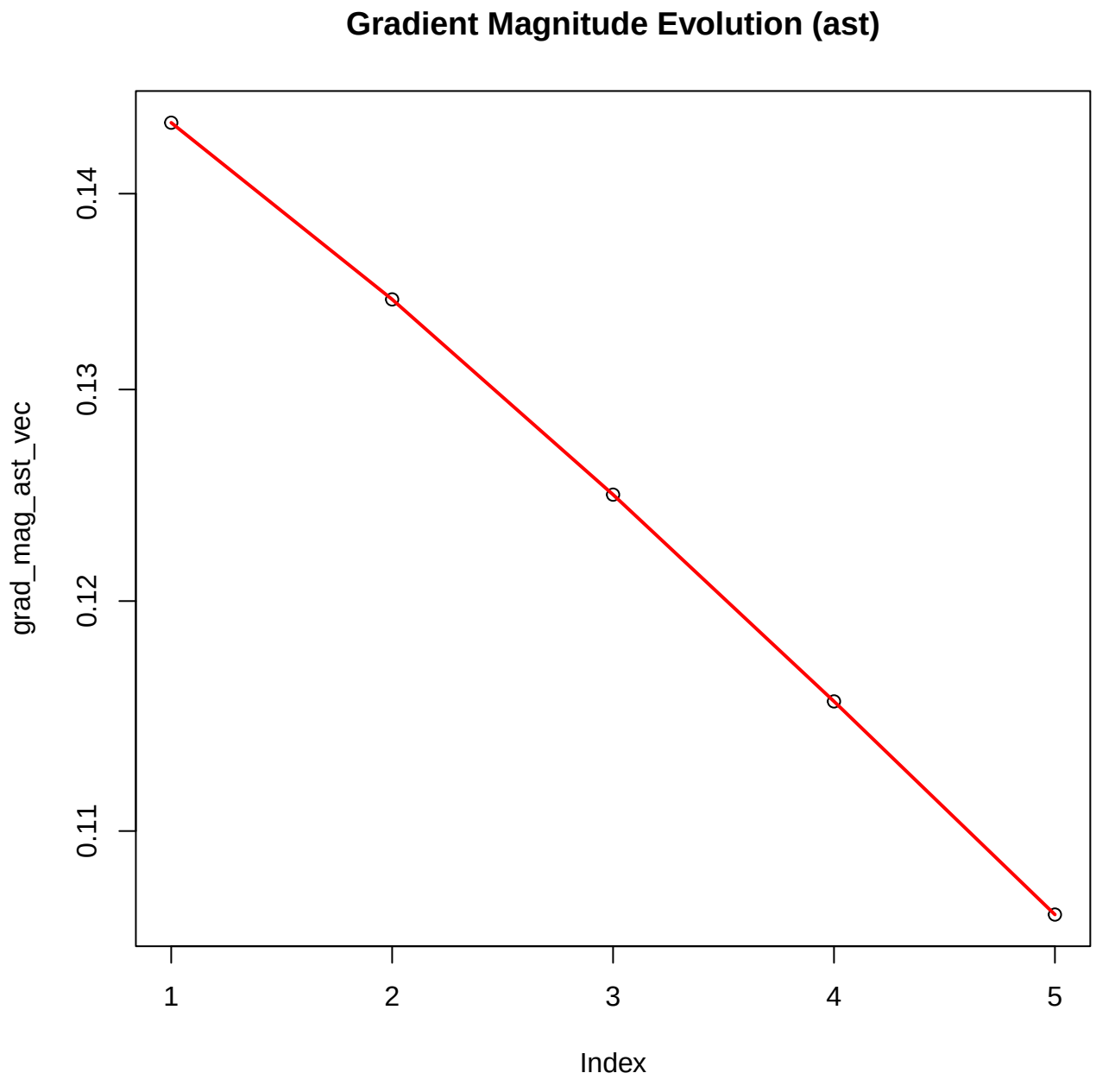
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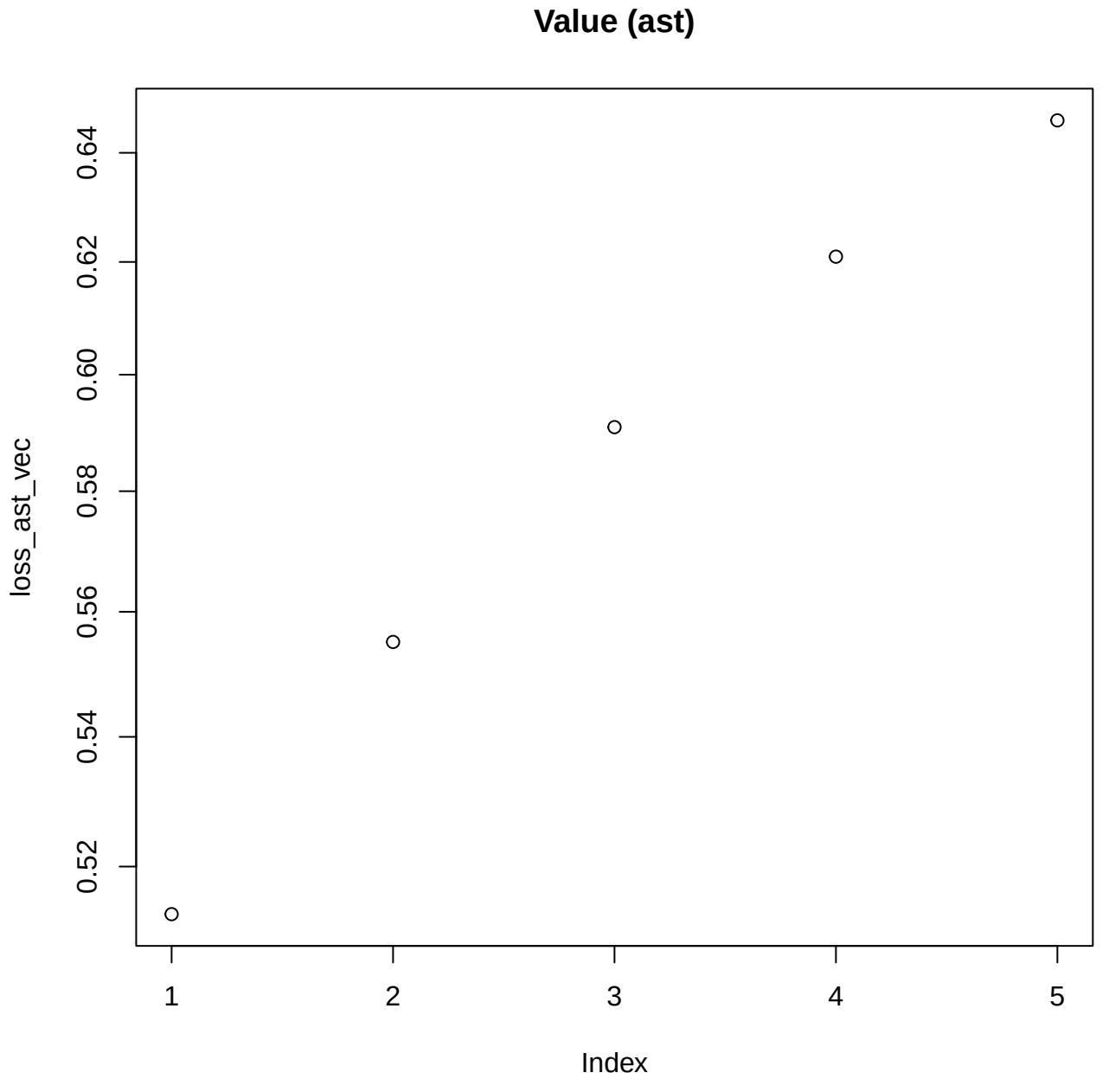
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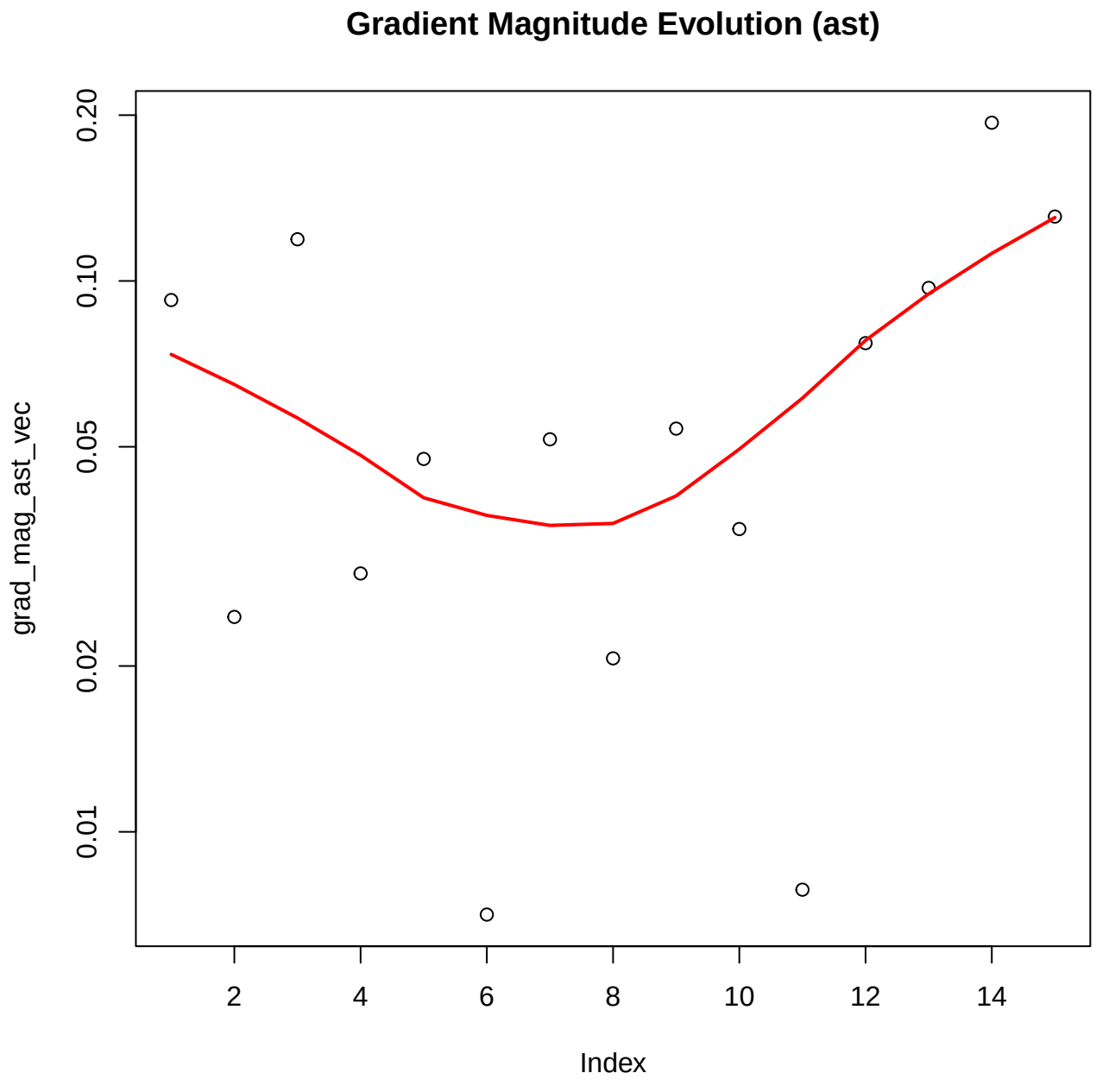
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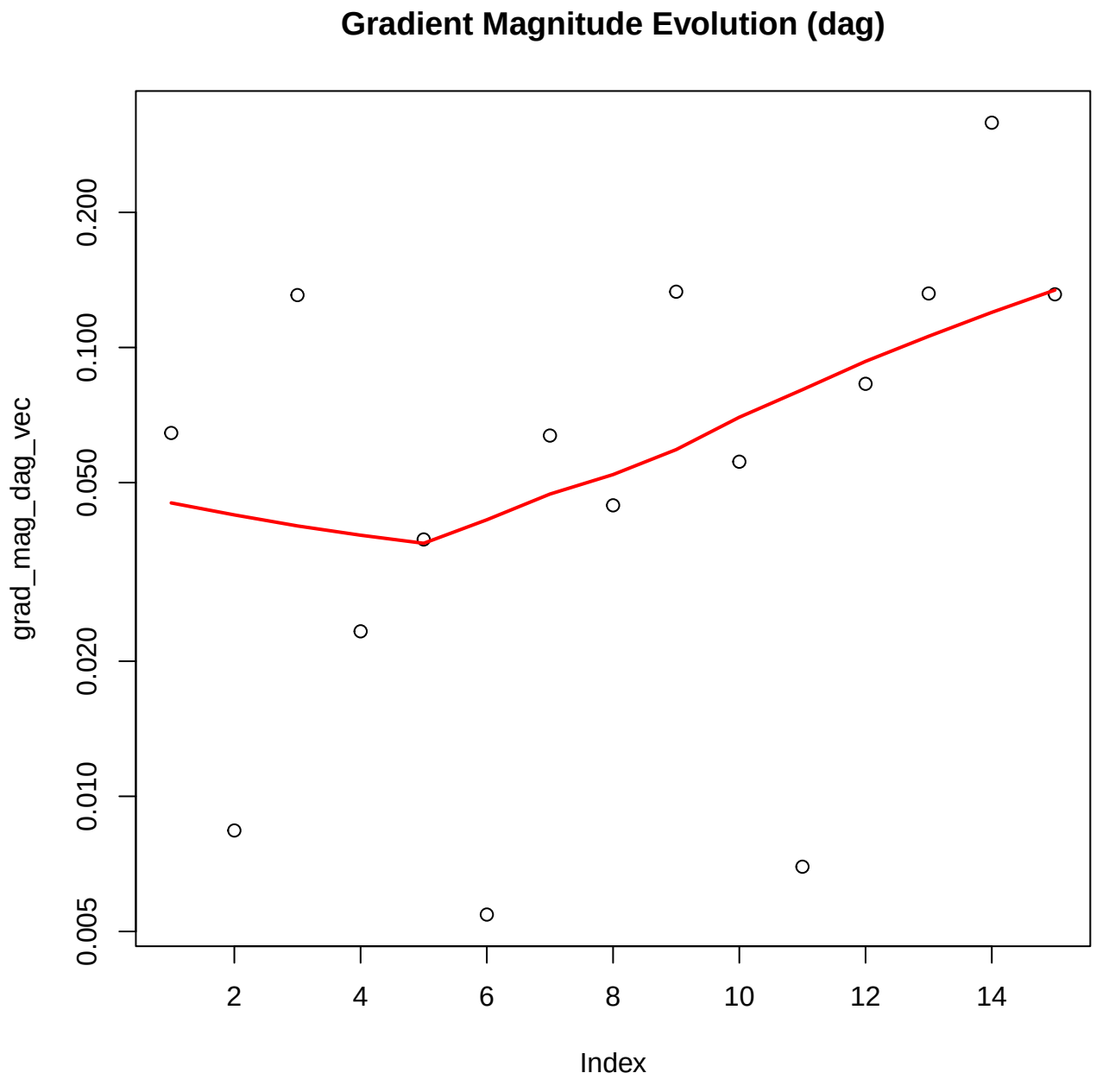
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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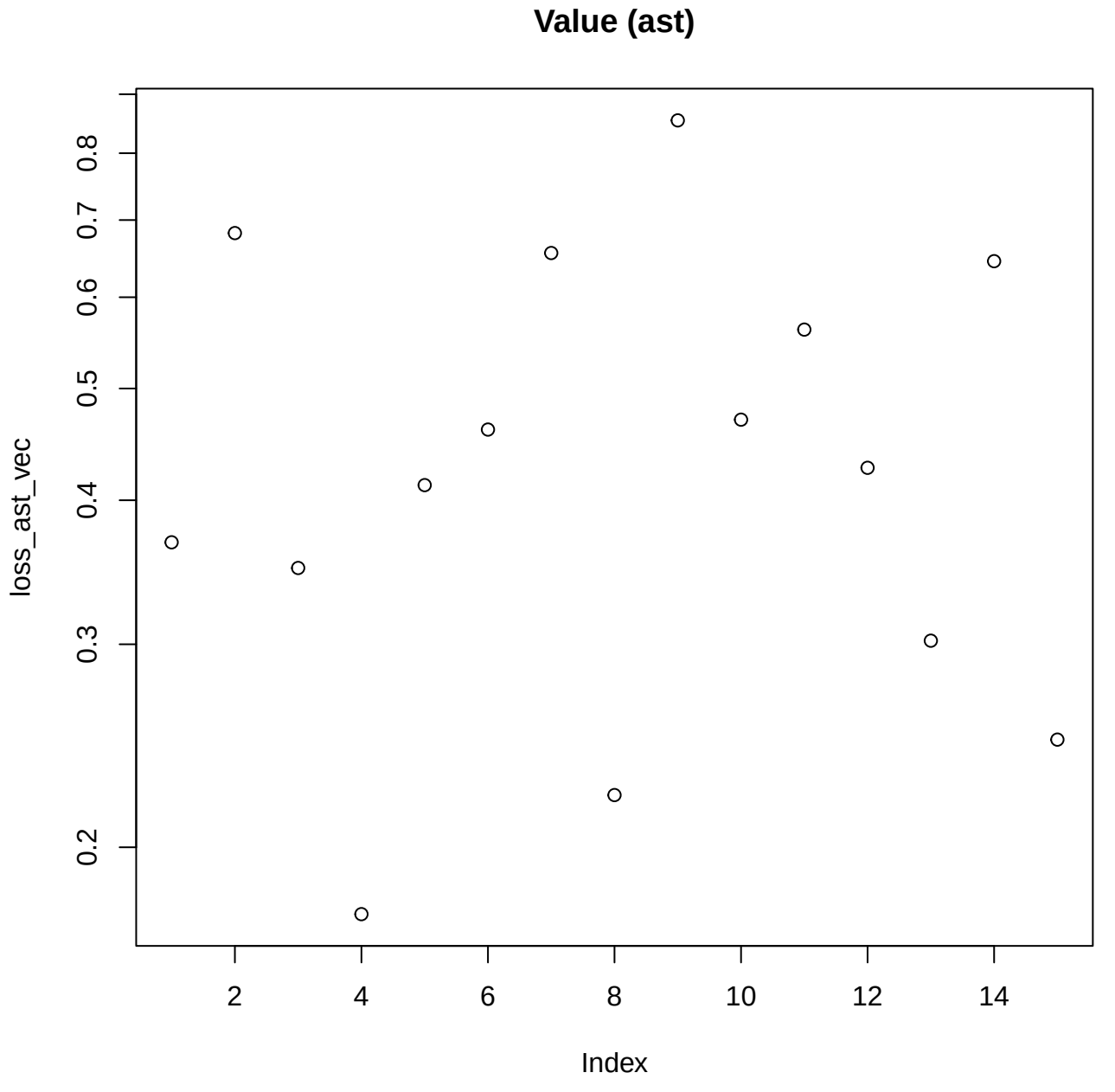
Value (ast)
loss_ast_vec
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0.6
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10
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14
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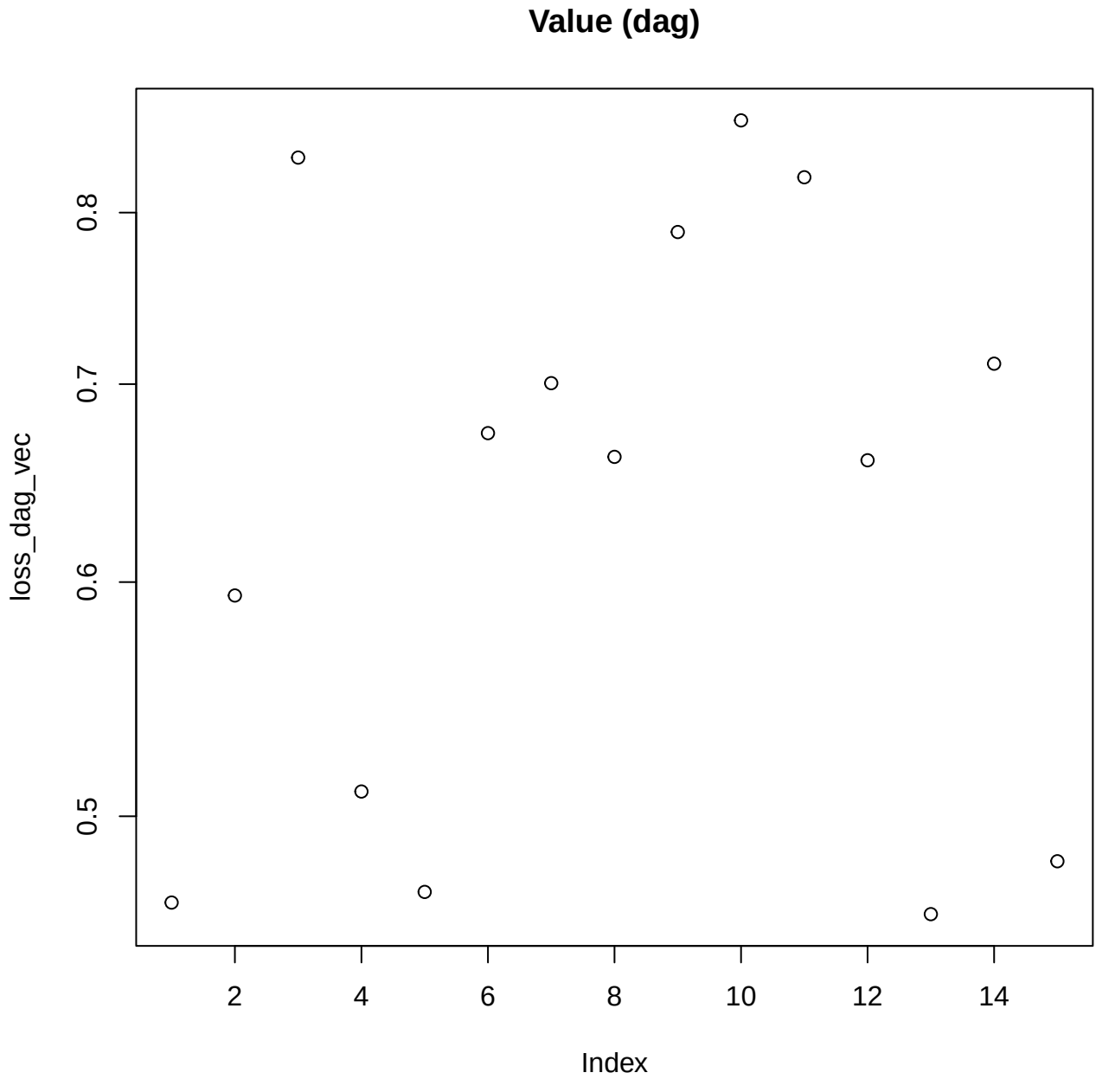
Value (dag)
loss_dag_vec
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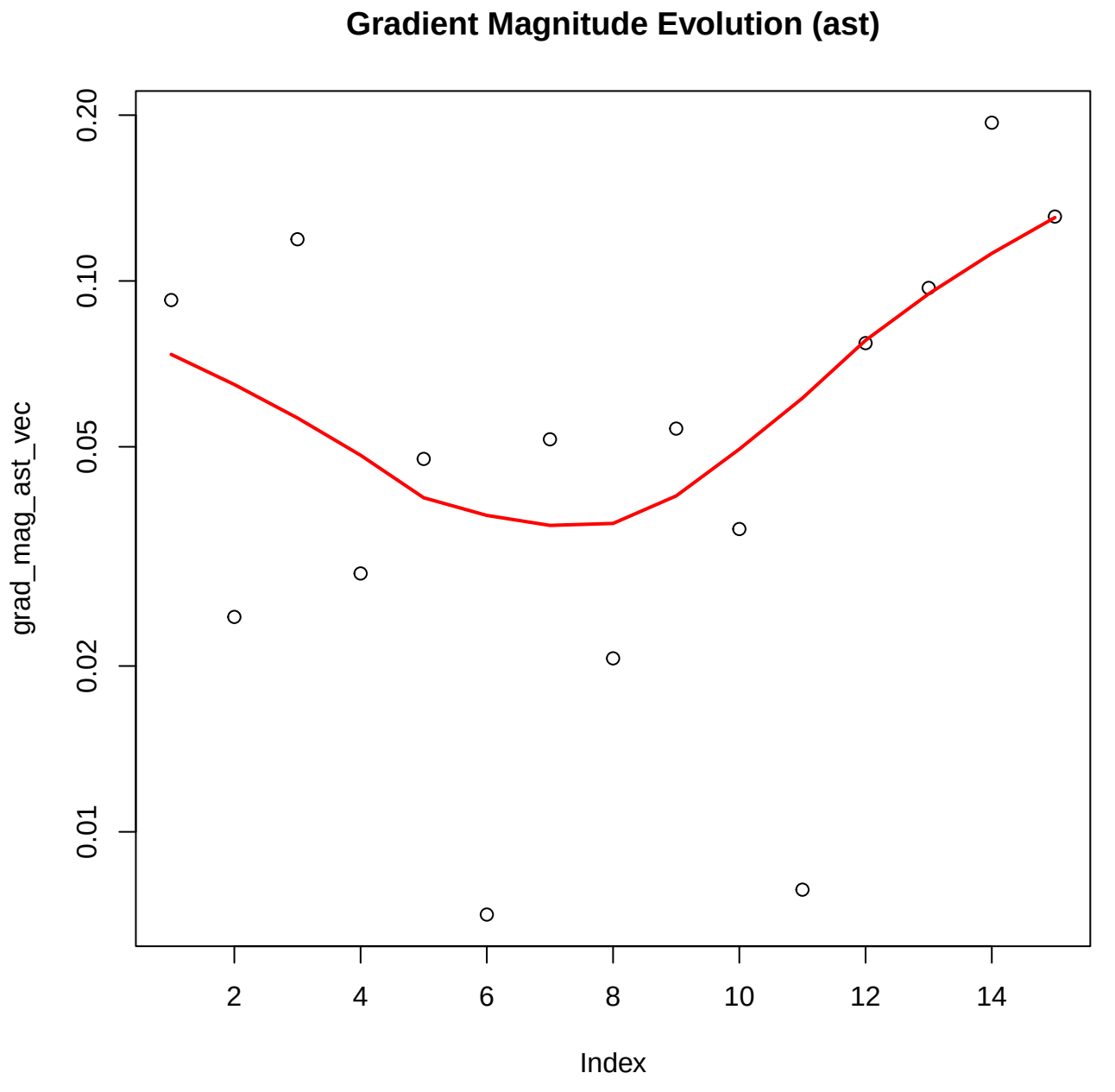
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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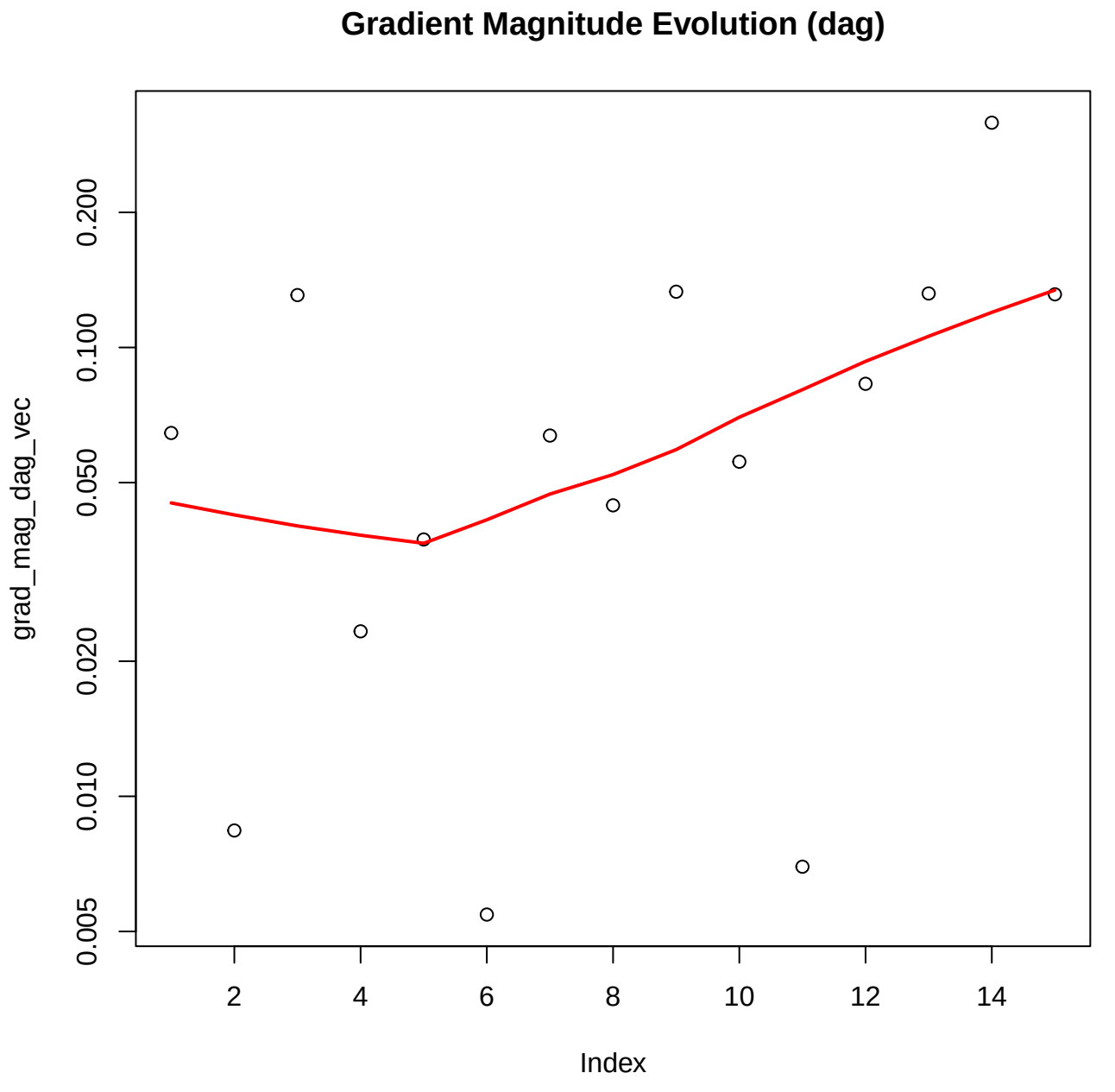
Gradient Magnitude Evolution (dag)
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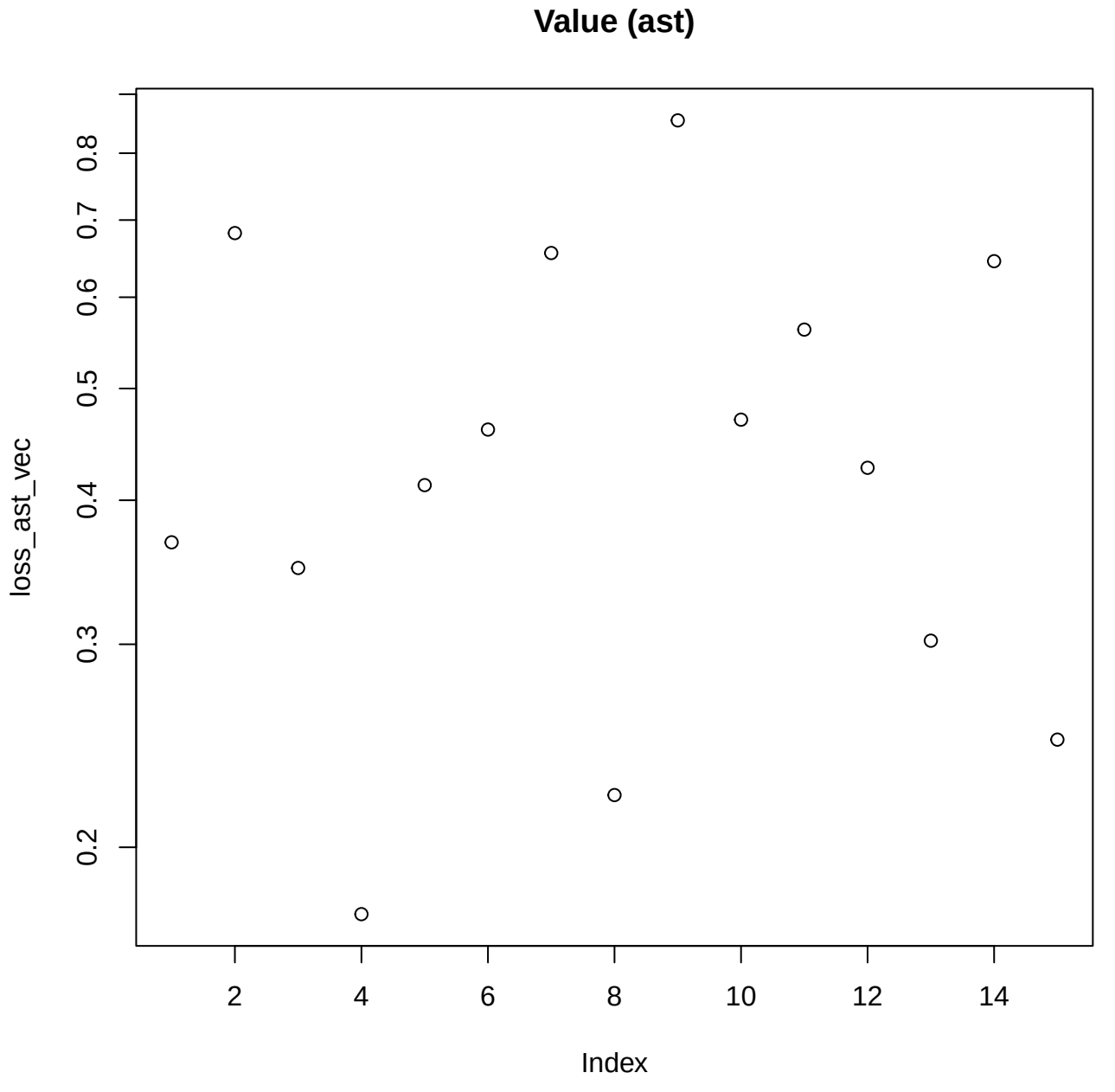
Value (ast)
loss_ast_vec
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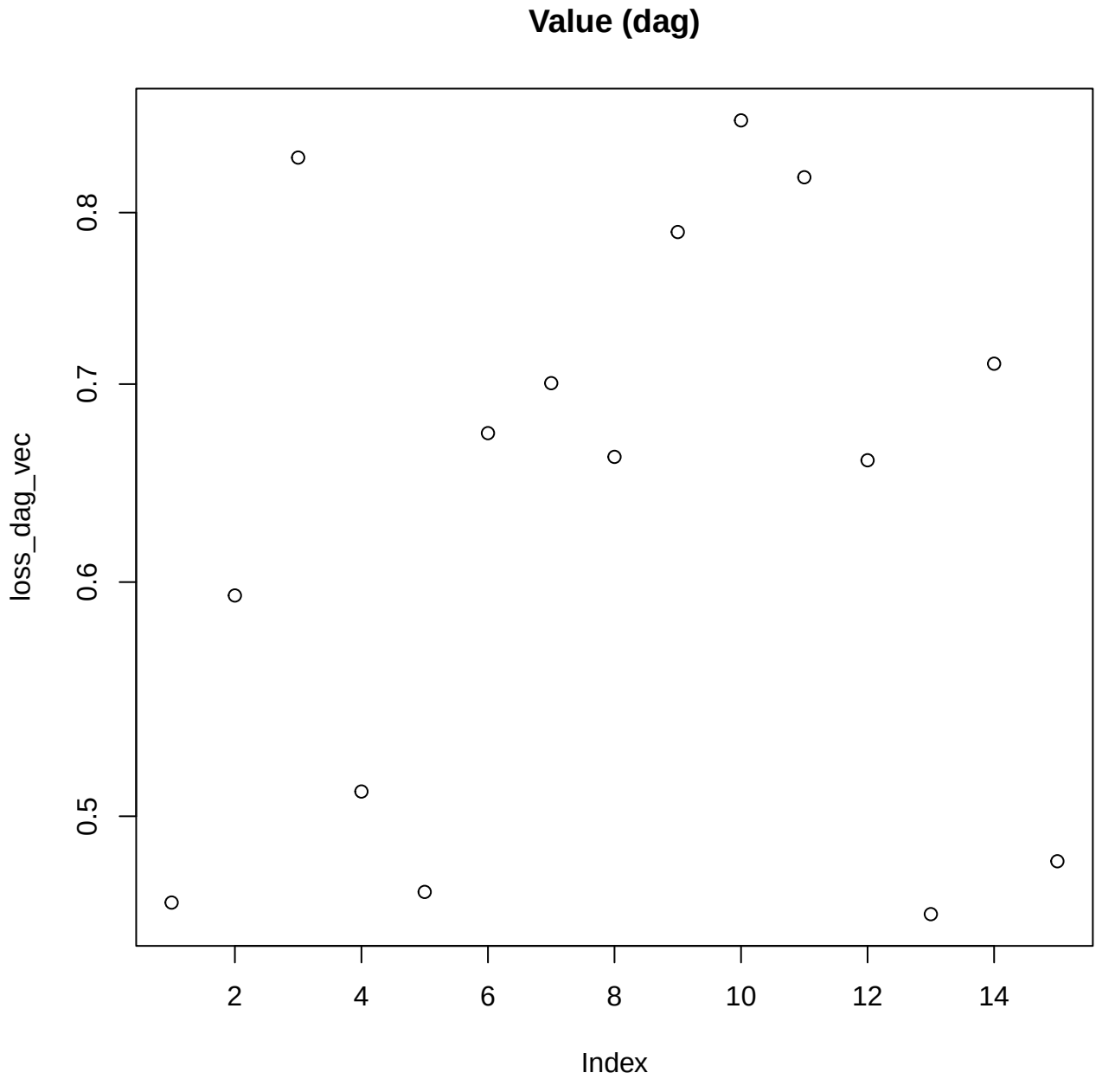
Value (dag)
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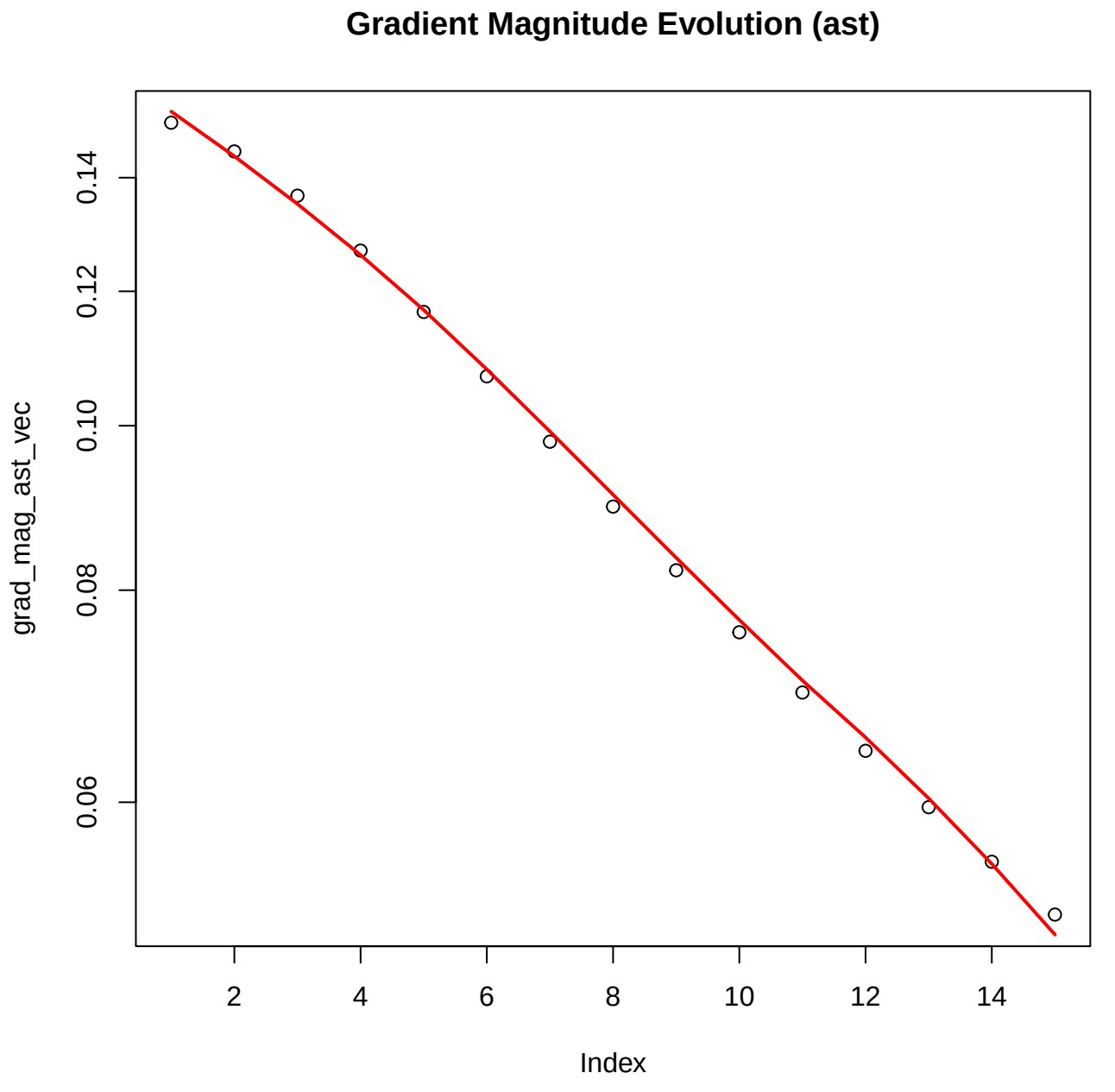
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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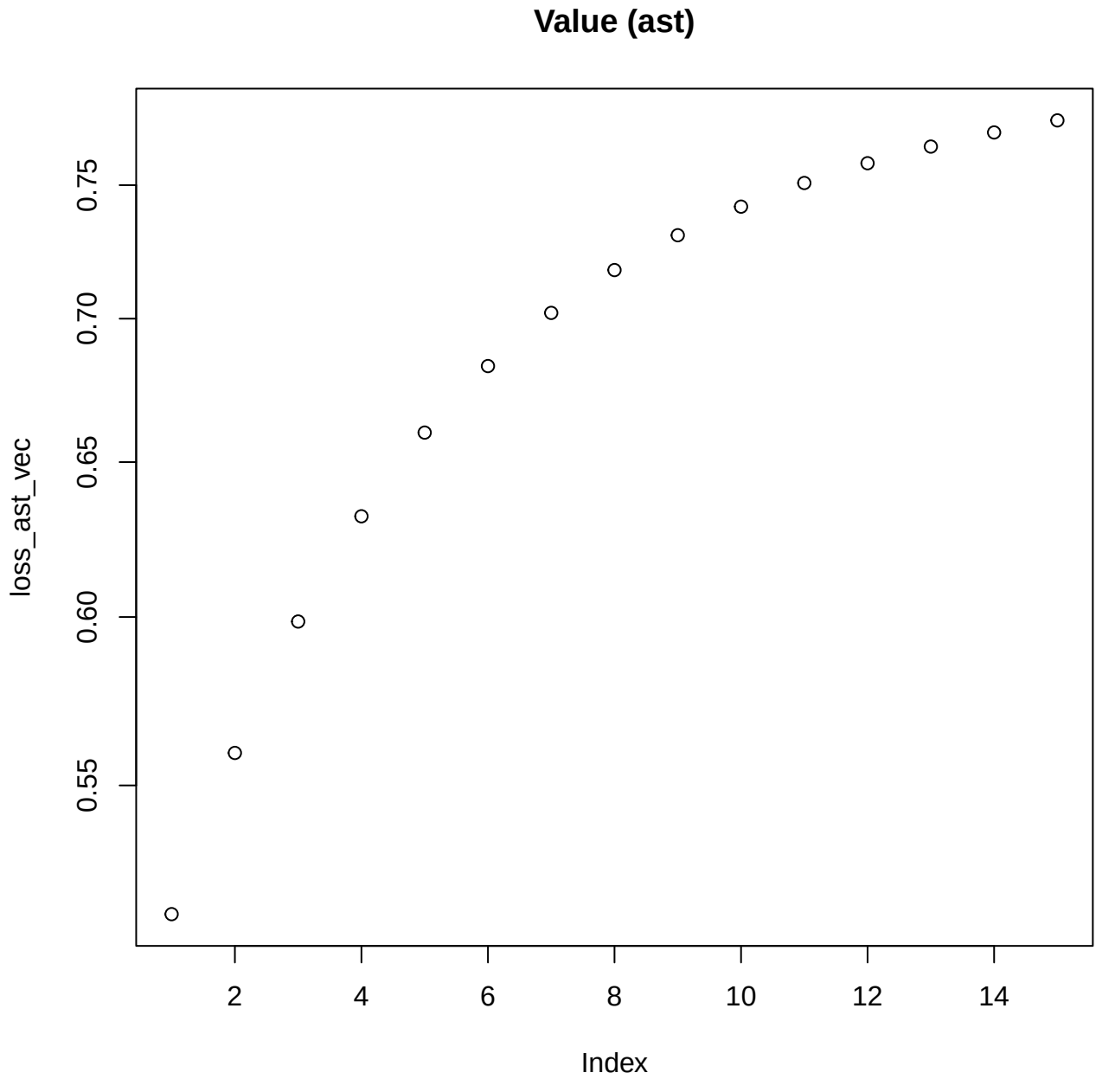
Value (ast)
loss_ast_vec
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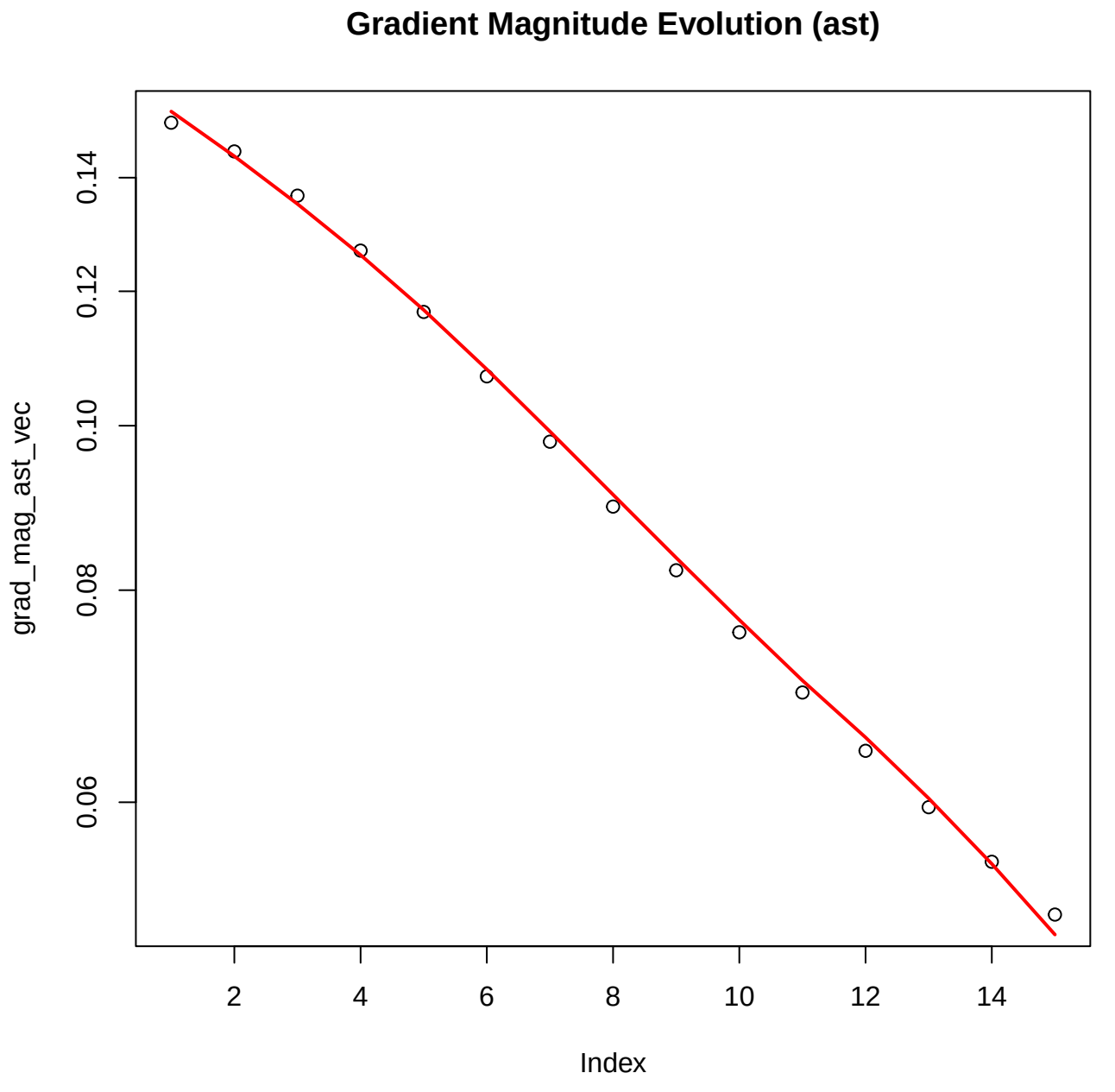
14
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.08
0.10
0.12
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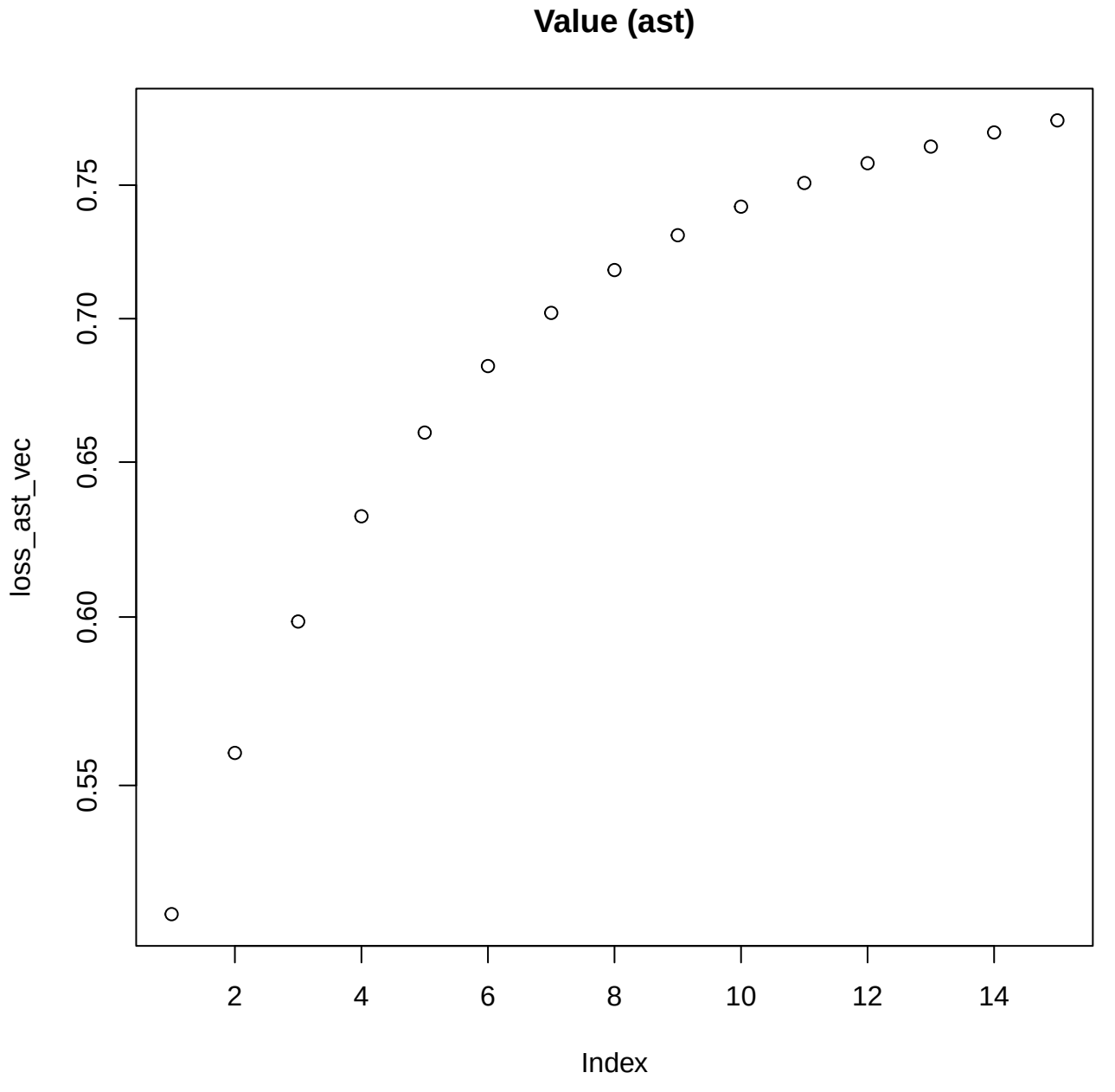
Value (ast)
loss_ast_vec
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0.60
0.65
0.70
0.75
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12
14
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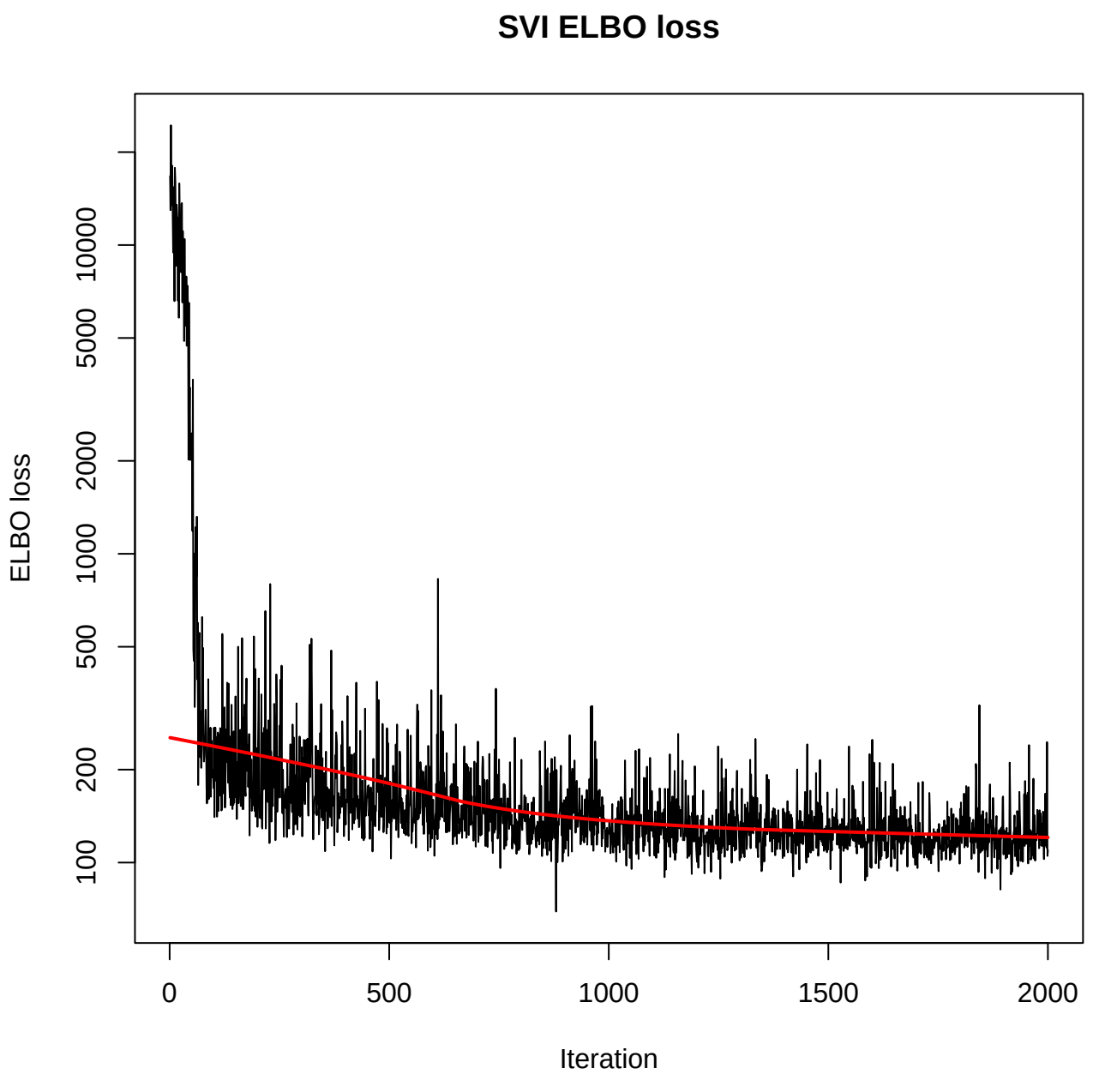
SVI ELBO loss
ELBO loss
10000
5000
2000
1000
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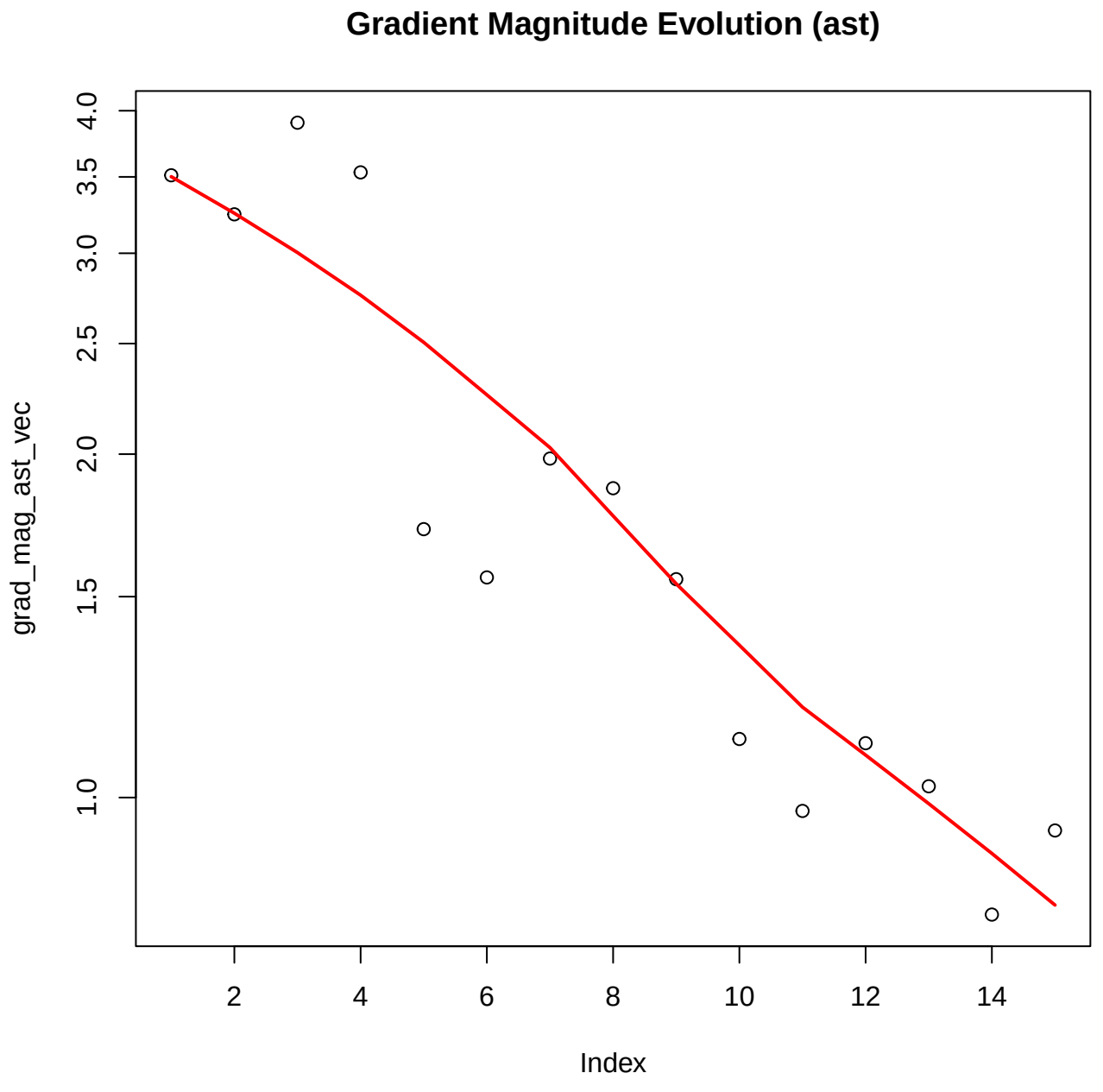
200
100
0
500
1000
1500
2000
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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3.0
2.5
2.0
1.5
1.0
2
4
6
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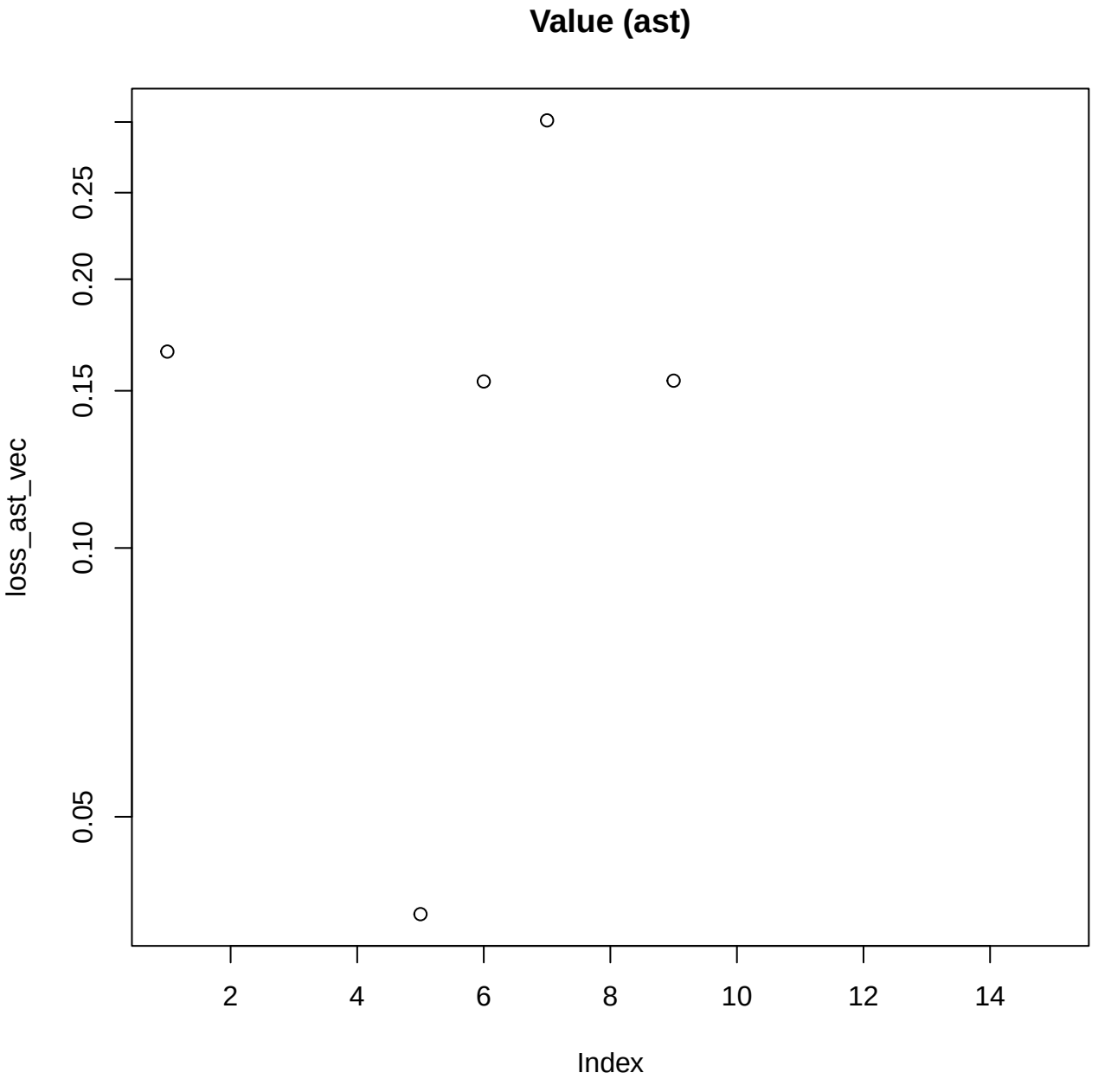
12
14
Index


Value (ast)
loss_ast_vec
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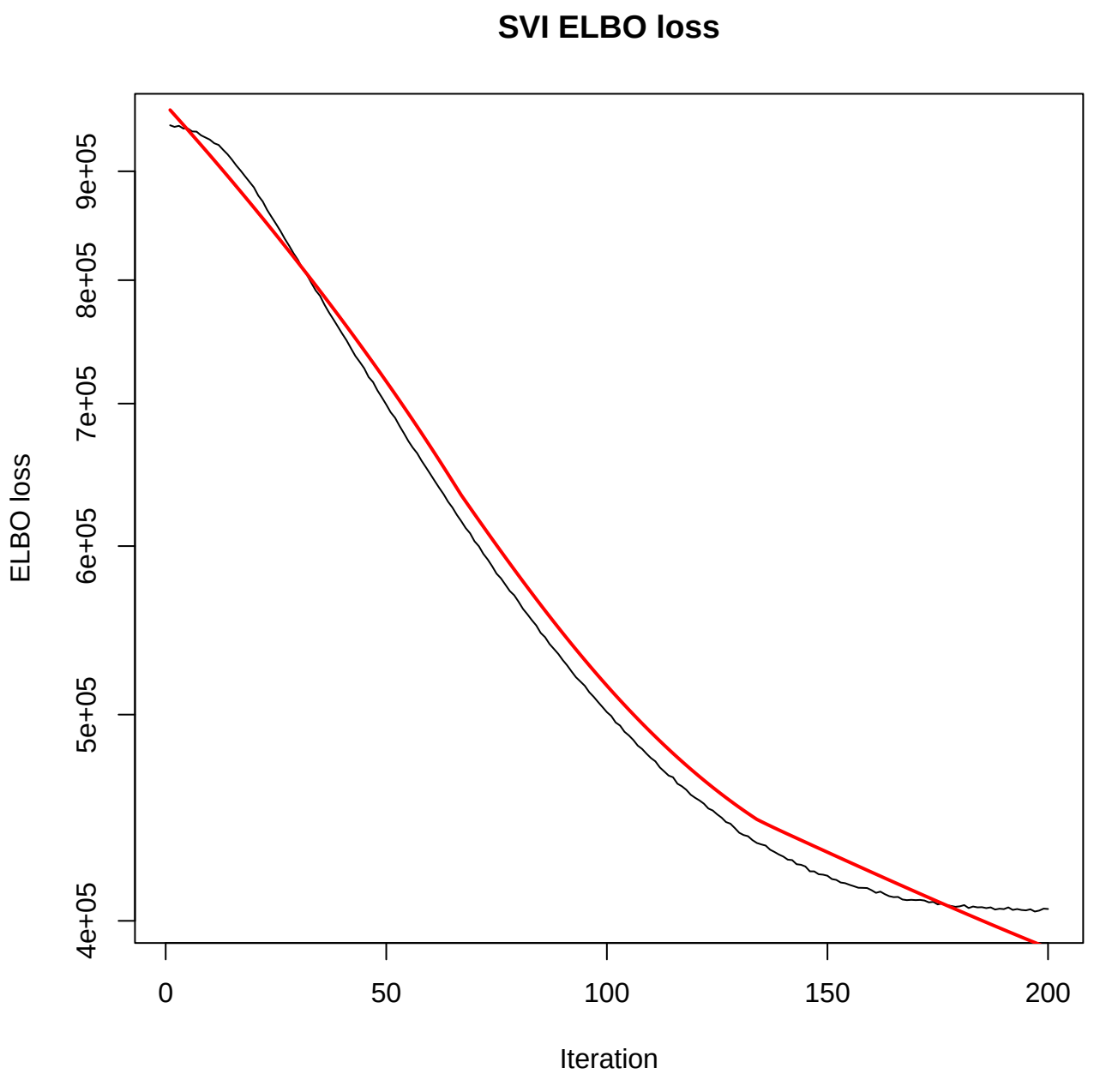
0.10
0.15
0.20
0.25
2
4
6
8
10
12
14
Index


SVI ELBO loss
ELBO loss
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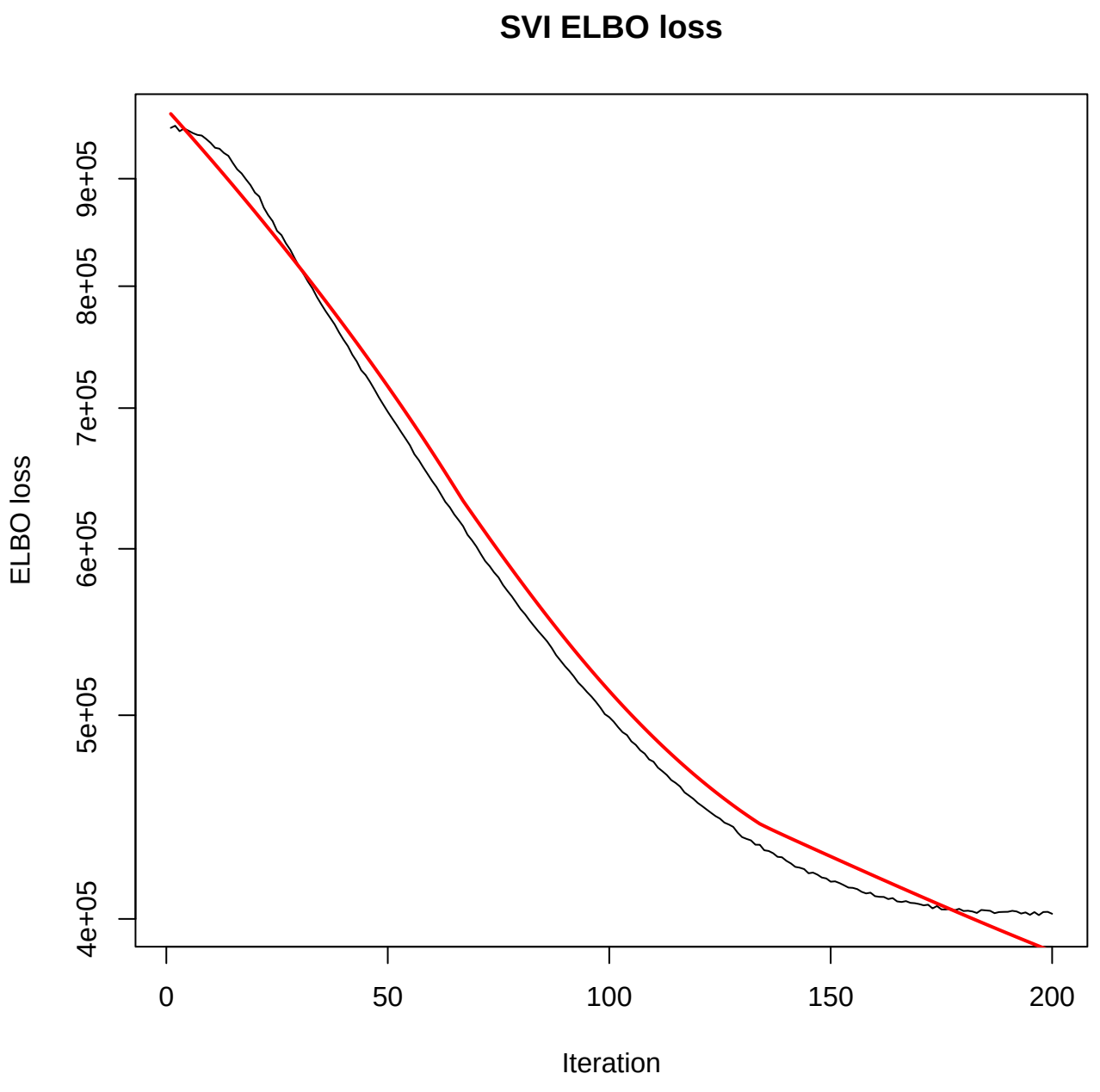
8e+05
7e+05
6e+05
5e+05
4e+05
0
50
100
150
200
Iteration


SVI ELBO loss
ELBO loss
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8e+05
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5e+05
4e+05
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50
100
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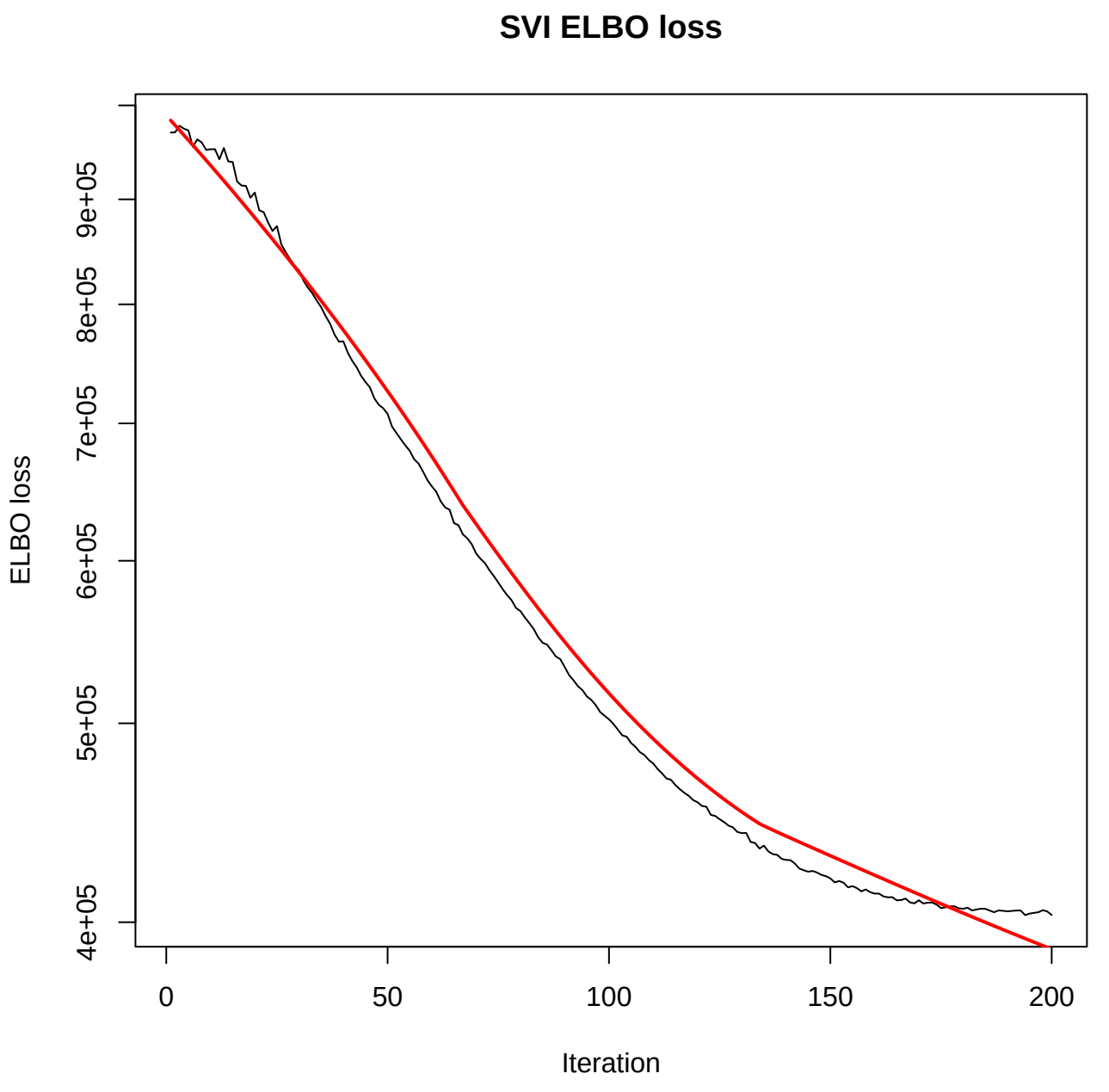
200
Iteration


SVI ELBO loss
ELBO loss
9e+05
8e+05
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5e+05
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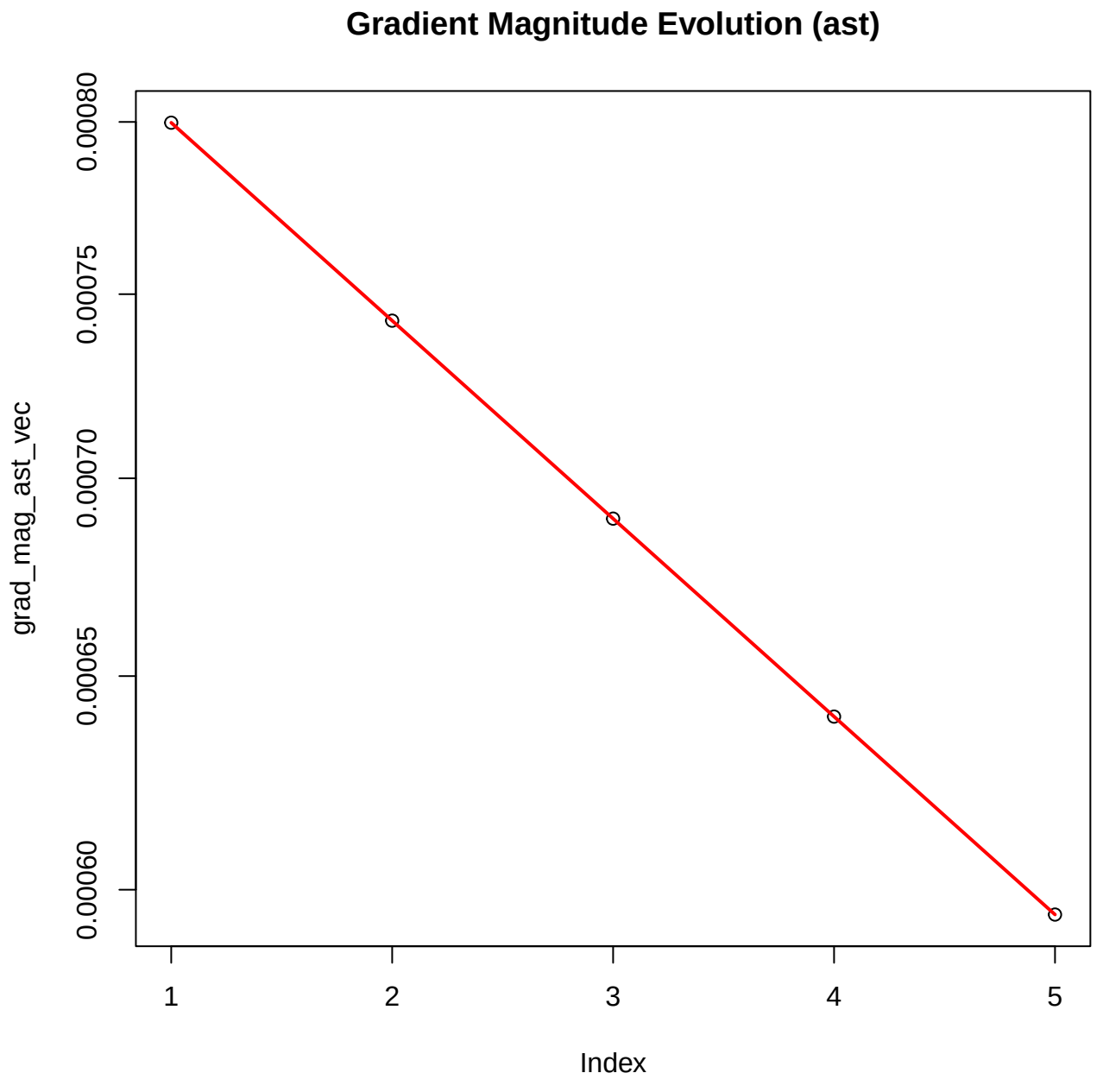
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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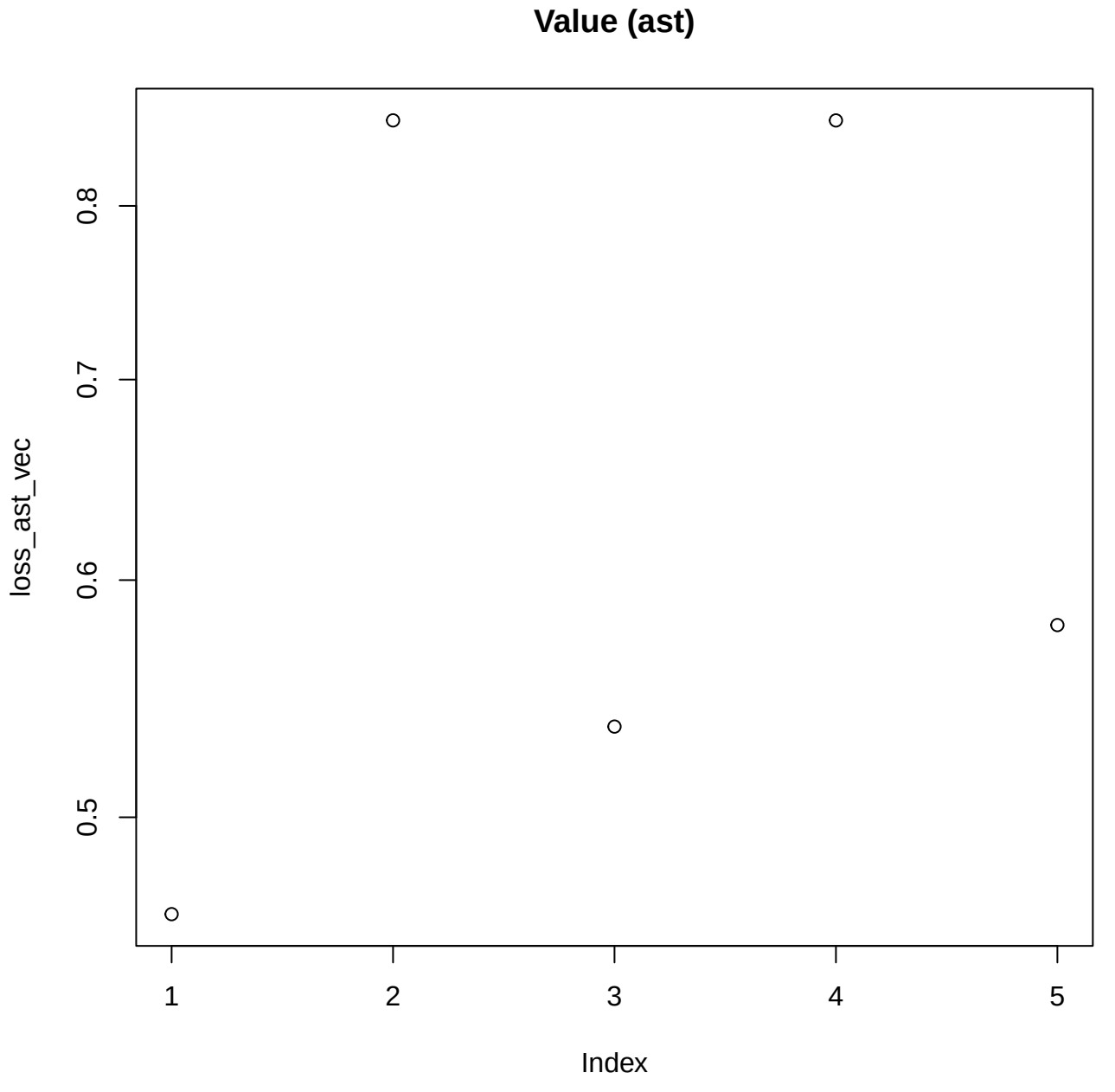
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.5
0.6
0.7
0.8
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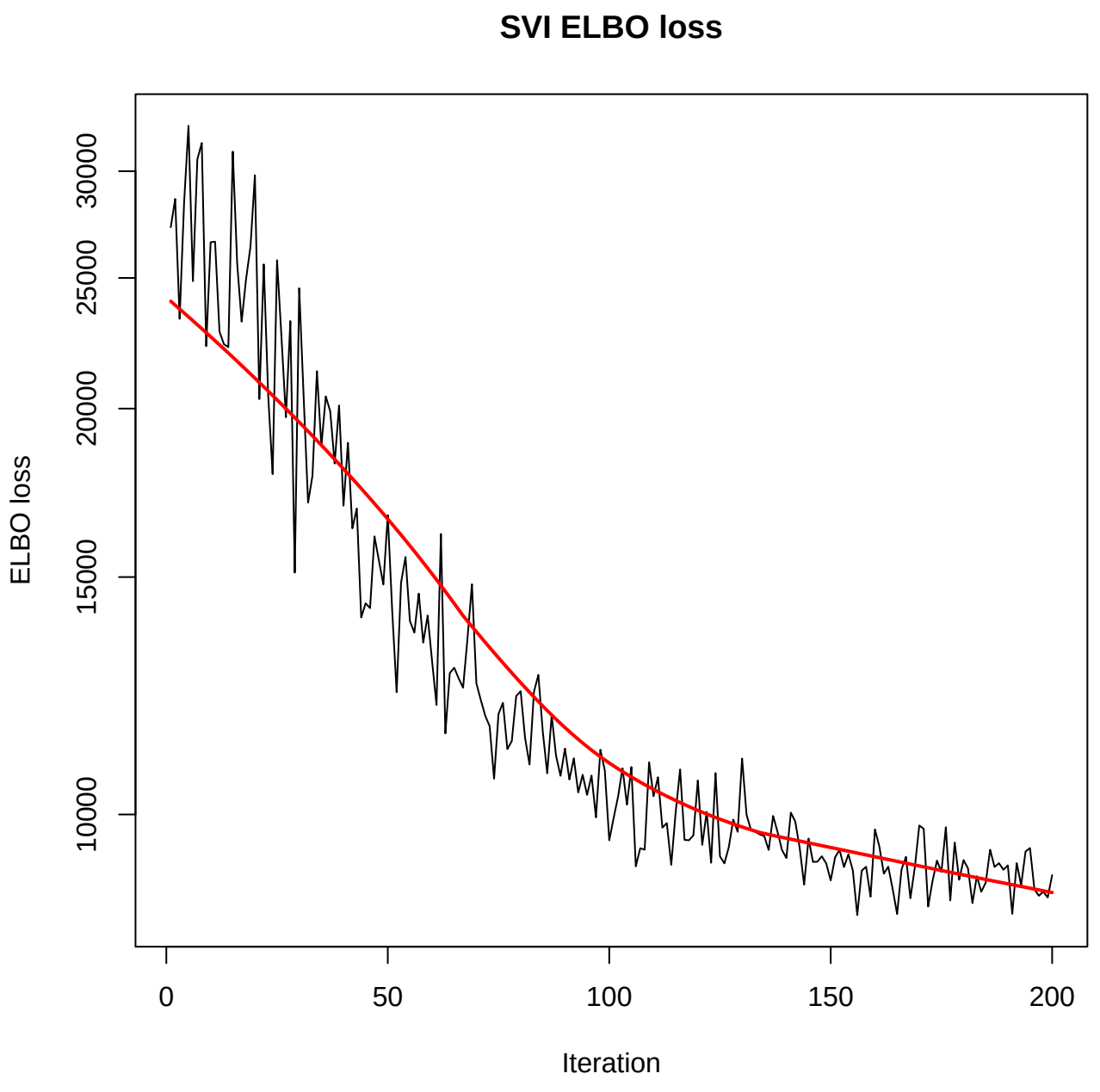
5
Index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ELBO loss
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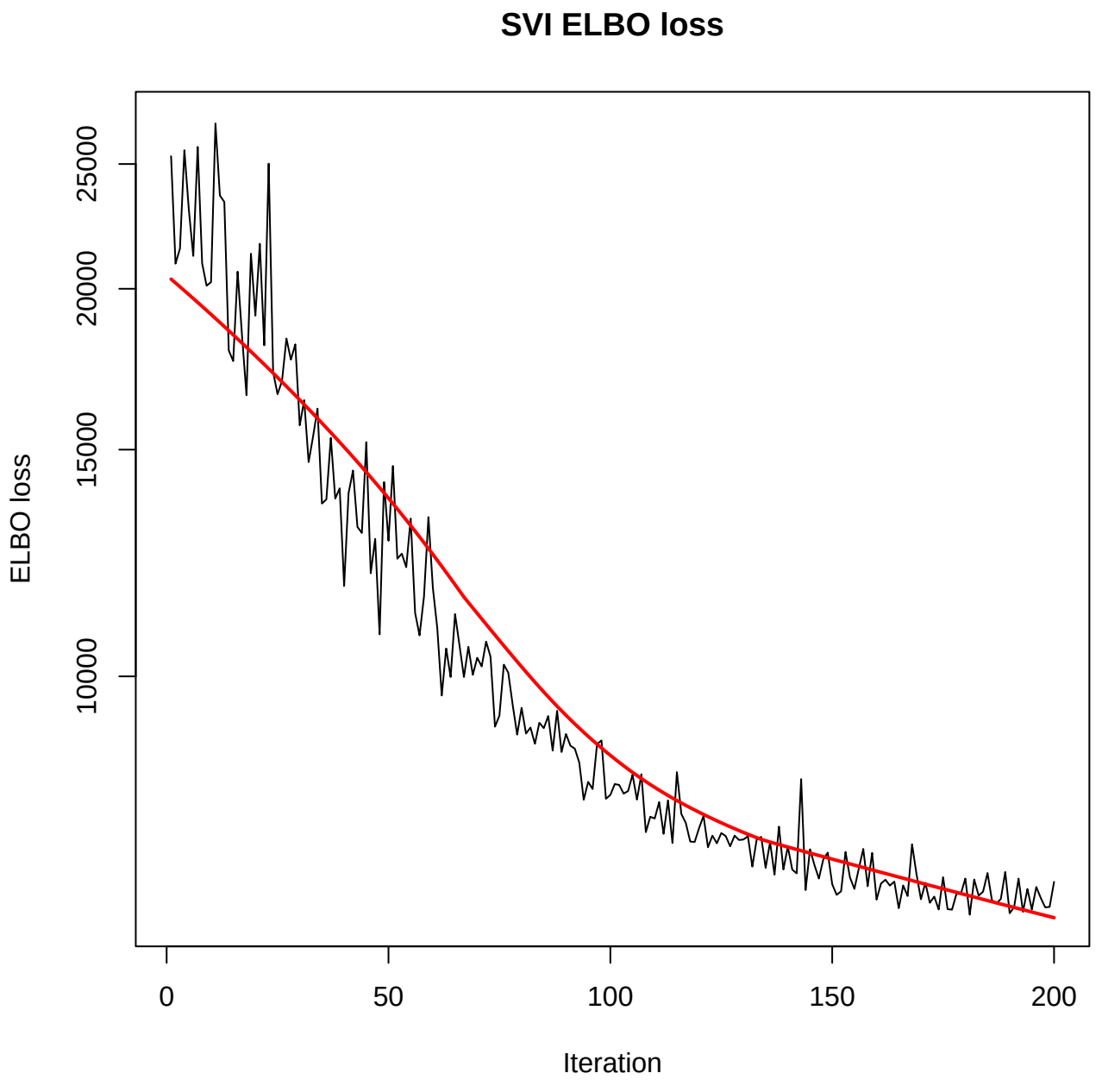
200
Iteration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ELBO loss
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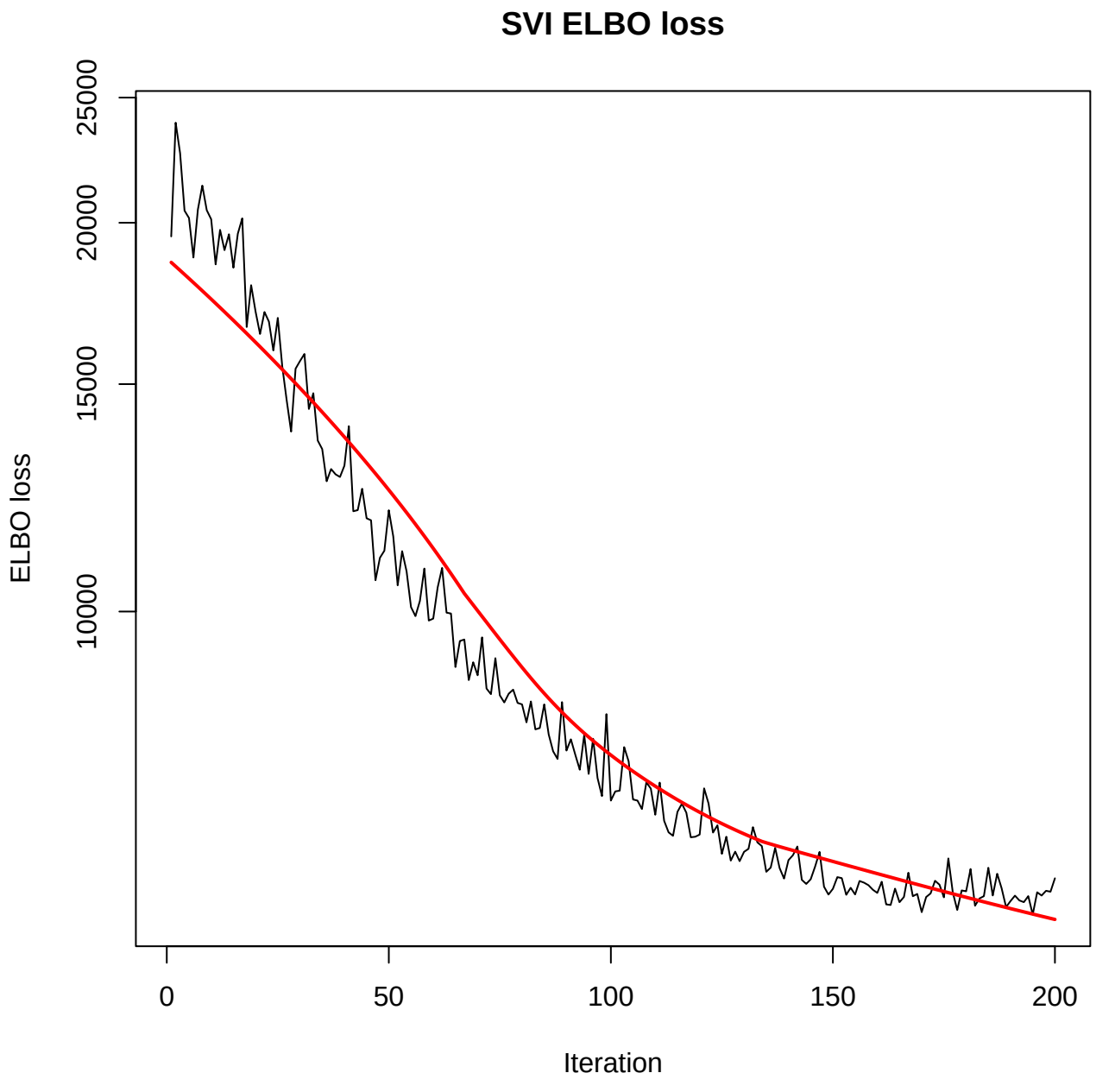
Iteration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ELBO loss
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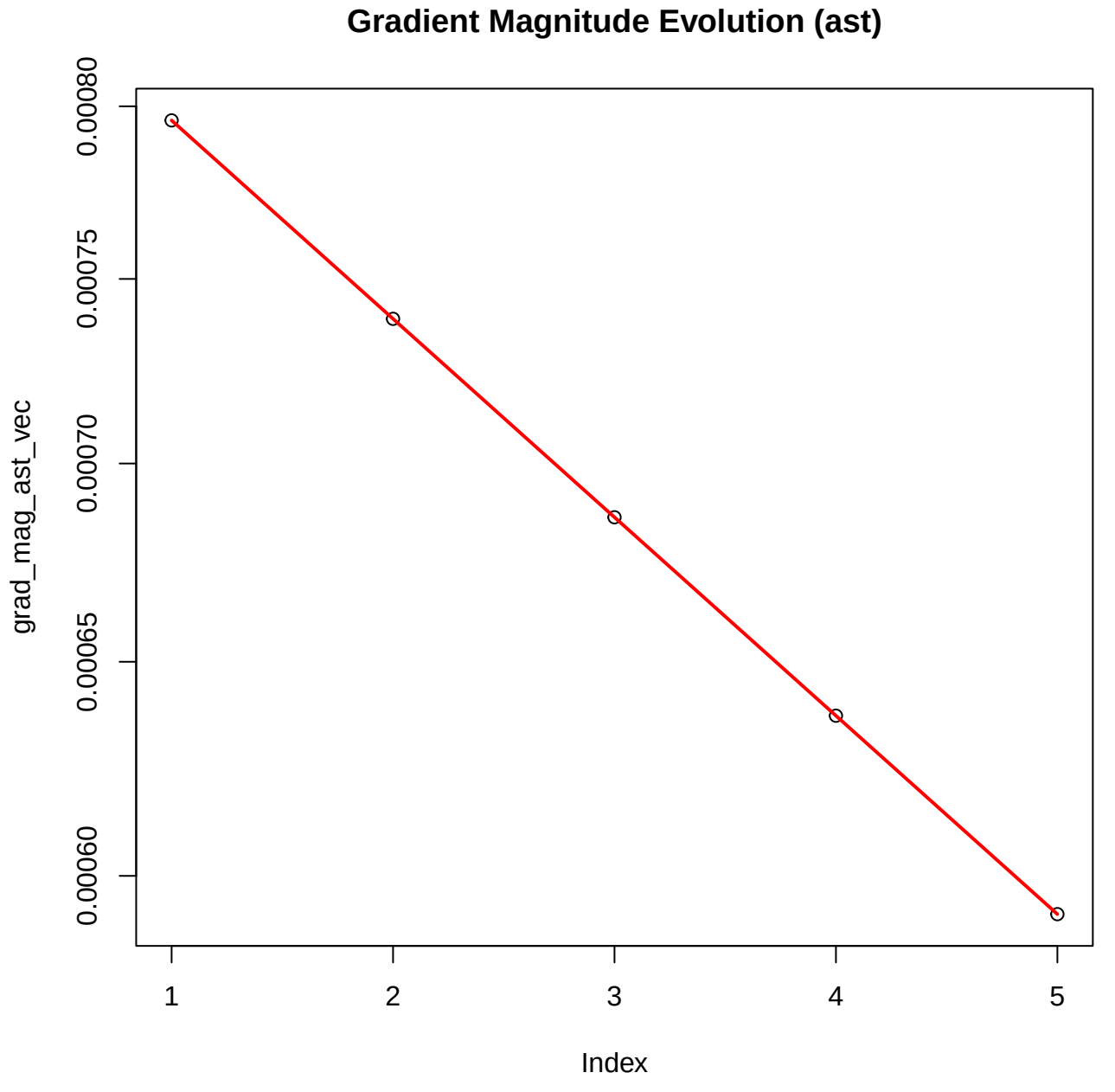
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.00075
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3
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5
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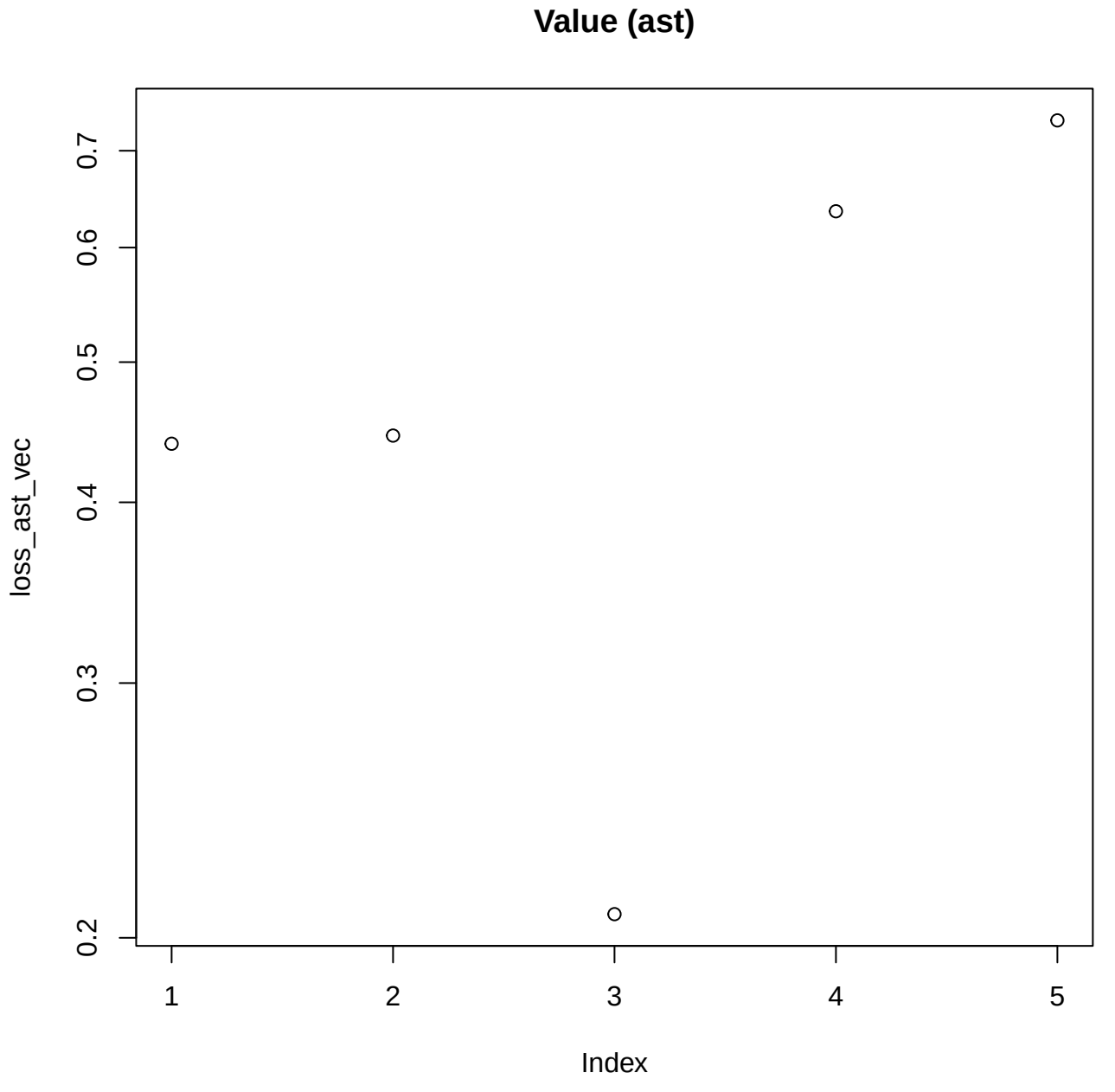
Value (ast)
loss_ast_vec
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0.5
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0.7
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2
3
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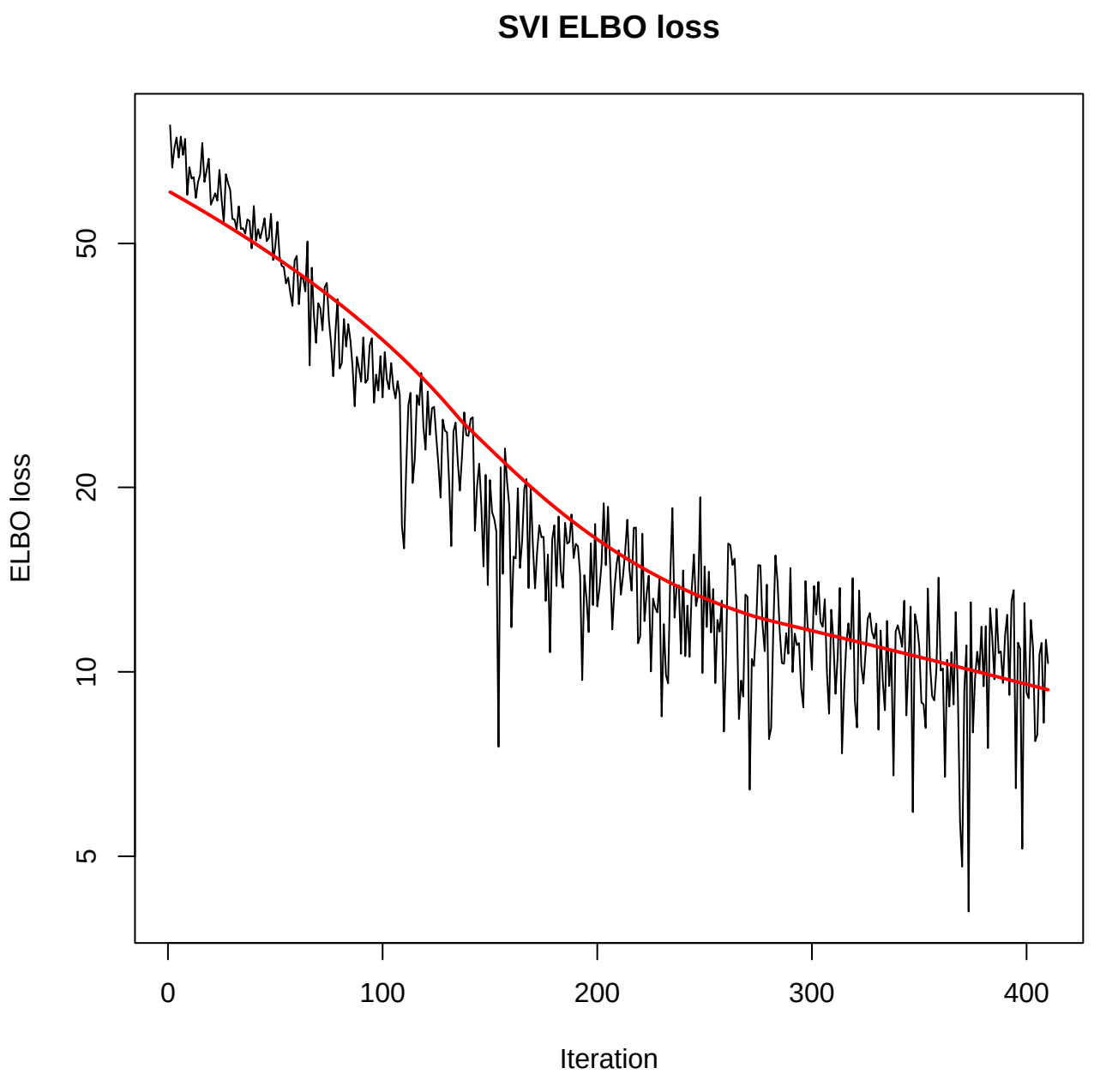
5
Index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ELBO loss
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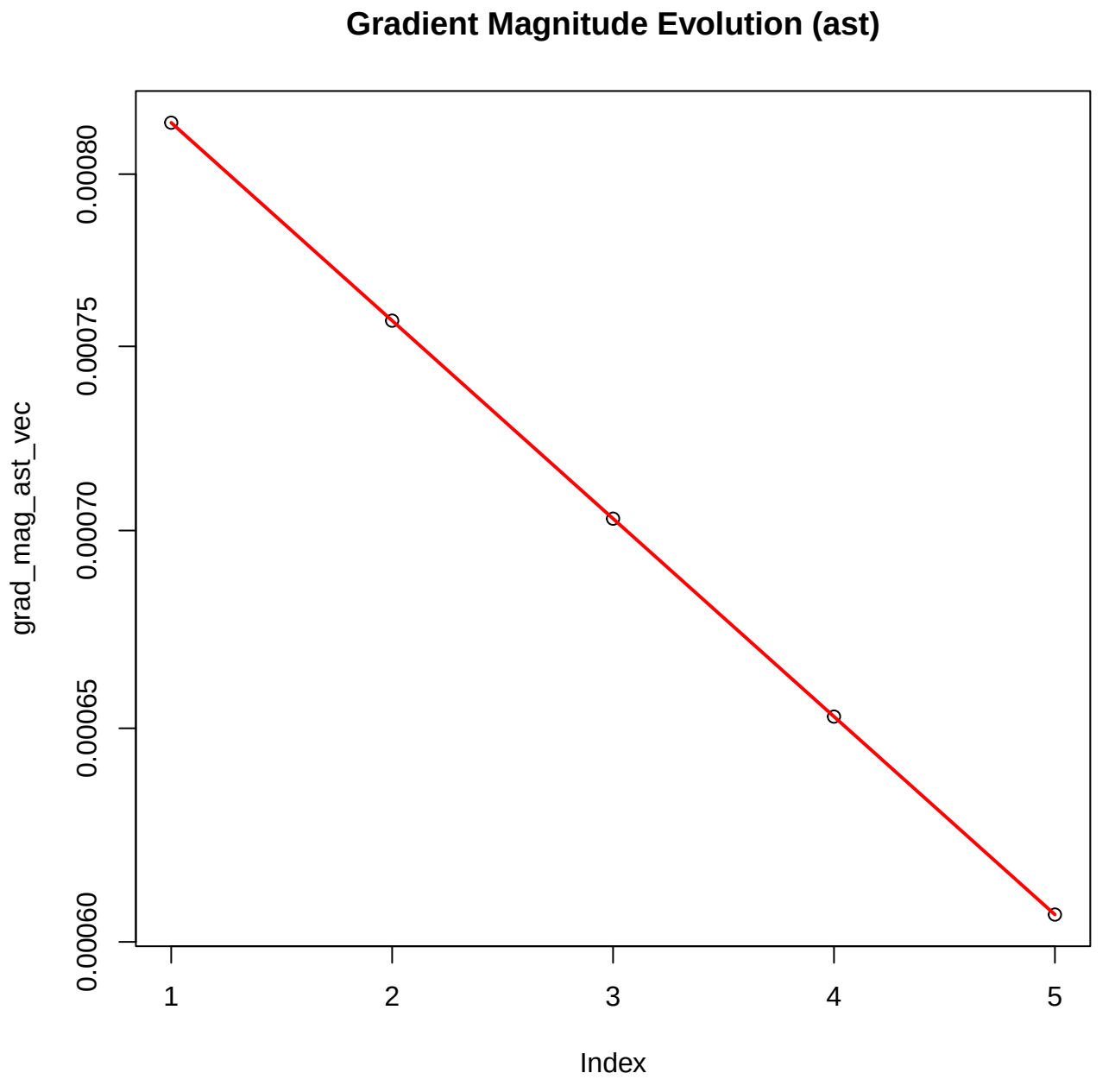
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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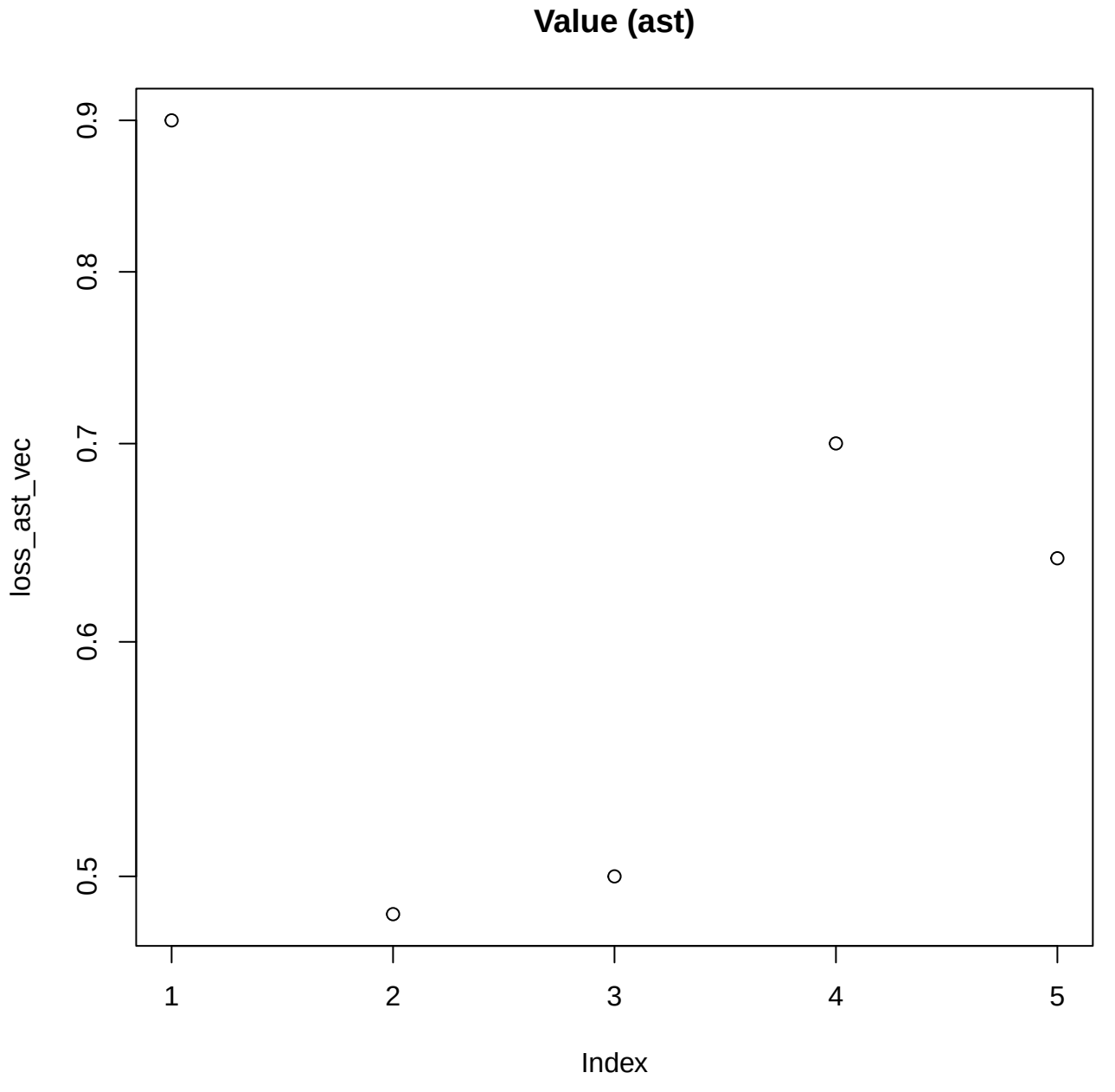
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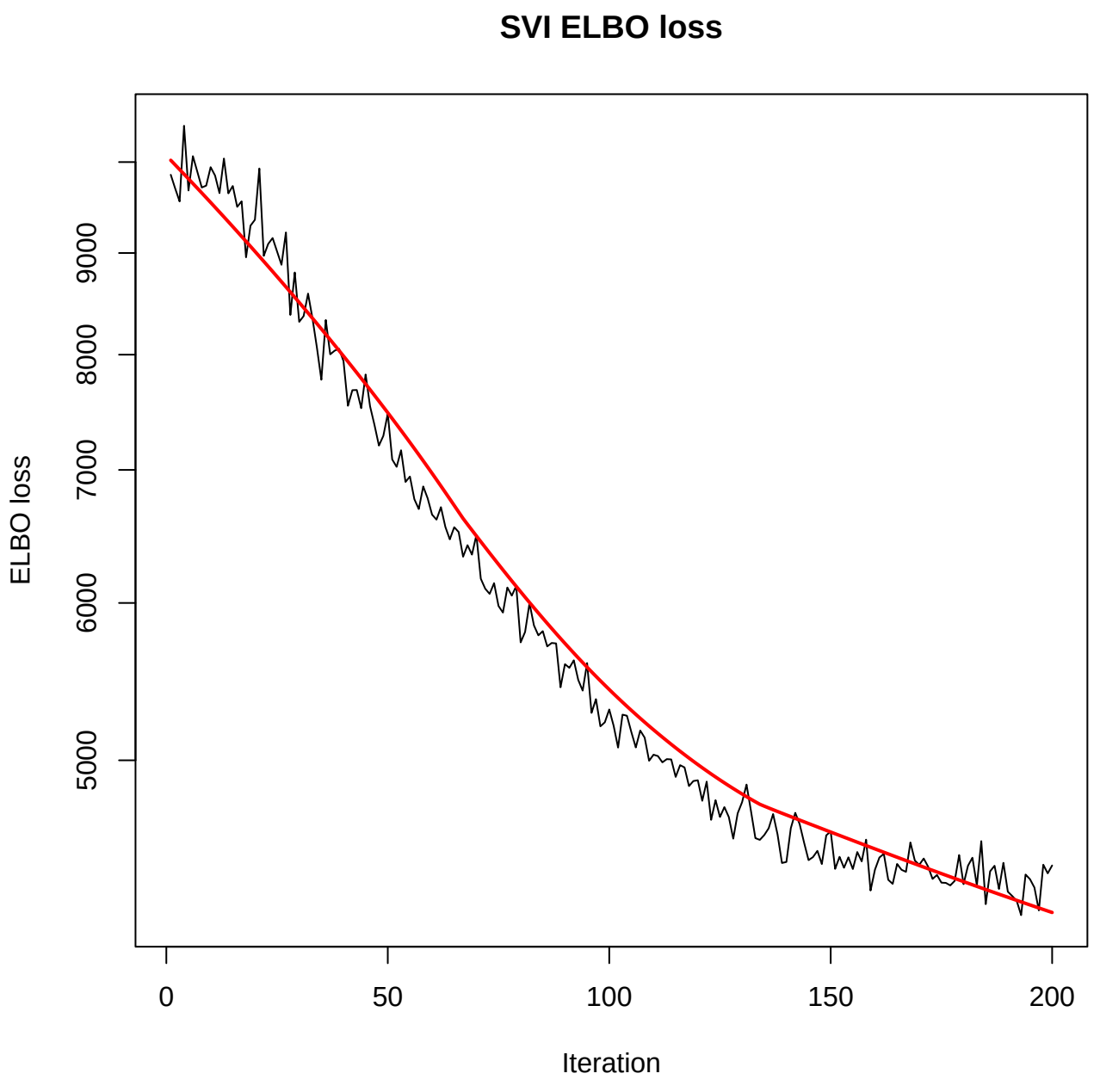
4
5
Index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ELBO loss
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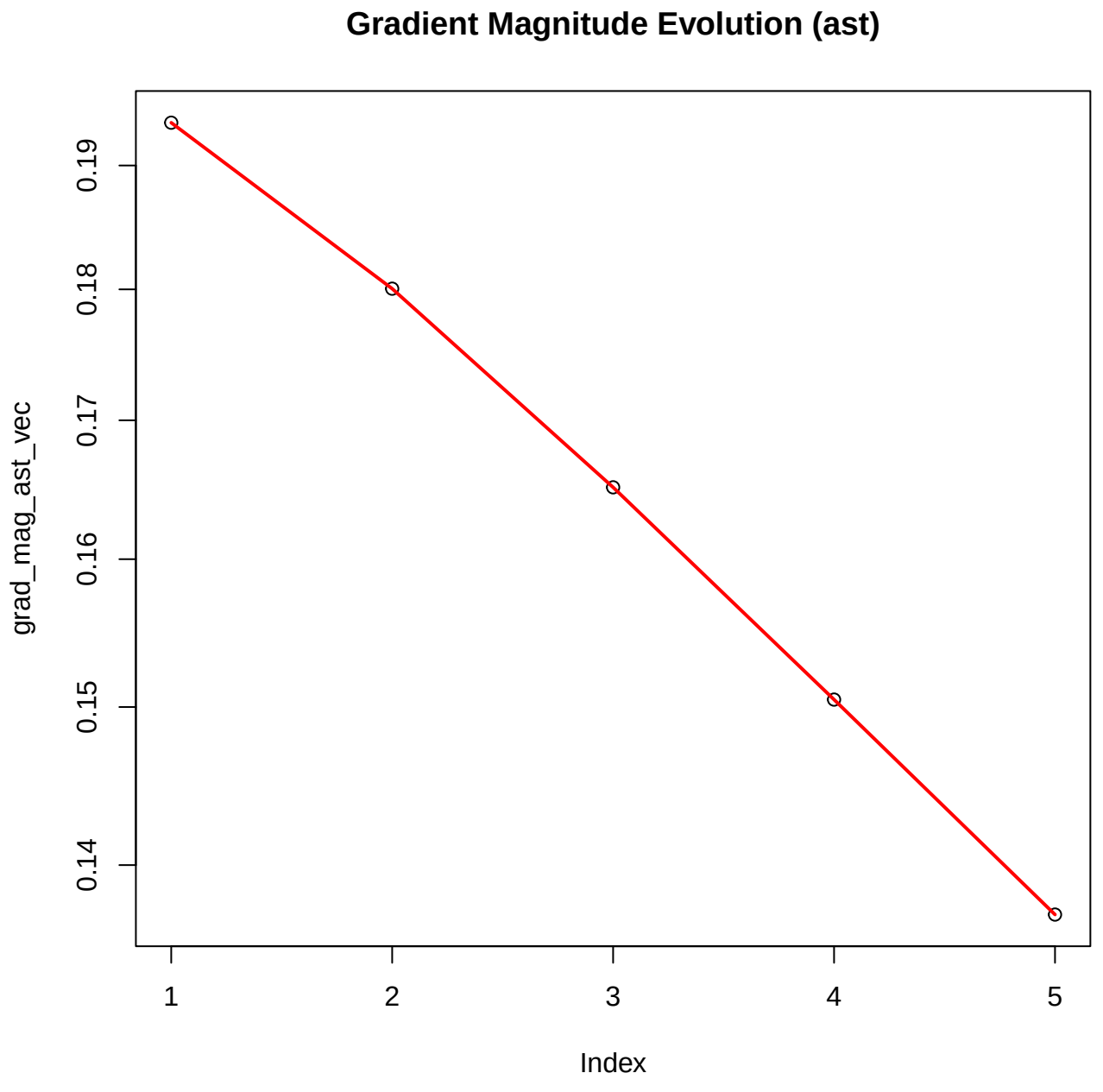
200
Iteration


Gradient Magnitude Evolution (ast)
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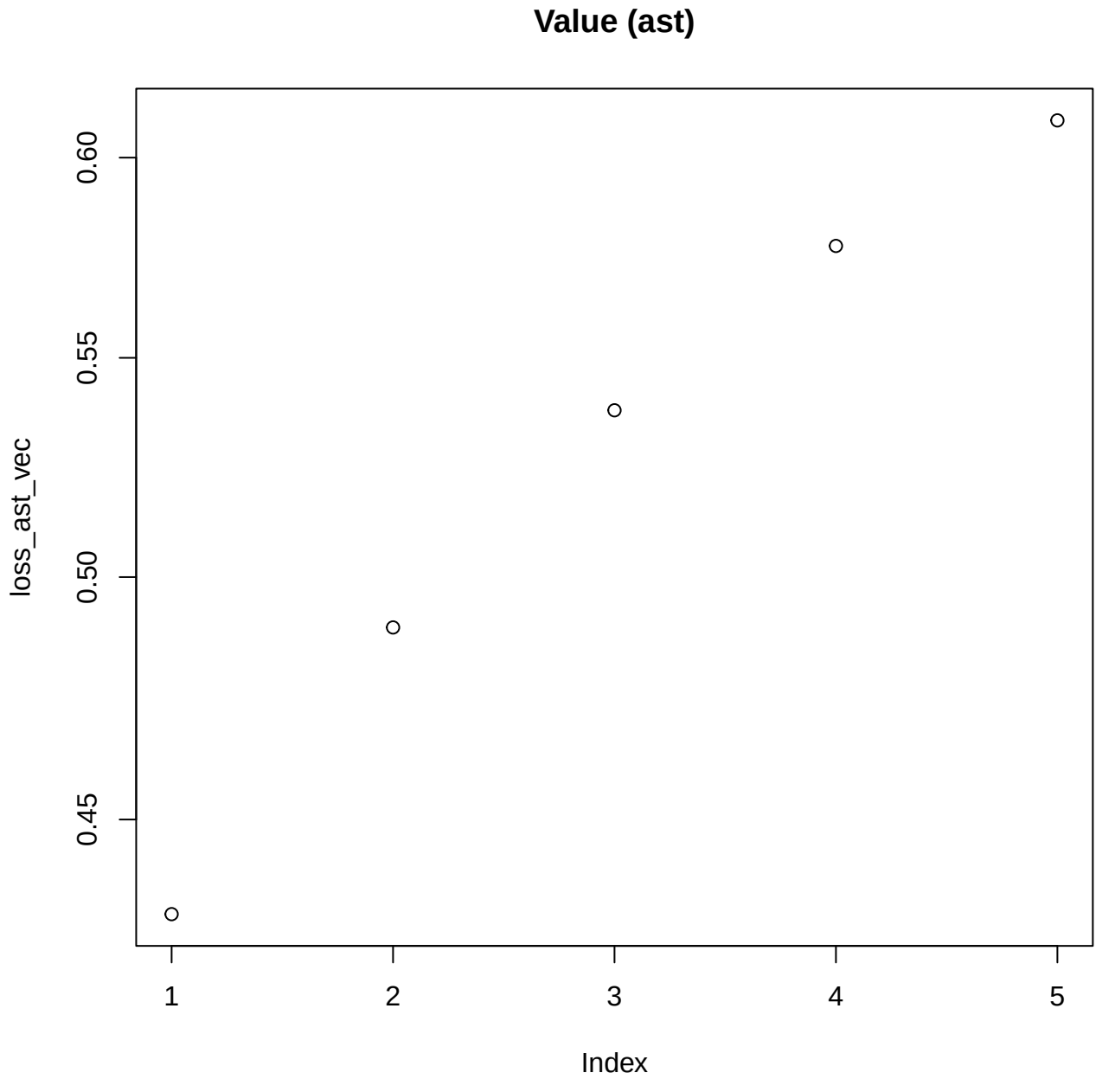
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0.60
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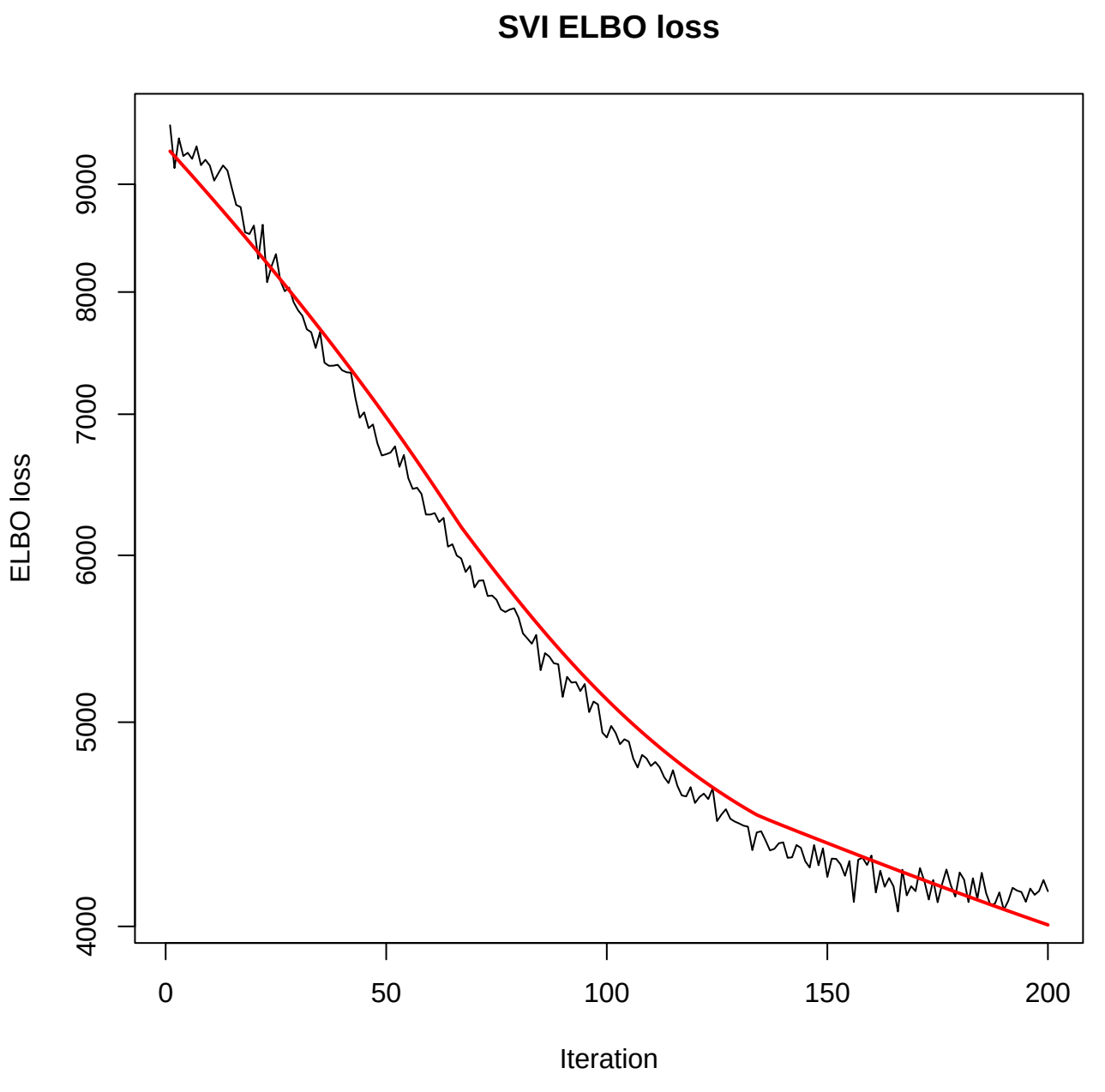
5
Index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ELBO loss
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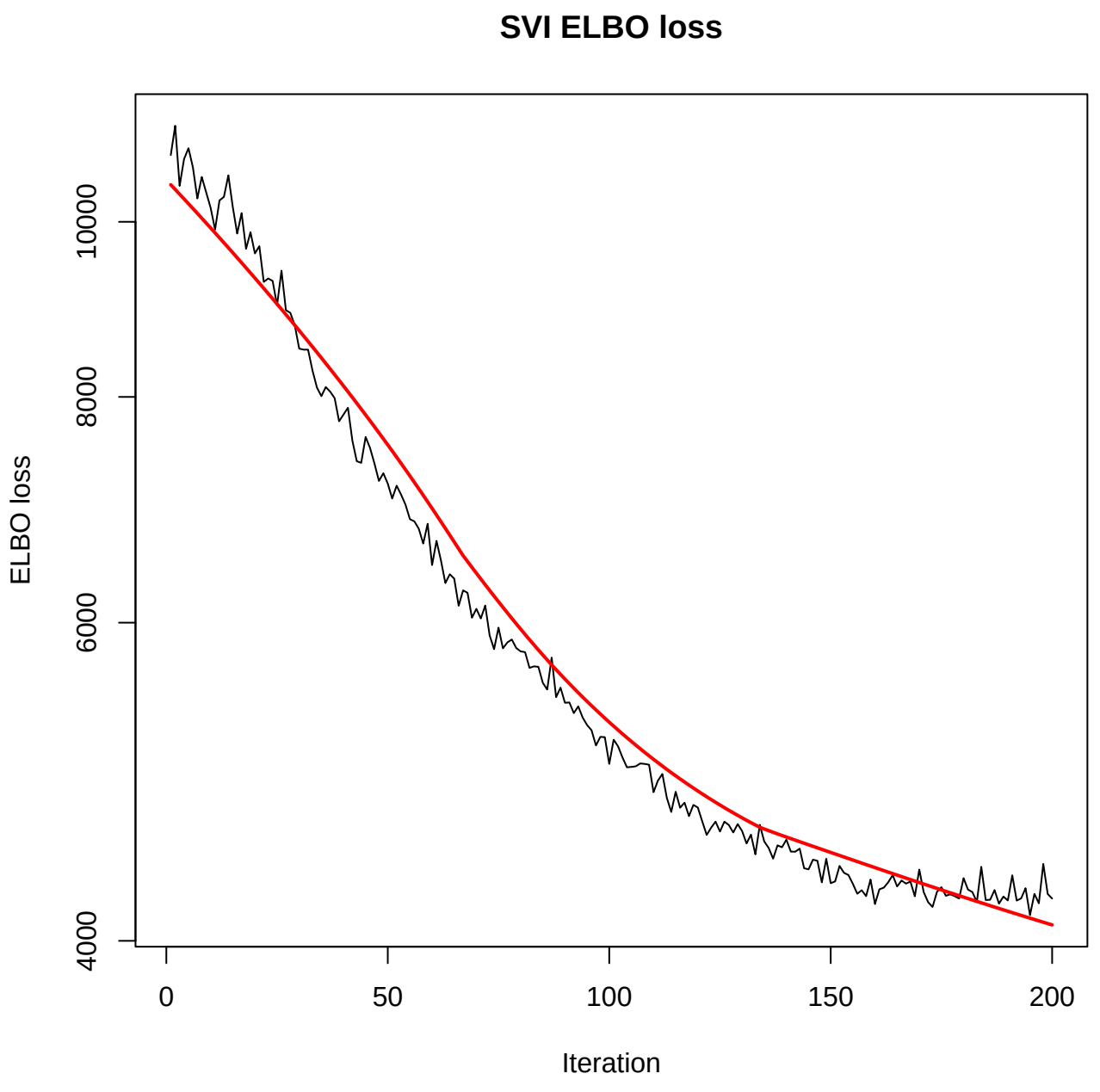
100
150
200
Iteration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ELBO loss
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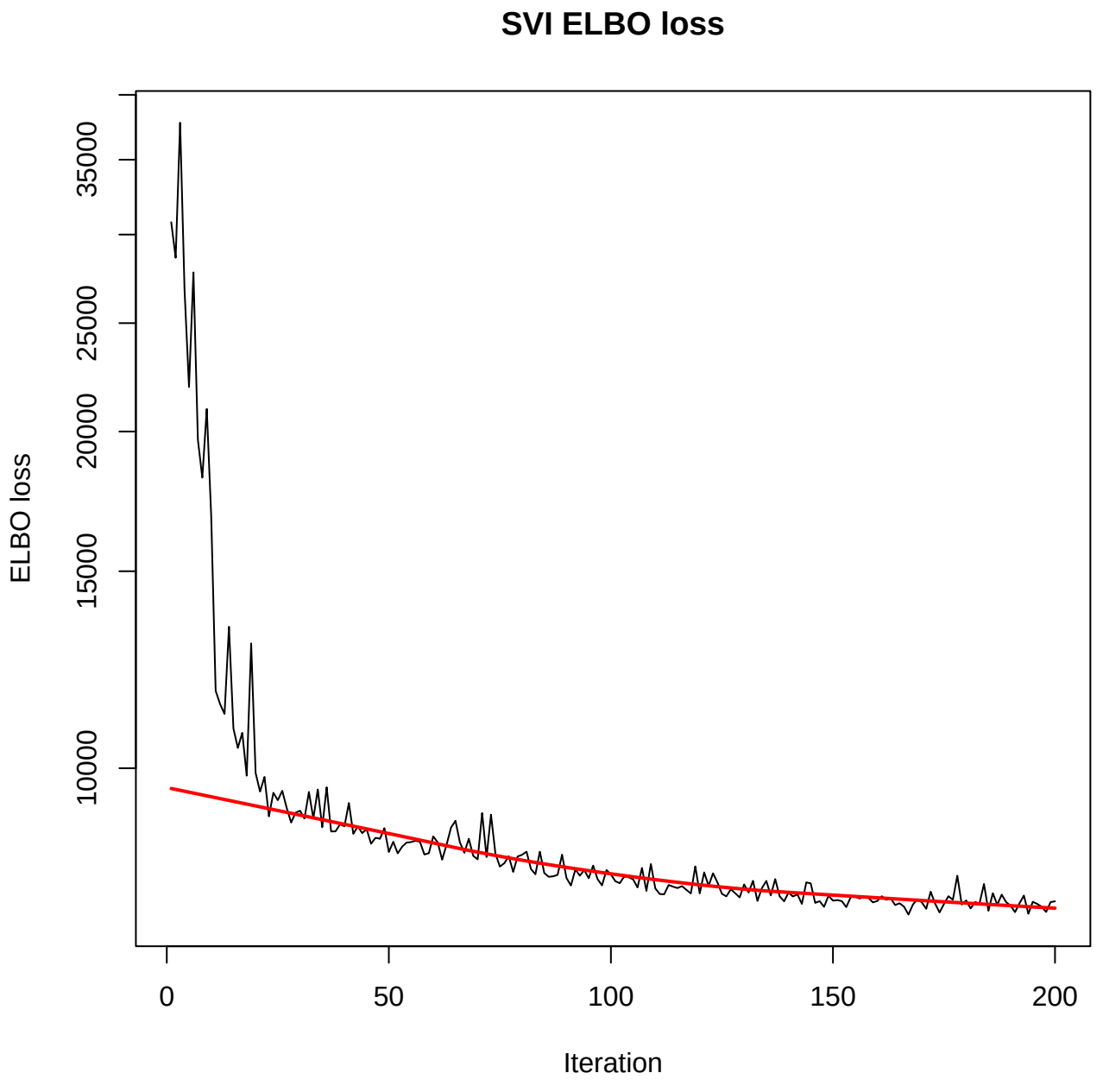
Iteration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ELBO loss
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100
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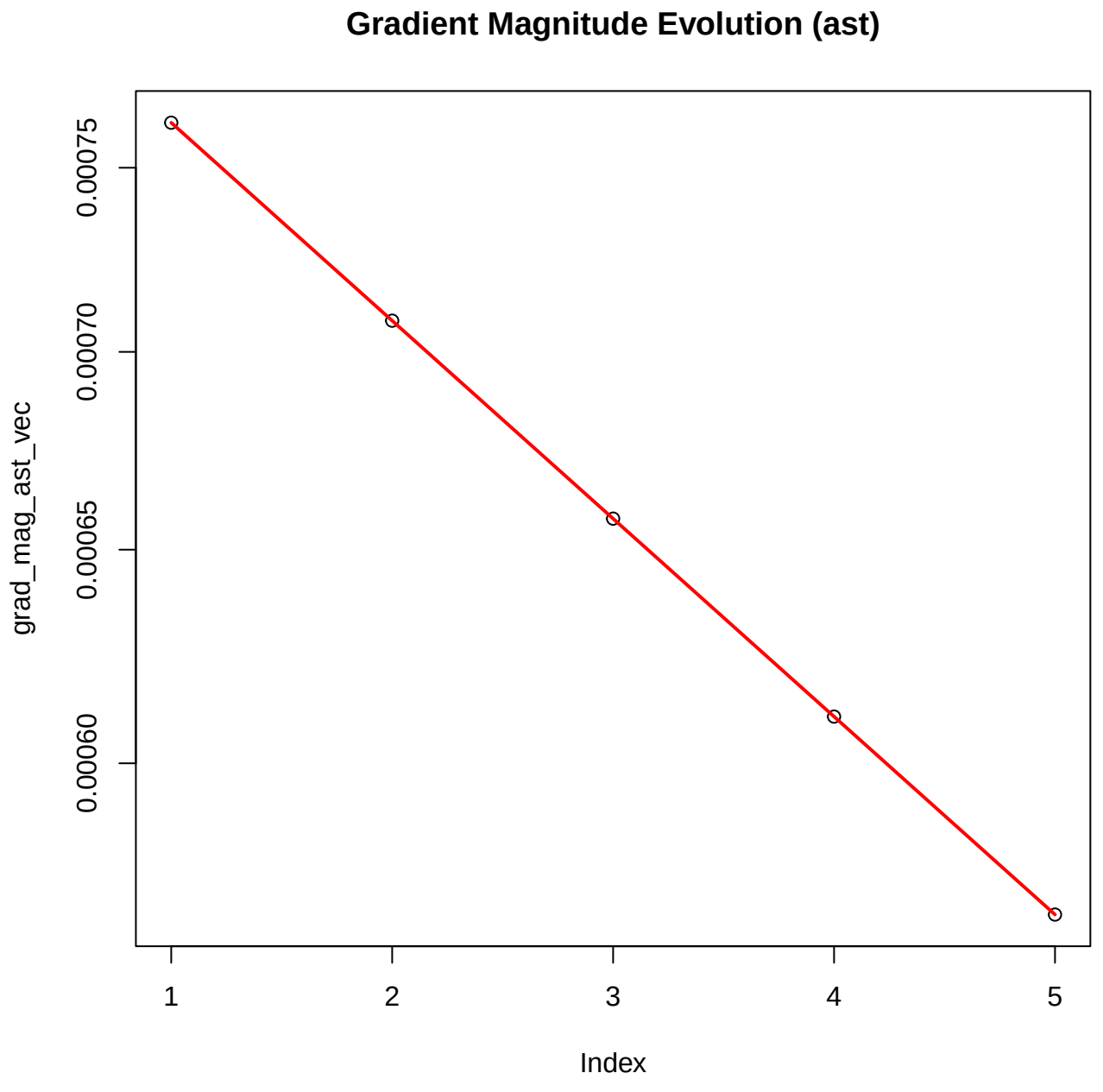
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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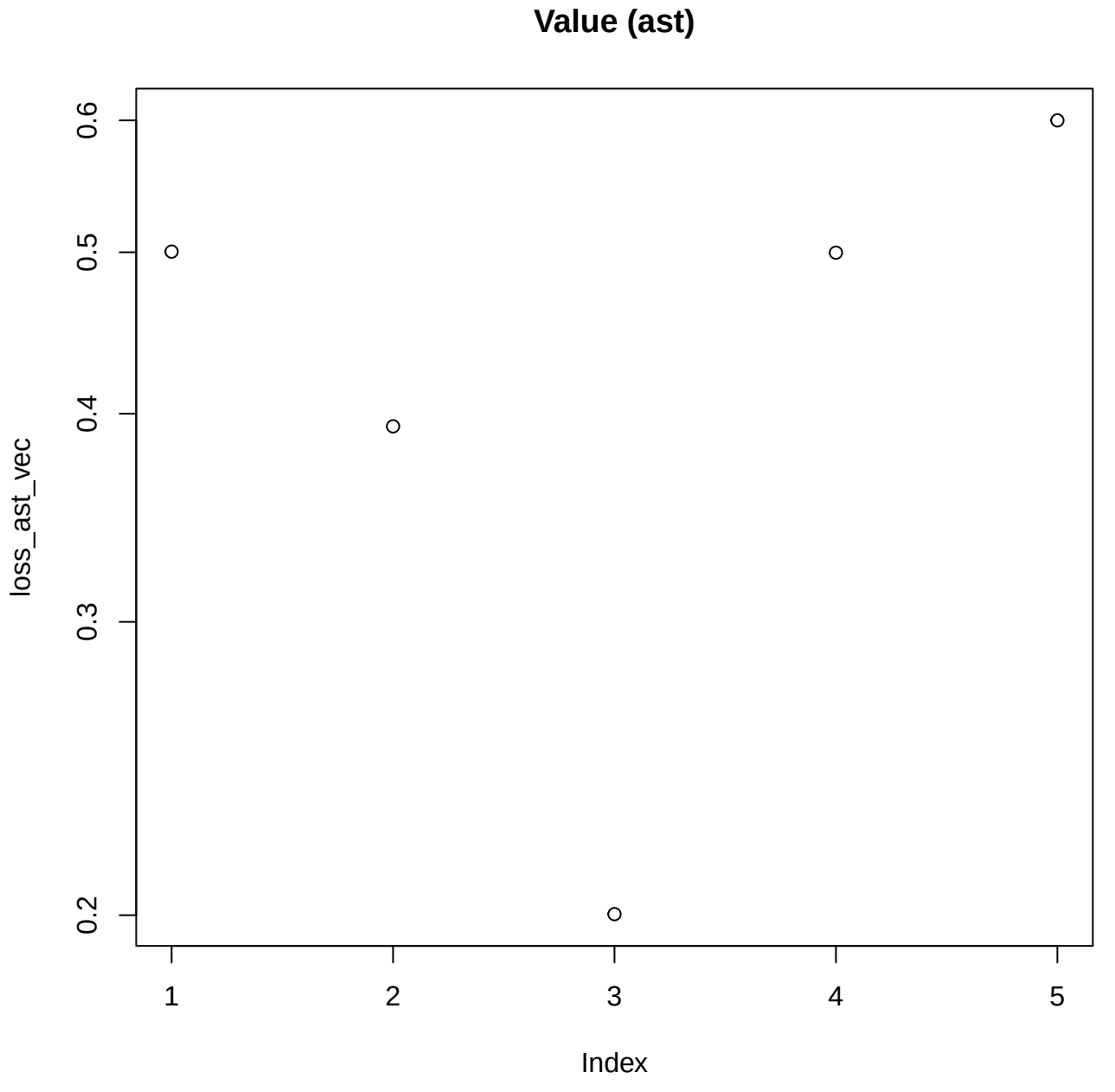
0.00070
0.00075
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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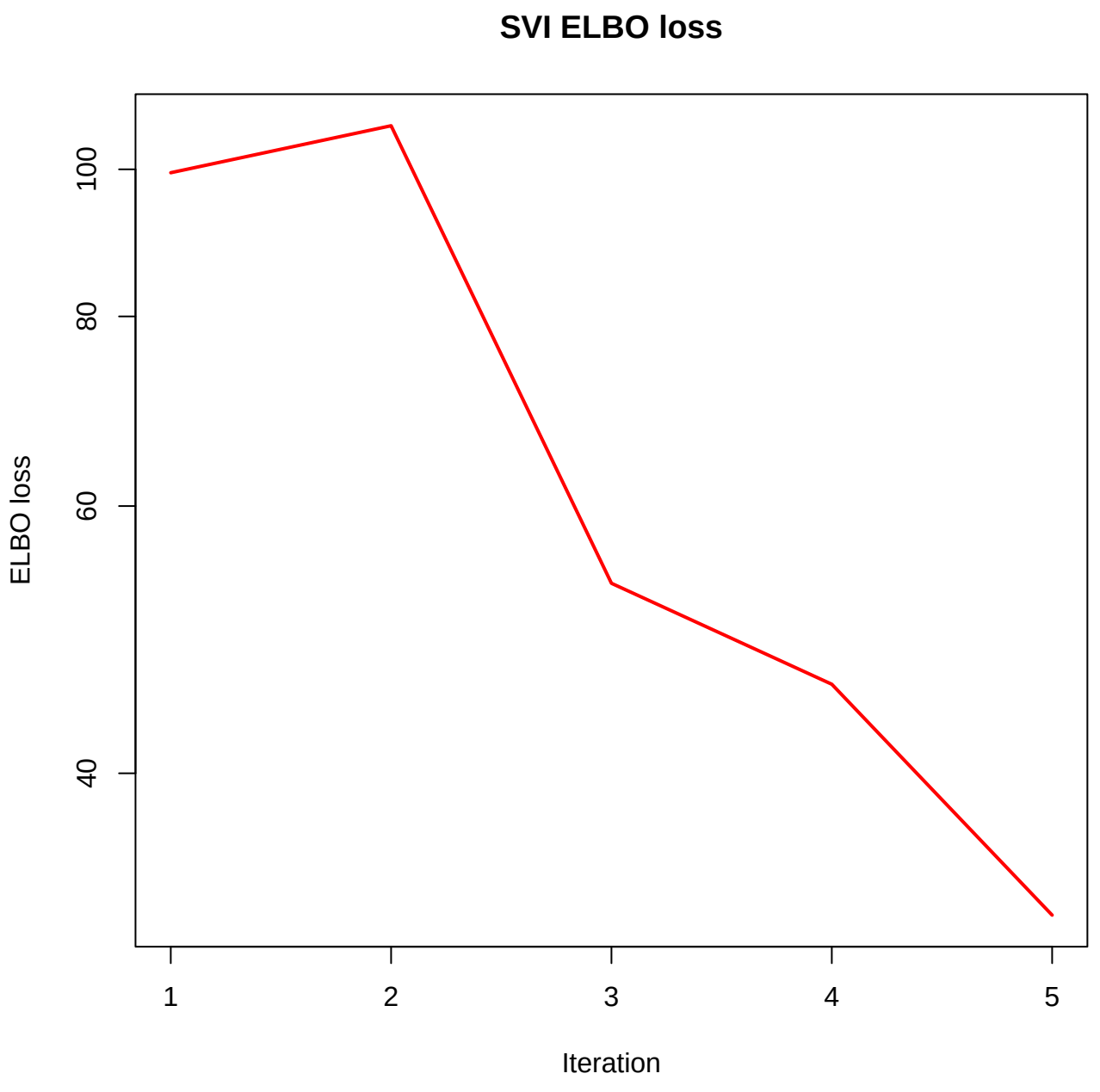
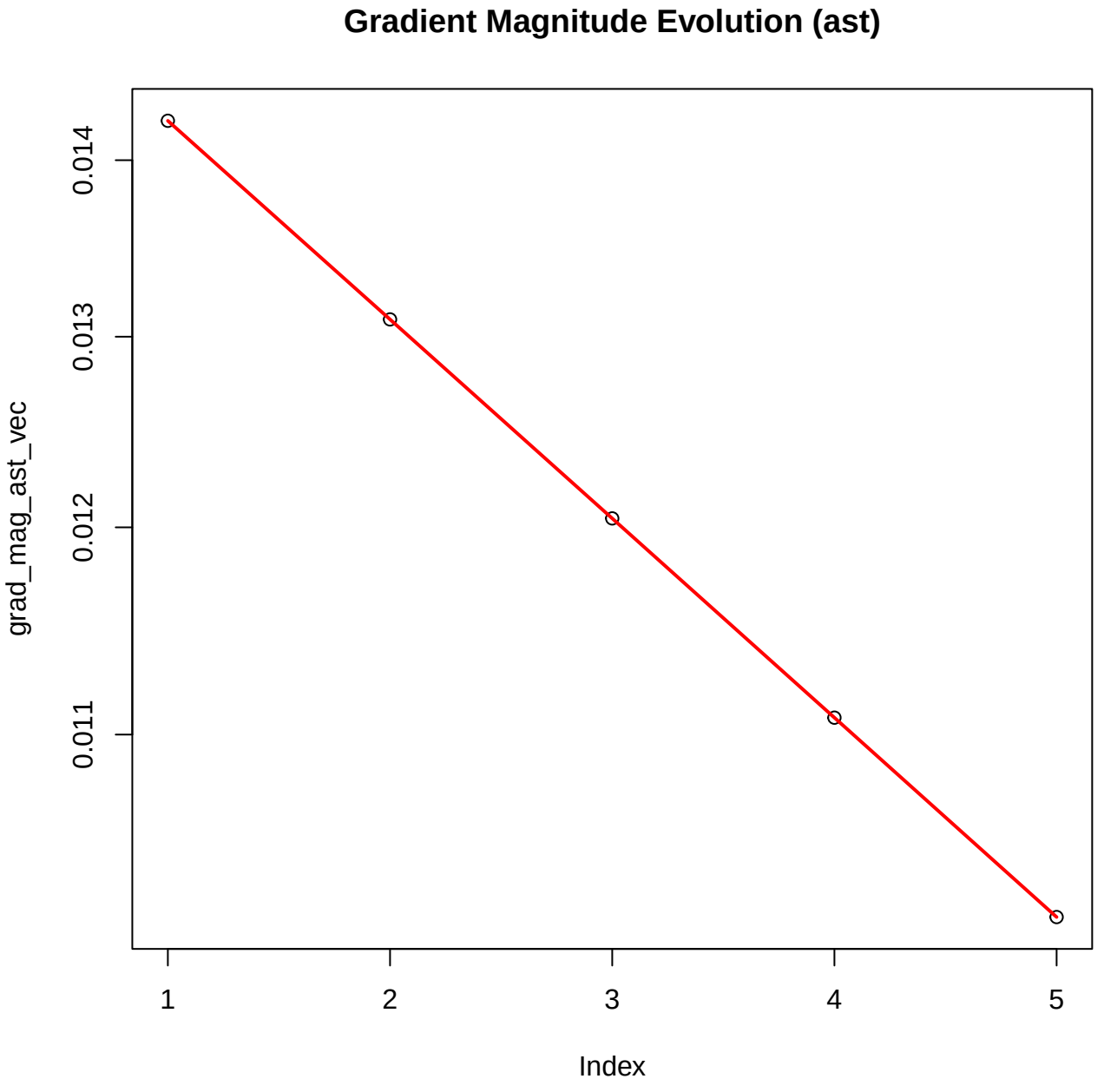
0.5
0.6
1
2
3
4
5
Index


SVI ELBO loss
ELBO loss
100
80
60
40
1
2
3
4
5
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.014
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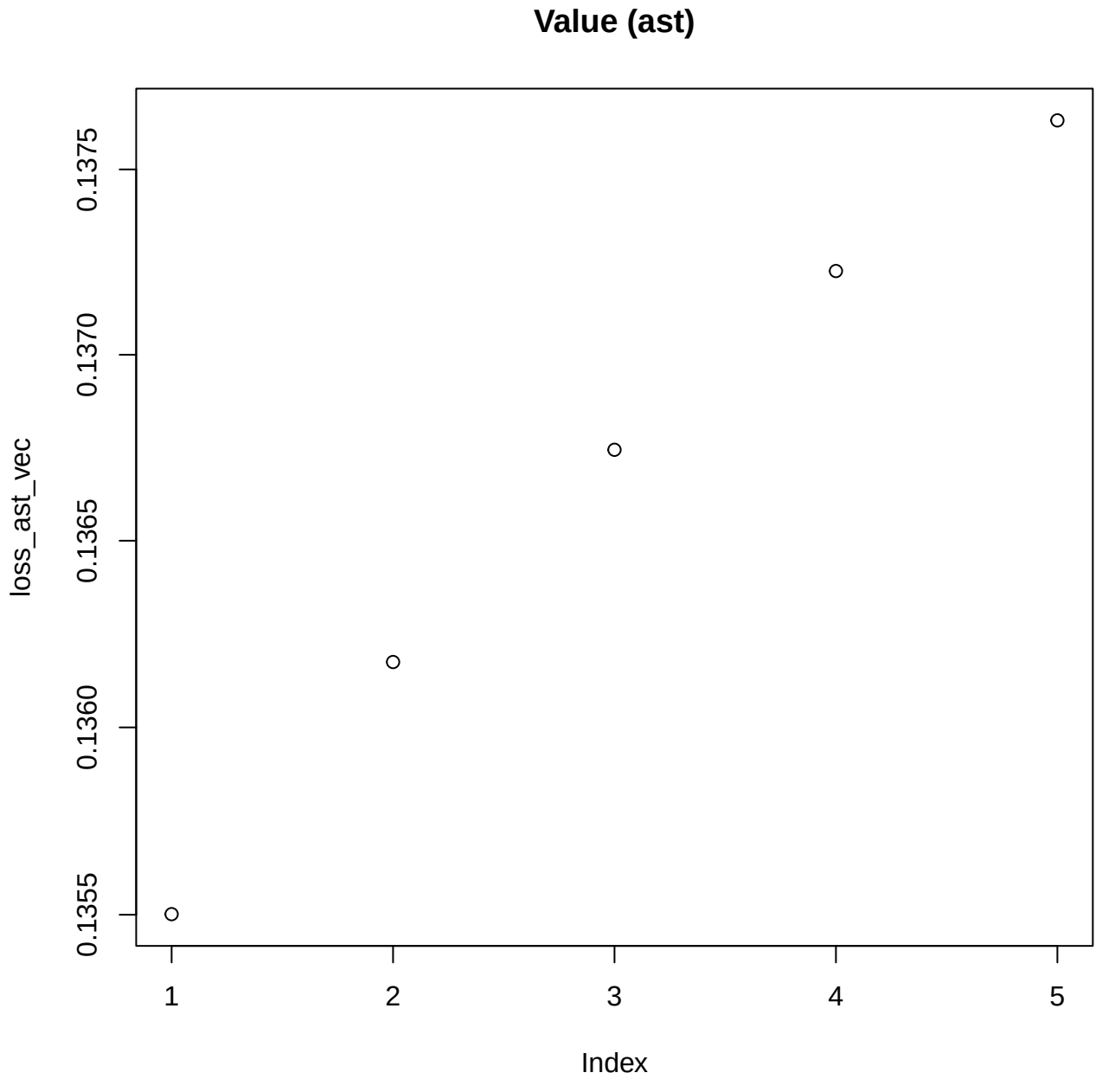
5
Index


Value (ast)
loss_ast_vec
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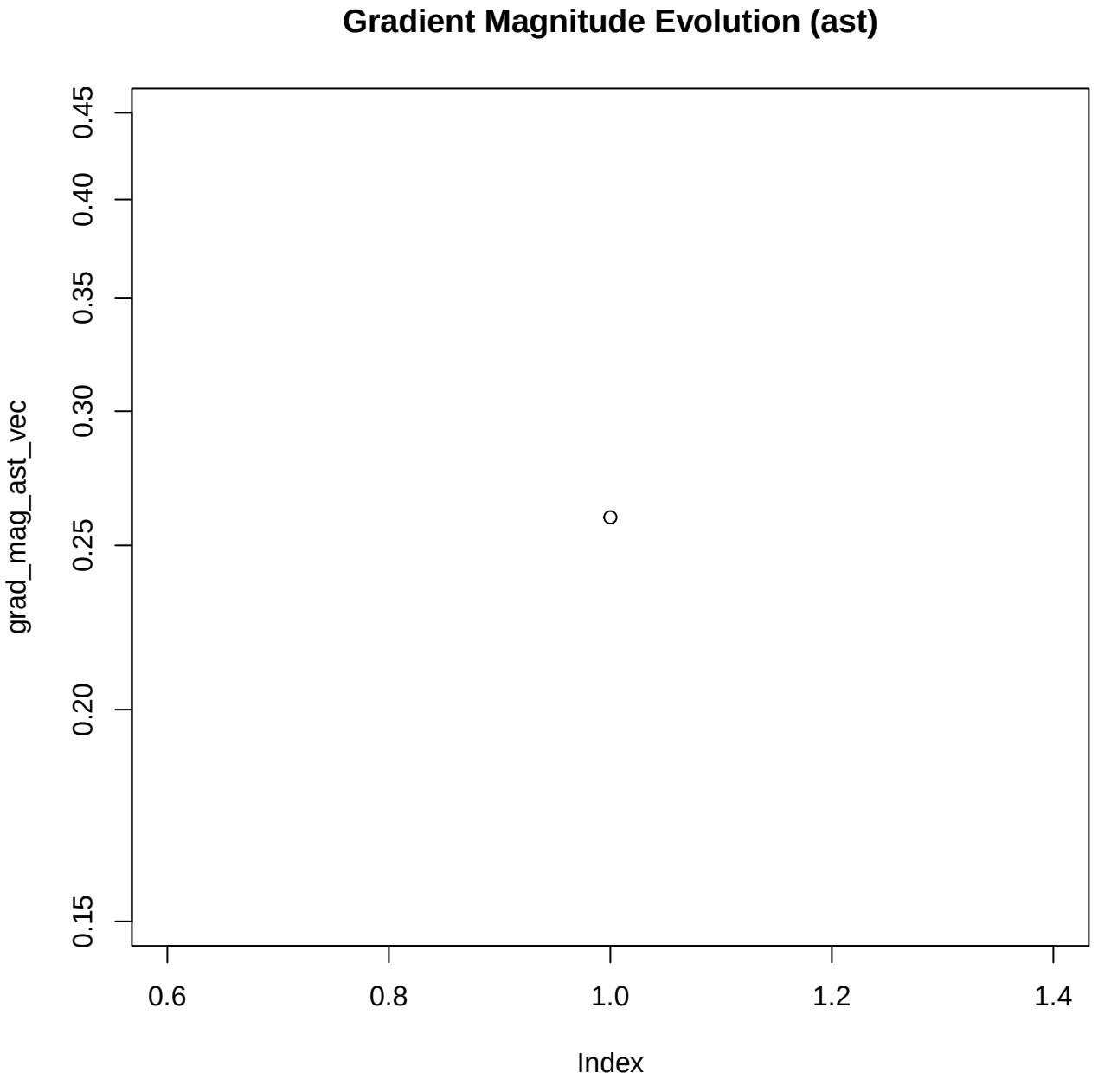
0.1370
0.1375
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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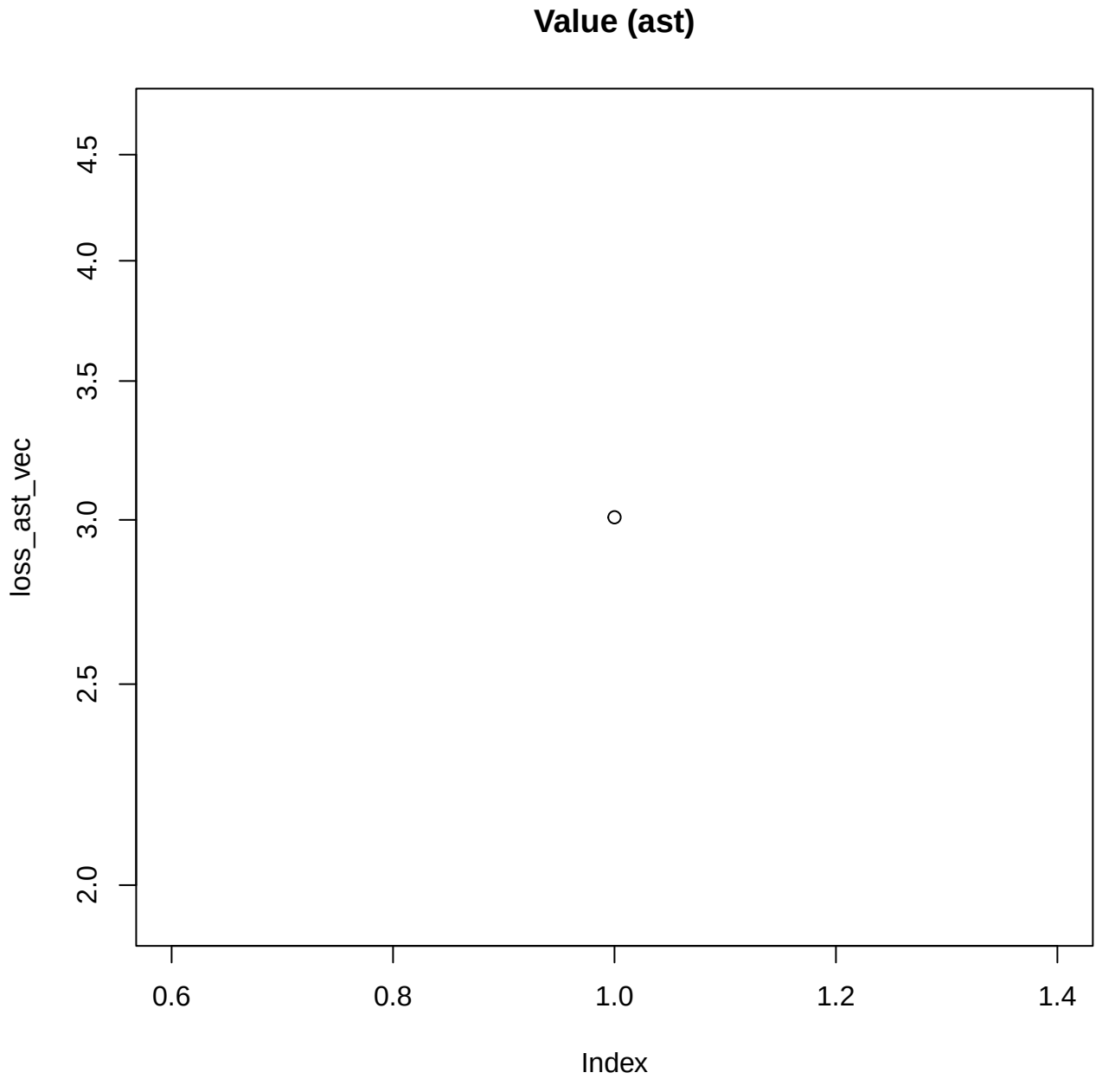
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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4.0
3.5
3.0
2.5
2.0
0.6
0.8
1.0
1.2
1.4
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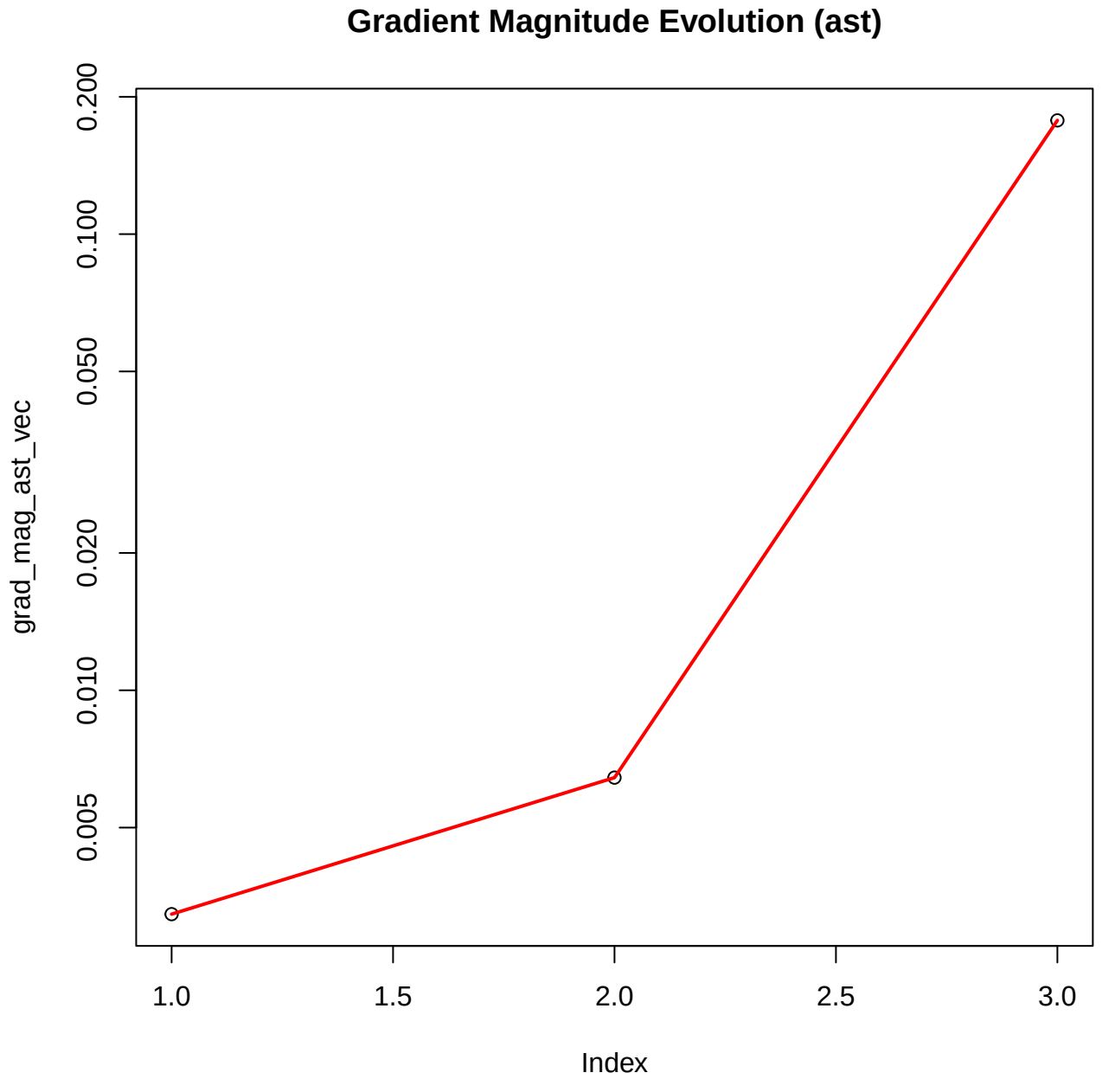
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
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2.5
3.0
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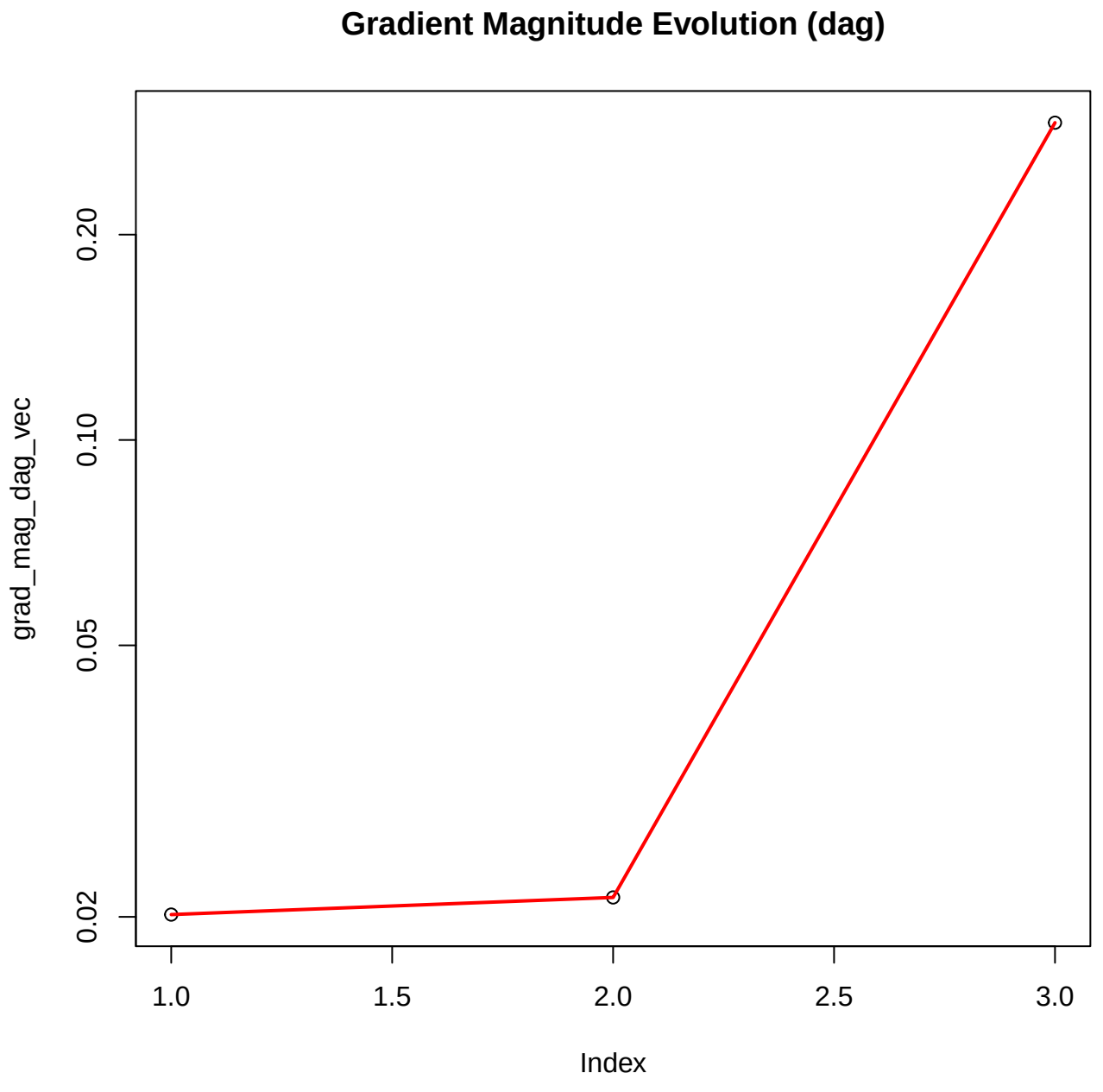
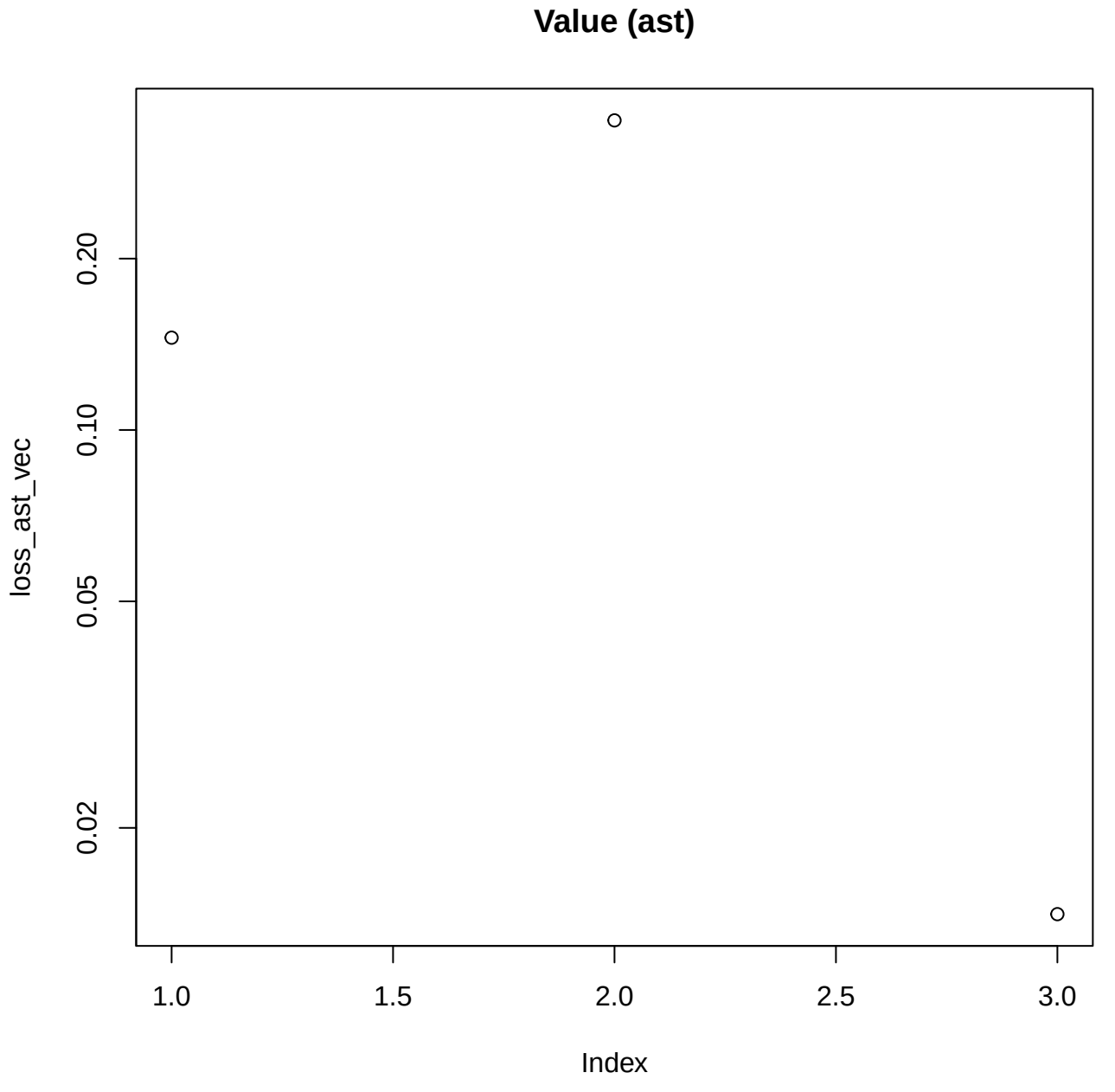

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index


Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
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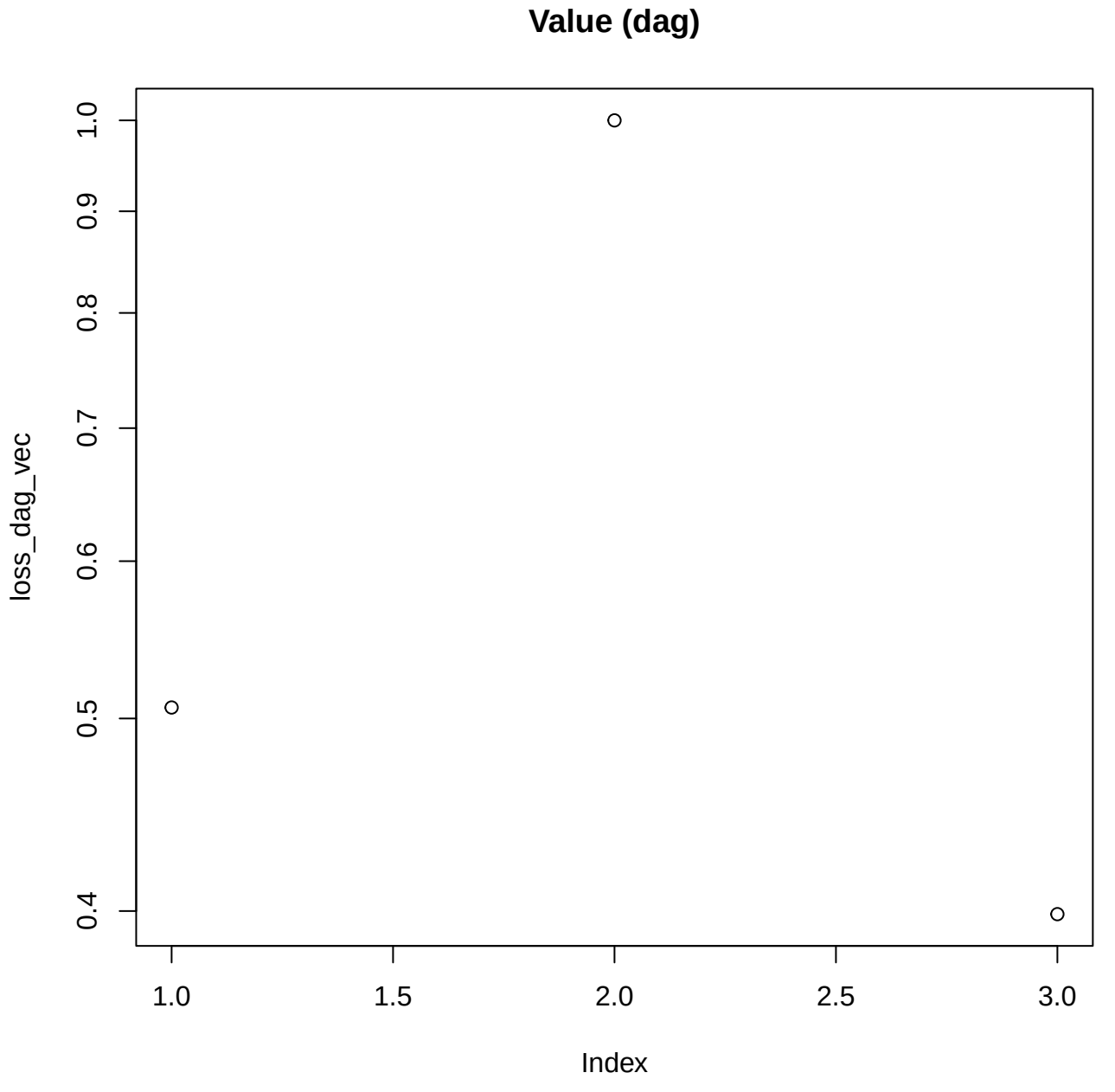
1.5
2.0
2.5
3.0
Index


Value (dag)
loss_dag_vec
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0.9
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0.7
0.6
0.5
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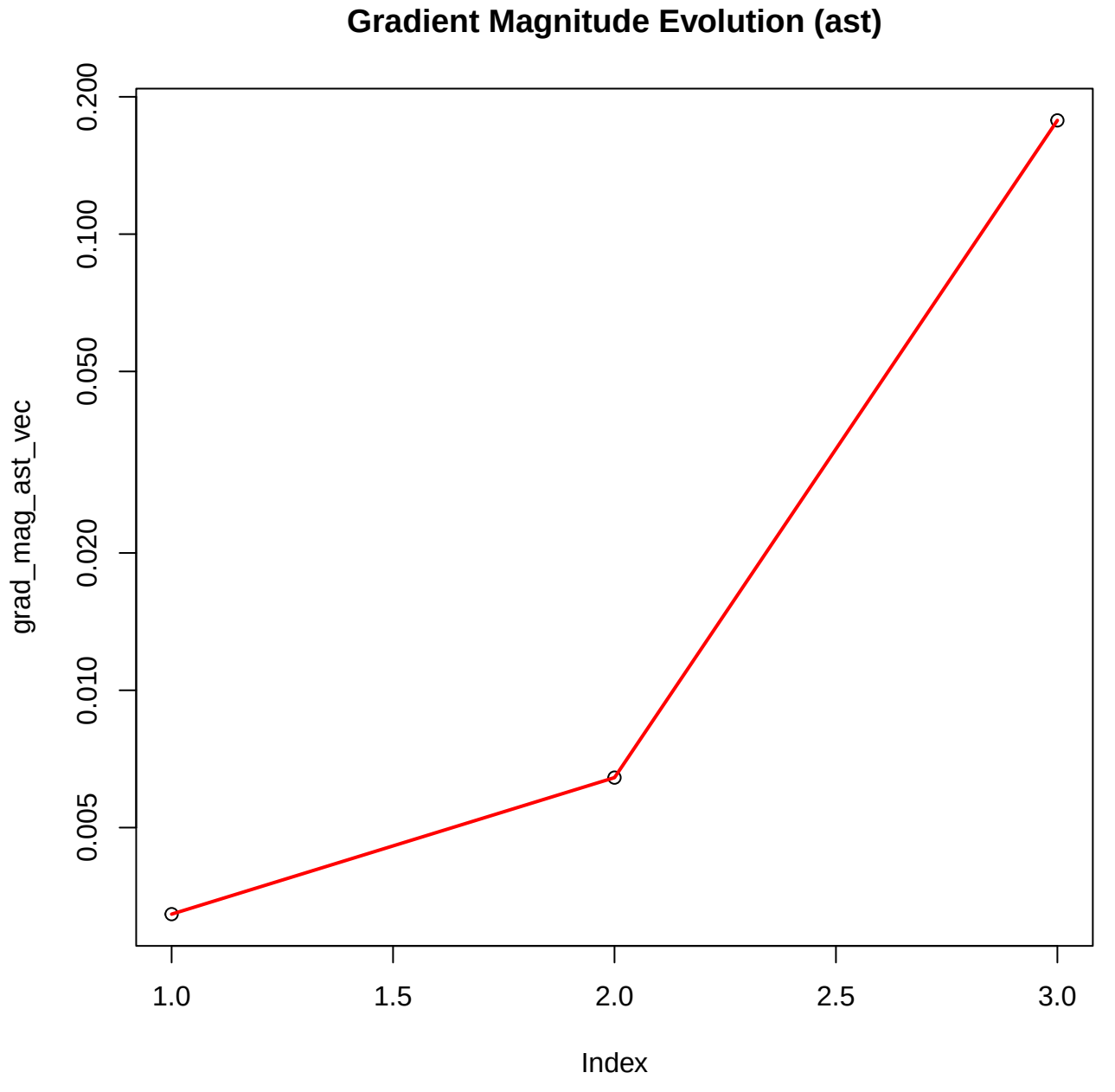
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
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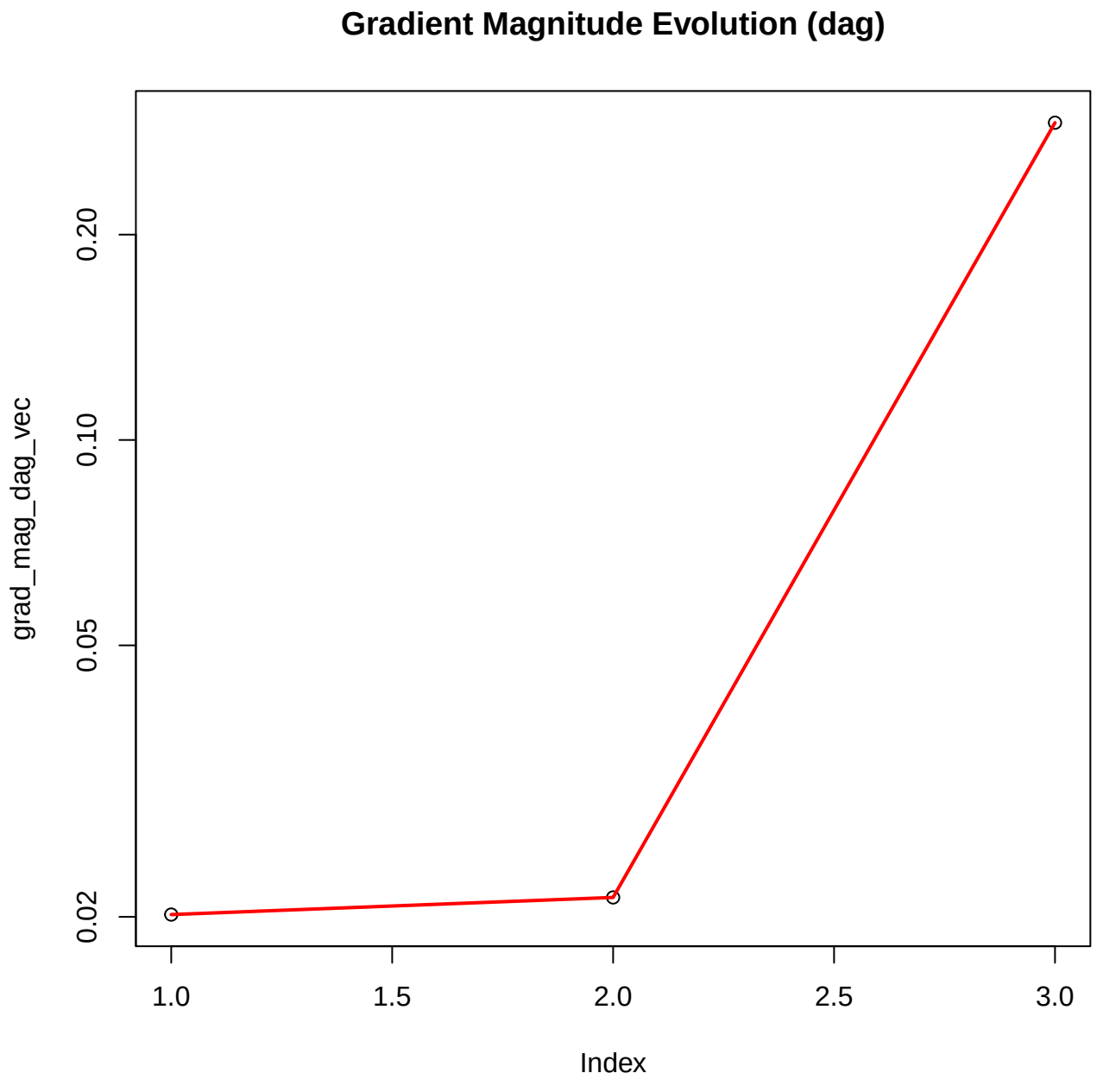
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
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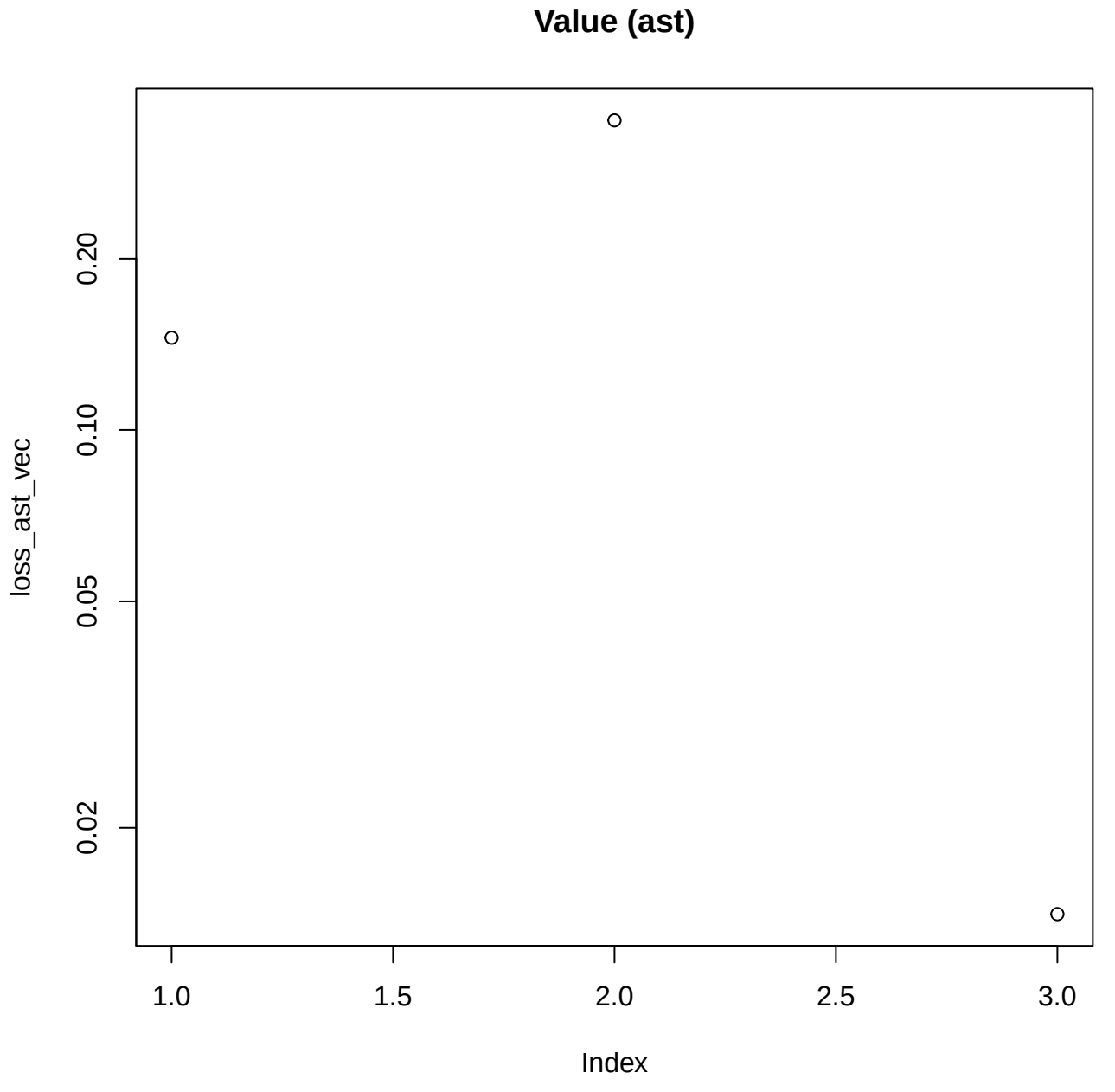
1.5
2.0
2.5
3.0
Index


Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
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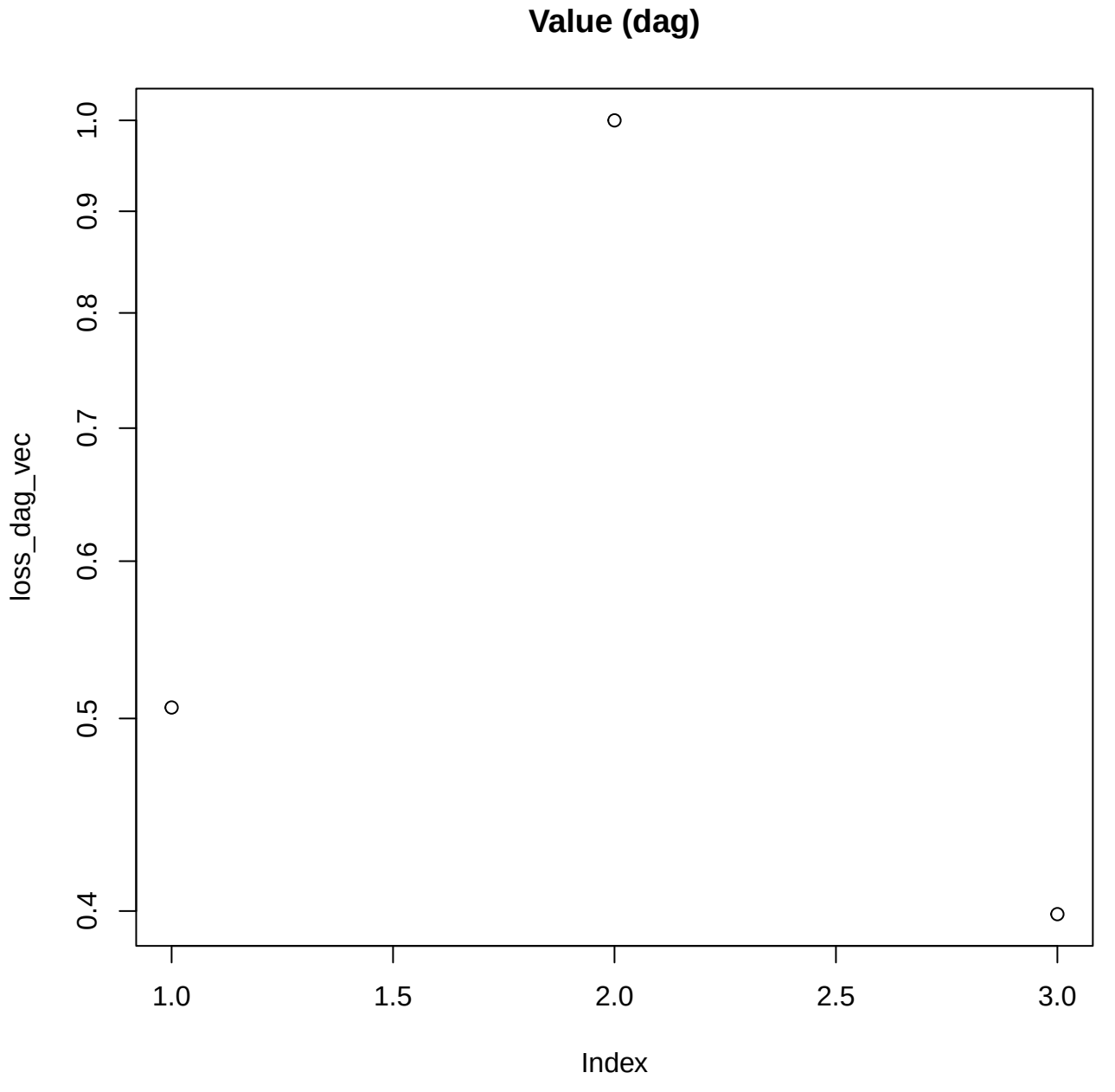
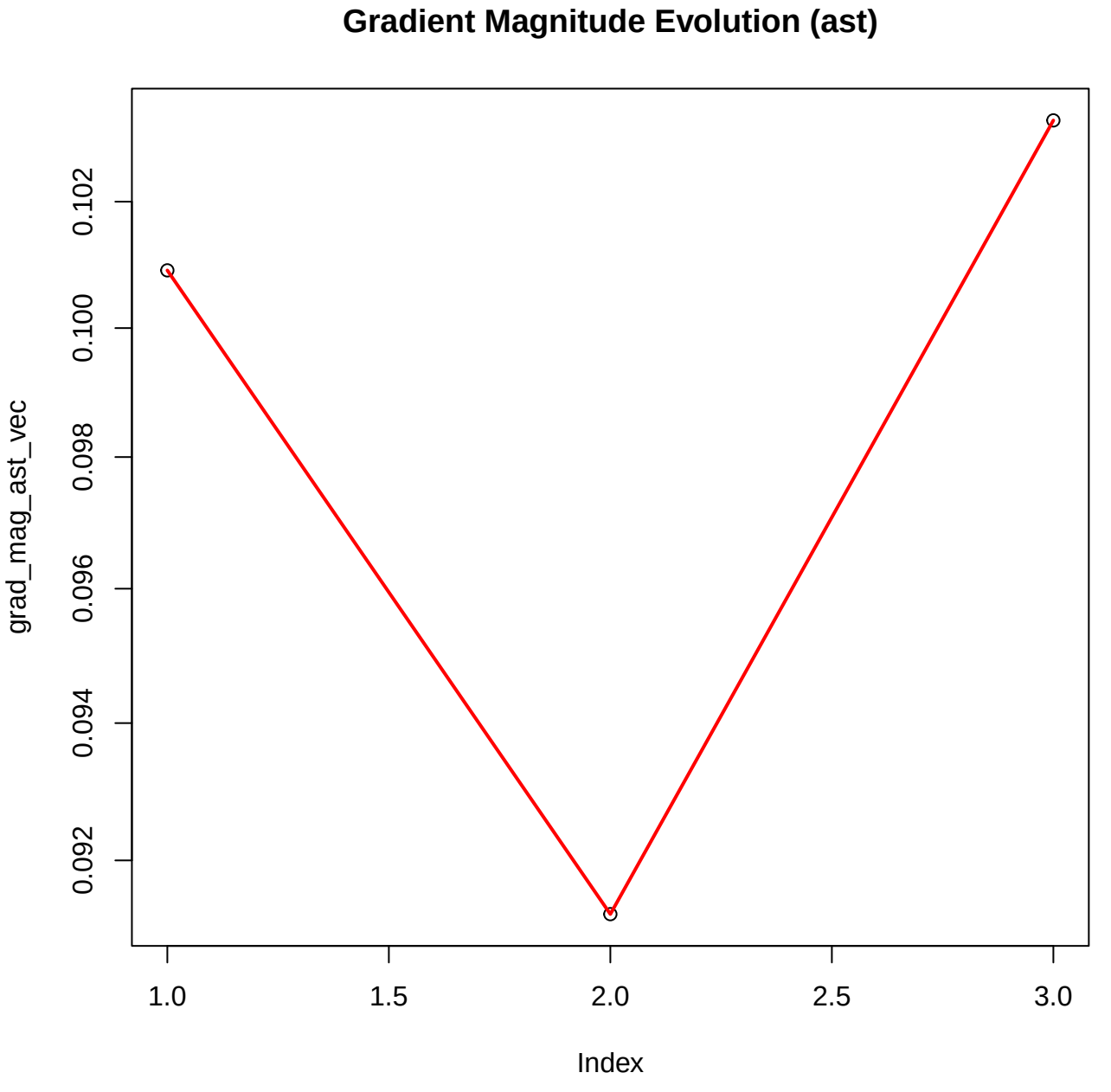
1.5
2.0
2.5
3.0
Index


Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.094
0.096
0.098
0.100
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2.0
2.5
3.0
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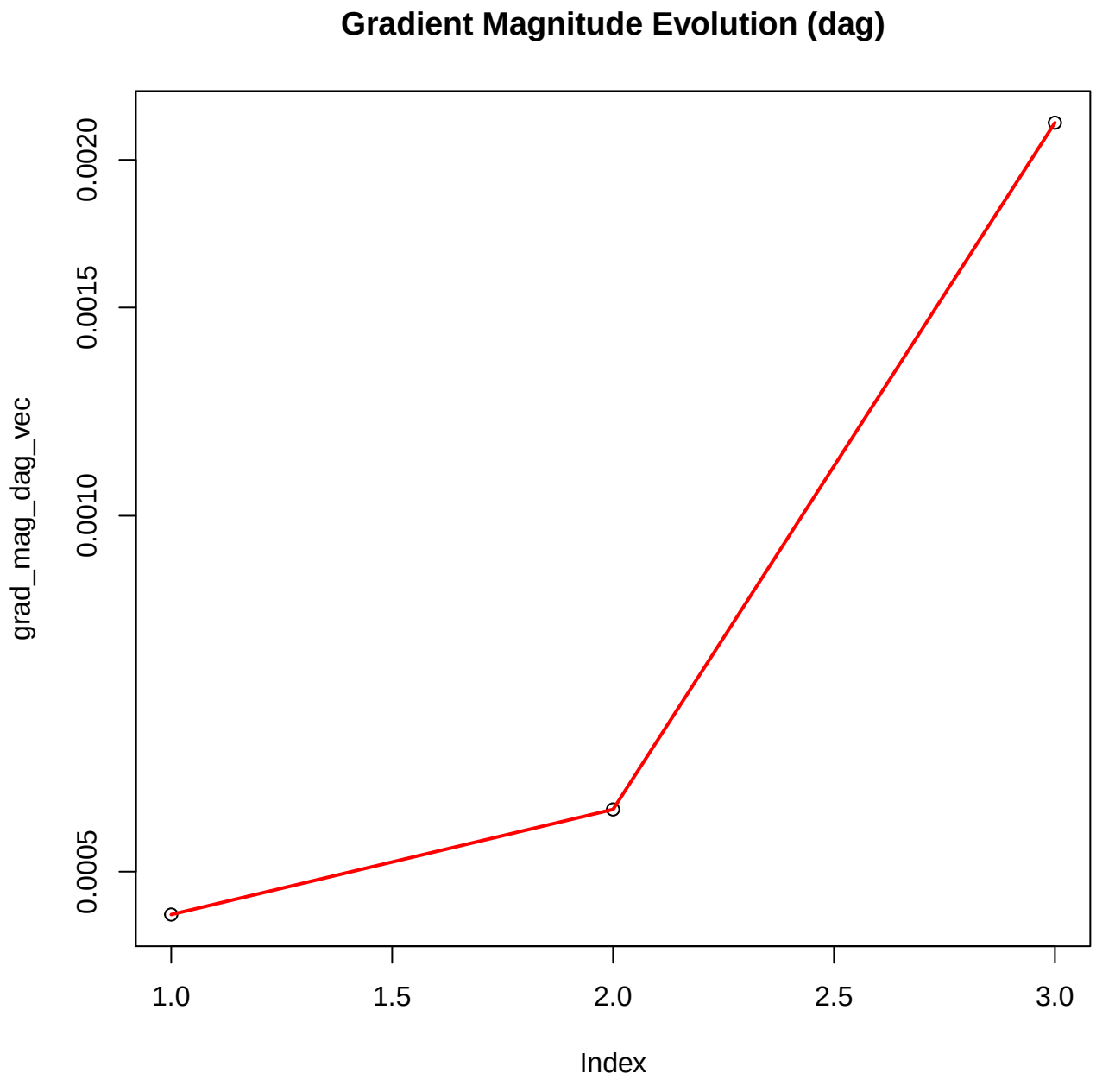
Gradient Magnitude Evolution (dag)
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0.0010
0.0015
0.0020
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2.5
3.0
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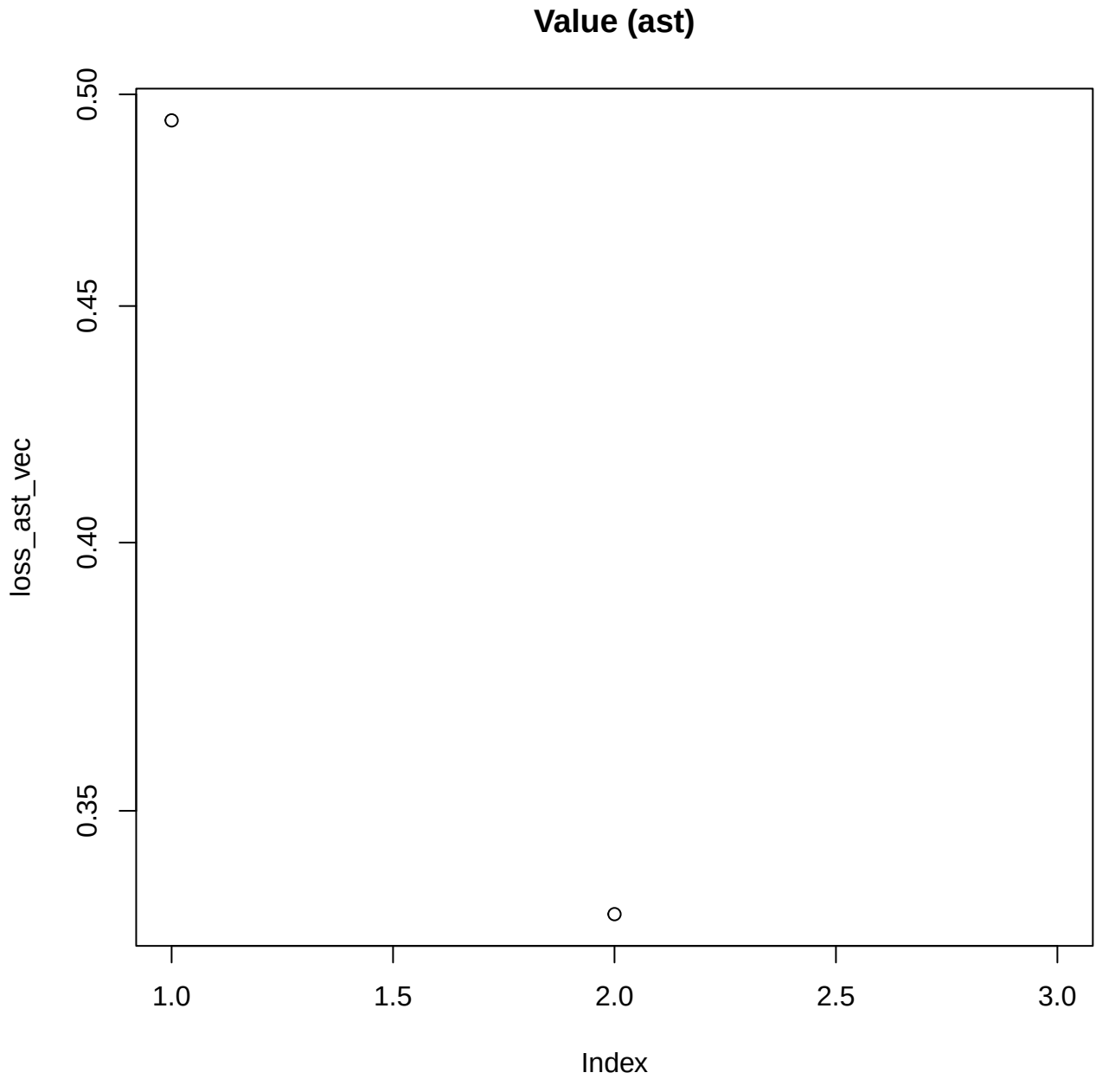
Value (ast)
loss_ast_vec
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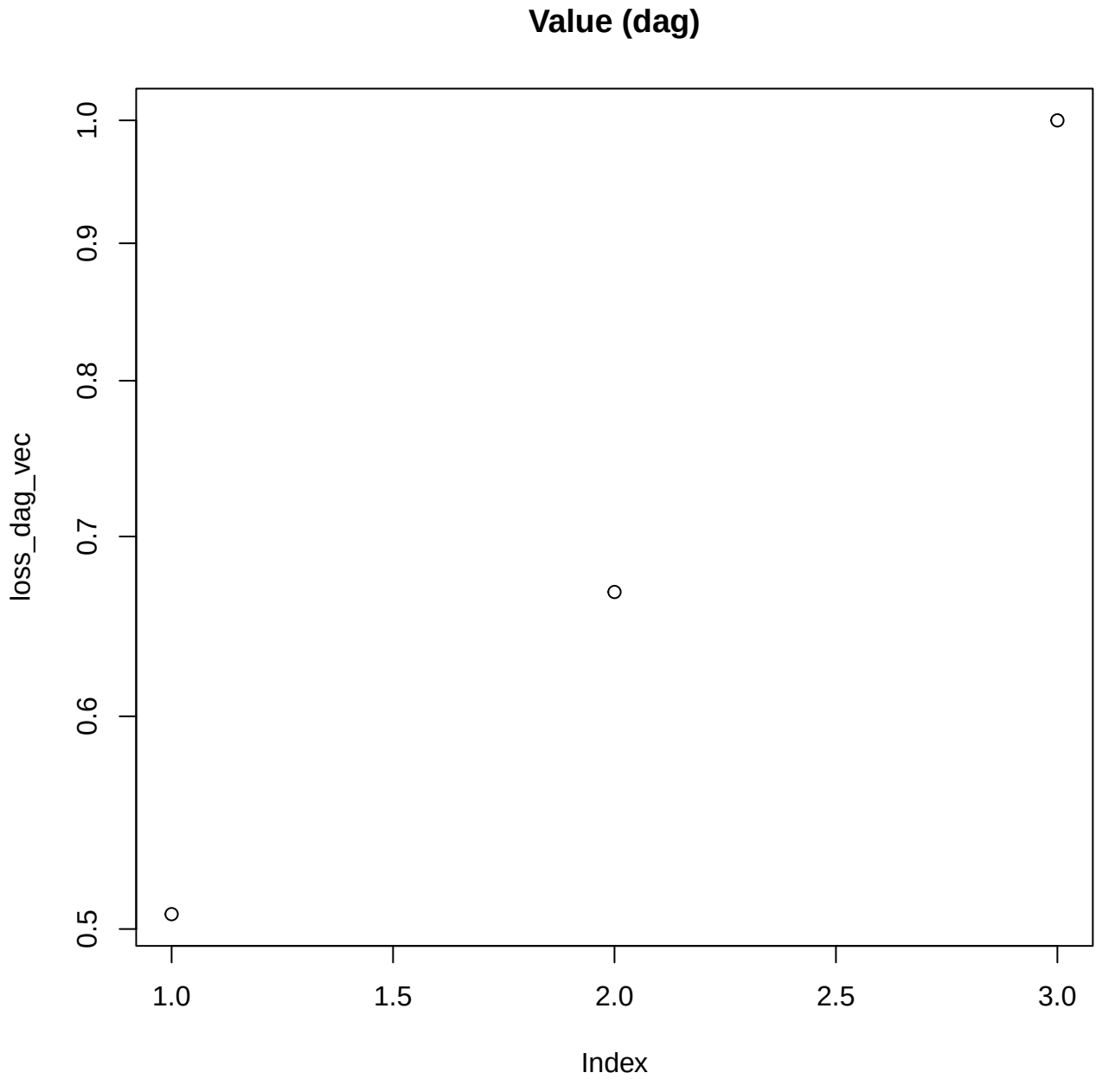
0.45
0.40
0.35
1.0
1.5
2.0
2.5
3.0
Index


Value (dag)
loss_dag_vec
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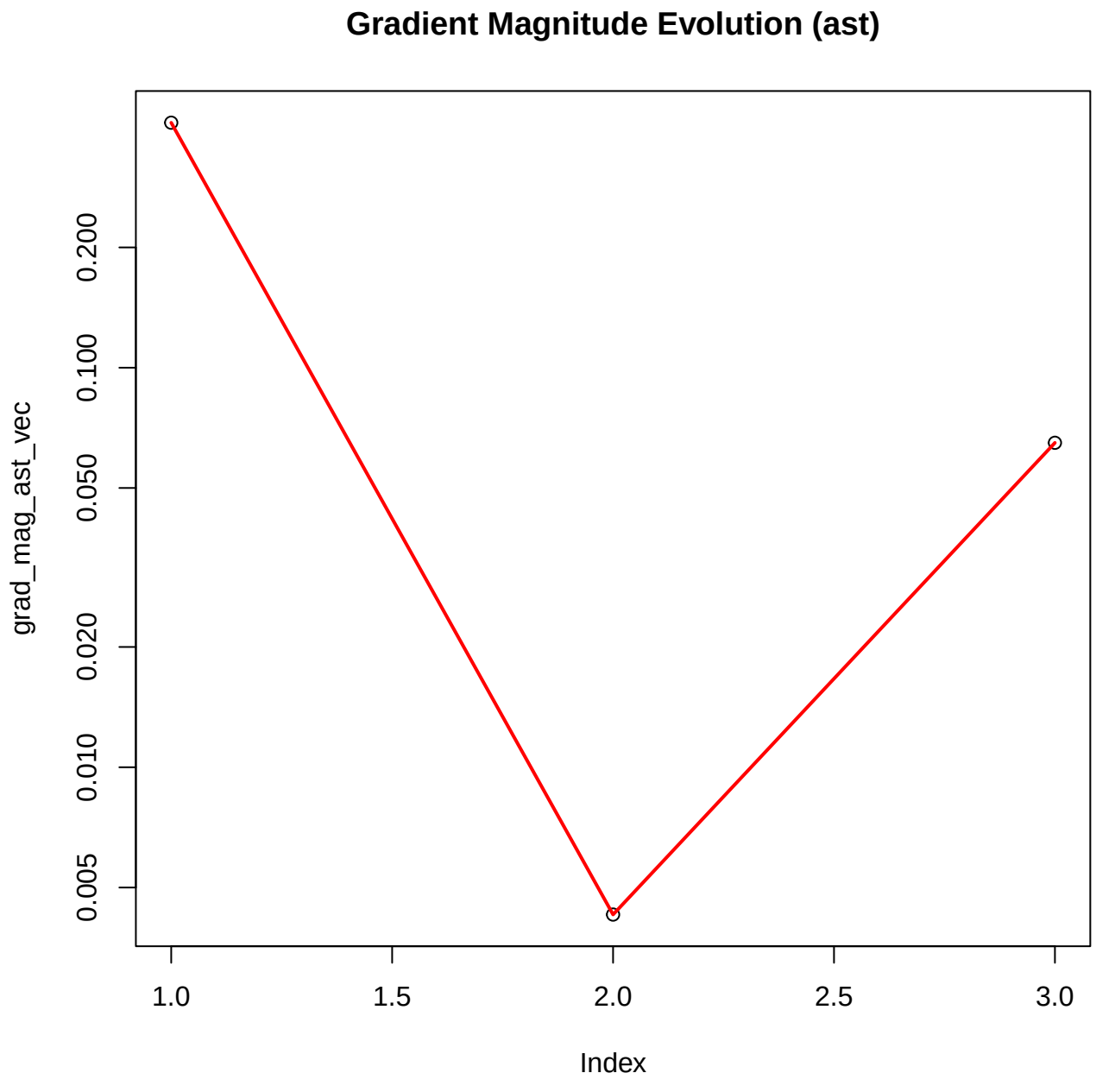
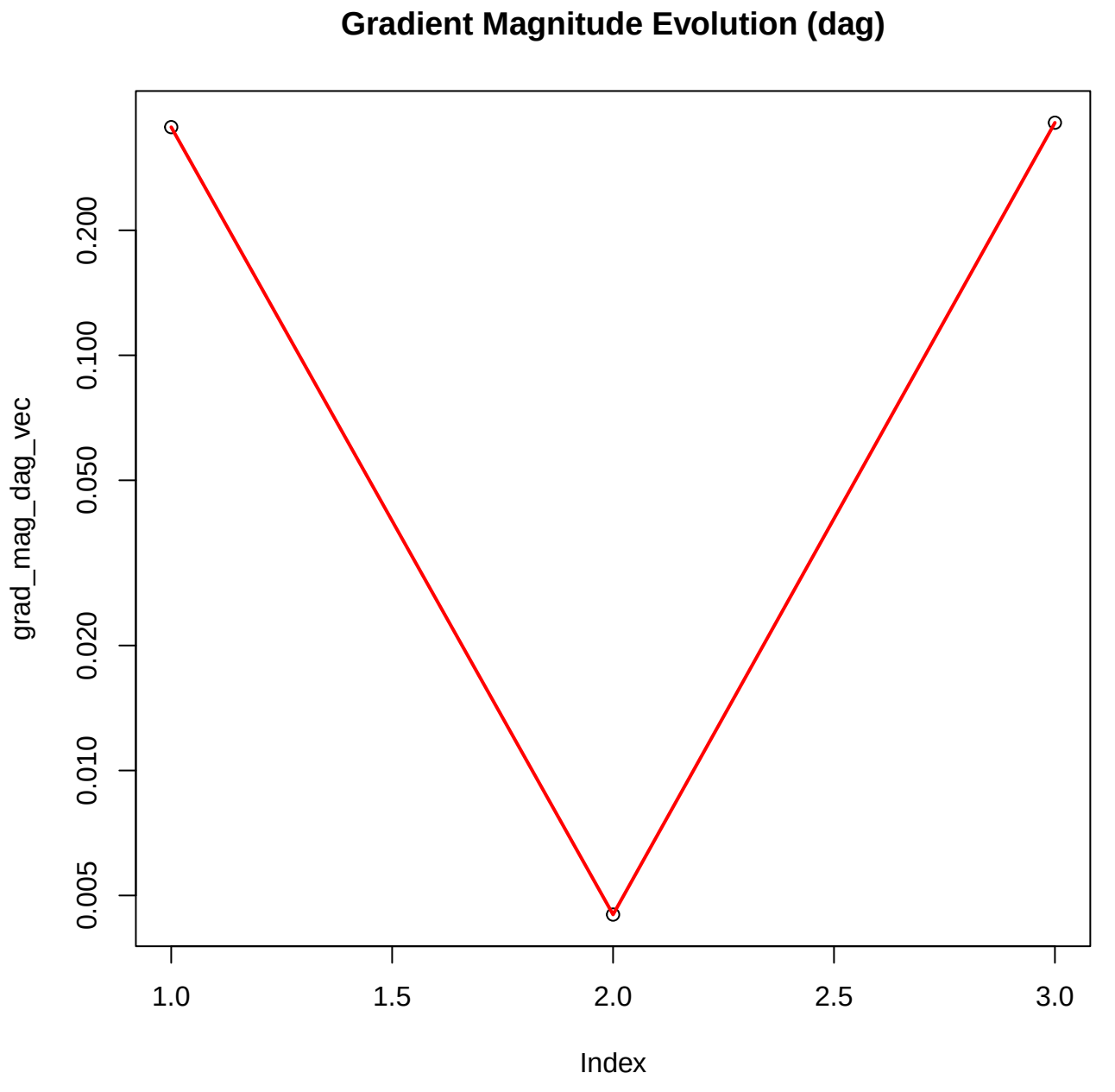
0.6
0.7
0.8
0.9
1.0
1.0
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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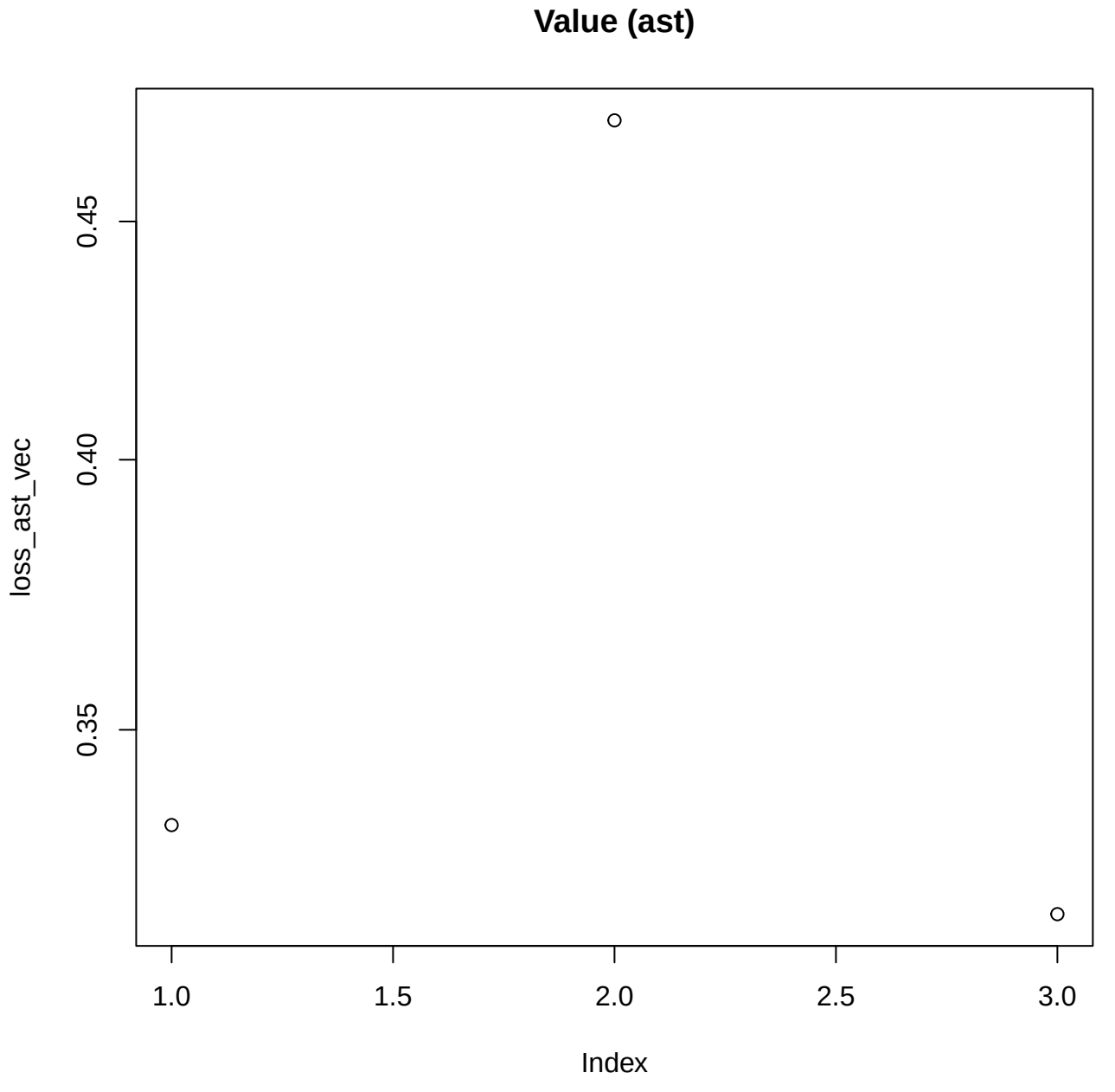
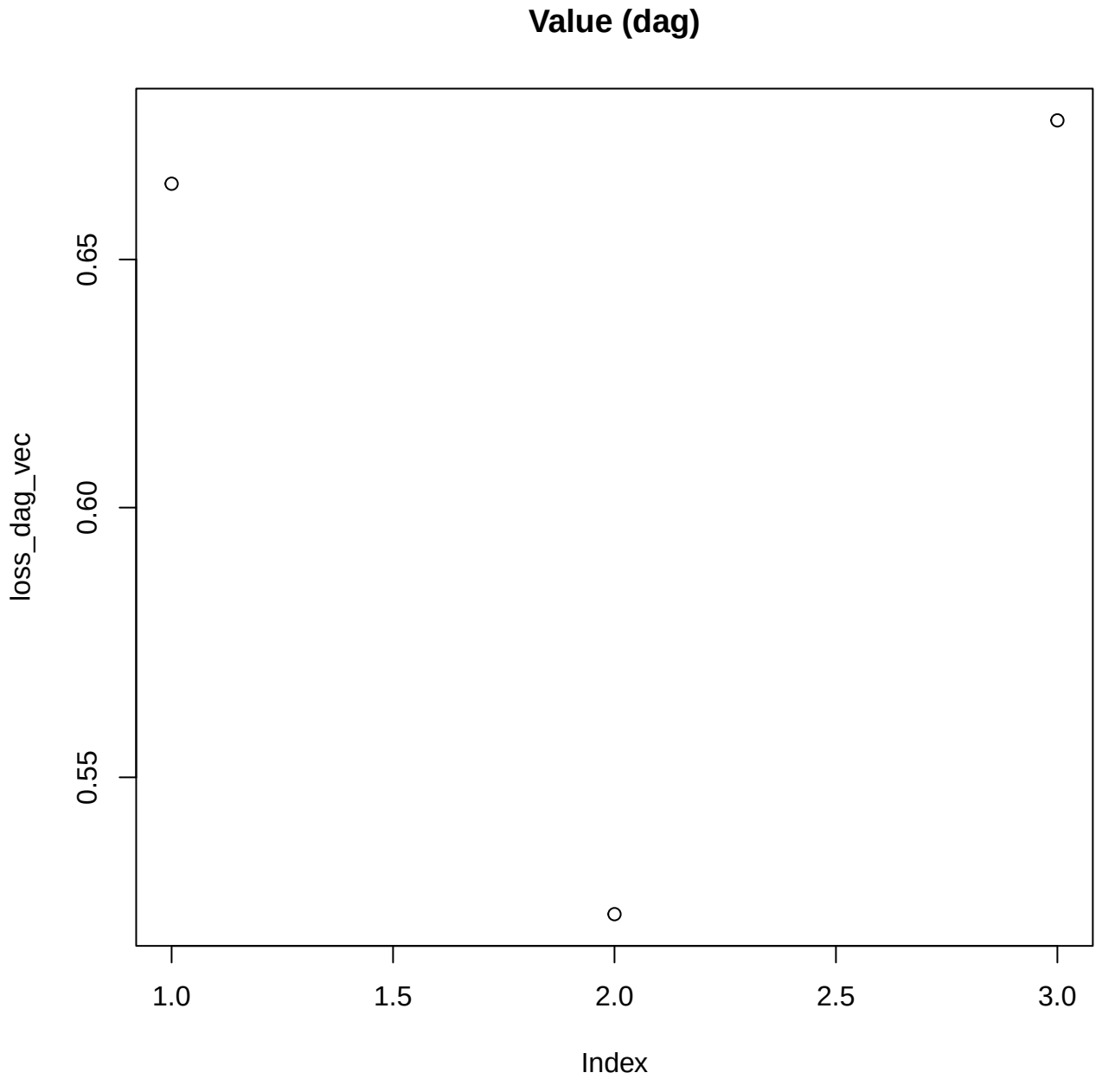
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index


Value (ast)
loss_ast_vec
0.35
0.40
0.45
1.0
1.5
2.0
2.5
3.0
Index


Value (dag)
loss_dag_vec
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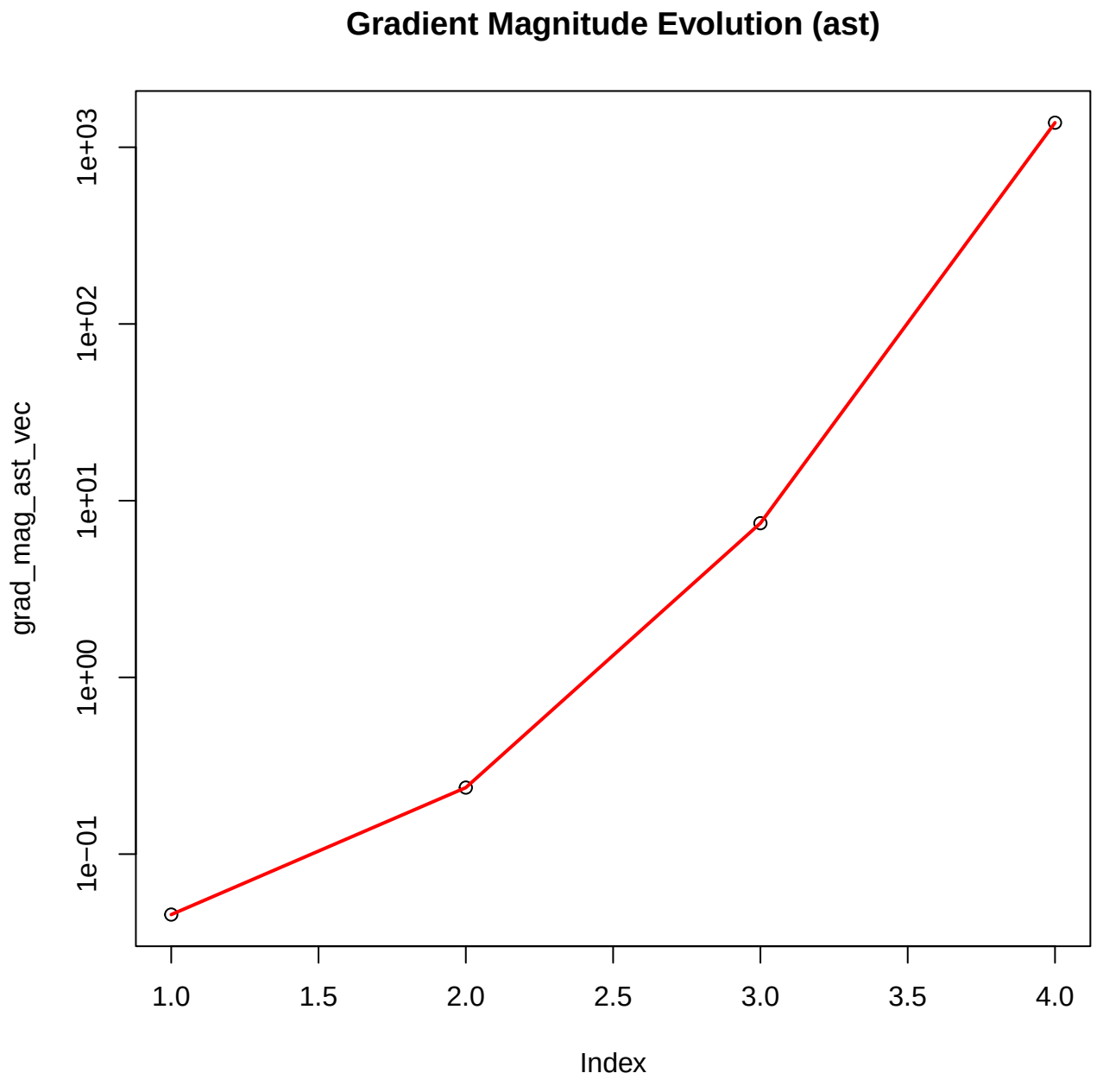
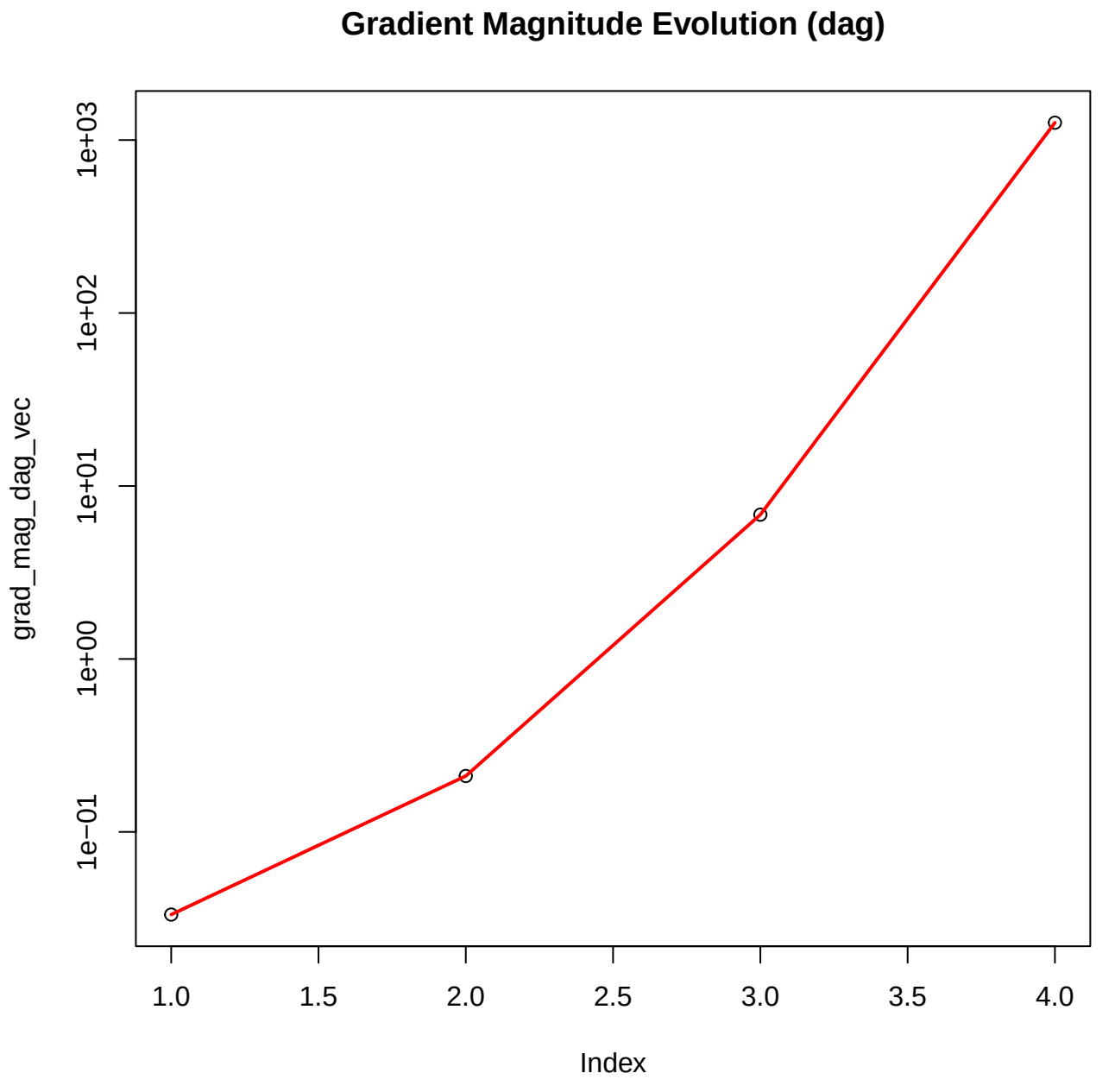
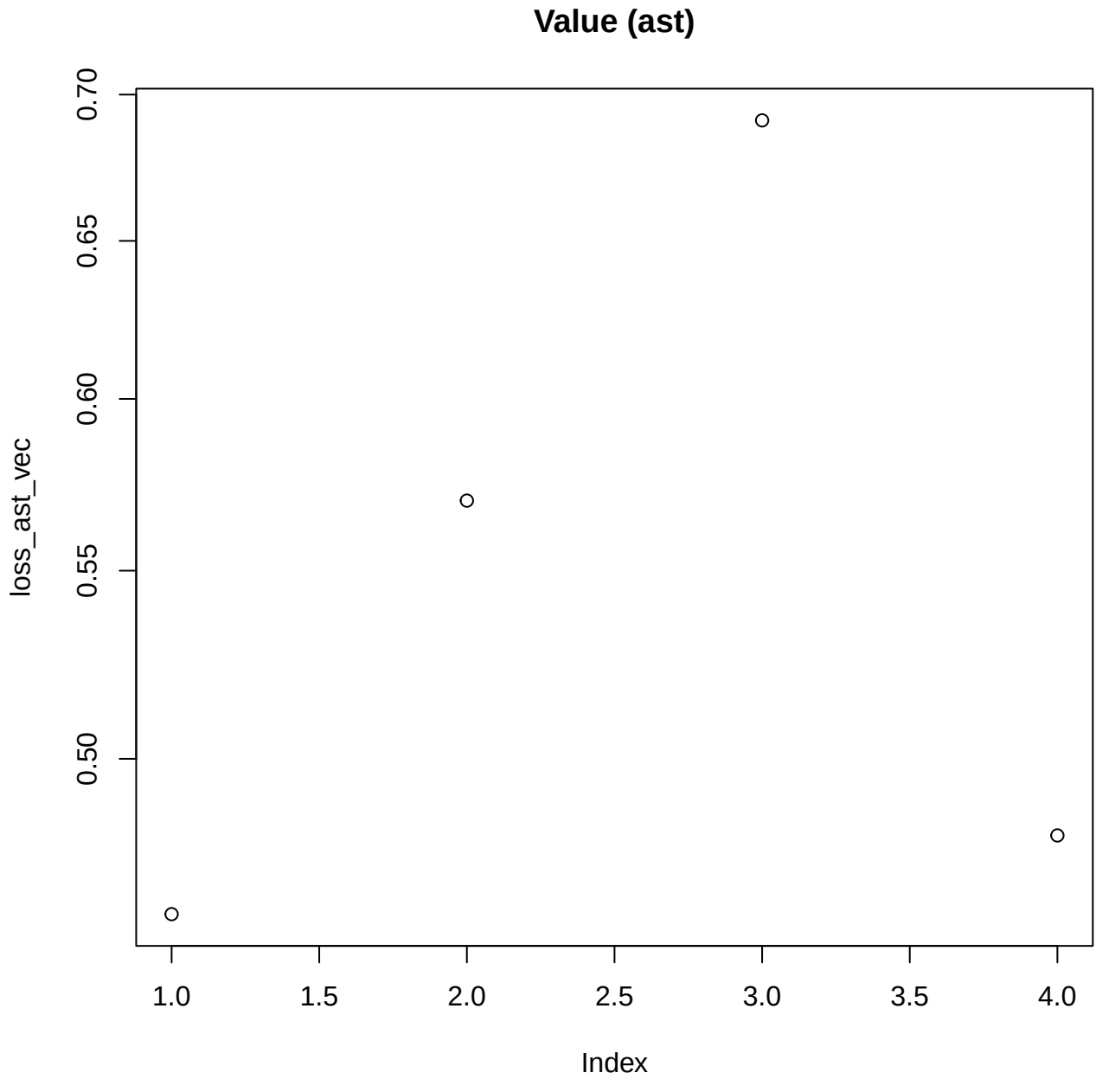
0.60
0.65
1.0
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index


Value (ast)
loss_ast_vec
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0.65
0.70
1.0
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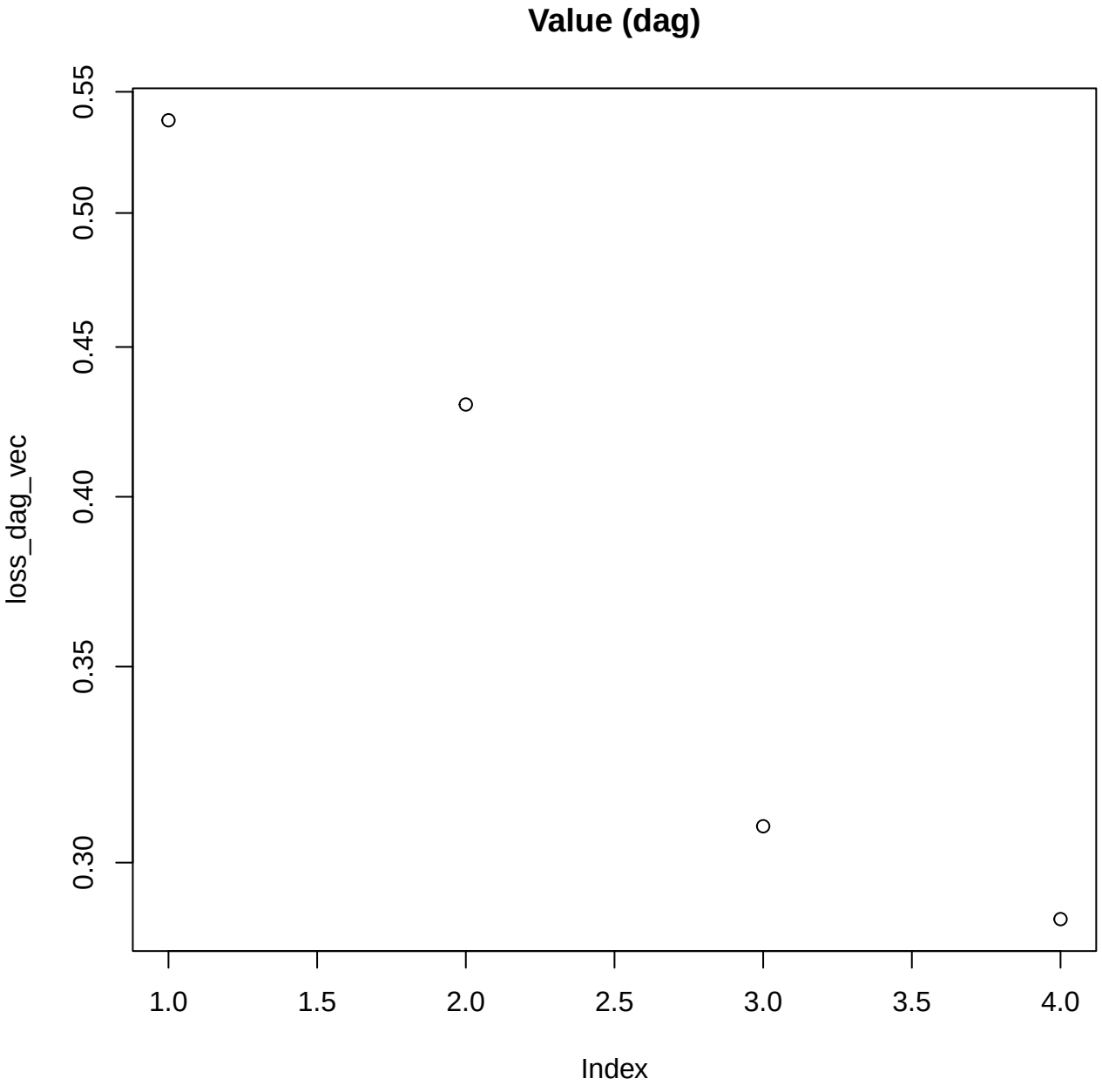
2.5
3.0
3.5
4.0
Index


Value (dag)
loss_dag_vec
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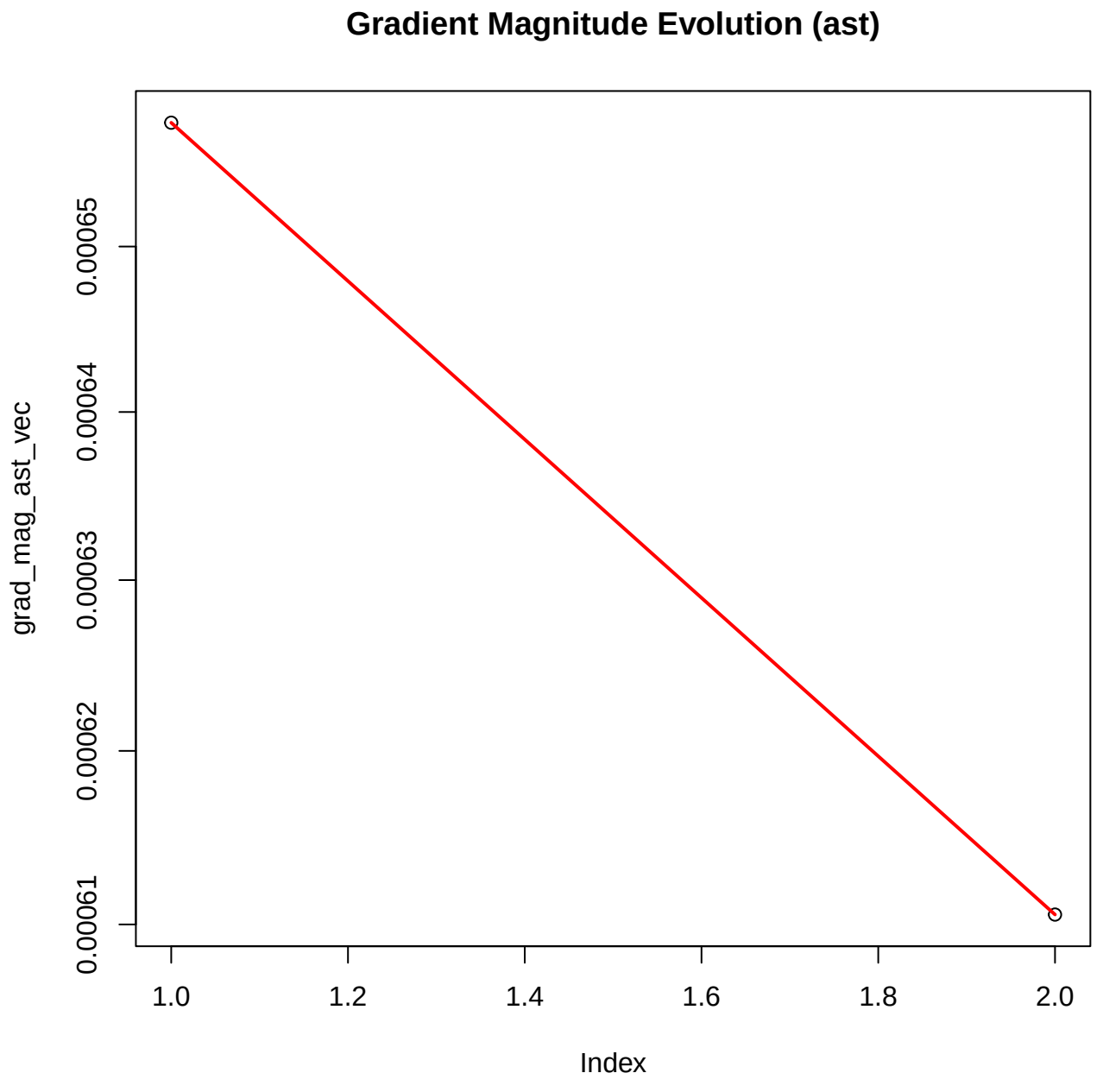
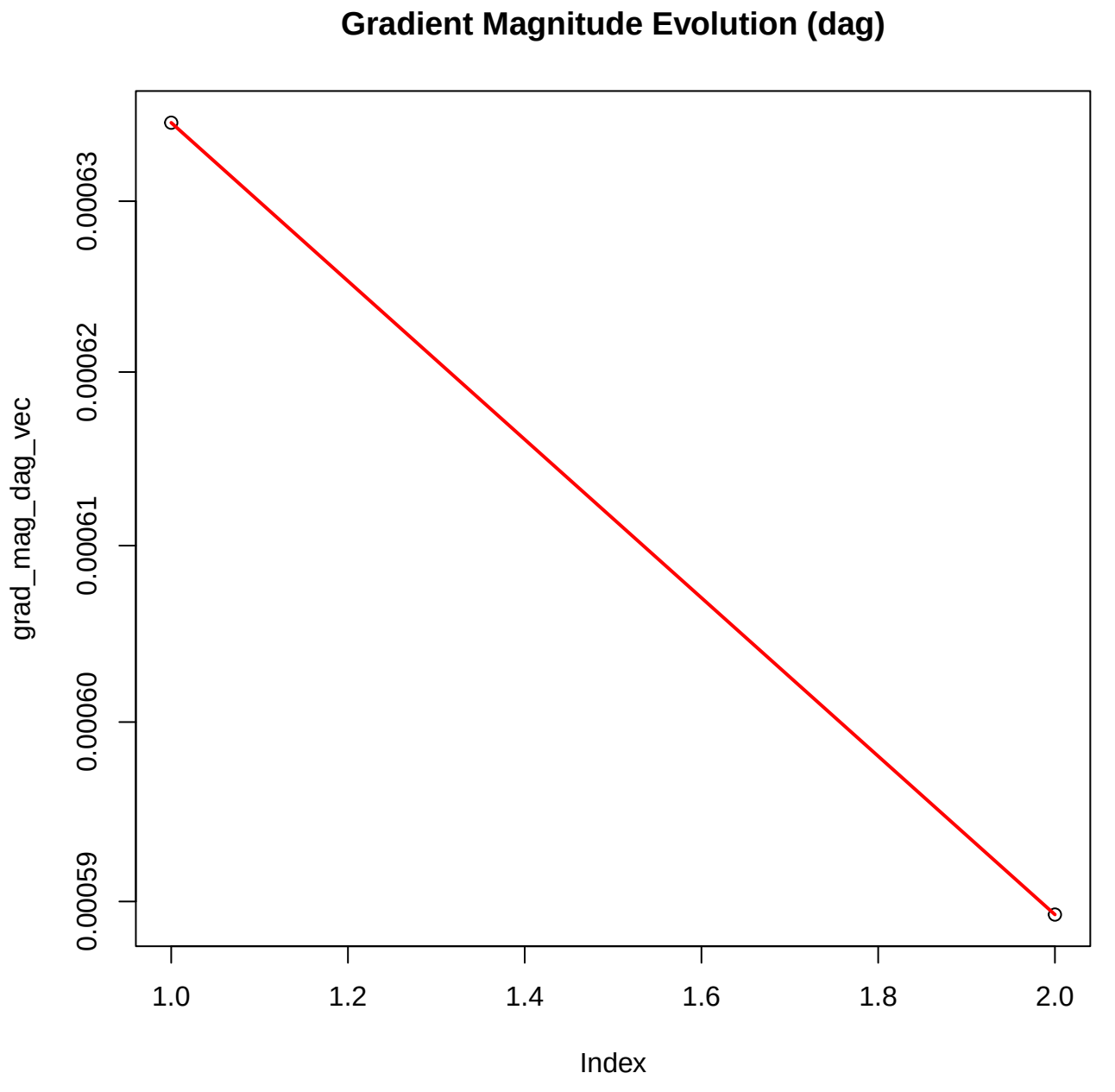
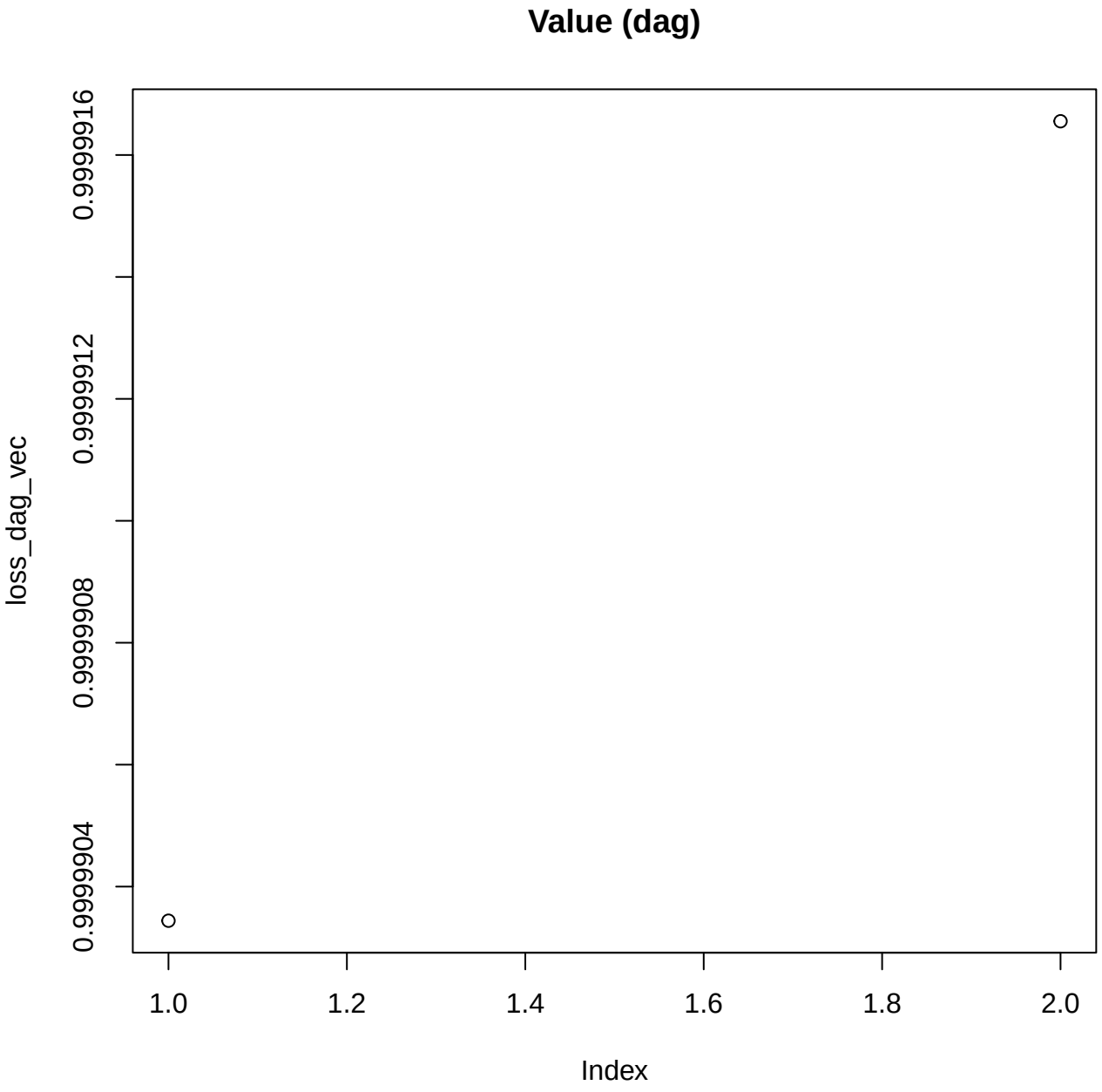
0.50
0.45
0.40
0.35
0.30
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
1.0
1.2
1.4
1.6
1.8
2.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.00059
0.00060
0.00061
0.00062
0.00063
1.0
1.2
1.4
1.6
1.8
2.0
Index


Value (dag)
loss_dag_vec
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0.9999912
0.9999916
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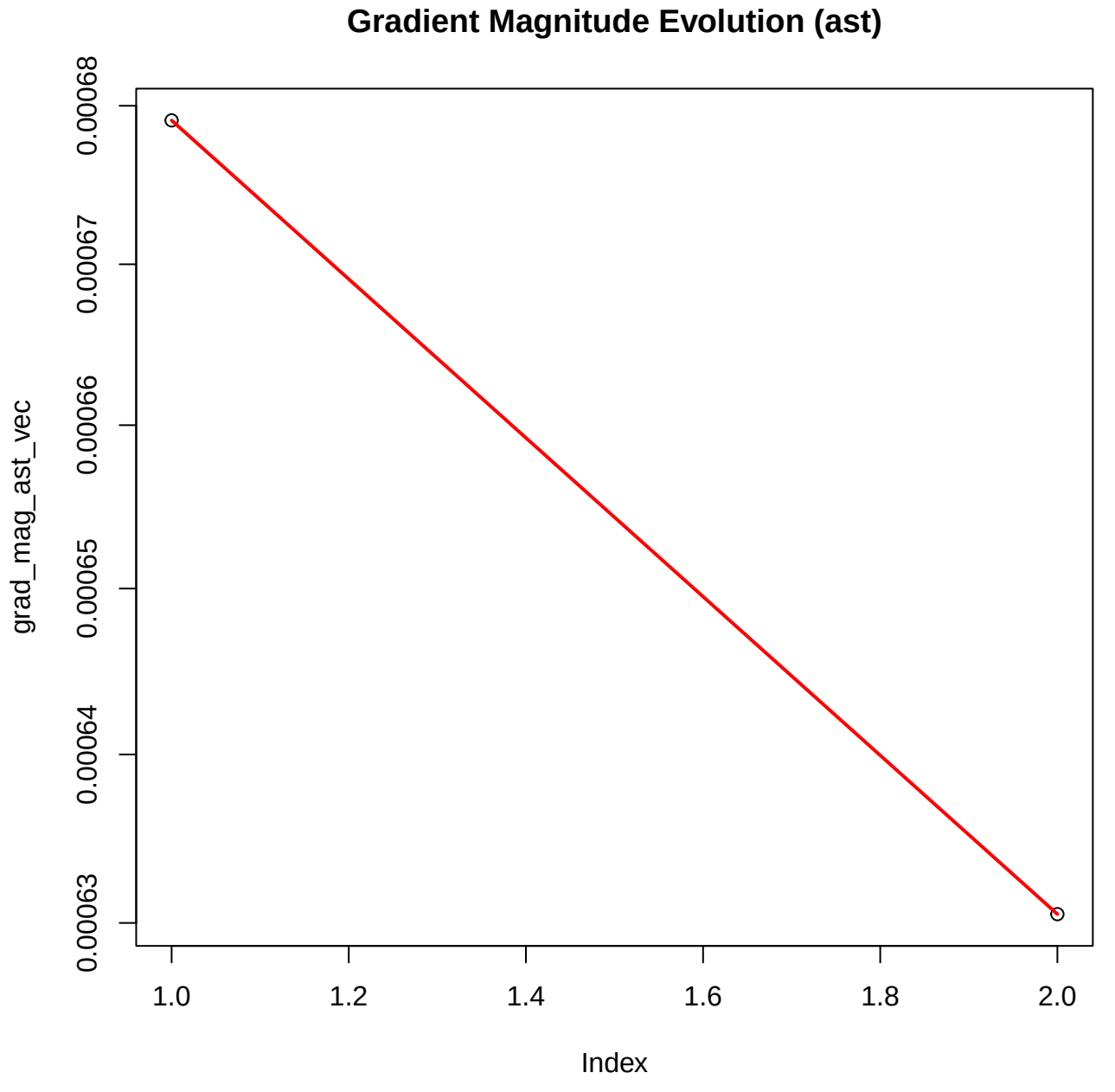
1.4
1.6
1.8
2.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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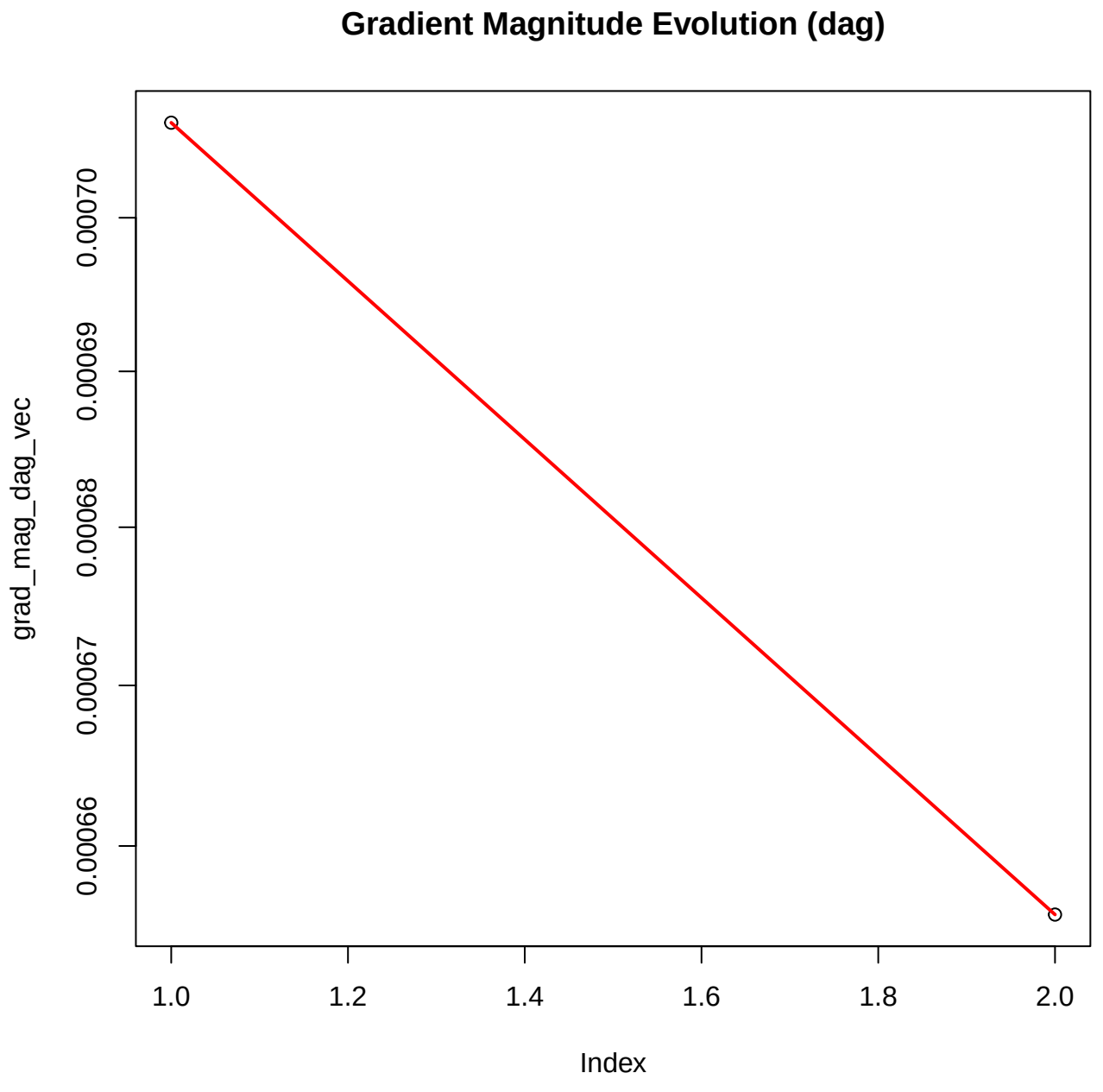
0.00064
0.00065
0.00066
0.00067
0.00068
1.0
1.2
1.4
1.6
1.8
2.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.00067
0.00068
0.00069
0.00070
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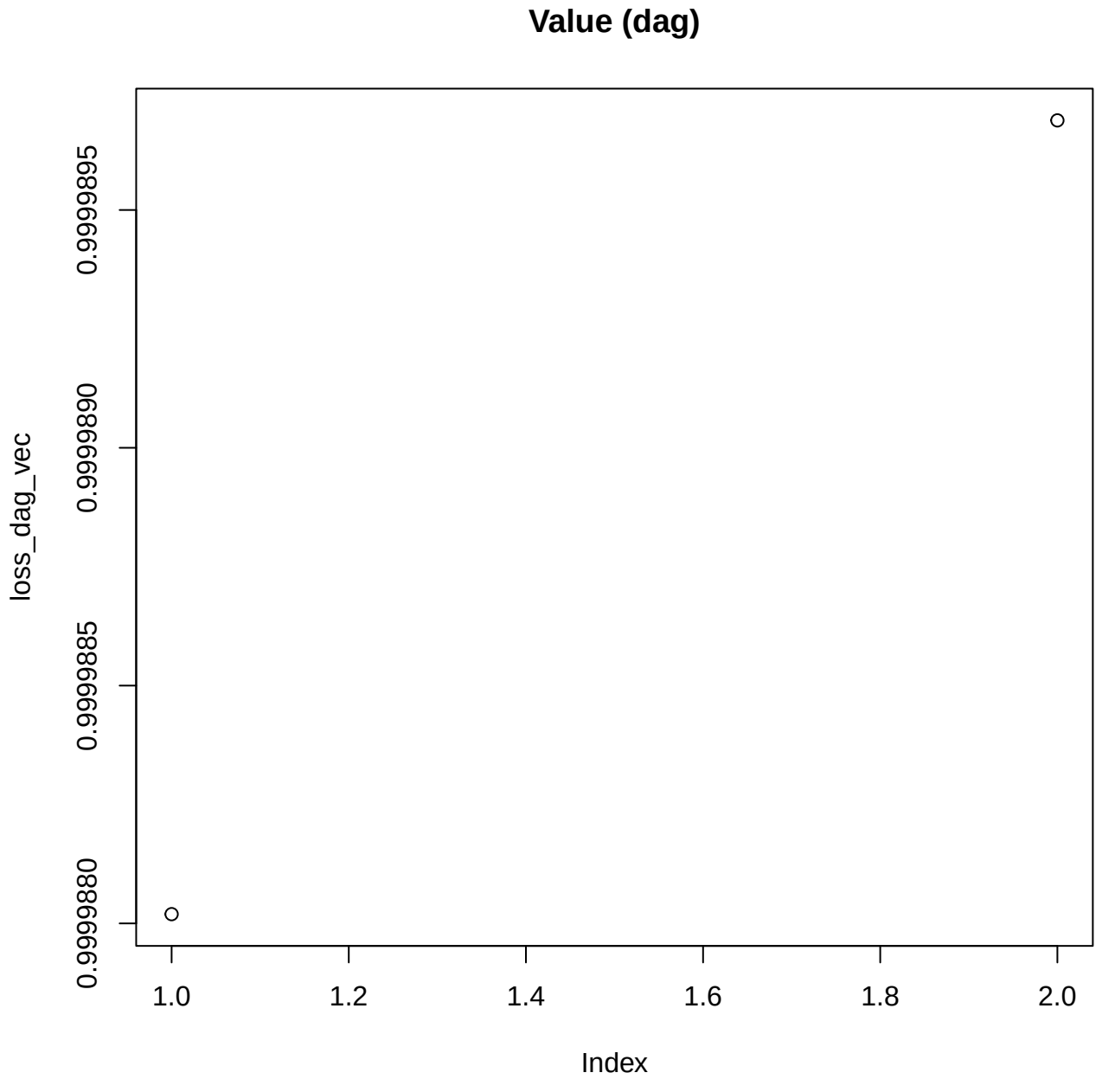
1.4
1.6
1.8
2.0
Index


Value (dag)
loss_dag_vec
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0.9999895
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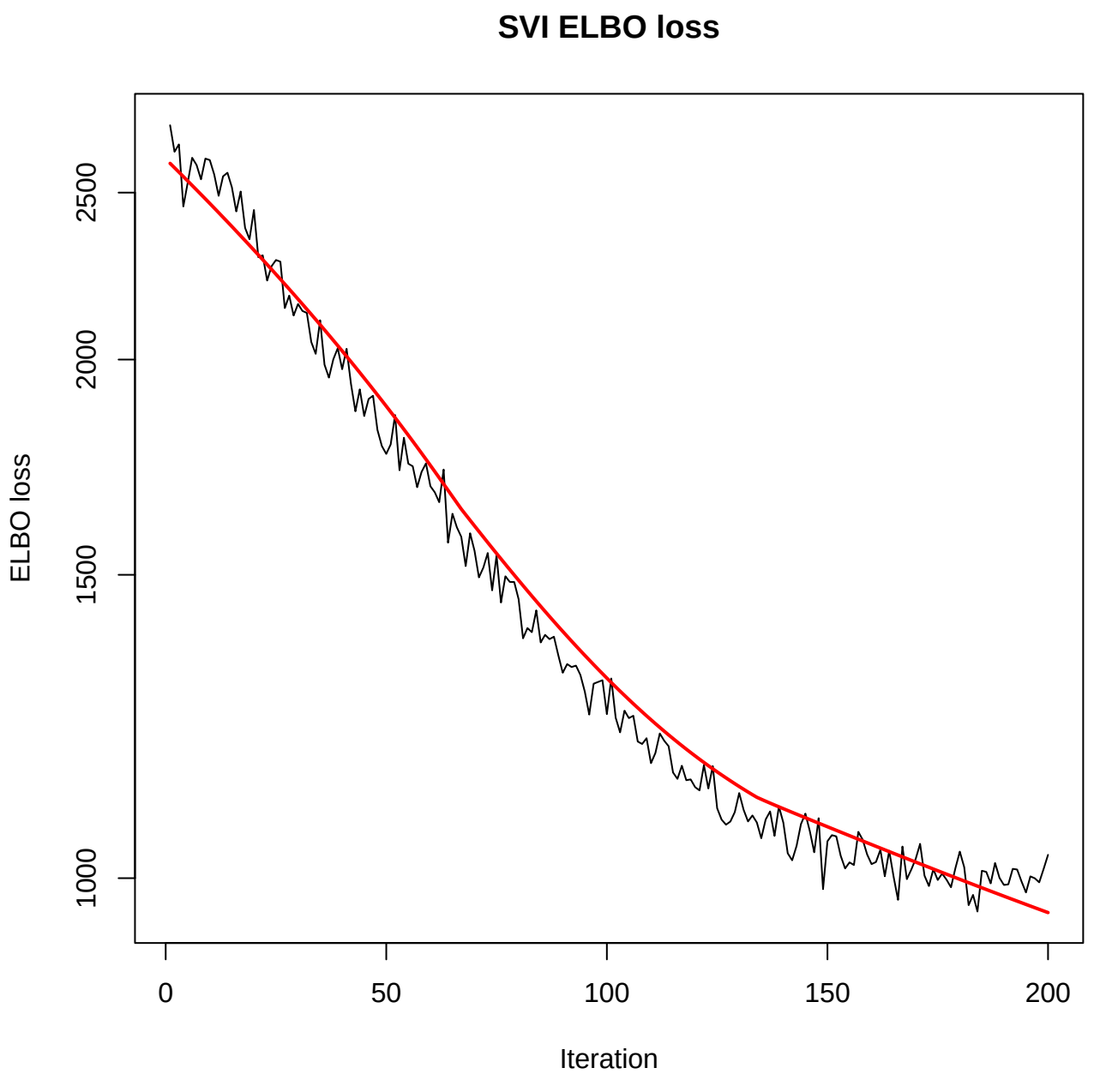
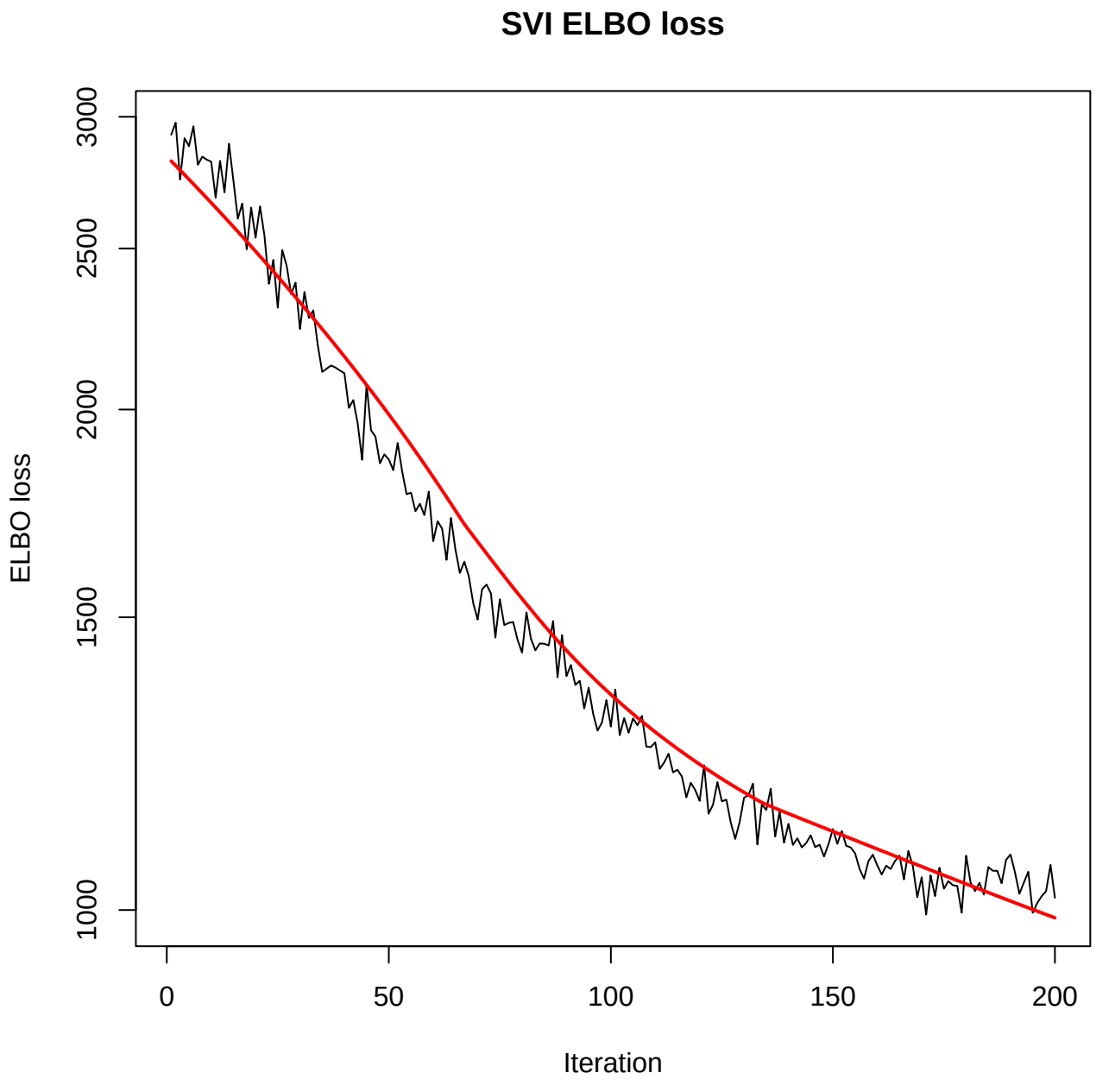
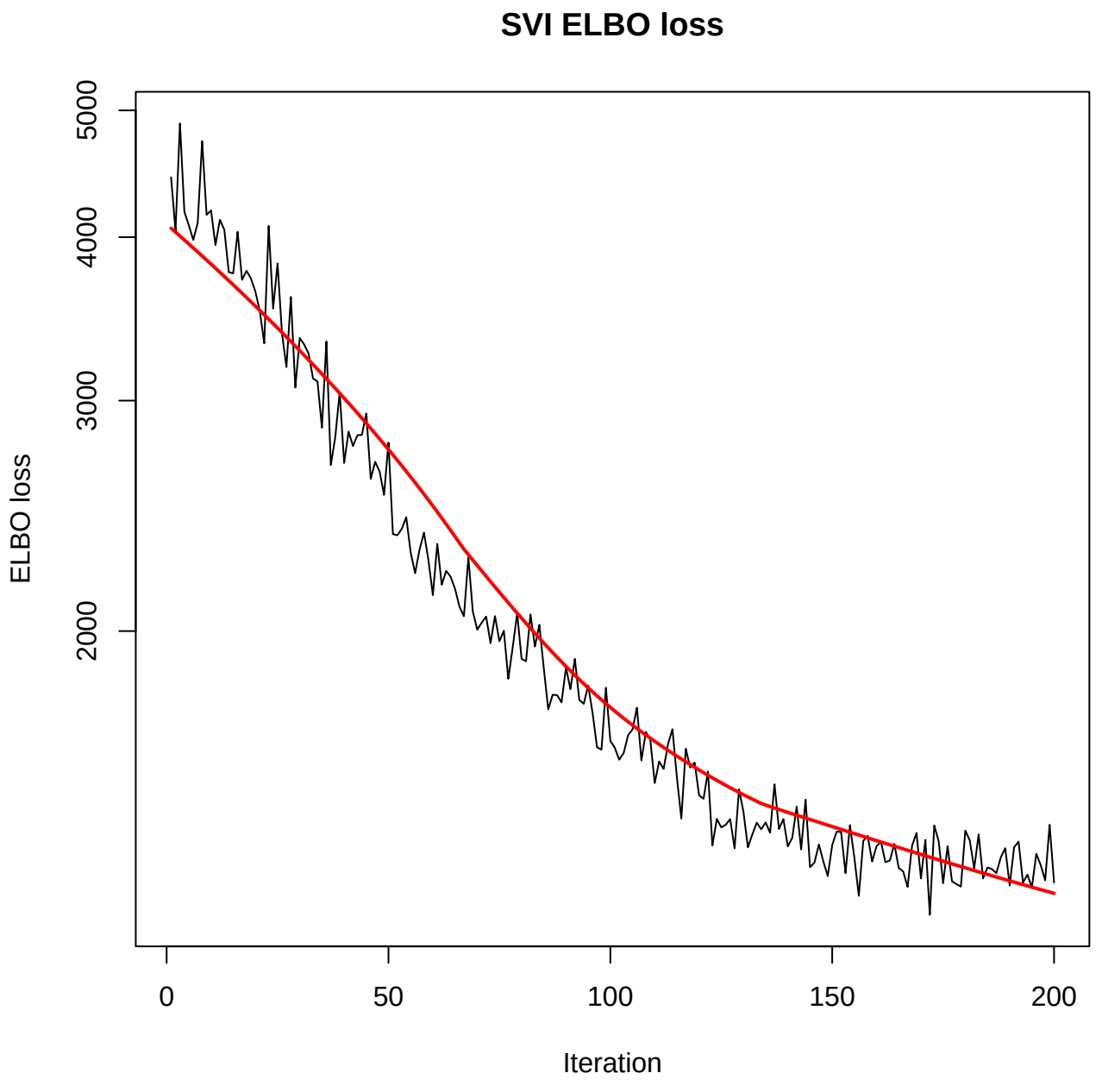
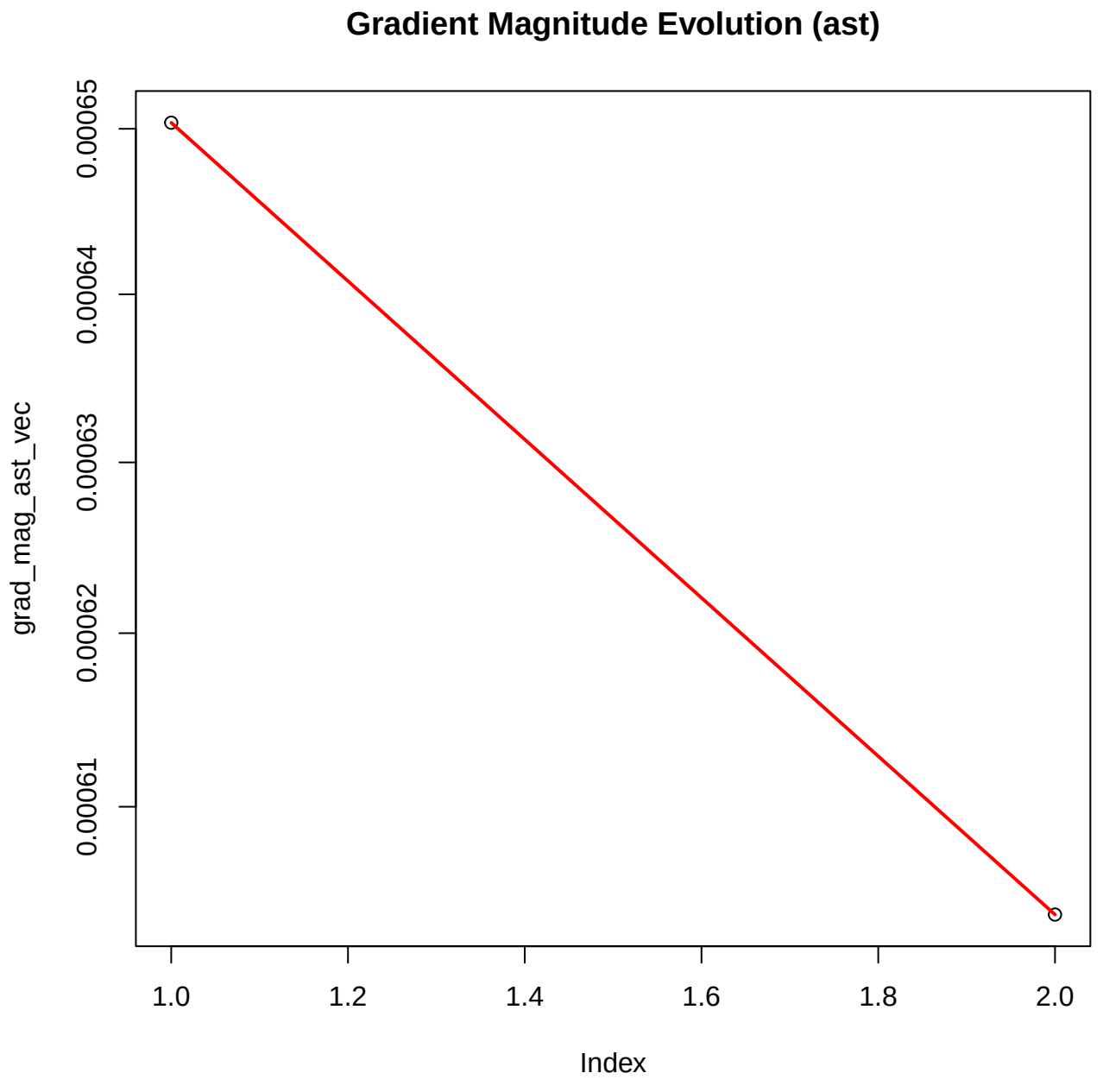
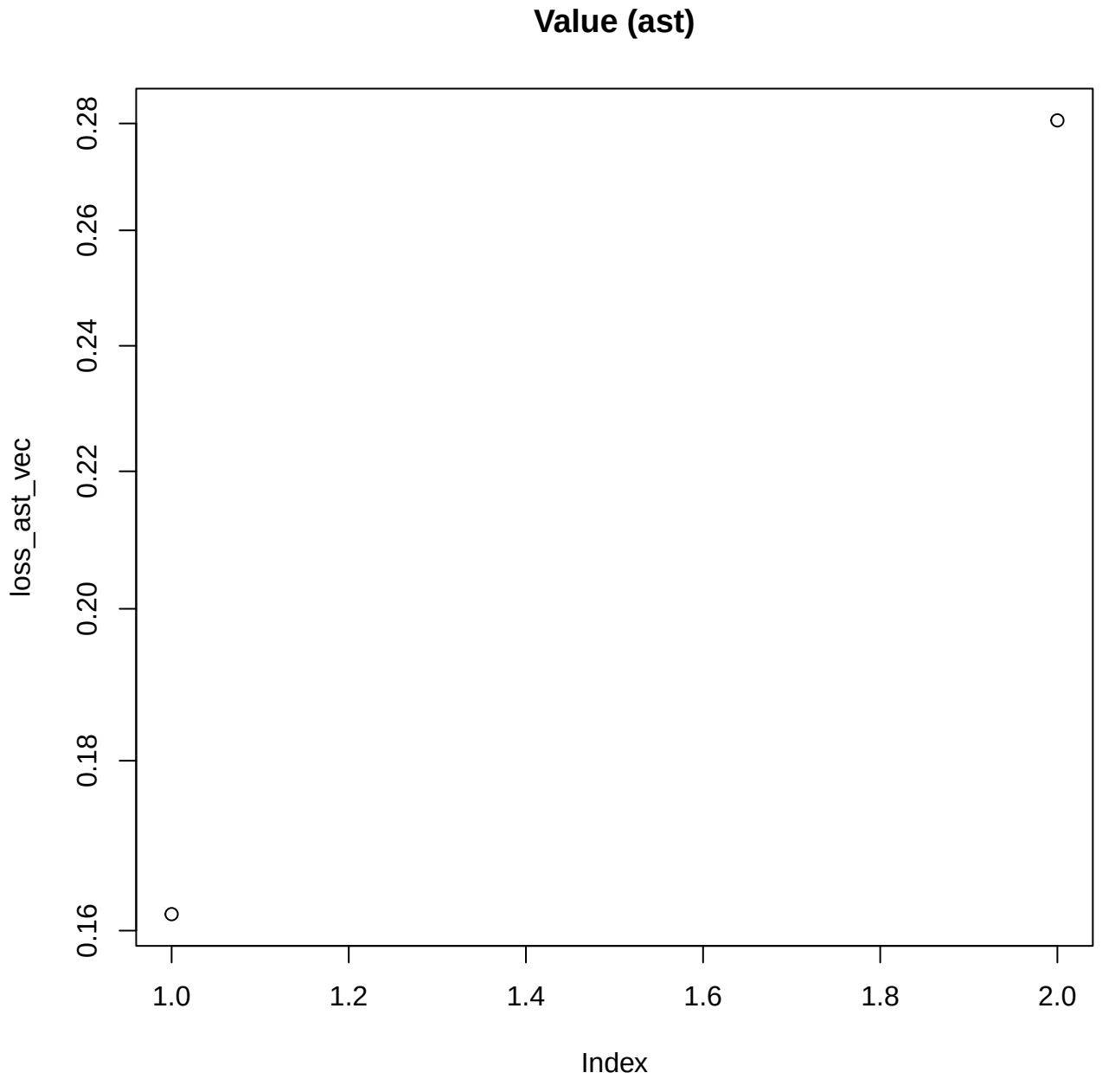
1.4
1.6
1.8
2.0
Index


SVI ELBO loss
ELBO loss
2500
2000
1500
1000
0
50
100
150
200
Iteration


SVI ELBO loss
ELBO loss
3000
2500
2000
1500
1000
0
50
100
150
200
Iteration


SVI ELBO loss
ELBO loss
5000
4000
3000
2000
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
1.0
1.2
1.4
1.6
1.8
2.0
Index


Value (ast)
loss_ast_vec
0.16
0.18
0.20
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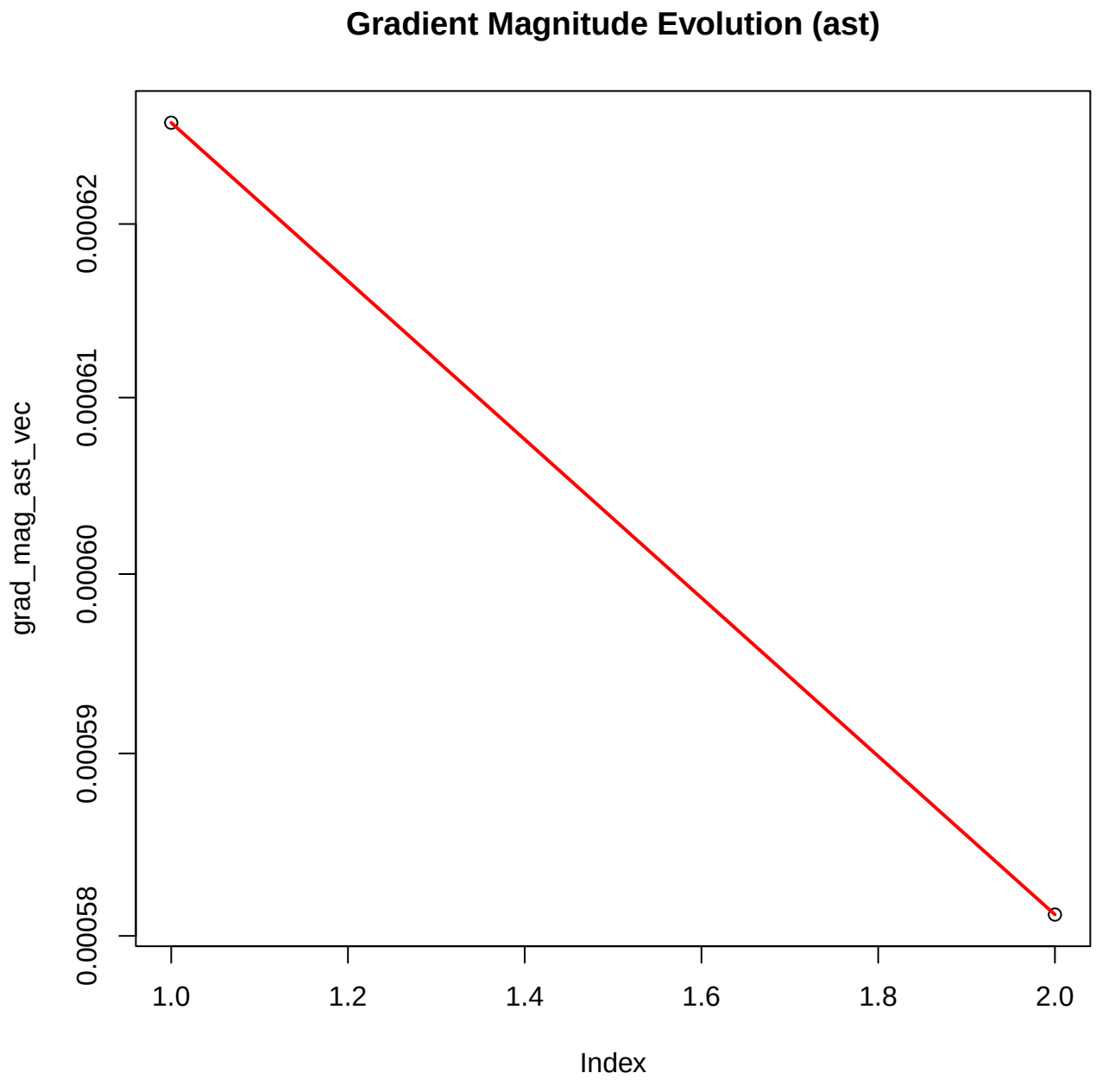
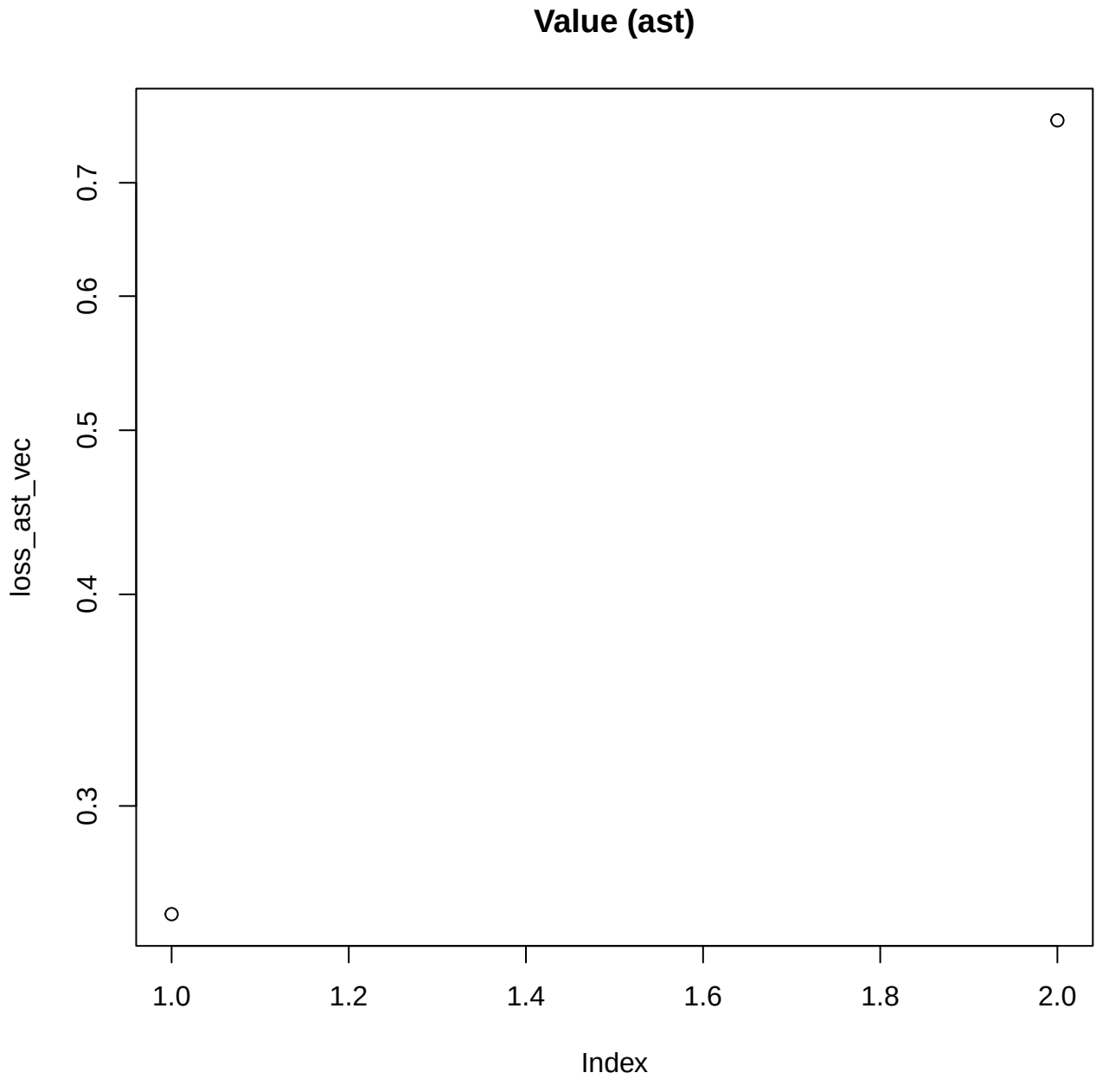
0.24
0.26
0.28
1.0
1.2
1.4
1.6
1.8
2.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00058
0.00059
0.00060
0.00061
0.00062
1.0
1.2
1.4
1.6
1.8
2.0
Index


Value (ast)
loss_ast_vec
0.3
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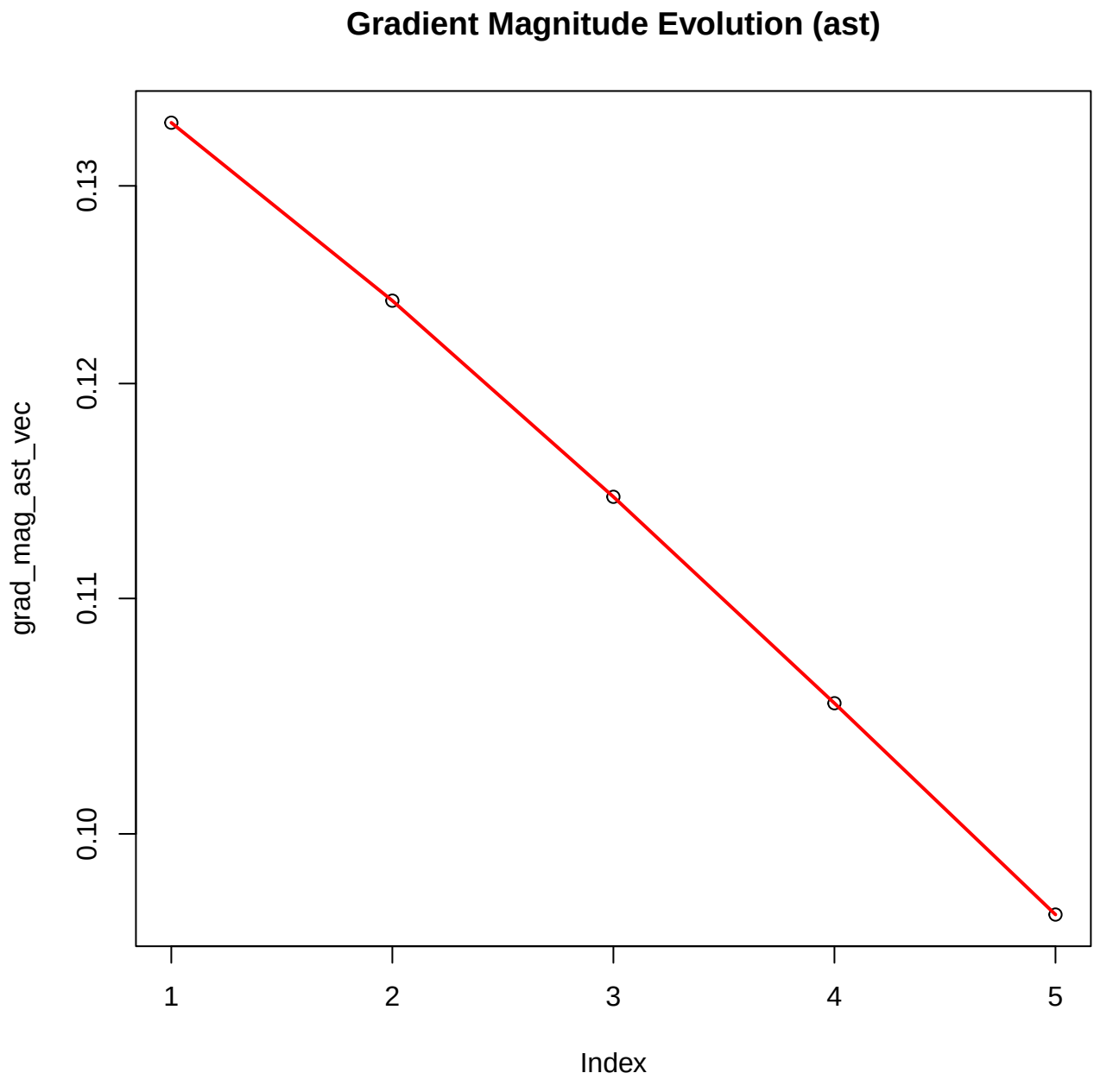
0.5
0.6
0.7
1.0
1.2
1.4
1.6
1.8
2.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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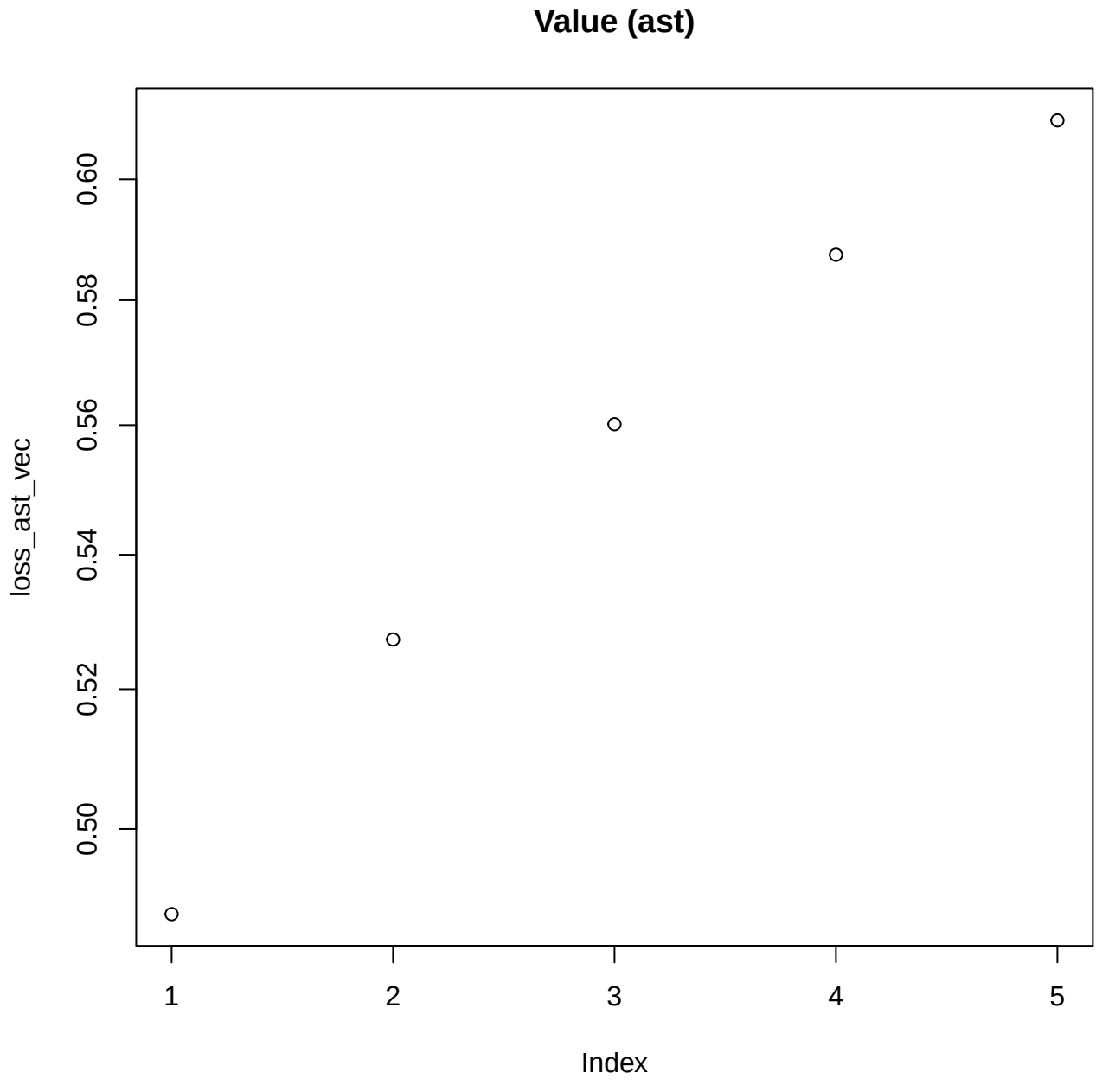
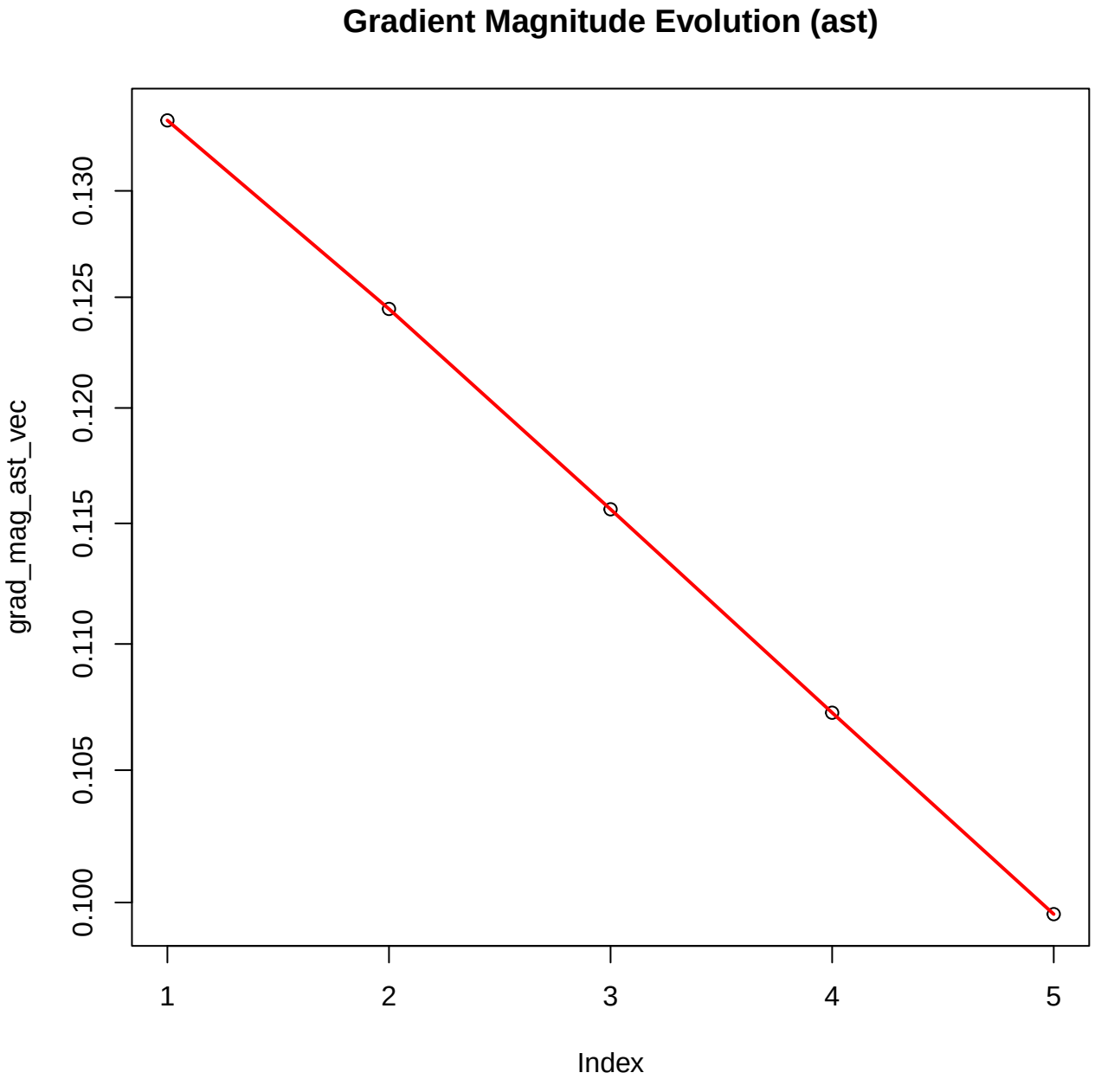
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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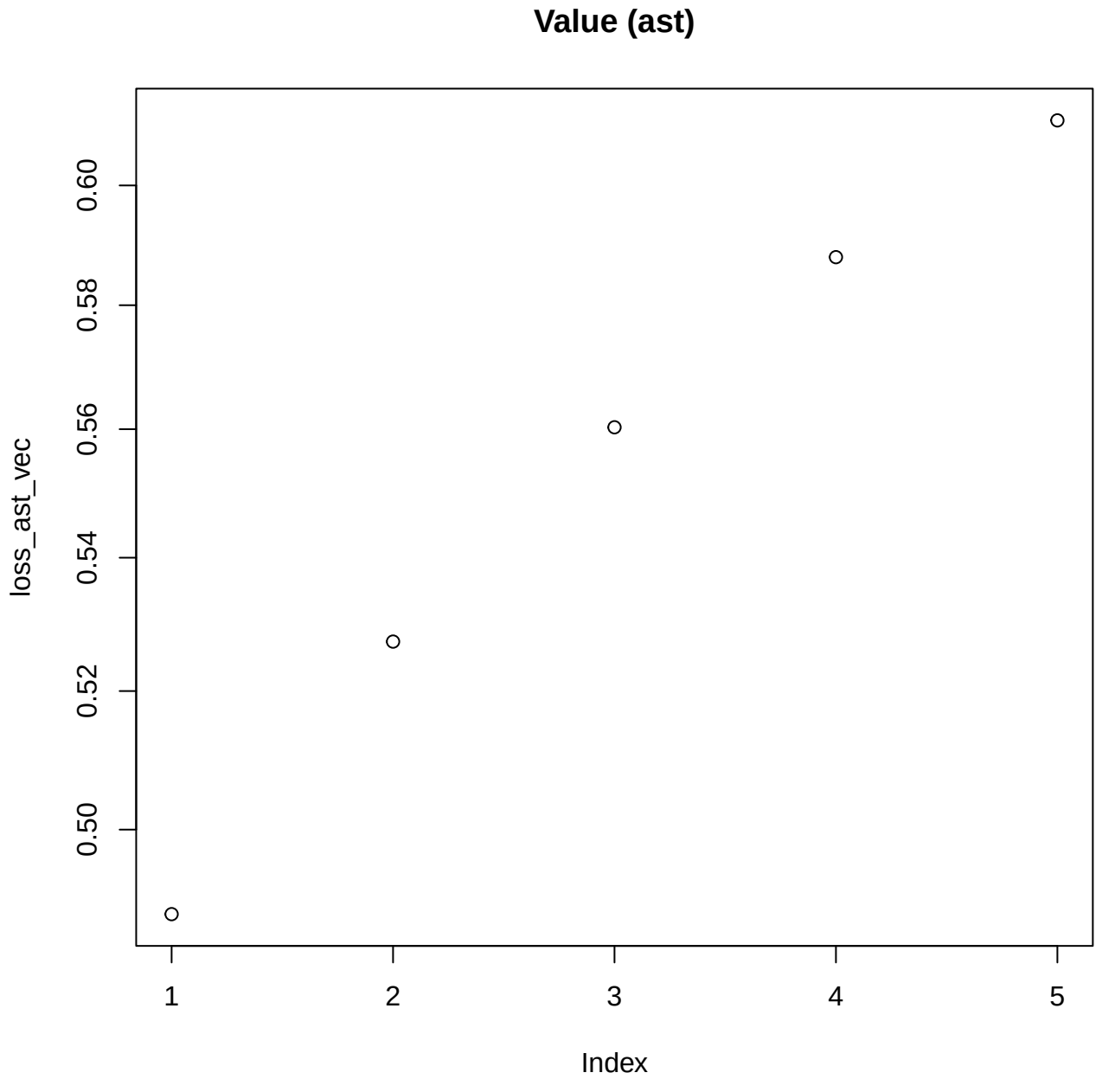
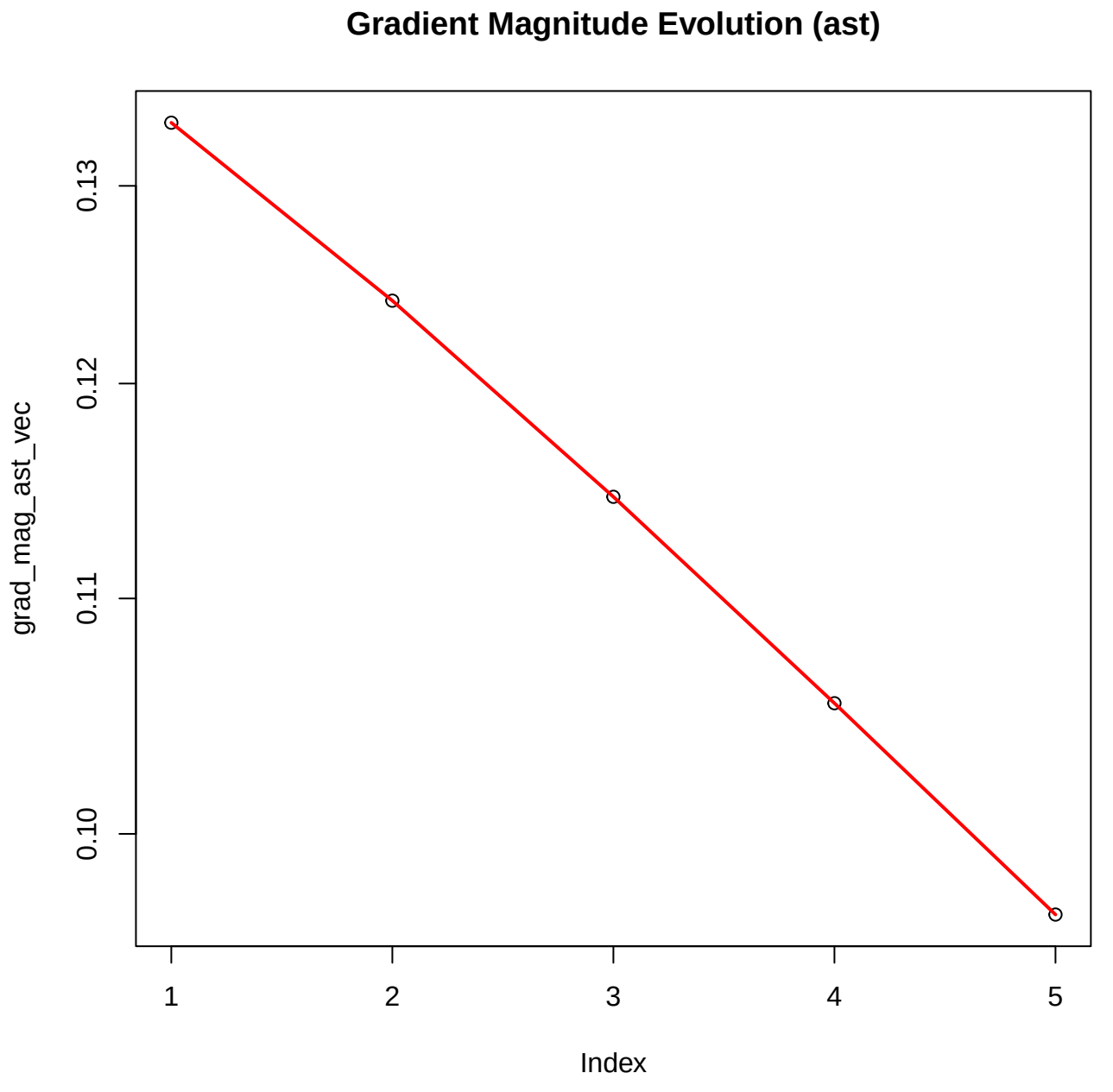
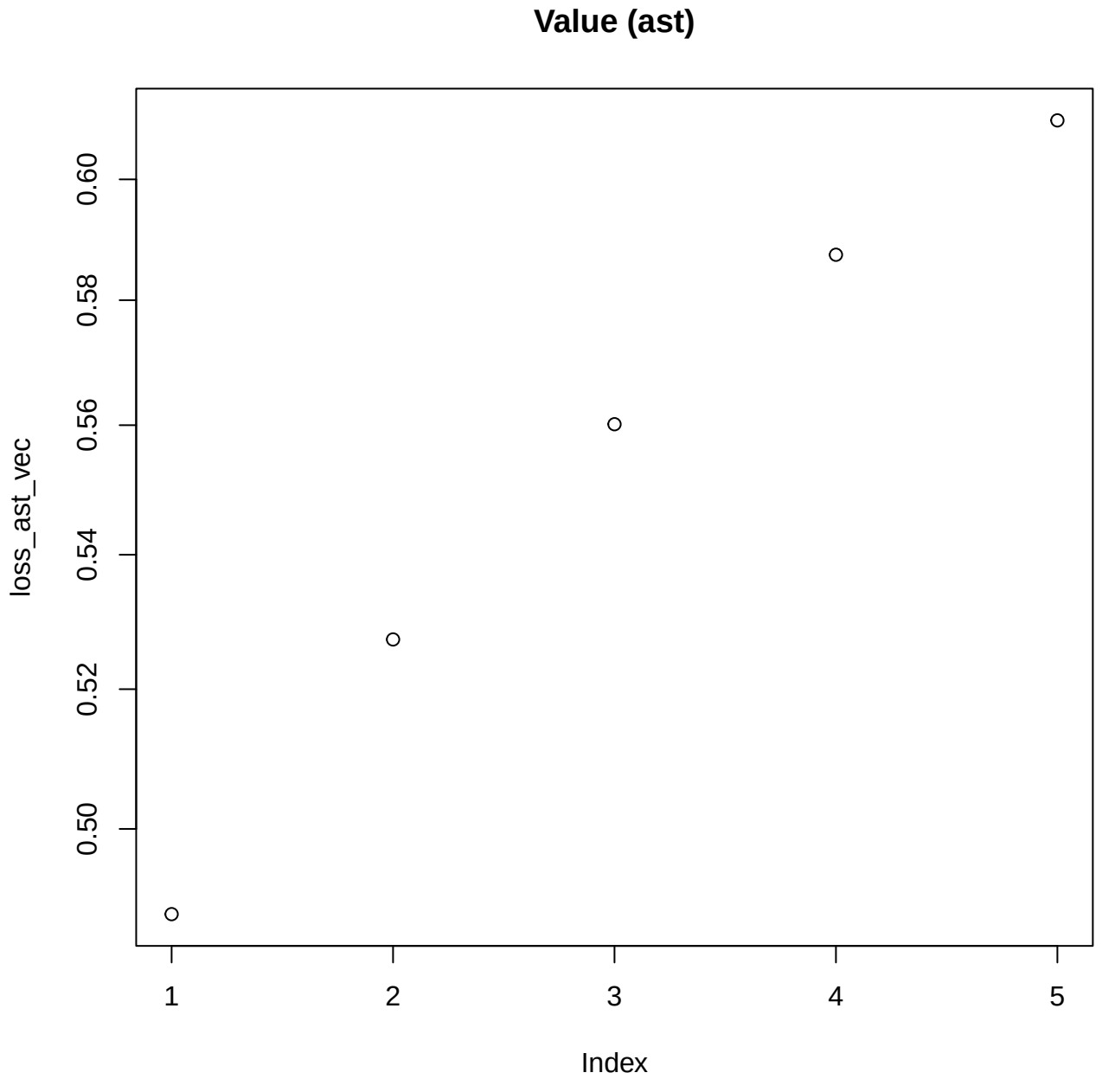
0.105
0.110
0.115
0.120
0.125
0.130
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
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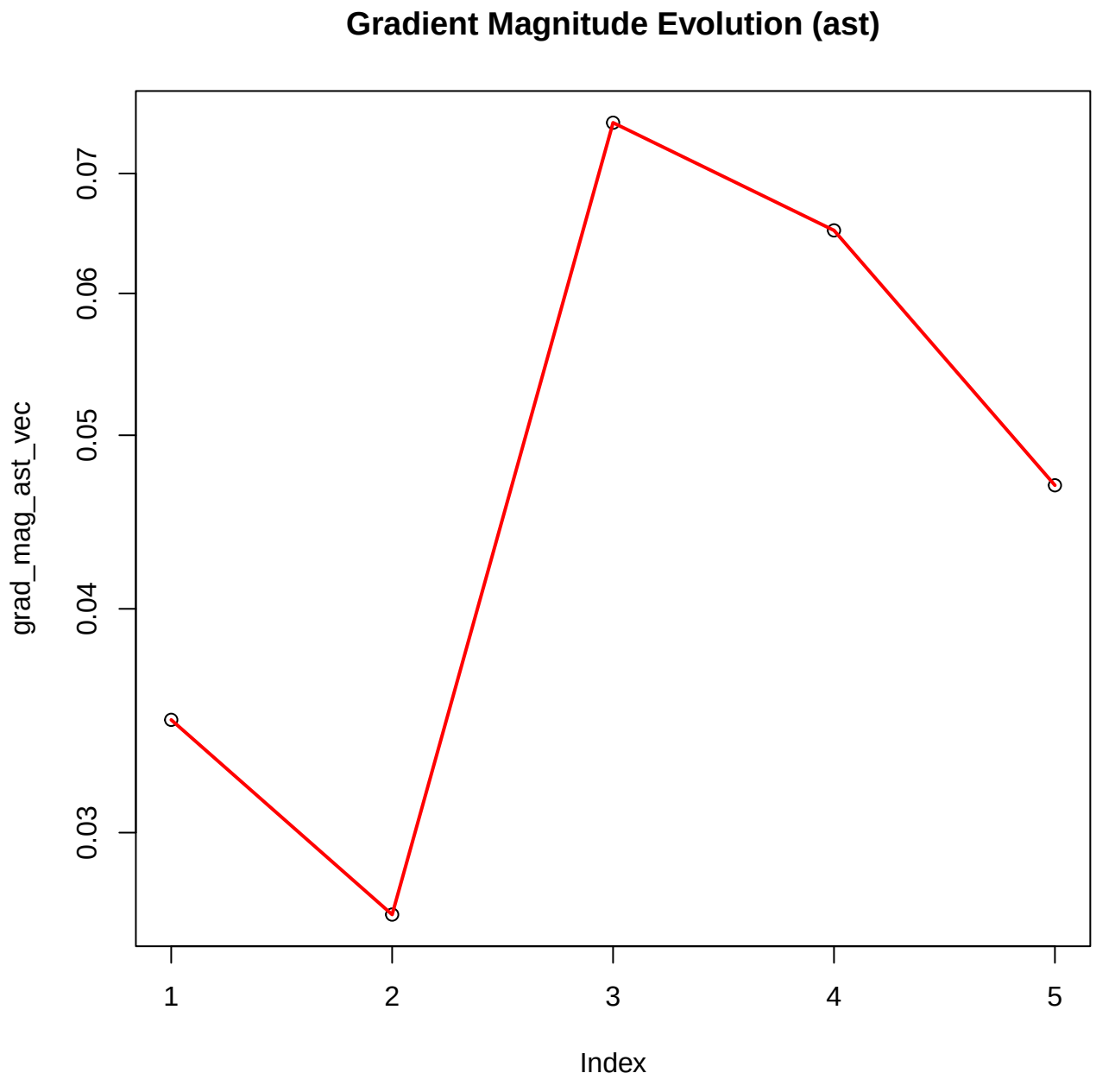
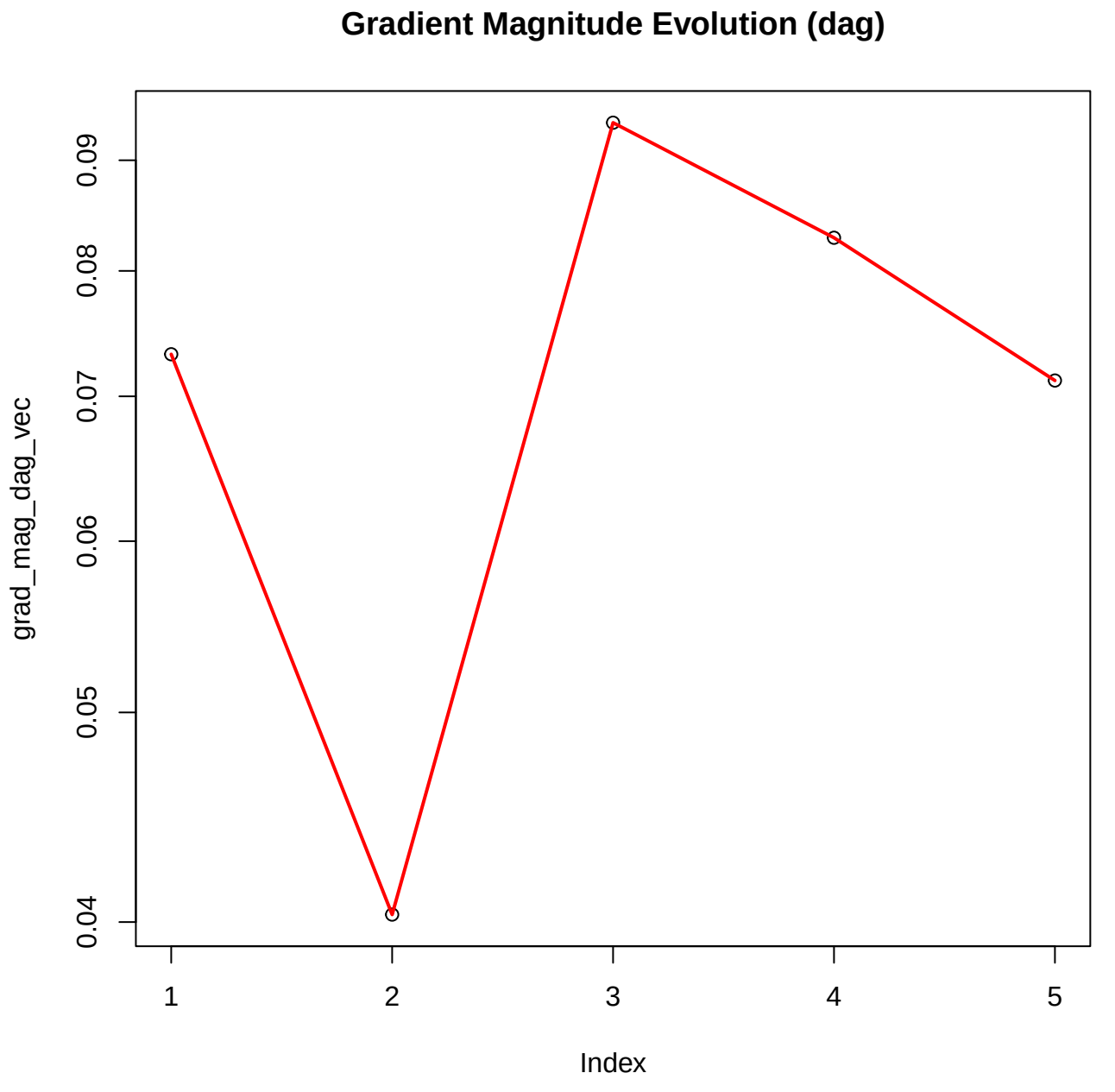
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.03
0.04
0.05
0.06
0.07
1
2
3
4
5
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.05
0.06
0.07
0.08
0.09
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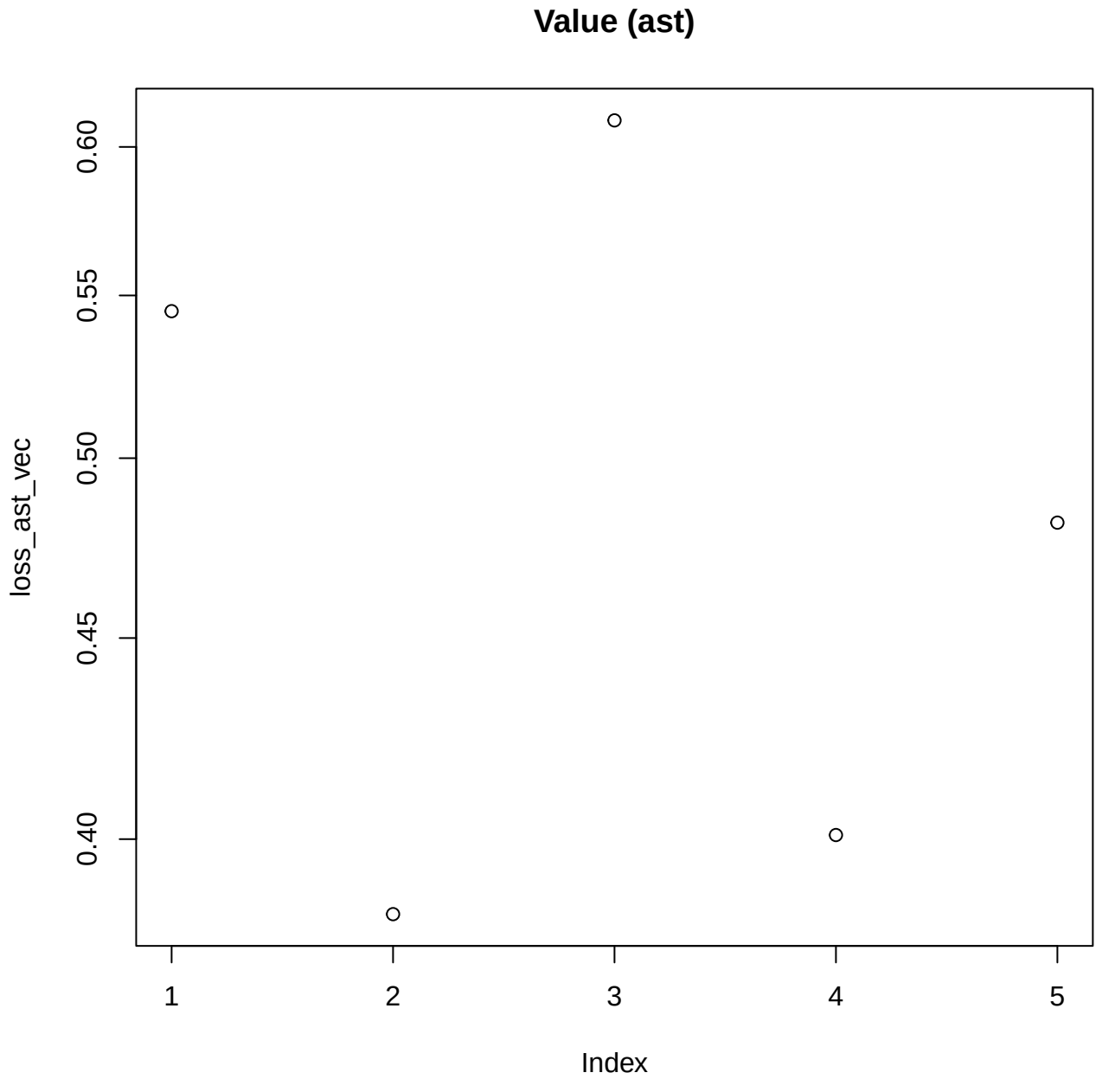
2
3
4
5
Index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loss_ast_vec
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0.50
0.55
0.60
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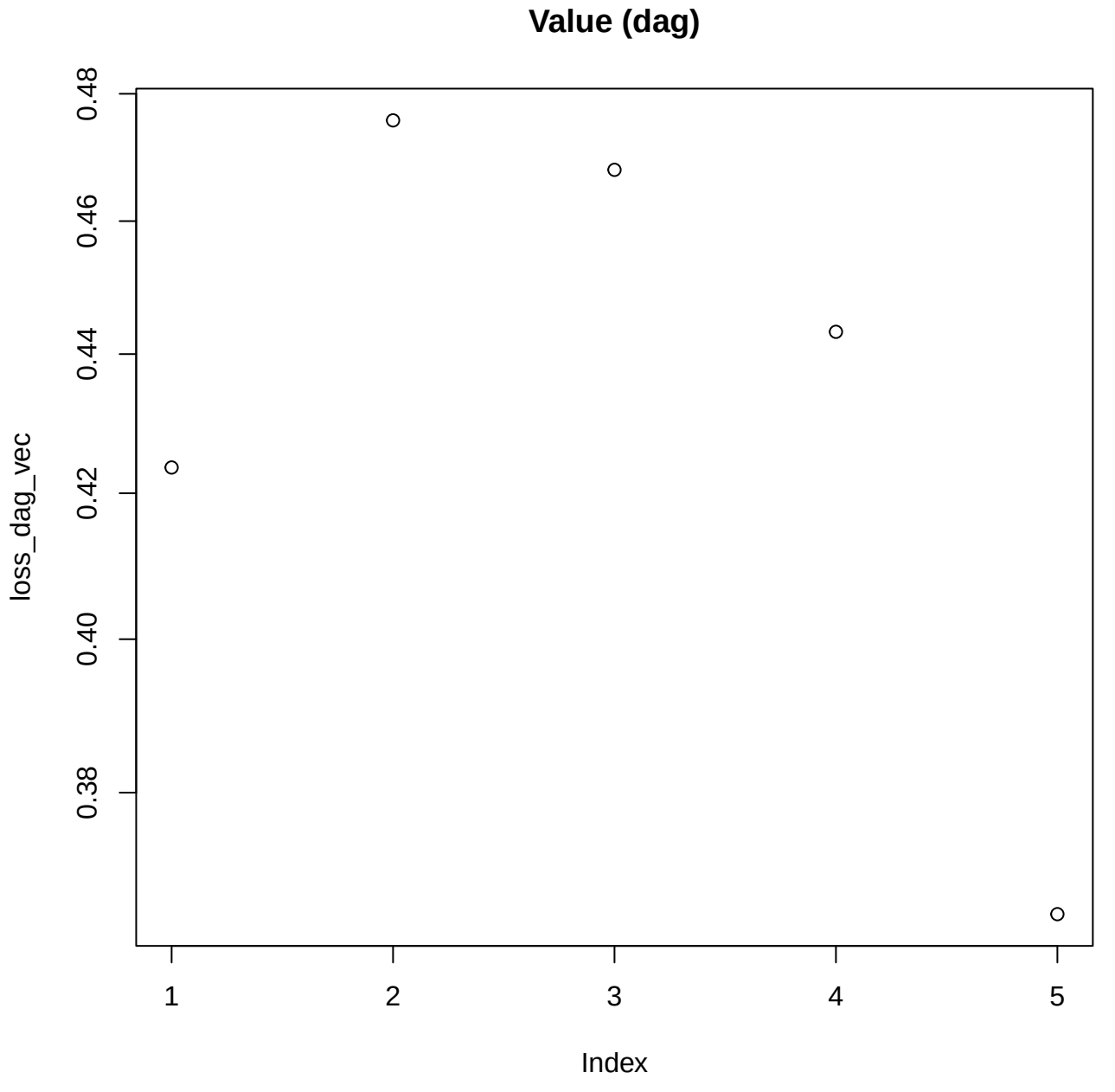
3
4
5
Index


Value (dag)
loss_dag_vec
0.38
0.40
0.42
0.44
0.46
0.48
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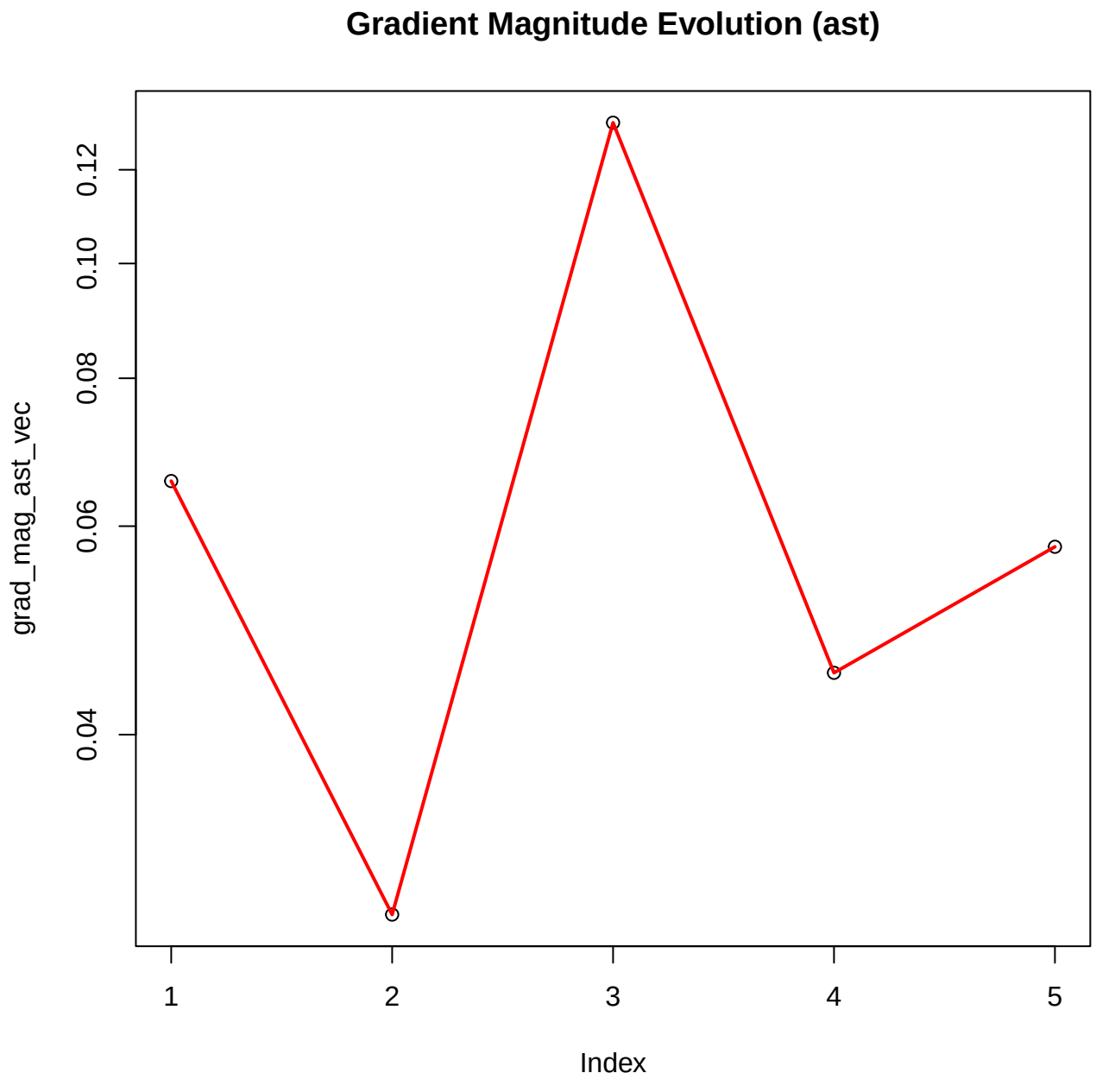
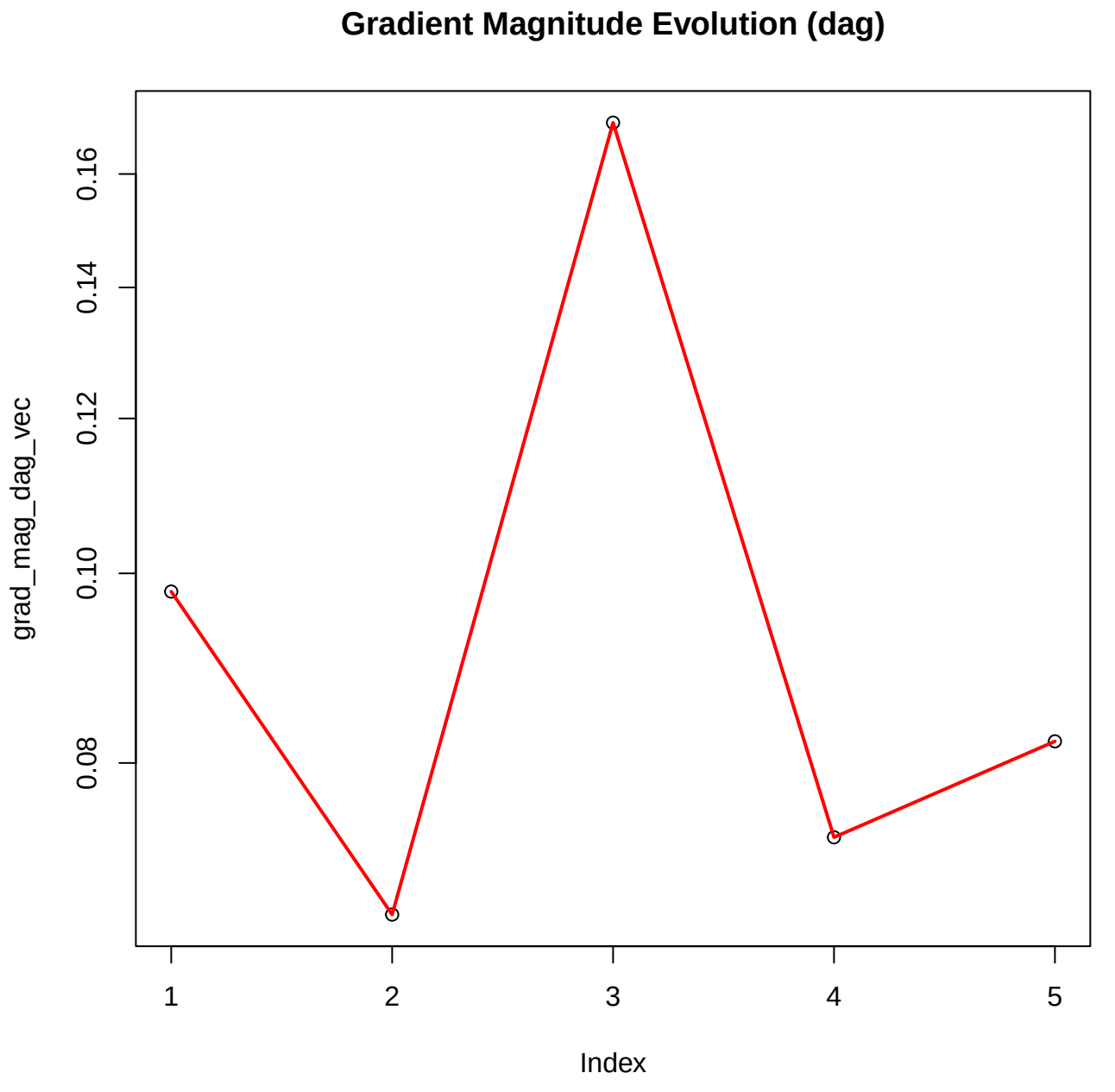
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.04
0.06
0.08
0.10
0.12
1
2
3
4
5
Index


Gradient Magnitude Evolution (dag)
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0.12
0.14
0.16
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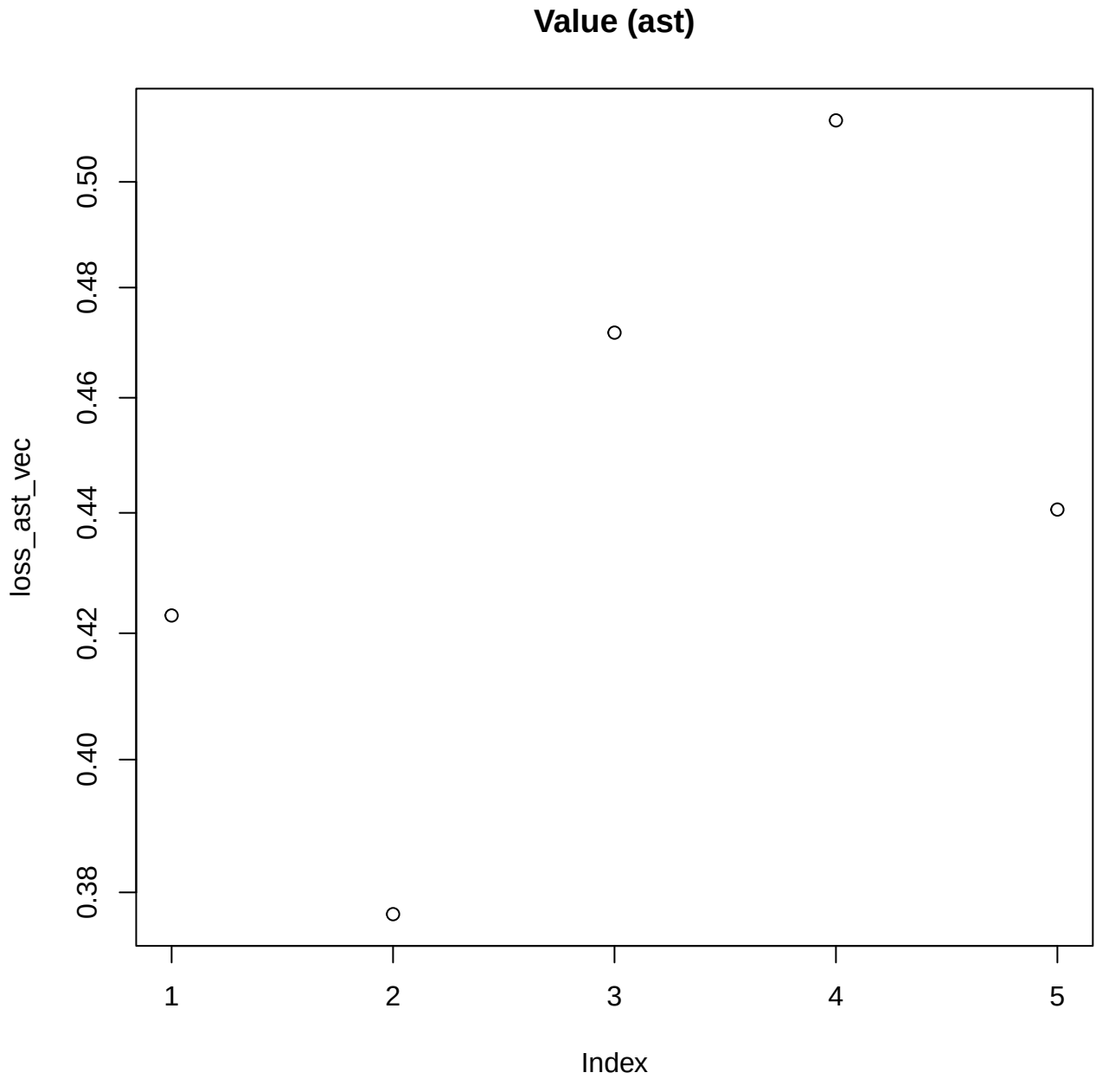
2
3
4
5
Index


Value (ast)
loss_ast_vec
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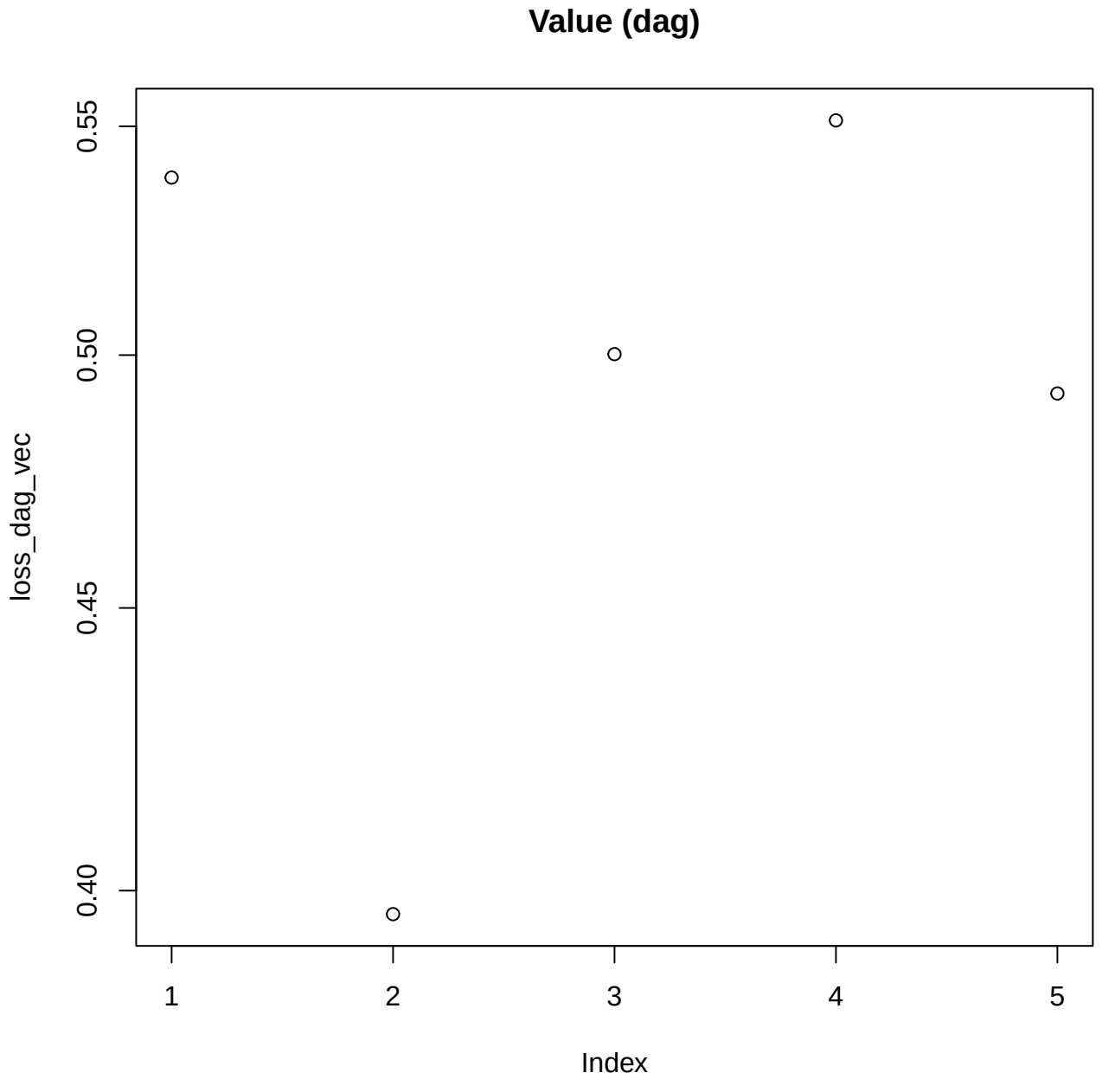
0.40
0.42
0.44
0.46
0.48
0.50
1
2
3
4
5
Index


Value (dag)
loss_dag_vec
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0.45
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0.55
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2
3
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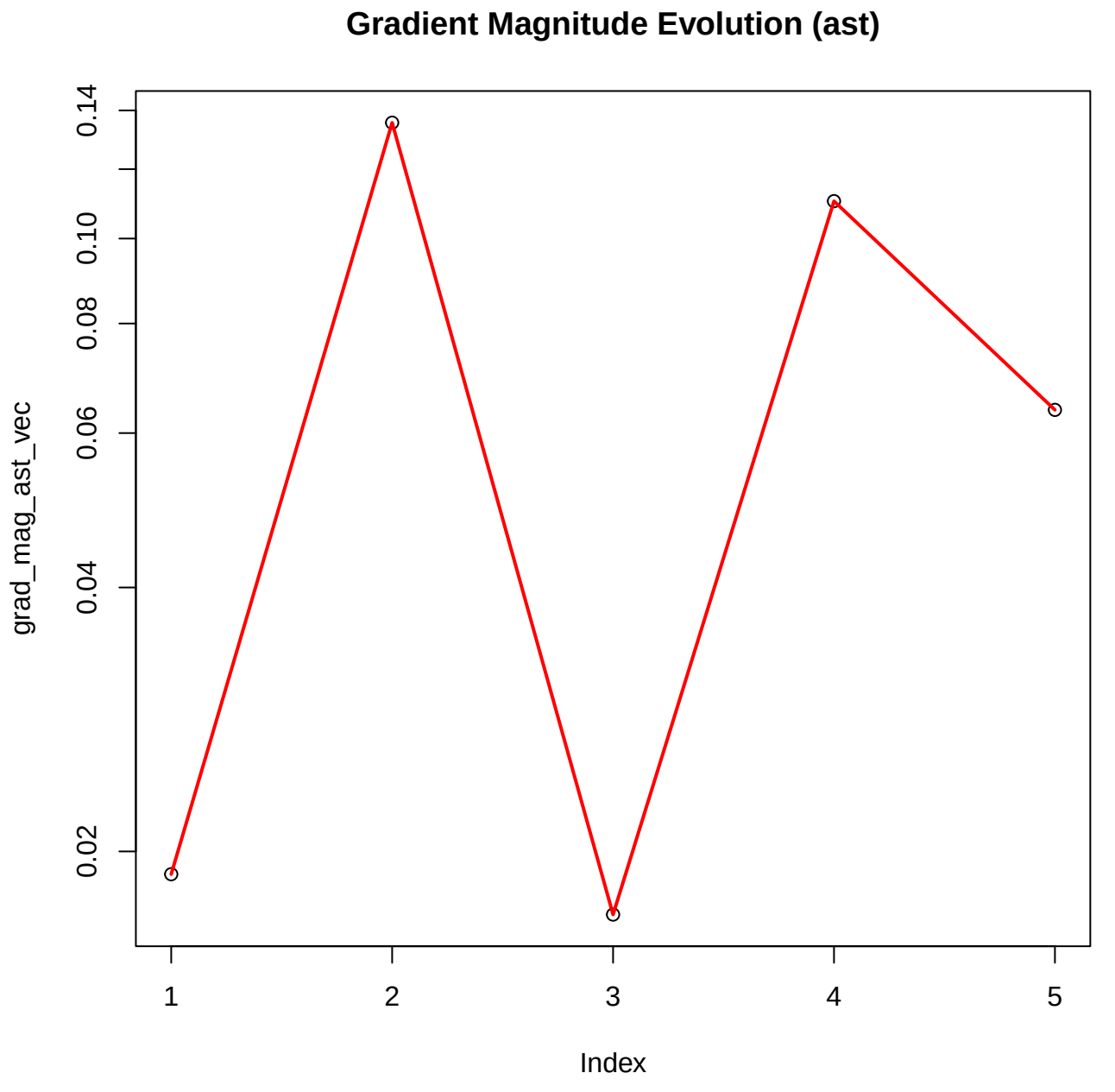
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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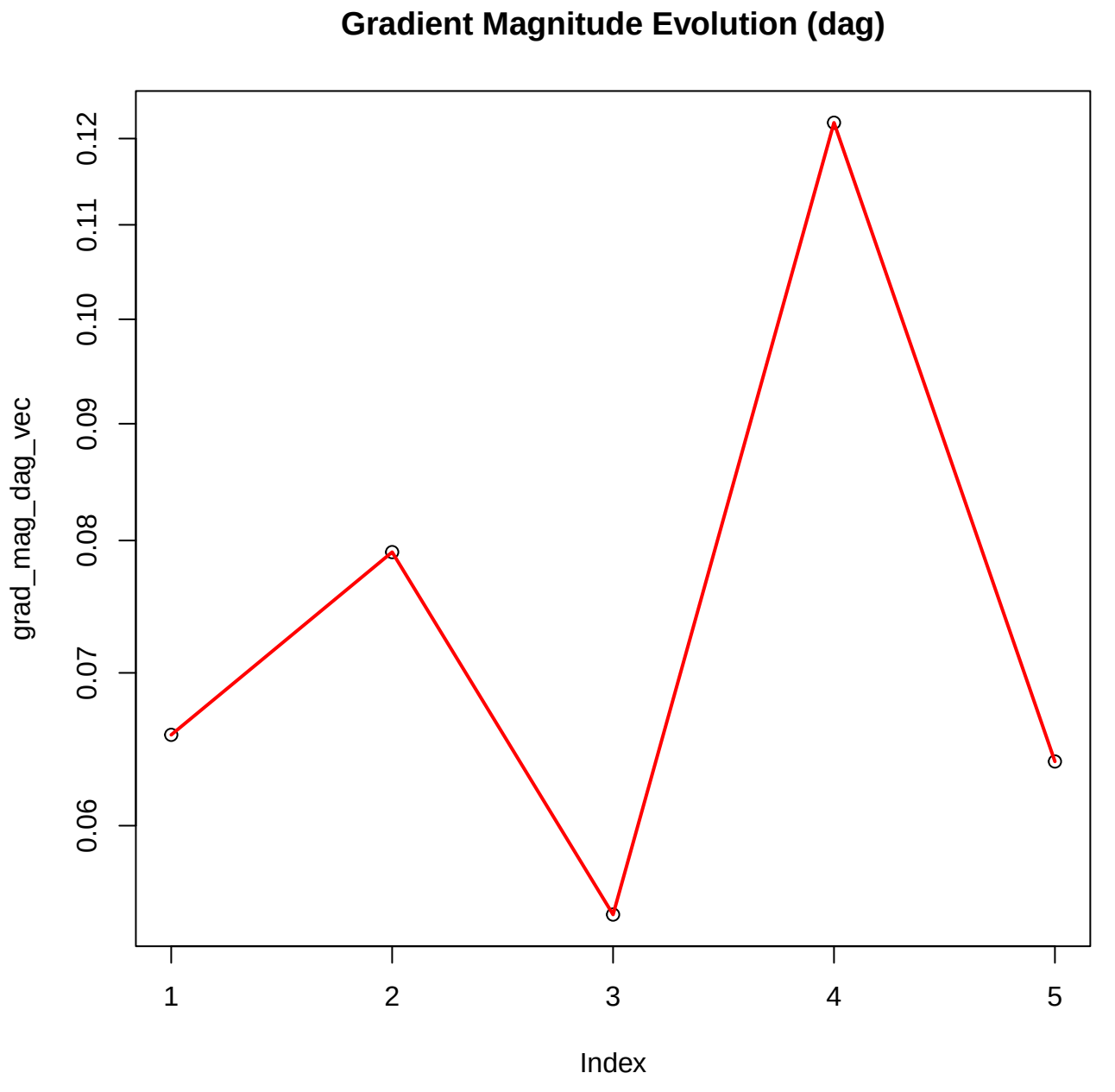
0.06
0.08
0.10
0.14
1
2
3
4
5
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.09
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0.12
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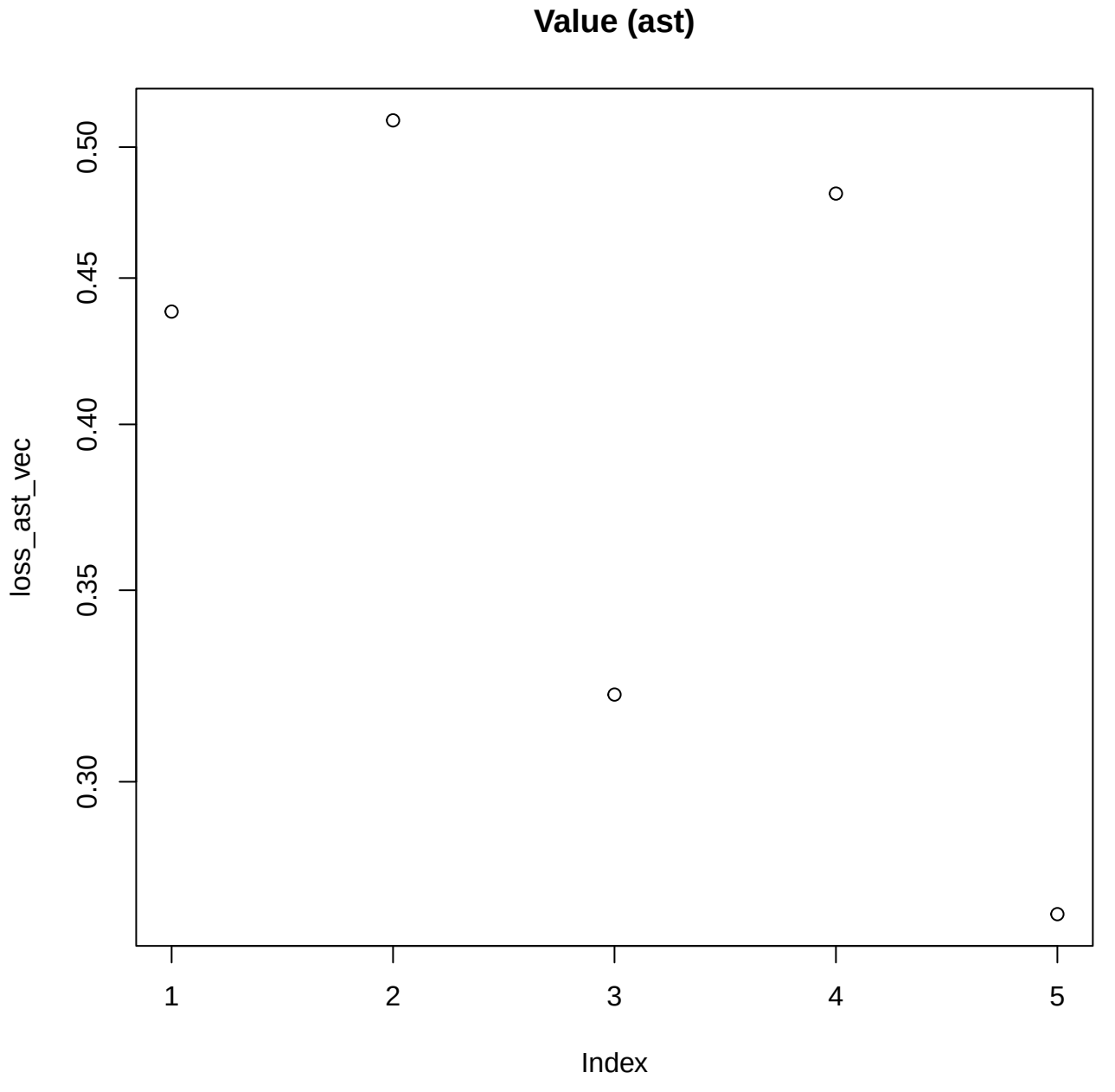
3
4
5
Index


Value (ast)
loss_ast_vec
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0.50
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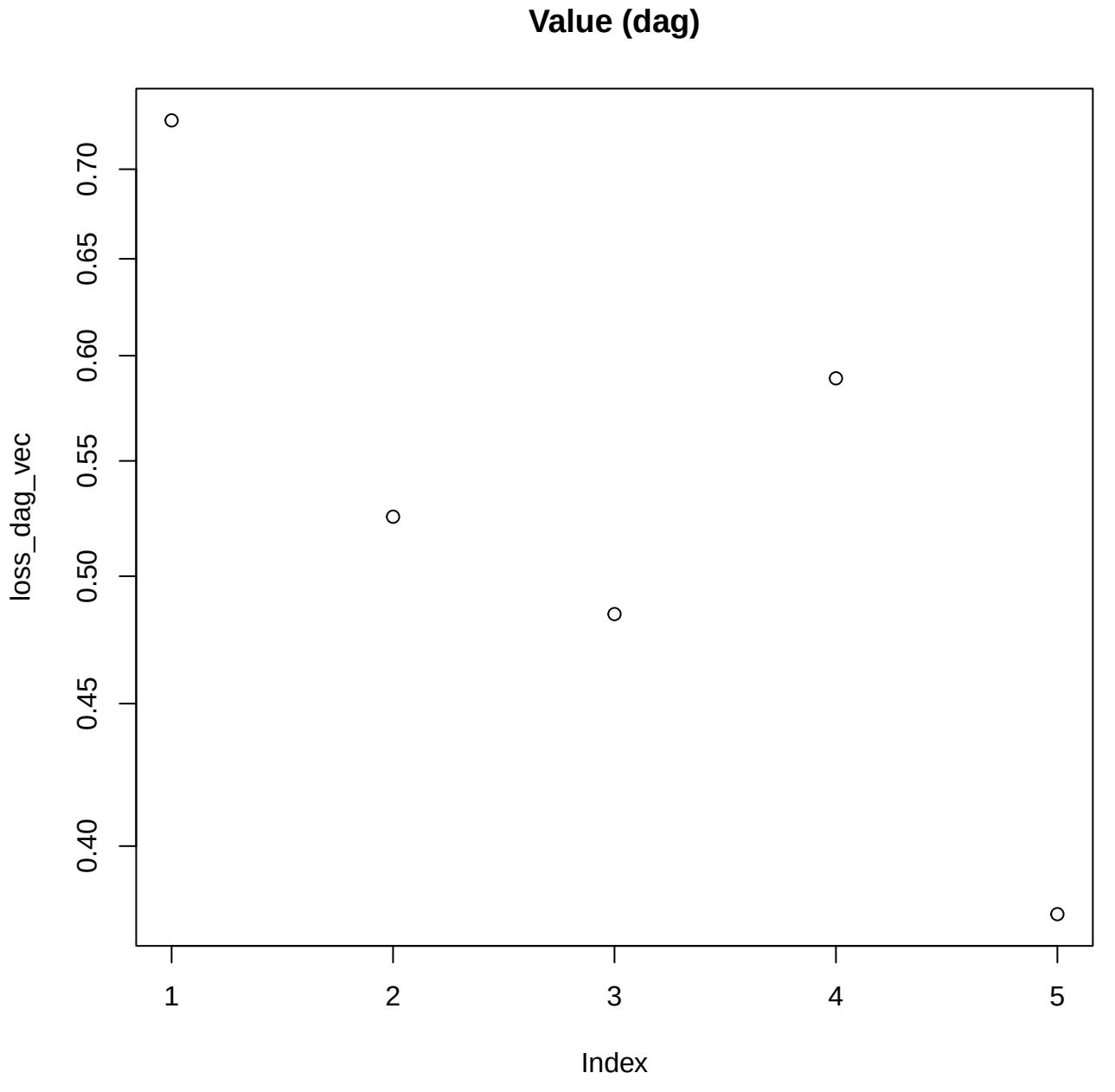
2
3
4
5
Index


Value (dag)
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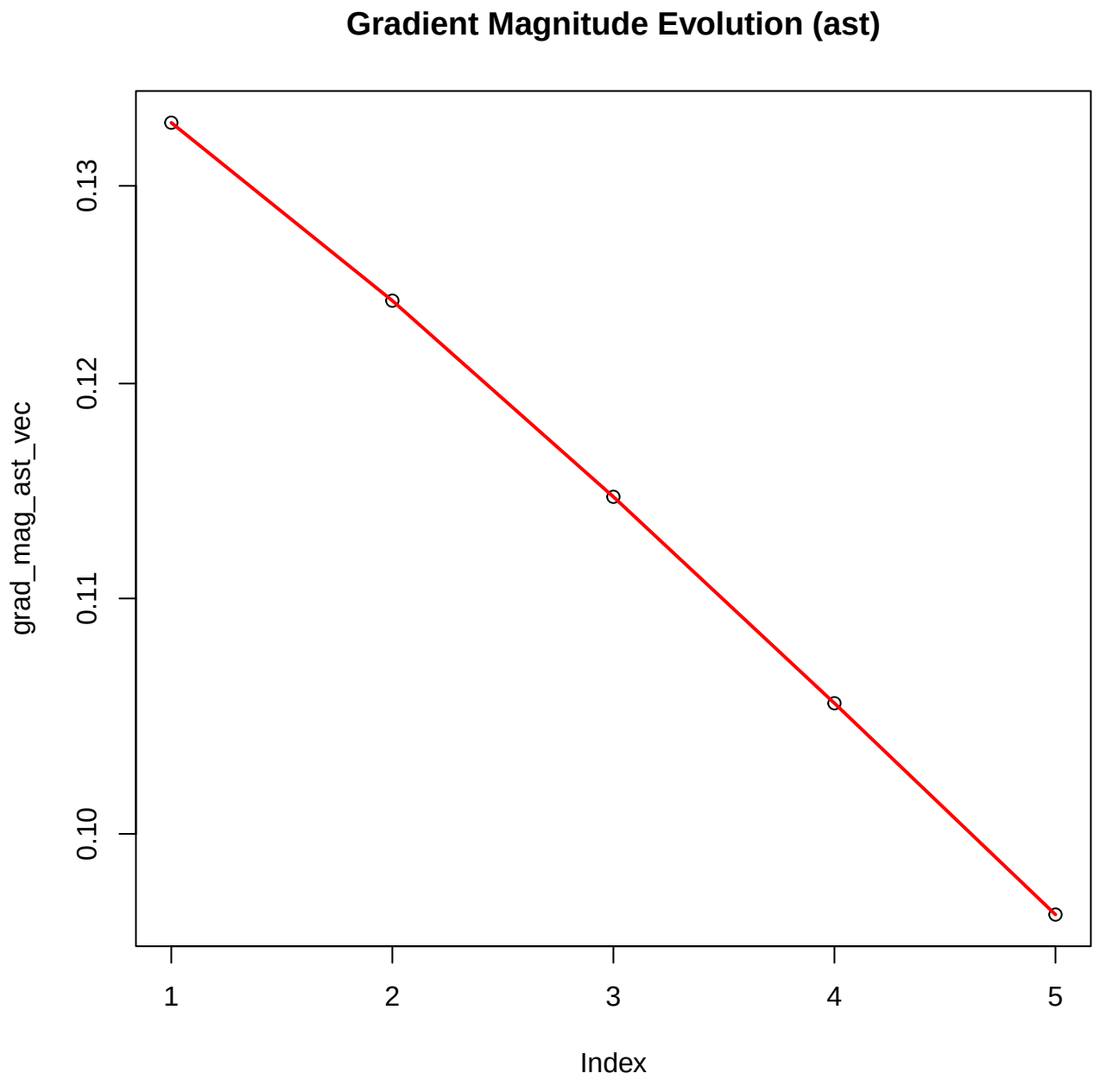
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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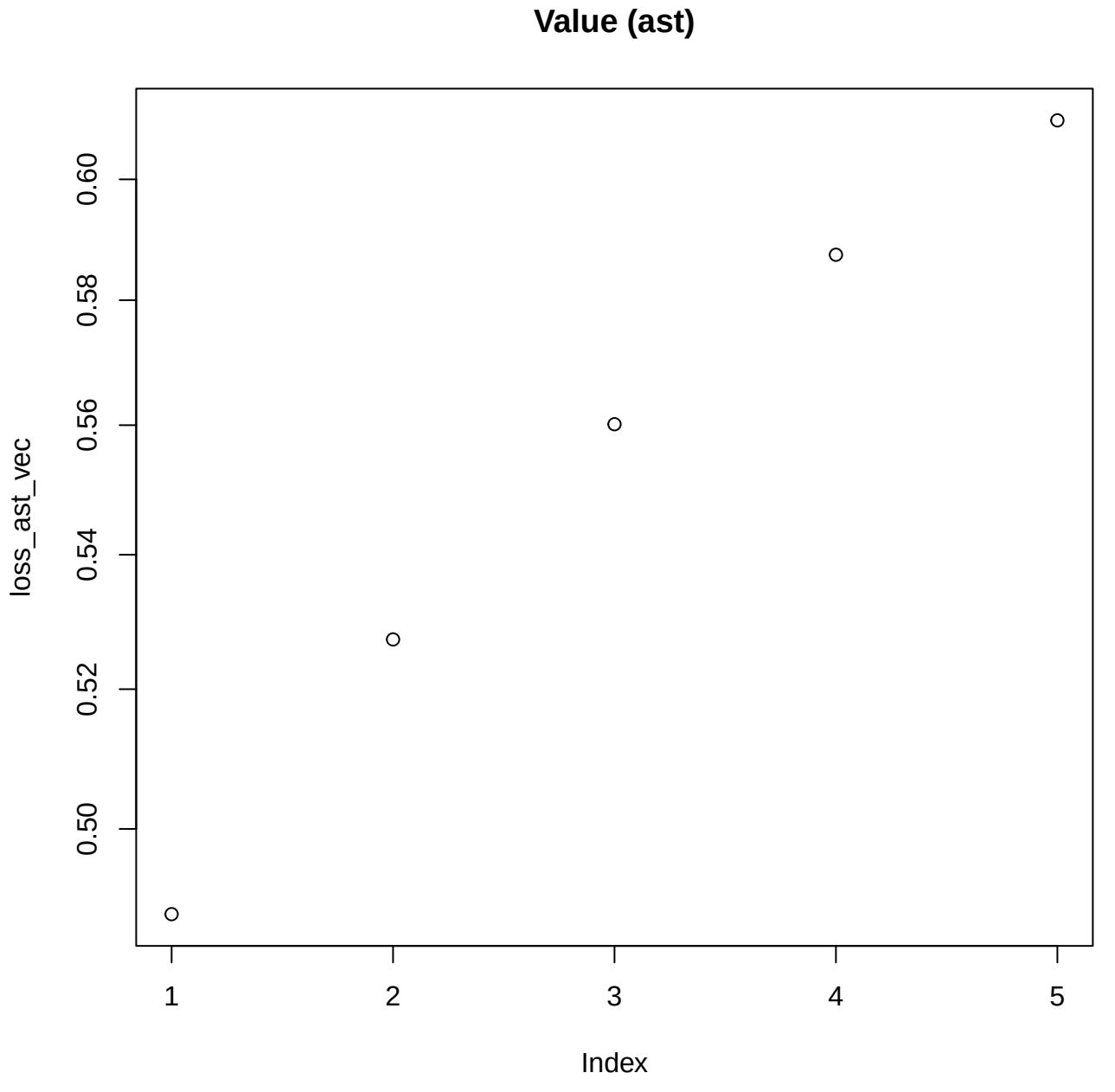
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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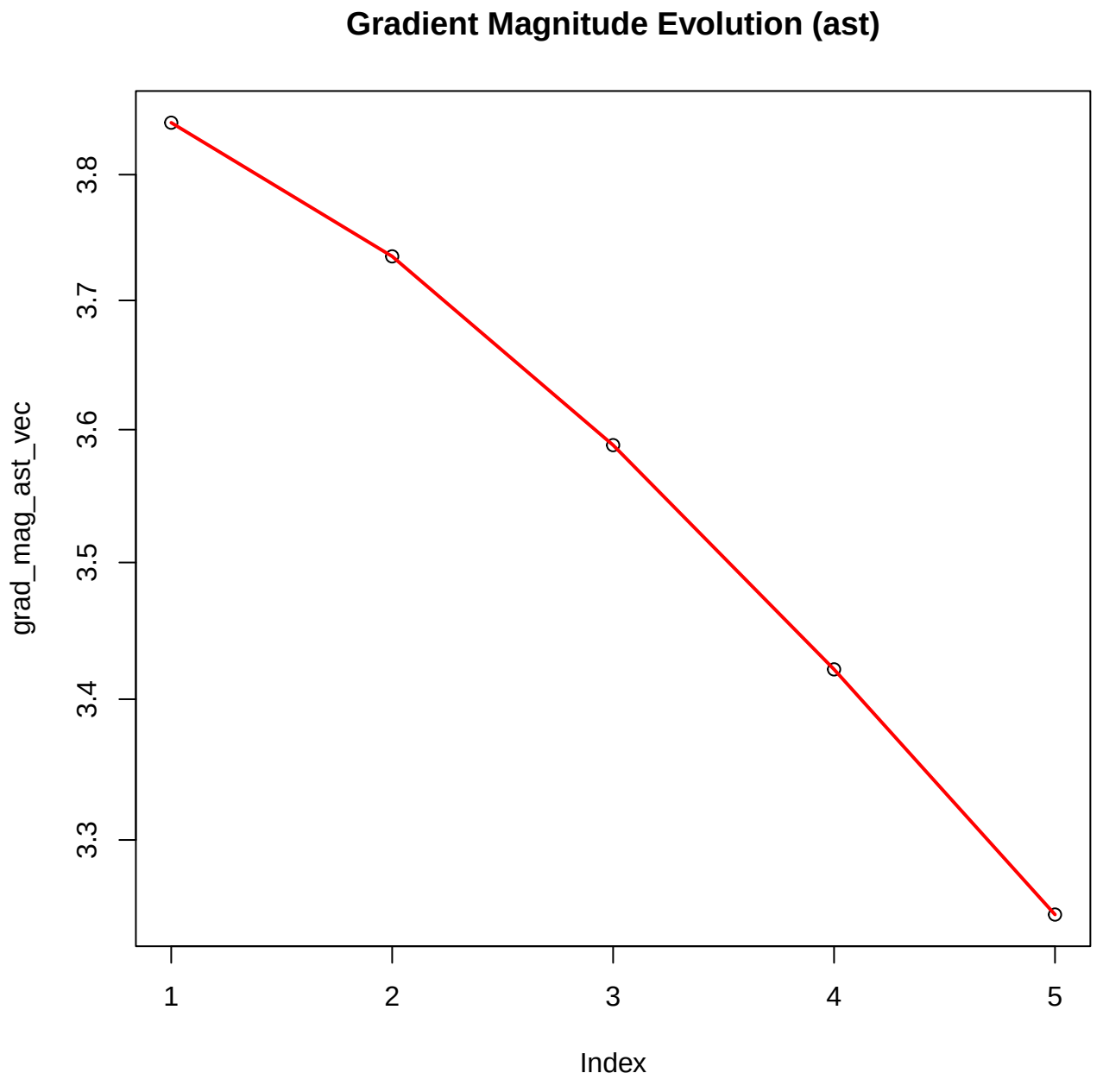
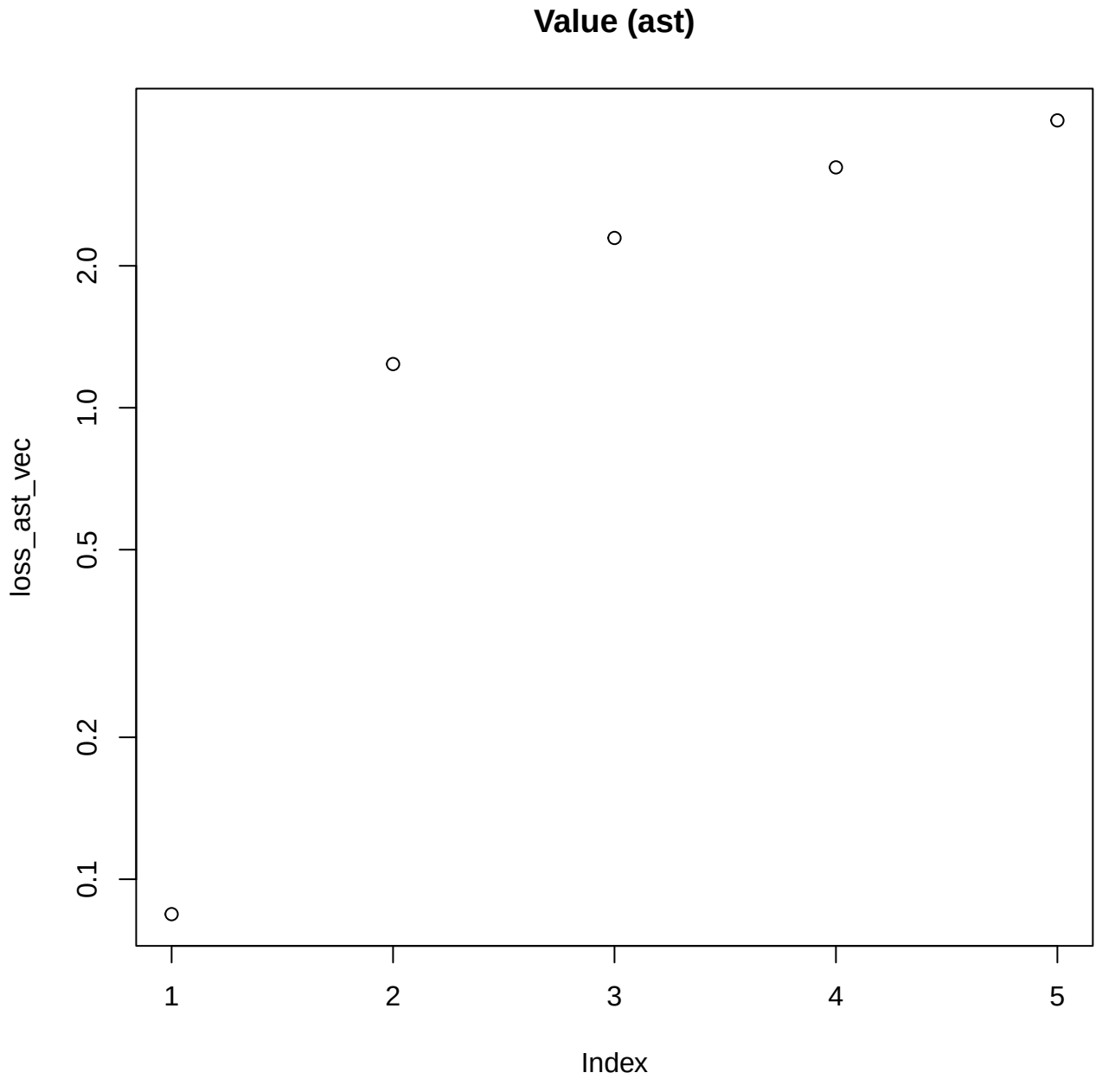
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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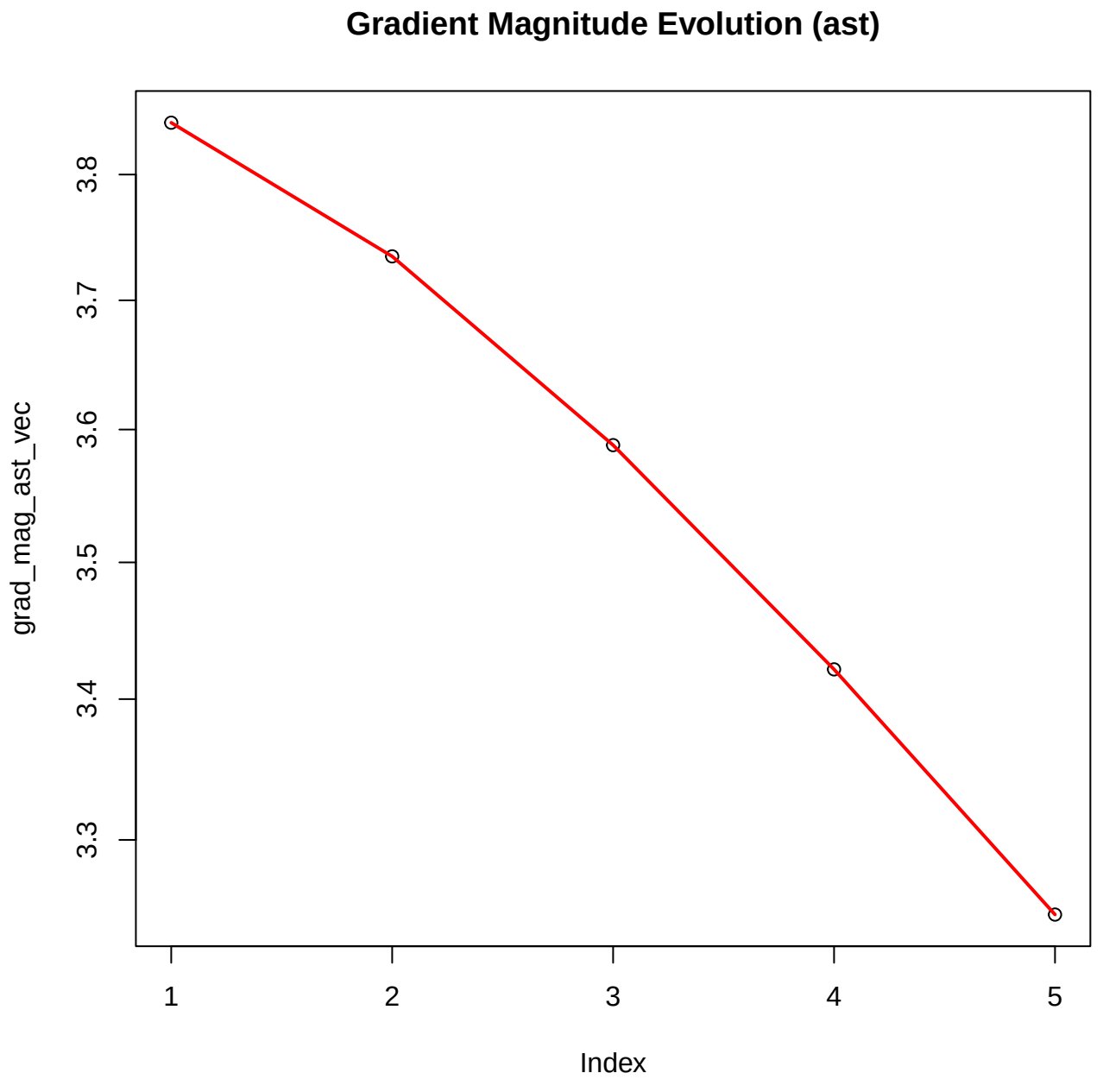
0.2
0.5
1.0
2.0
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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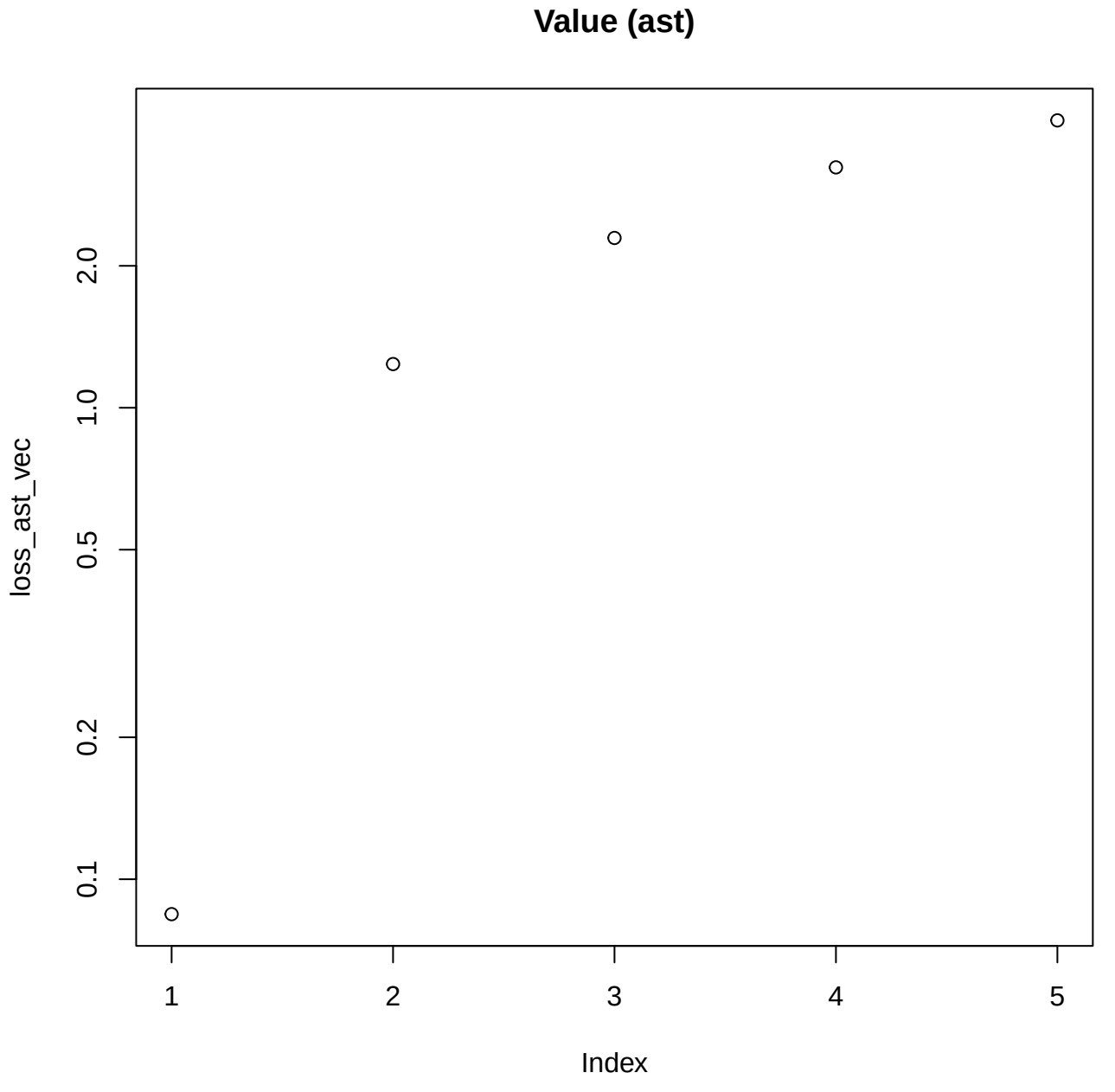
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.5
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2.0
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2
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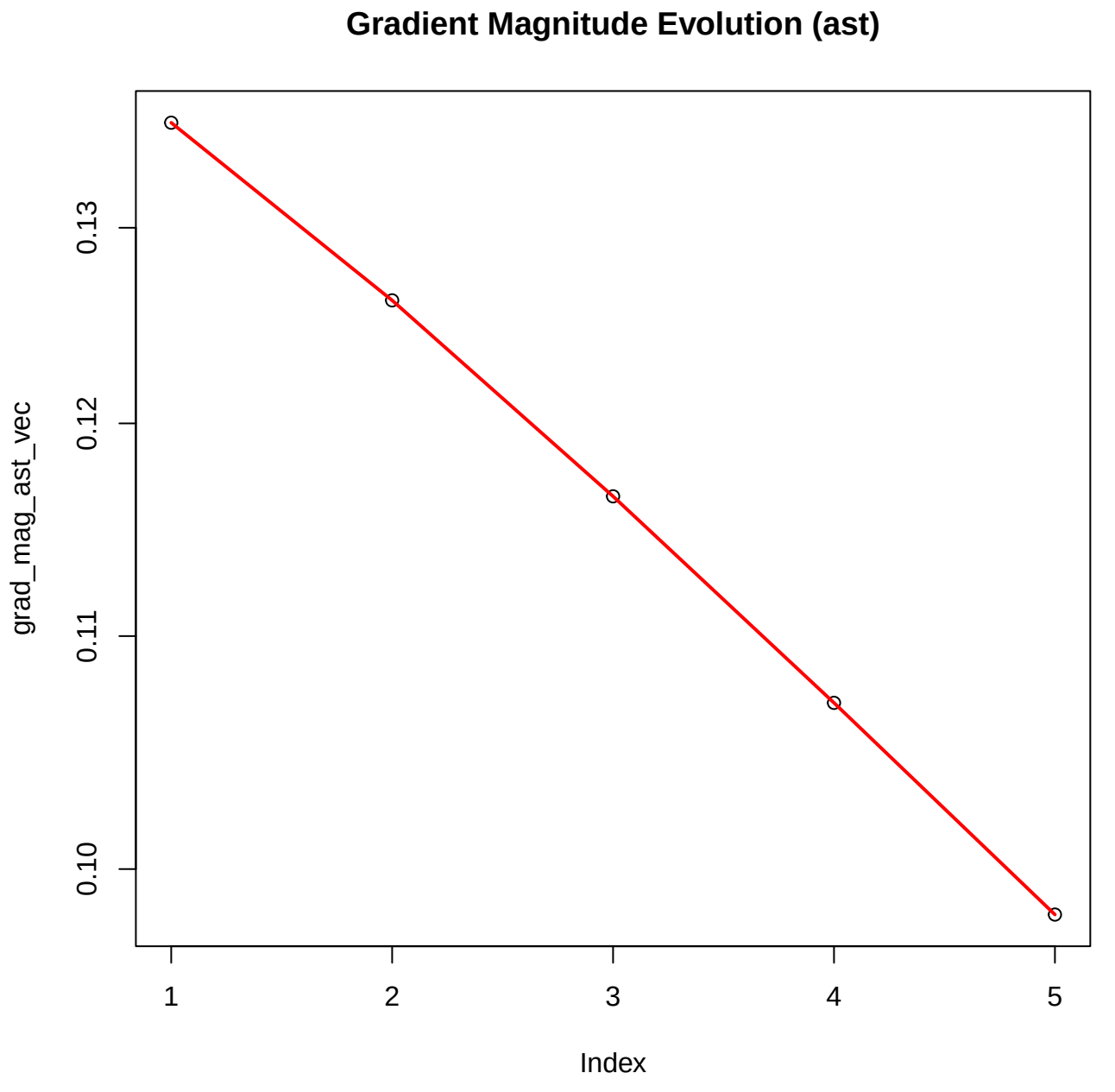
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.12
0.13
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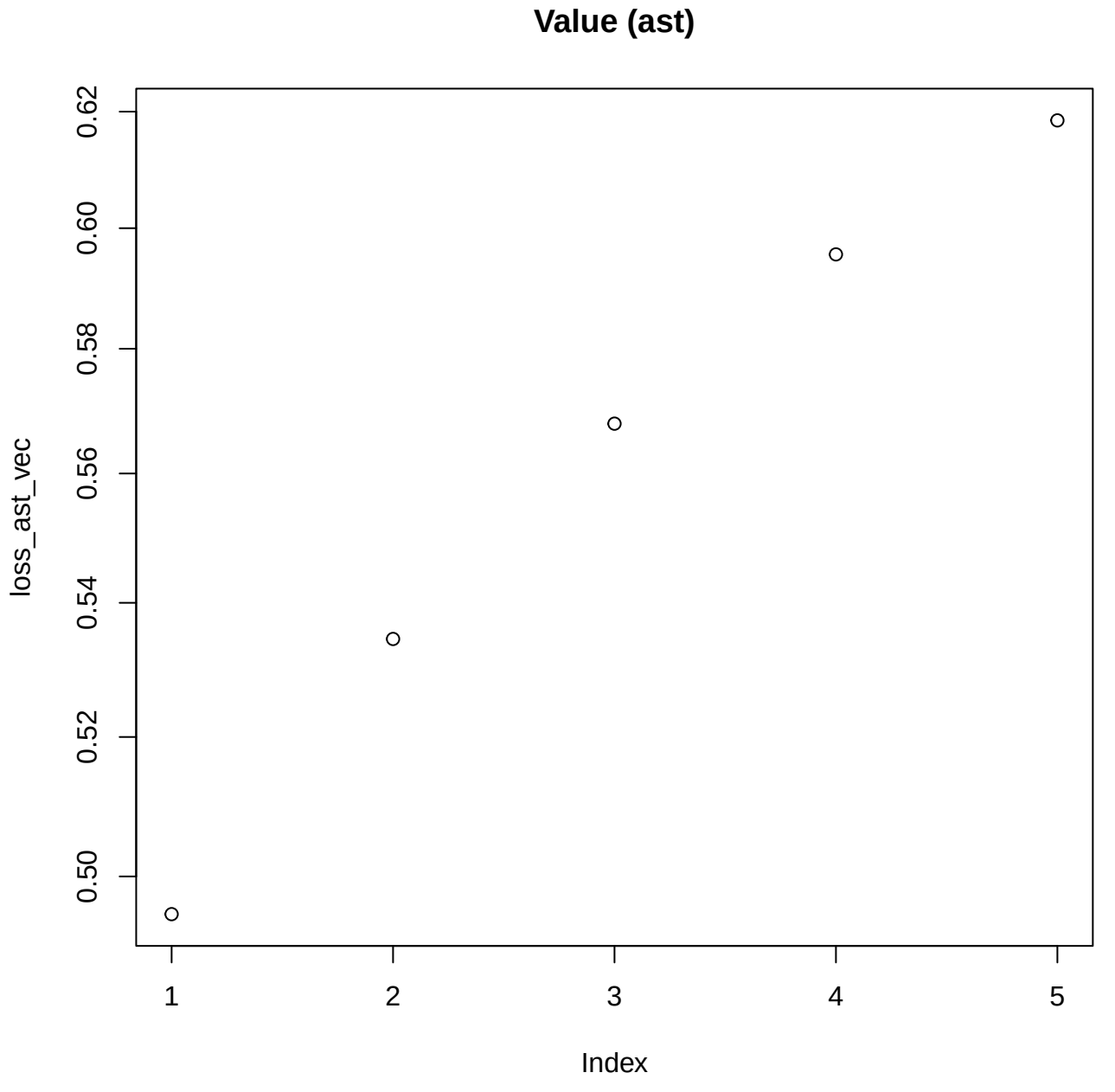
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.62
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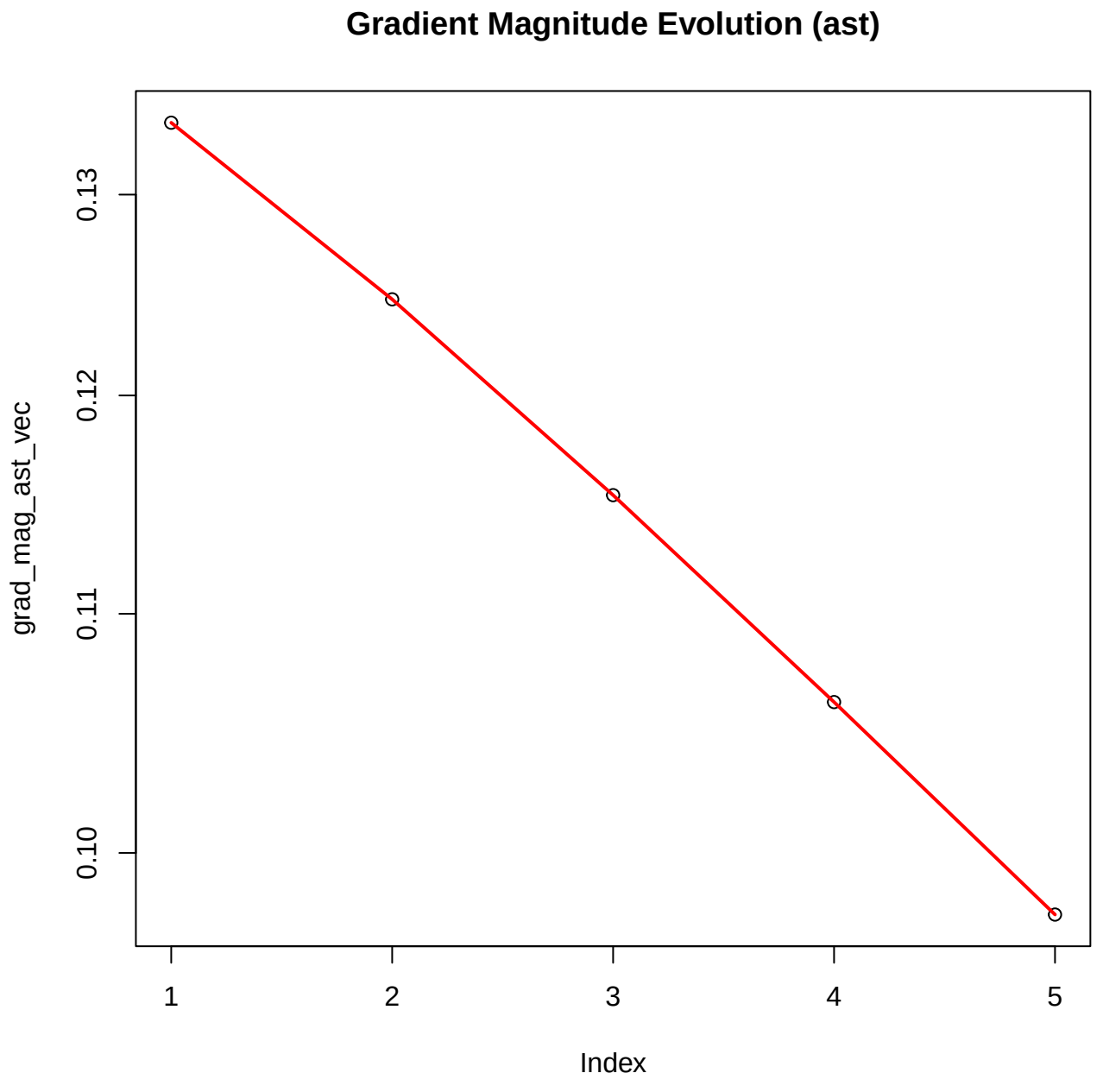
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.11
0.12
0.13
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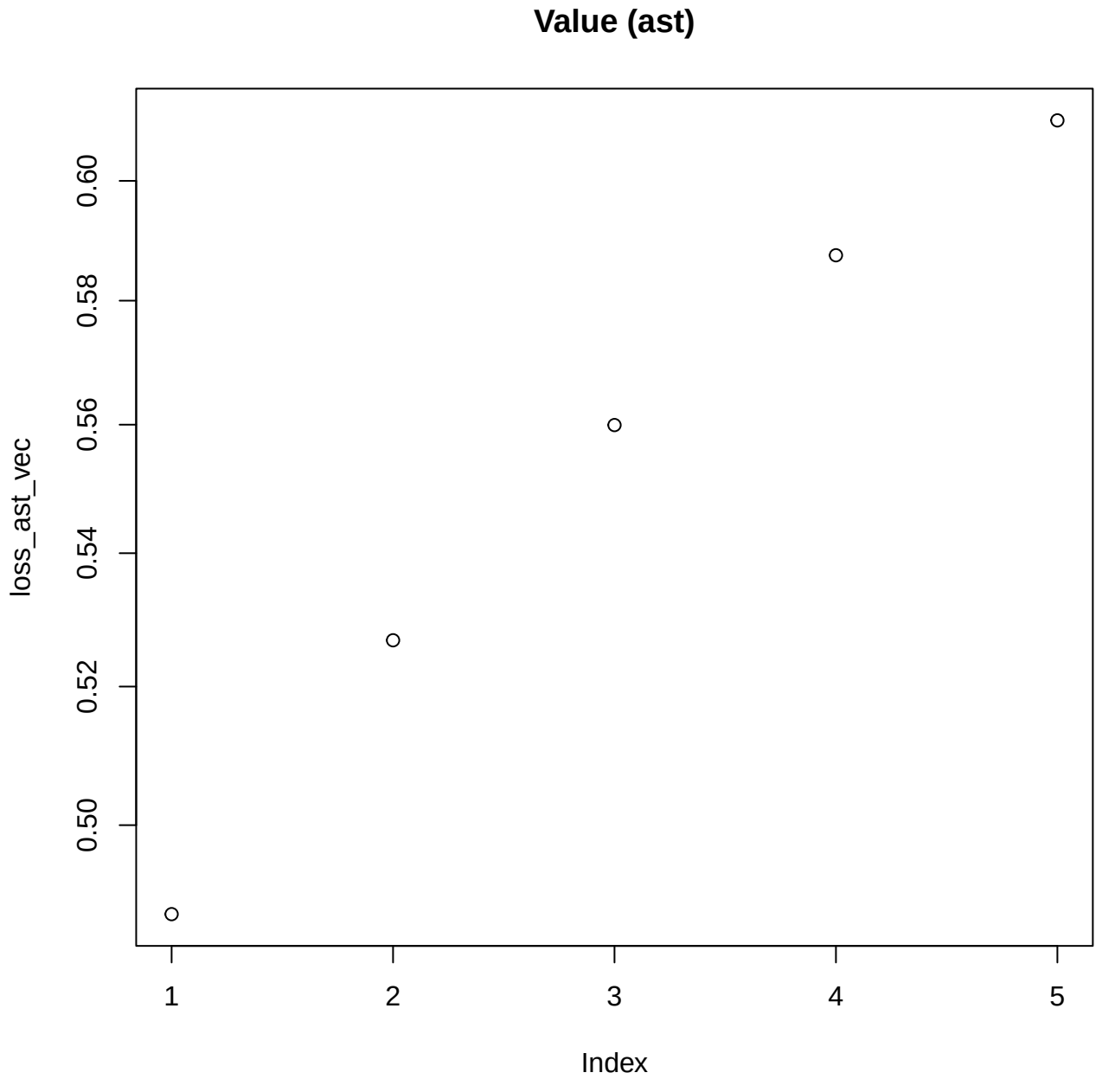
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.54
0.56
0.58
0.60
1
2
3
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5
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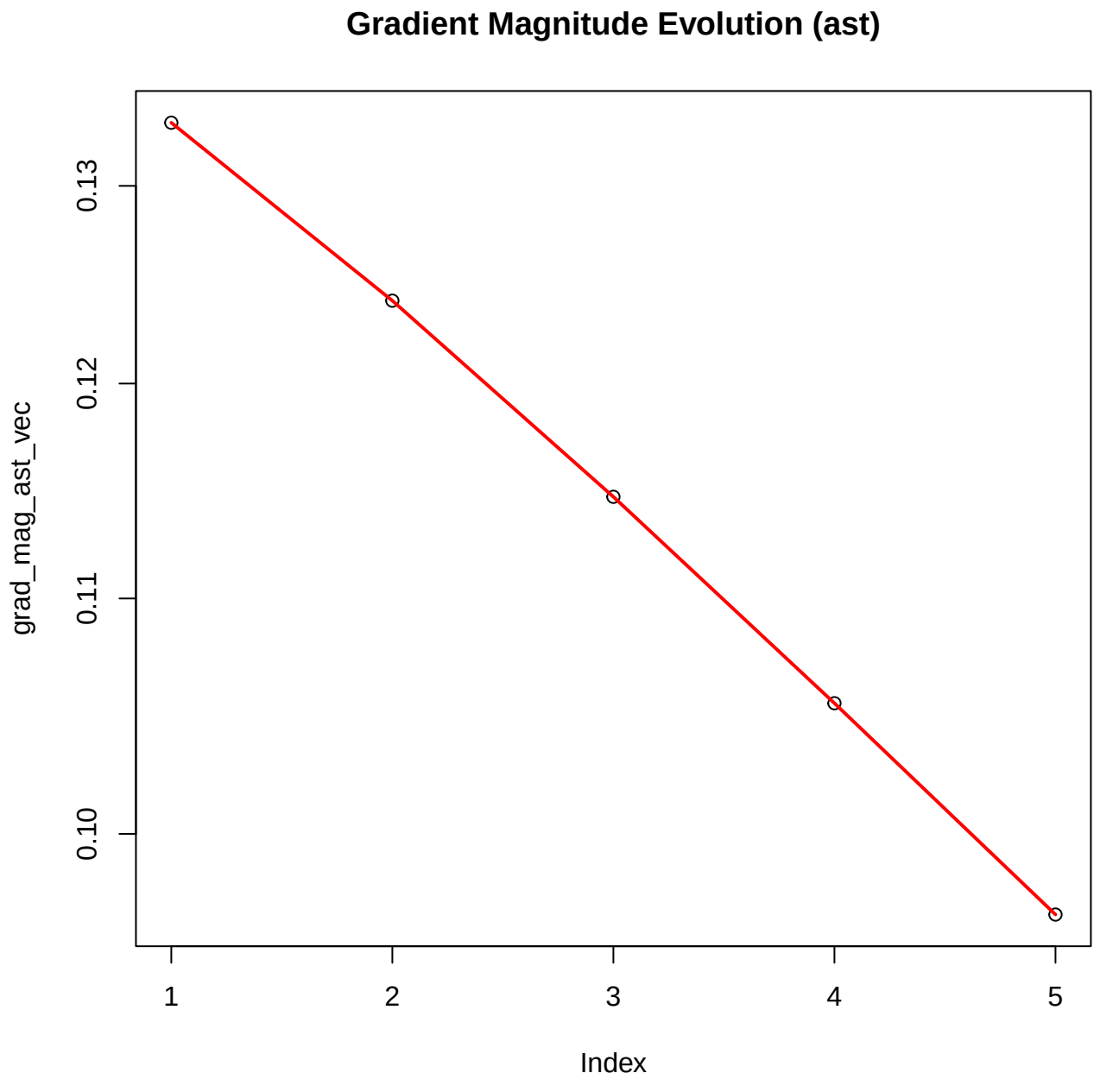
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.11
0.12
0.13
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2
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4
5
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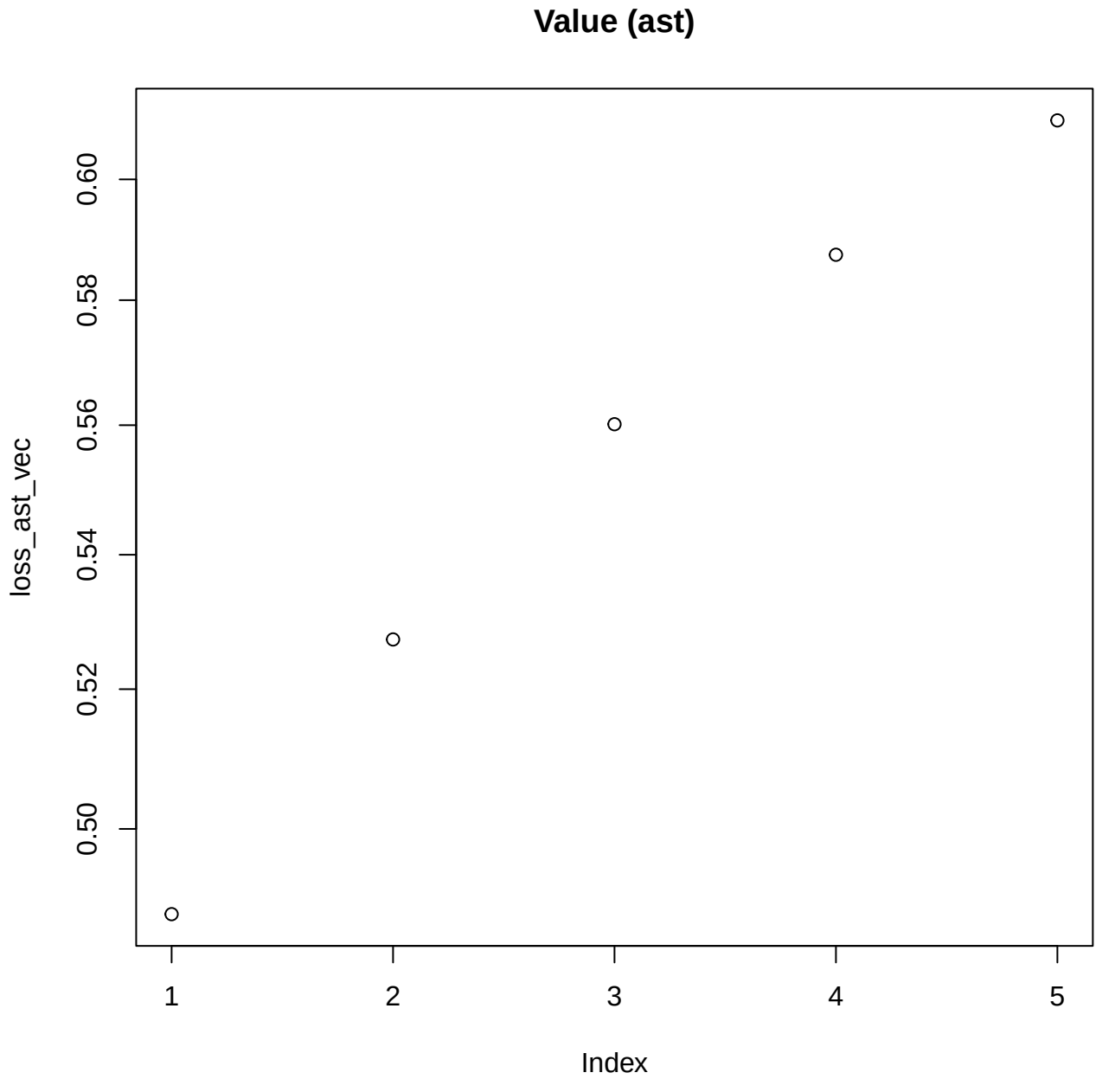
Value (ast)
loss_ast_vec
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0.54
0.56
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0.60
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3
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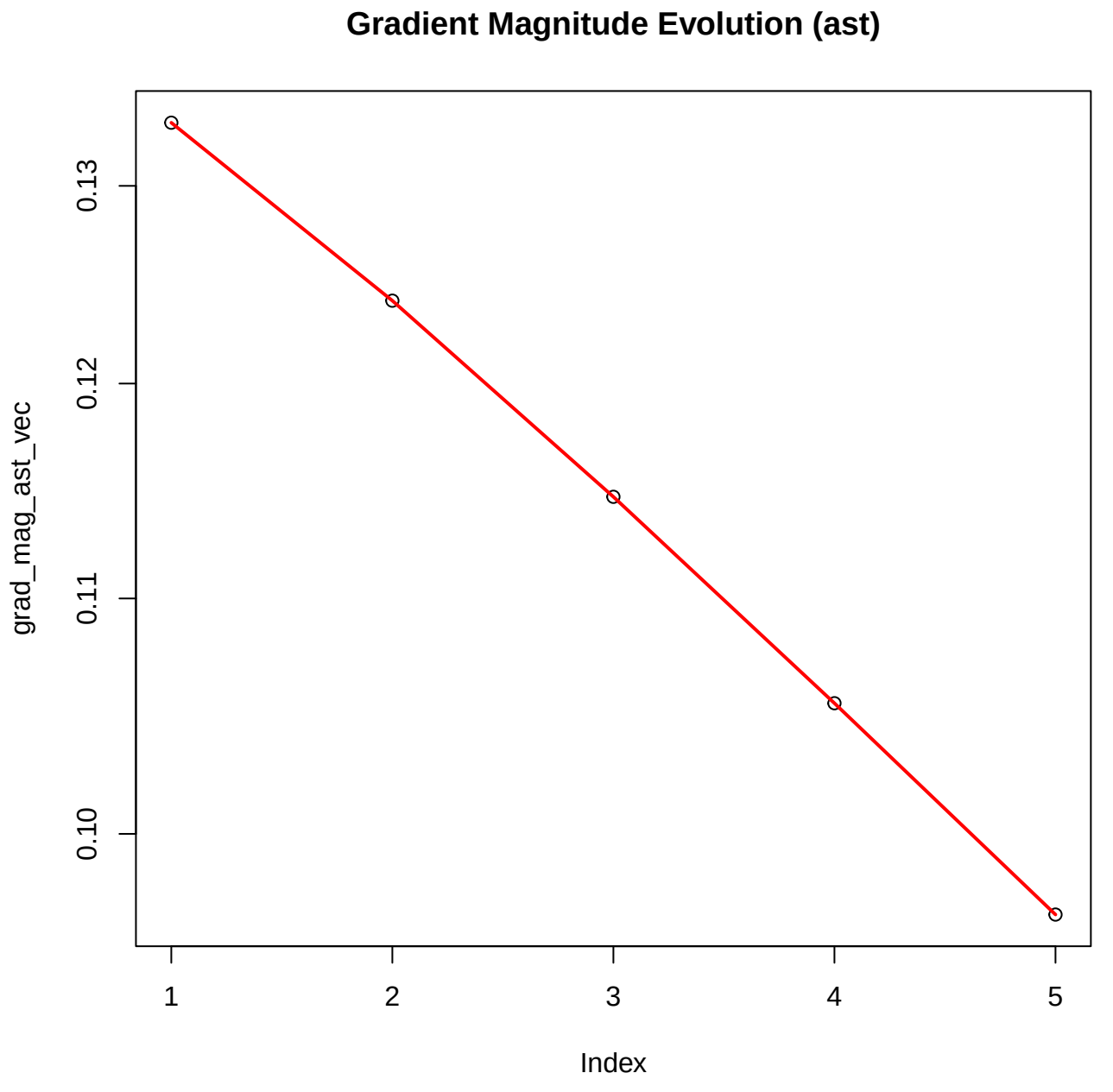
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
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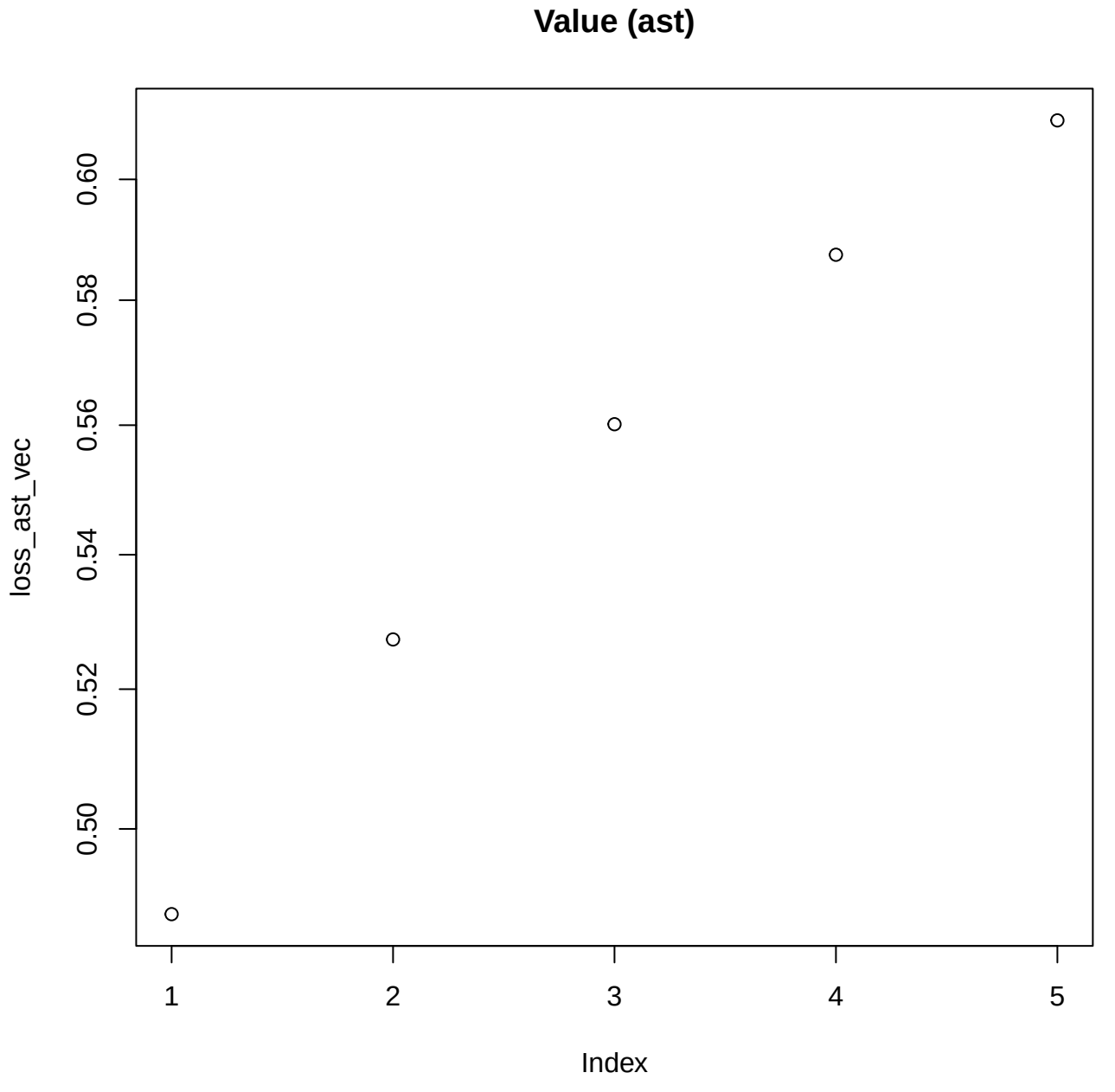
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.54
0.56
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0.60
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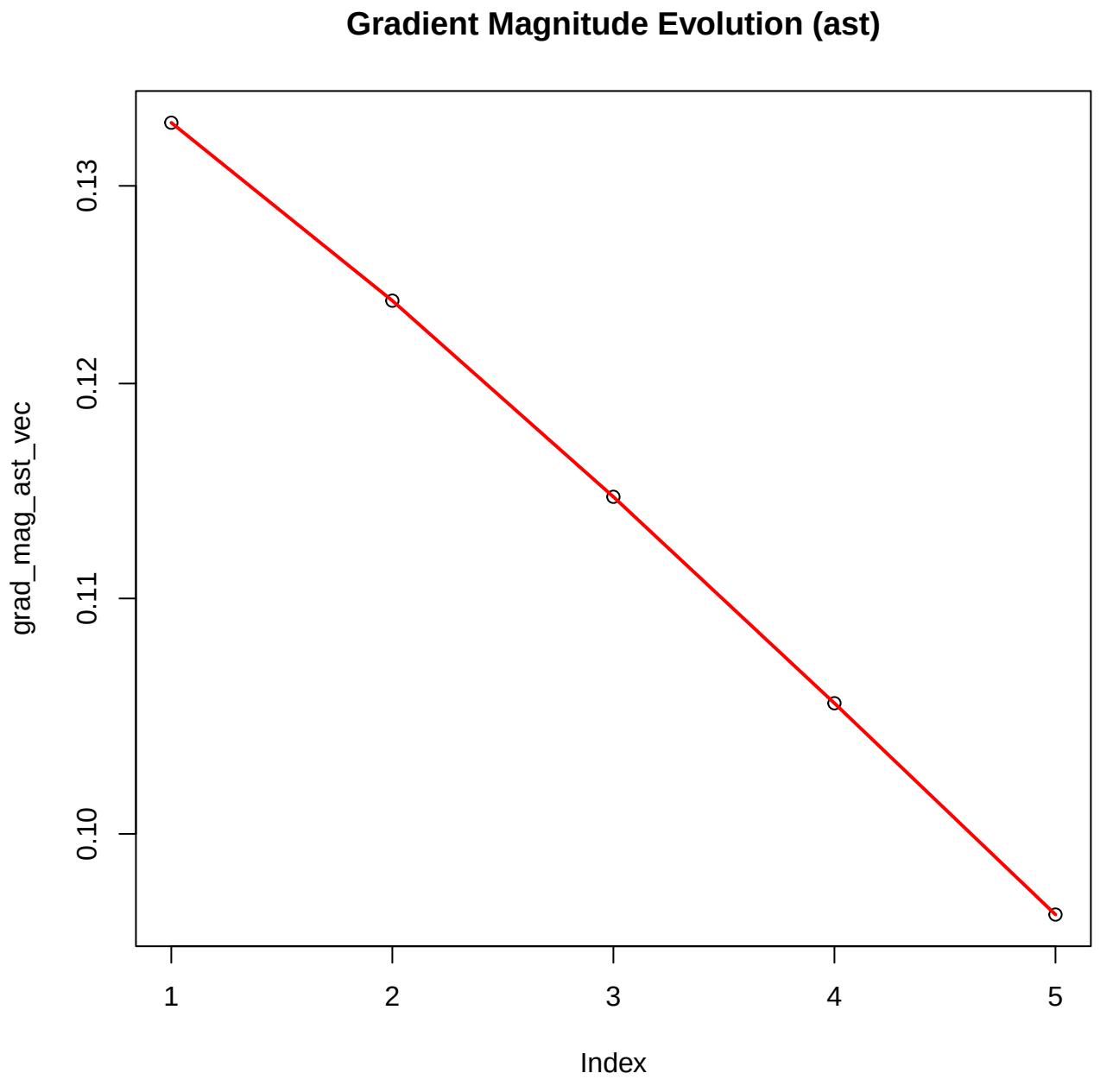
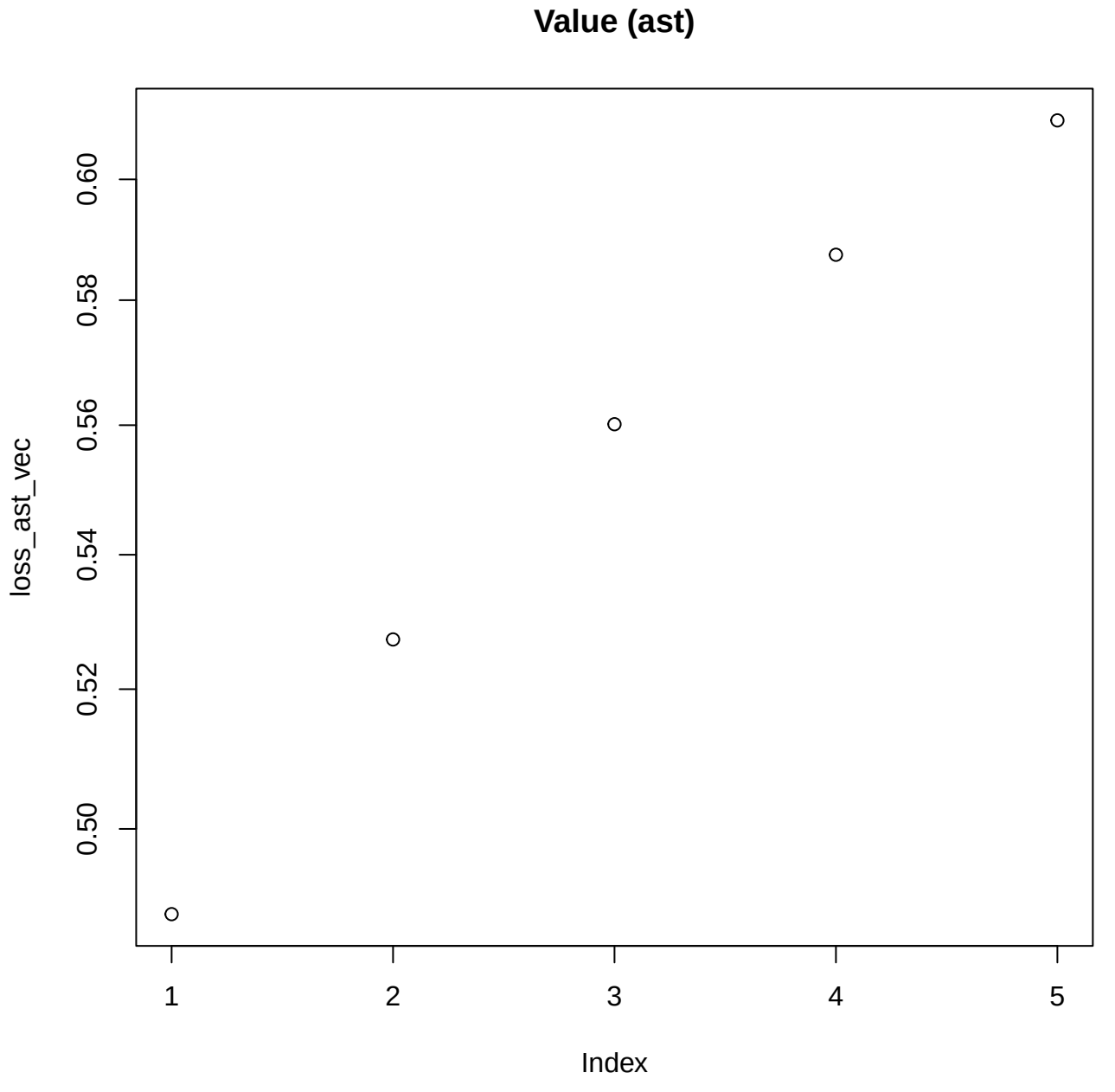

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
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0.60
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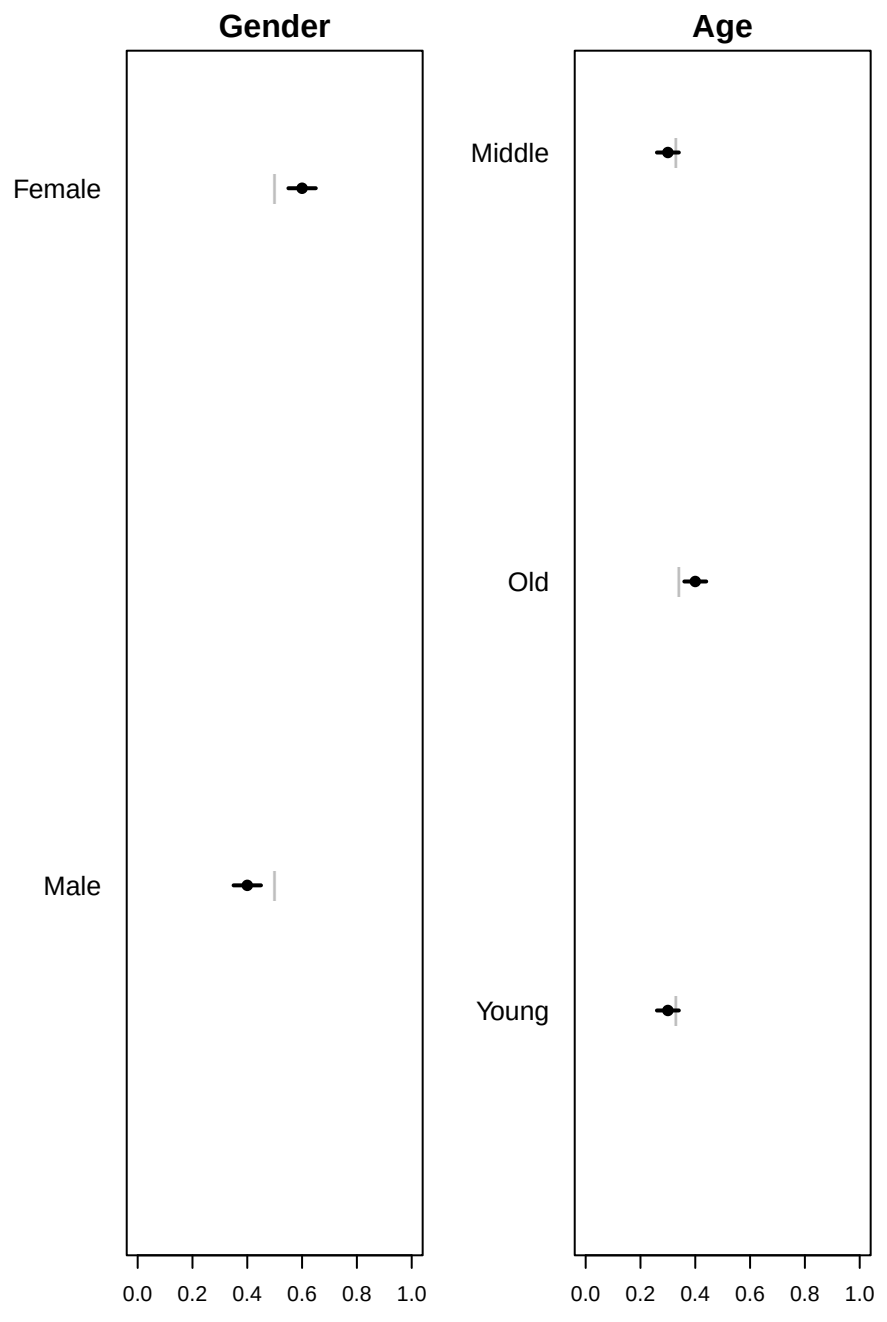
3
4
5
Index


Gender
Female
Male
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0.2
0.4
0.6
0.8
1.0
Age
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Old
Young
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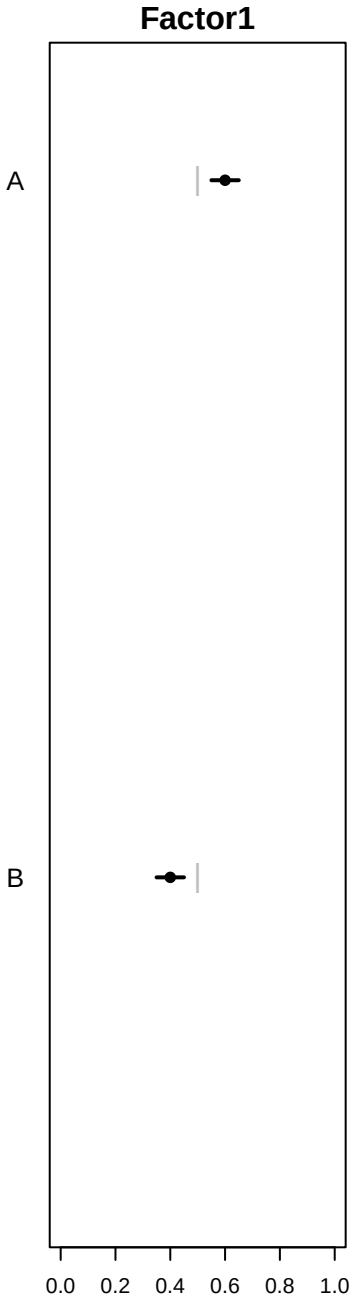
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
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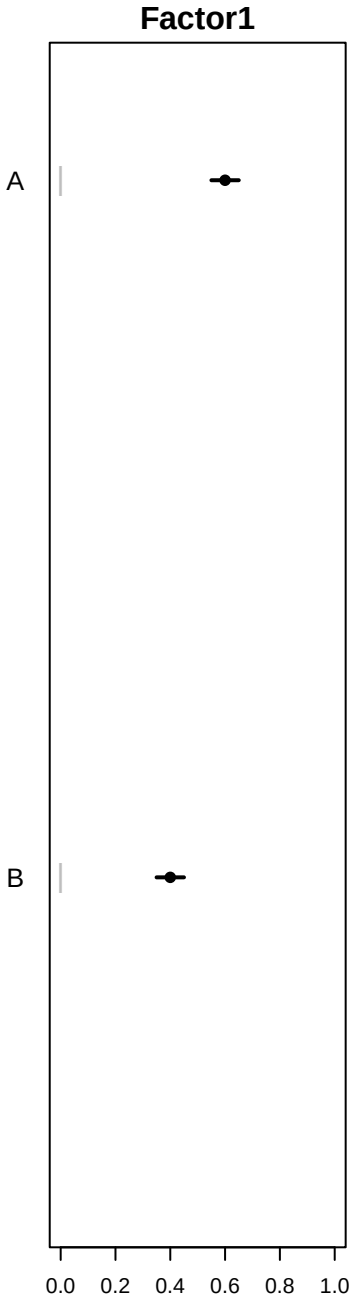
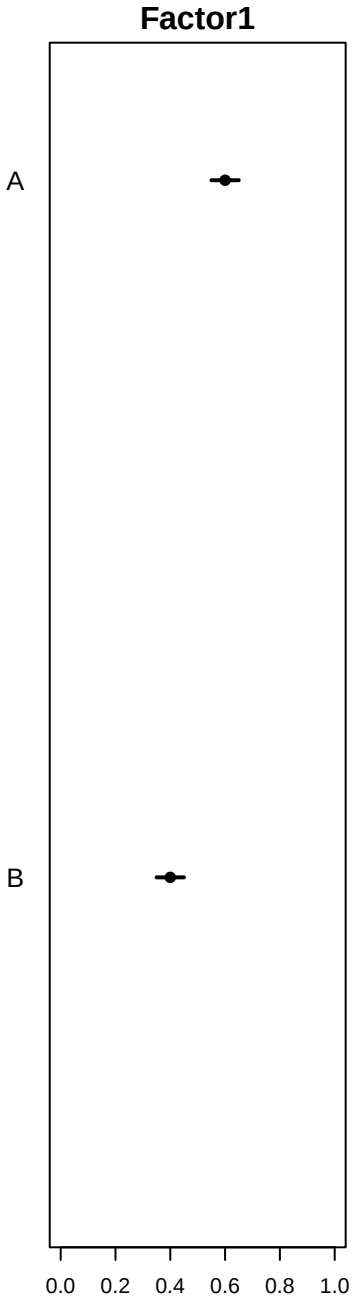
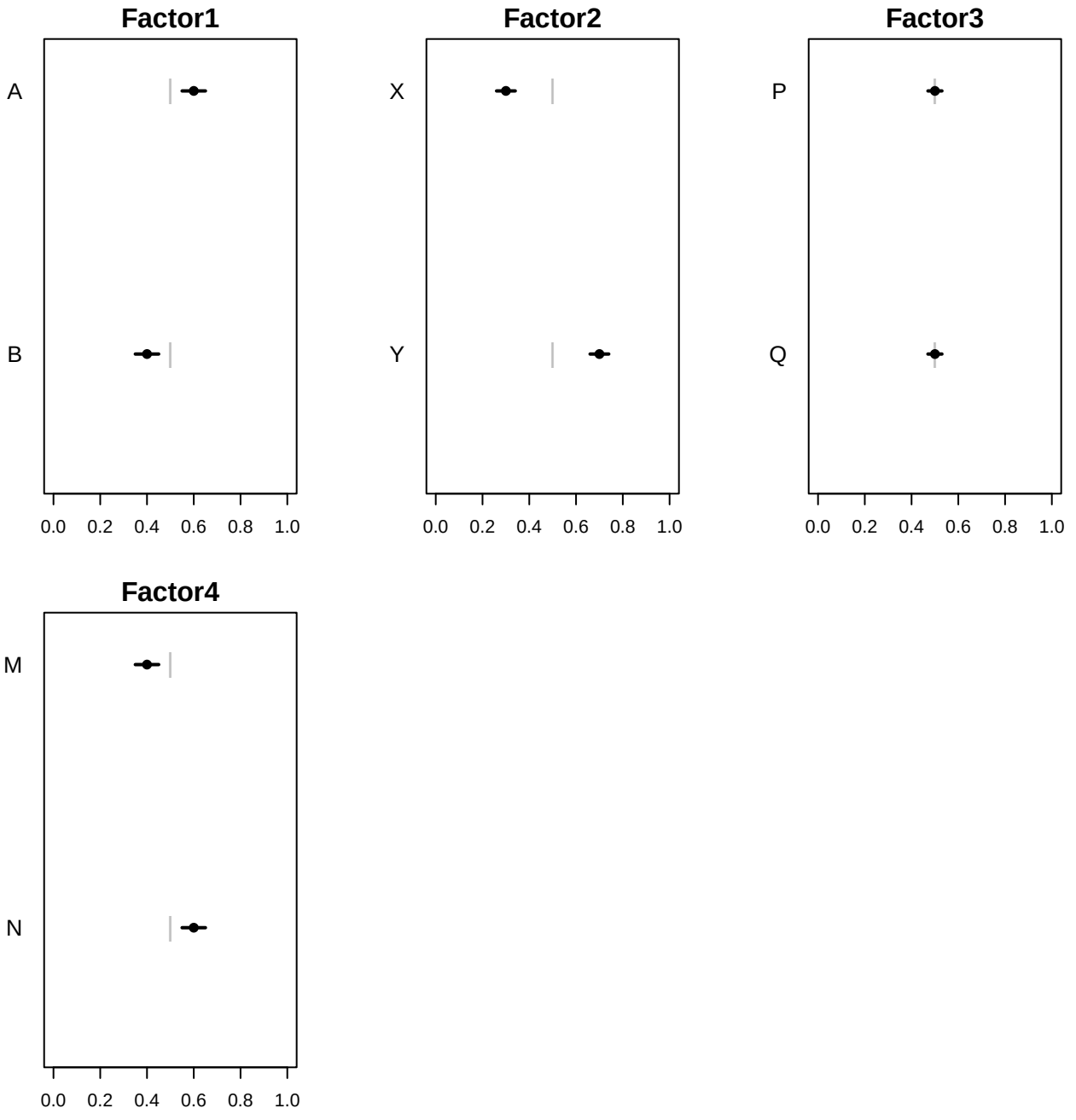
0.6
0.8
1.0


Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
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0.2
0.4
0.6
0.8
1.0
Factor2
X
Y
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0.2
0.4
0.6
0.8
1.0
Factor3
P
Q
0.0
0.2
0.4
0.6
0.8
1.0
Factor4
M
N
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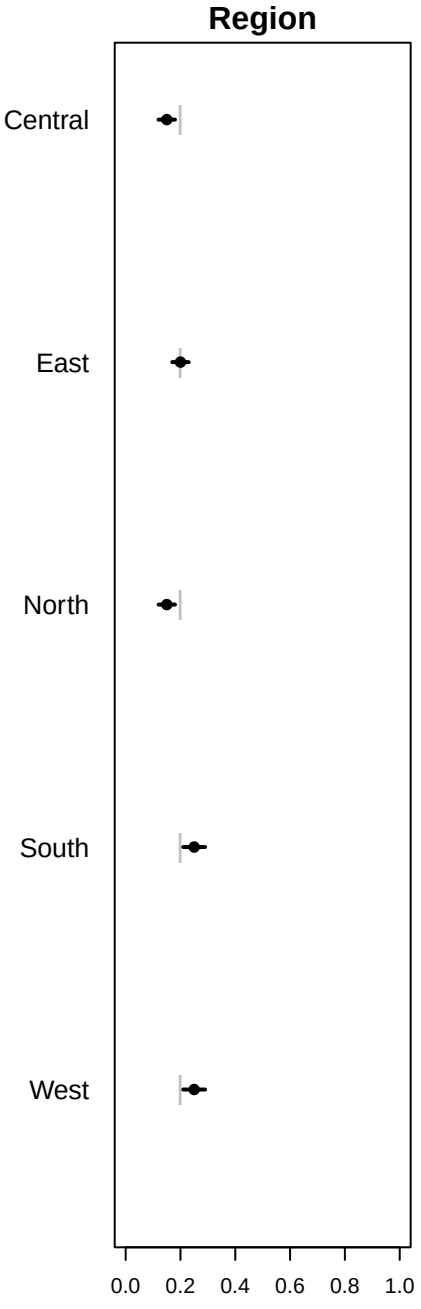
0.2
0.4
0.6
0.8
1.0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East
North
South
West
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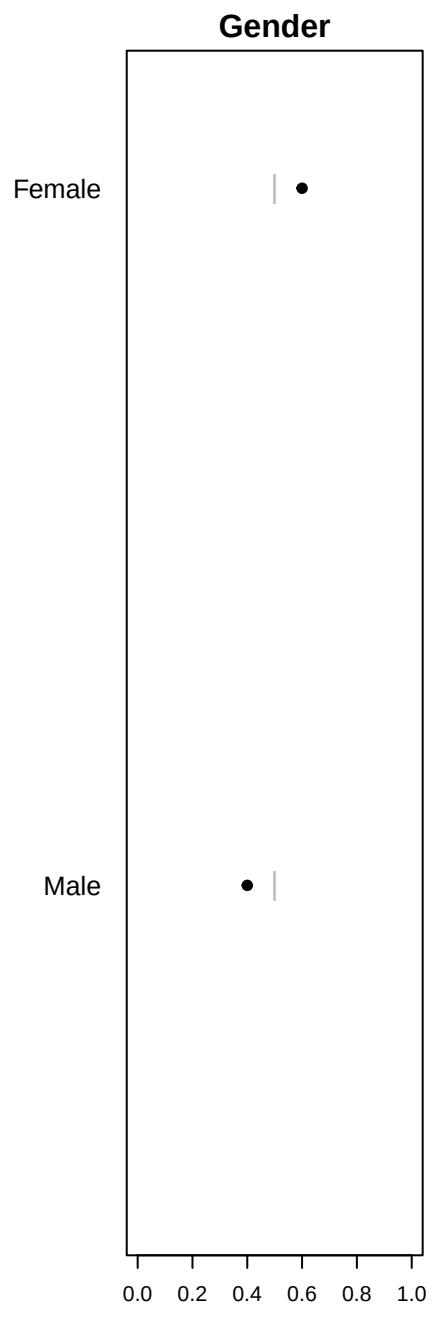
Gender
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Male
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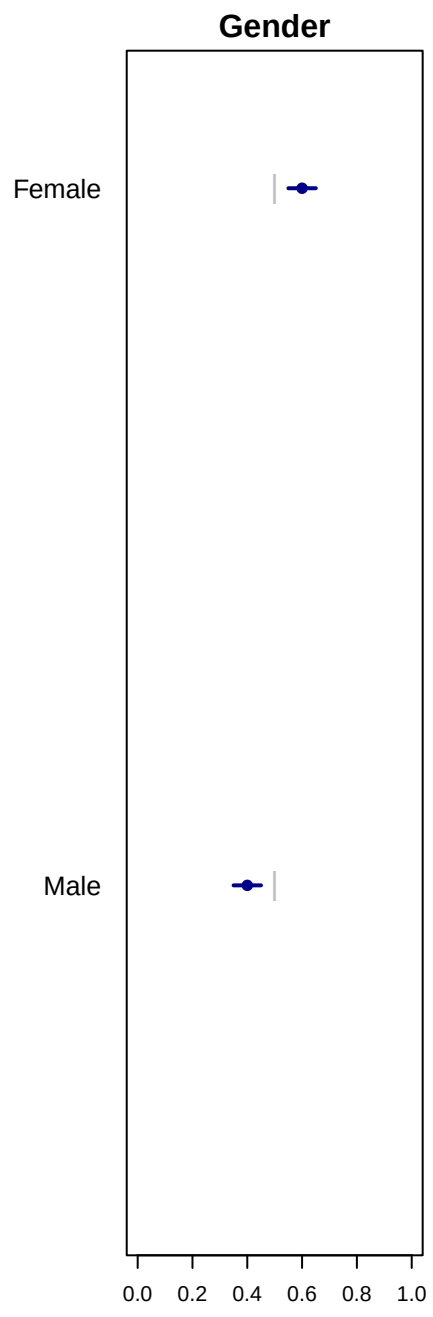
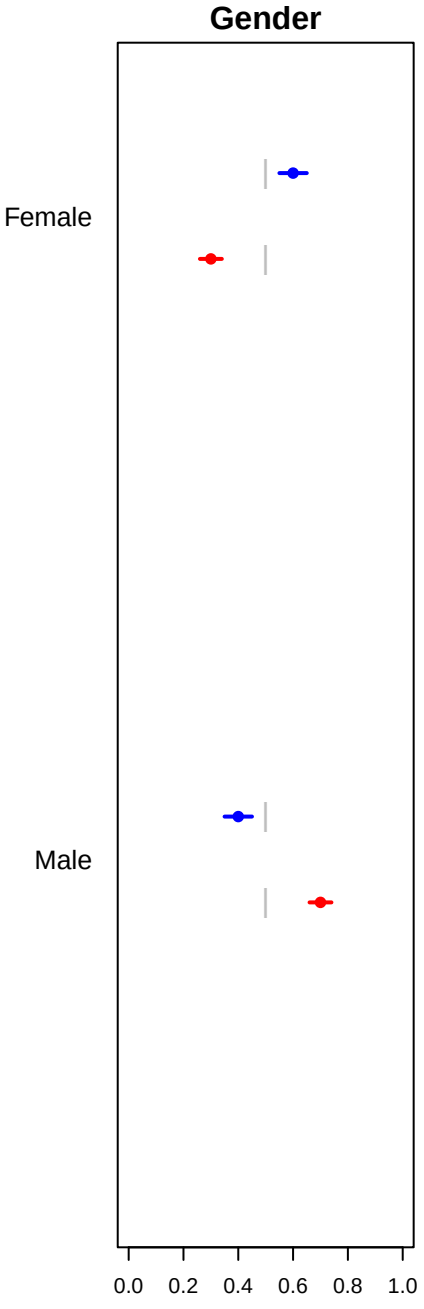
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
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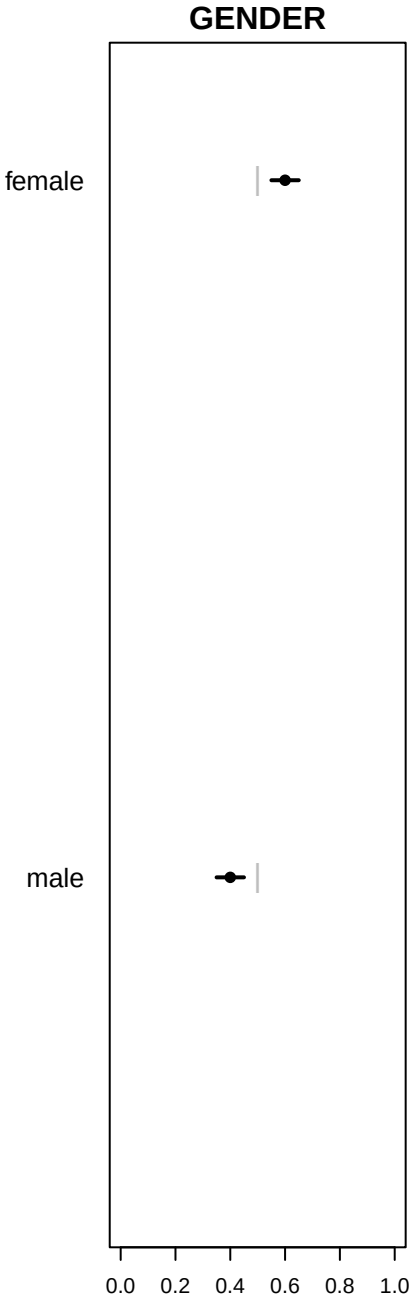
0.2
0.4
0.6
0.8
1.0


GENDER
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male
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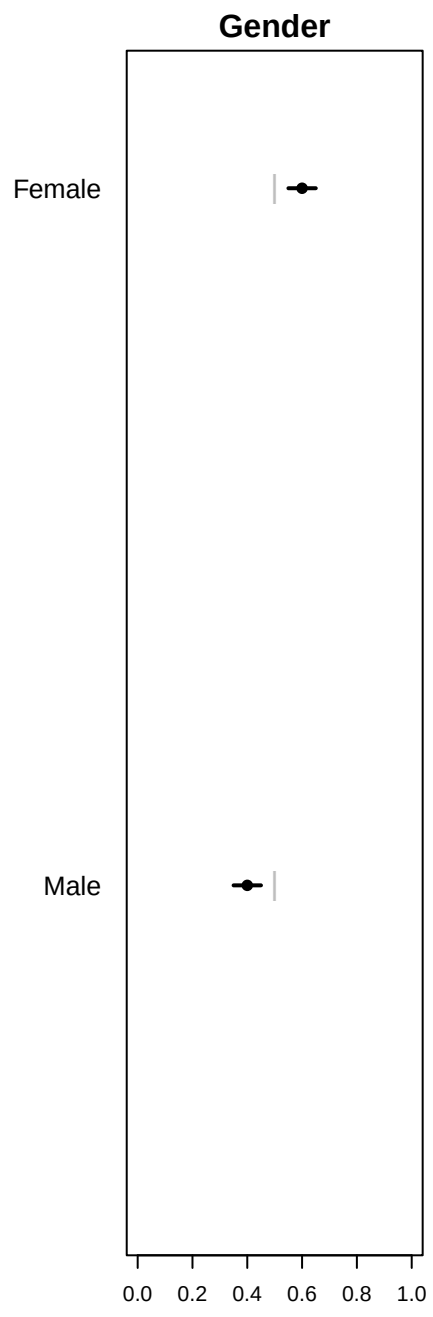
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0